Neurocritical Care Board Review: 250 Practice Questions with Explanations

250 single-best-answer questions written to the content of *The Practice of Neurocritical Care*, 2nd Edition (Neurocritical Care Society, 2021), organized by chapter. Answers with explanations follow each question.

Contents

Part I: General Critical Care

Airway, Mechanical Ventilation, and Respiratory Disease

Question 1

Airway, Mechanical Ventilation, and Respiratory Disease · Control of breathing / abnormal respiratory patterns · easy

A 68-year-old man with atrial fibrillation presents with vertigo, dysphagia, and hiccups. Blood pressure is 148/82 mmHg, heart rate 78, SpO_2 94% on room air. On examination he has right-sided Horner syndrome, right palatal weakness, and crossed sensory loss. Over the next several hours the nurse notes that his breathing has become completely irregular, with breaths of varying depth separated by unpredictable pauses, without any crescendo-decrescendo pattern. Which lesion location best accounts for this breathing pattern?

- **A.** Midbrain tectum
- **B.** Bilateral cerebral hemispheres
- **C.** Dorsolateral pons at the parabrachial nucleus
- **D.** Ventral medulla involving the pre-Botzinger complex
- **E.** Cervical spinal cord above C3

Answer: D. Ventral medulla involving the pre-Botzinger complex

Irregularly irregular breathing with random pauses and variable depth is ataxic breathing, which the chapter attributes to medullary lesions involving the pre-Botzinger complex of the ventral medulla, the region that generates respiratory rhythm. Bilateral hemispheric, diencephalic, or bilateral pontine lesions instead produce Cheyne-Stokes respirations, a crescendo-decrescendo pattern, so answer B is wrong. Pontine lesions produce central neurogenic hyperventilation and apneustic breathing, not ataxic breathing, so answer C is wrong. The midbrain does not participate in respiratory control, making answer A wrong. A high cervical cord lesion above C3-C5 causes inadequate ventilation from loss of phrenic output rather than an irregular central rhythm, so answer E is wrong.

Key point:
ataxic breathing localizes to the medulla and the pre-Botzinger complex, while Cheyne-Stokes localizes to bilateral hemispheric, diencephalic, or bilateral pontine injury.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 2

Question 2

Airway, Mechanical Ventilation, and Respiratory Disease · Neuromuscular respiratory failure: FVC and NIF thresholds · moderate

A 34-year-old woman weighing 60 kg is admitted with ascending weakness and areflexia five days after a diarrheal illness. She is now unable to lift her head off the pillow, has a nasal voice, and is pooling oral secretions. Blood pressure is 132/78 mmHg, heart rate 104, respiratory rate 26, SpO_2 96% on 2 L nasal cannula. Bedside testing shows a forced vital capacity of 0.9 L and a negative inspiratory force of -15 cmH_2O. Arterial blood gas shows pH 7.34, $PaCO_2$ 48 mmHg, PaO_2 82 mmHg. Which of the following is the most appropriate next step?

- **A.** Continue observation with forced vital capacity measurements every 4 hours
- **B.** Perform endotracheal intubation and initiate mechanical ventilation
- **C.** Start bilevel noninvasive positive pressure ventilation
- **D.** Start high-flow nasal cannula oxygen at 40 L/min
- **E.** Begin incentive spirometry and chest physiotherapy

Answer: B. Perform endotracheal intubation and initiate mechanical ventilation

Her forced vital capacity of 0.9 L is below 1 L and is only 15 mL/kg, below the 20 to 25 mL/kg threshold, and her negative inspiratory force of -15 cmH_2O is greater than -20 cmH_2O, so by the chapter's criteria she has significant respiratory muscle weakness. Noninvasive ventilation is an option in neuromuscular respiratory failure only in patients without excessive secretions or bulbar impairment, and she has both a nasal voice and pooling secretions, so answer C is wrong. High-flow nasal cannula and incentive spirometry with chest physiotherapy do not support ventilation in a patient who is already hypercapnic and cannot generate an effective cough, making answers D and E wrong. Continued observation is unsafe because hypoxia occurs late in neuromuscular failure, since shunting at the lung bases initially preserves oxygen saturation, so a normal SpO_2 is falsely reassuring and answer A is wrong. Neck flexor weakness, as demonstrated by her inability to lift her head, correlates with diaphragmatic weakness and is an additional sign of impending failure.

Key point:
forced vital capacity below 1 L or below 20 to 25 mL/kg, or a negative inspiratory force greater than -20 cmH_2O, signals significant respiratory weakness, and bulbar impairment or copious secretions rules out noninvasive ventilation.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 3

Question 3

Airway, Mechanical Ventilation, and Respiratory Disease · Induction agents for intubation · moderate

A 58-year-old man with a Hunt and Hess grade IV subarachnoid hemorrhage requires urgent intubation for airway protection. Blood pressure is 88/50 mmHg on low-dose norepinephrine, heart rate 96, SpO_2 93% on a nonrebreather. His history includes a myocardial infarction with drug-eluting stent placement two months ago and a documented anaphylactic reaction to eggs and soy. Which induction agent is most appropriate?

- **A.** Propofol 1.5 mg/kg intravenously
- **B.** Midazolam 0.2 mg/kg intravenously
- **C.** Etomidate 0.3 mg/kg intravenously
- **D.** Ketamine 2 mg/kg intravenously
- **E.** Fentanyl 3 mcg/kg intravenously as the sole induction agent

Answer: C. Etomidate 0.3 mg/kg intravenously

Etomidate at 0.3 mg/kg intravenously confers hemodynamic stability while decreasing cerebral blood flow and cerebral metabolic rate, which is exactly what is needed in a hypotensive patient with an unsecured aneurysm. Propofol at 1.0 to 2.5 mg/kg causes hypotension and bradycardia and is contraindicated in egg or soy allergy, so answer A is wrong. Midazolam at 0.2 to 0.3 mg/kg also causes hypotension and has a long duration of action of 2 to 6 hours, which obscures serial neurologic examination, so answer B is wrong. Ketamine at 1.5 to 2.0 mg/kg is a reasonable choice in hemodynamically unstable patients but must be used with caution in acute coronary syndrome and ischemic heart disease, which this patient has, so answer D is wrong. Fentanyl at 1 to 3 mcg/kg is a premedication used to blunt the sympathetic response and can itself cause hypotension; it is not an induction agent, so answer E is wrong. The main tradeoff with etomidate is adrenal insufficiency, myoclonus, and a lowered seizure threshold.

Key point:
etomidate 0.3 mg/kg is the induction agent of choice when hemodynamic stability matters, and propofol is absolutely avoided in egg or soy allergy.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 7

Question 4

Airway, Mechanical Ventilation, and Respiratory Disease · Neuromuscular blocking agents · moderate

A 71-year-old woman with acetylcholine receptor antibody positive myasthenia gravis presents in crisis after a urinary tract infection. She has ptosis, dysarthria, and paradoxical abdominal breathing. Blood pressure is 138/80 mmHg, heart rate 110, respiratory rate 32, SpO2 91% on 6 L nasal cannula. Forced vital capacity is 0.7 L and PaCO2 is 56 mmHg. She last ate two hours ago, and rapid sequence intubation is planned. Which neuromuscular blocking agent is most appropriate?

- **A.** Succinylcholine 1.5 mg/kg intravenously
- **B.** Succinylcholine 4 mg/kg intramuscularly
- **C.** Cisatracurium 0.15 mg/kg intravenously
- **D.** No paralytic; induce with propofol alone
- **E.** Rocuronium 1 mg/kg intravenously

Answer: E. Rocuronium 1 mg/kg intravenously

Rocuronium 1 mg/kg intravenously is the paralytic the chapter designates as ideal for patients with contraindications to succinylcholine, and its onset of 1 to 2 minutes is fast enough for rapid sequence intubation. Succinylcholine is explicitly to be avoided in myasthenia gravis and additionally carries risks of hyperkalemia, malignant hyperthermia, rhabdomyolysis, and increased intracranial and intraocular pressure, so answers A and B are wrong; the intramuscular route also has a delayed onset of 3 to 4 minutes, which is unsuitable here. Cisatracurium 0.15 to 0.2 mg/kg has a slower onset of 2 to 3 minutes and a duration of 20 to 35 minutes, making it a poorer fit for rapid sequence intubation in a patient at risk of aspirating recently ingested stomach contents, so answer C is wrong. Omitting the paralytic forfeits the aspiration protection that rapid sequence intubation is designed to provide, so answer D is wrong.

Key point:
in myasthenia gravis, avoid succinylcholine and use rocuronium 1 mg/kg intravenously for rapid sequence intubation.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 8

Question 5

Airway, Mechanical Ventilation, and Respiratory Disease · Hyperventilation, cerebral blood flow, and ICP · moderate

A 28-year-old man with severe traumatic brain injury has an intracranial pressure of 32 mmHg despite sedation, hyperosmolar therapy, and cerebrospinal fluid drainage. The respiratory rate is increased on the ventilator, and over the next several hours his $PaCO_2$ falls to 25 mmHg. His intracranial pressure decreases to 18 mmHg and his cerebral perfusion pressure improves from 58 to 72 mmHg, and the team plans to maintain these settings overnight. Which of the following best describes the principal risk of this strategy?

A. Hypocarbia causes cerebral vasoconstriction, increasing the volume of hypoperfused brain and decreasing total cerebral blood flow despite the improved intracranial pressure and cerebral perfusion pressure numbers

B. Hypocarbia causes cerebral vasodilation, increasing cerebral blood volume and producing rebound intracranial hypertension

C. The resulting respiratory acidosis directly worsens cytotoxic cerebral edema

D. The increased respiratory rate raises intrathoracic pressure, impairing jugular venous drainage and raising intracranial pressure

E. Sustained hypocarbia causes barotrauma and pneumothorax before it affects cerebral perfusion

Answer: A. Hypocarbia causes cerebral vasoconstriction, increasing the volume of hypoperfused brain and decreasing total cerebral blood flow despite the improved intracranial pressure and cerebral perfusion pressure numbers

Hypocarbia produces cerebral vasoconstriction, and traumatic brain injury patients who underwent hyperventilation had increased volumes of hypoperfused brain and a total decrease in cerebral blood flow even though their cerebral perfusion pressure and intracranial pressure measurements improved. For this reason hyperventilation should be avoided in patients with brain ischemia from any cause except as a brief rescue therapy for emergent intracranial hypertension, and it should be maintained only for a short period. Answer B inverts the physiology, since it is hypercarbia, not hypocarbia, that produces cerebral vasodilation. Hyperventilation produces respiratory alkalosis rather than acidosis, so answer C is wrong. Elevated intrathoracic pressure that impairs jugular venous drainage is a consequence of high positive end-expiratory pressure, not of an increased respiratory rate, so answer D is wrong. Barotrauma is not the dominant concern at these settings and does not precede the cerebral effects, so answer E is wrong.

Key point:
hyperventilation lowers intracranial pressure at the cost of cerebral blood flow and is a short-term rescue therapy only, never a maintenance strategy.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 8, 12

Question 6

Airway, Mechanical Ventilation, and Respiratory Disease · Readiness for a spontaneous breathing trial · hard

It is morning rounds in the neurocritical care unit and the team is selecting patients for a spontaneous breathing trial. Which of the following intubated patients meets criteria for a trial?

- **A.** A 24-year-old with diffuse axonal injury whose intracranial pressure is 26 mmHg on hypertonic saline, on PEEP of 5 cmH2O and FiO2 of 0.4
- **B.** A 61-year-old three days after coiling of a ruptured anterior communicating artery aneurysm, intracranial pressure 12 mmHg, RASS -1, SpO2 96% and PaO2 88 mmHg on FiO2 of 0.4, PEEP of 5 cmH2O, PaCO2 40 mmHg, heart rate 88, mean arterial pressure 78 mmHg off vasopressors, who opens his eyes but does not follow commands
- **C.** A 45-year-old in myasthenic crisis with a negative inspiratory force of -12 cmH2O on PEEP of 5 cmH2O and FiO2 of 0.4
- **D.** A 58-year-old with acute respiratory distress syndrome on PEEP of 12 cmH2O and FiO2 of 0.6
- **E.** A 70-year-old with septic shock and a mean arterial pressure of 55 mmHg on escalating norepinephrine

Answer: B. A 61-year-old three days after coiling of a ruptured anterior communicating artery aneurysm, intracranial pressure 12 mmHg, RASS -1, SpO2 96% and PaO2 88 mmHg on FiO2 of 0.4, PEEP of 5 cmH2O, PaCO2 40 mmHg, heart rate 88, mean arterial pressure 78 mmHg off vasopressors, who opens his eyes but does not follow commands

Readiness for a spontaneous breathing trial requires controlled intracranial pressure below 20 mmHg and a RASS of 0 to -2 if sedatives are being used, SpO2 above 90 percent with PaO2 above 60 mmHg on FiO2 of 0.4 or less, a normal or baseline PaCO2, PEEP of 8 cmH2O or less, respiratory rate below 35 breaths per minute, tidal volume above 5 mL/kg, and heart rate below 140 beats per minute with a mean arterial pressure above 60 mmHg on minimal or no vasopressors; the patient in option B satisfies every one of these, and inability to follow commands is not an exclusion. The patient in option A fails the neurologic criterion because the intracranial pressure of 26 mmHg exceeds 20 mmHg. The patient in option C fails the respiratory criteria because the negative inspiratory force must be at least -20 cmH2O in magnitude, that is more negative than -20, particularly when neuromuscular disease is present, and -12 does not meet that cutoff. The patient in option D fails on both PEEP above 8 cmH2O and FiO2 above 0.4. The patient in option E fails the cardiovascular criteria with a mean arterial pressure below 60 mmHg on rising vasopressor doses. Coma by itself is not a reason to delay weaning.

Key point:
the spontaneous breathing trial gates are intracranial pressure below 20 mmHg, PEEP of 8 or less, FiO2 of 0.4 or less, negative inspiratory force of at least -20 cmH2O in magnitude, and mean arterial pressure above 60 mmHg on minimal pressors, and failure to follow commands is not a disqualifier.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 10

Question 7

Airway, Mechanical Ventilation, and Respiratory Disease · Extubation decision in brain-injured patients · moderate

A 66-year-old man is on hospital day 6 after a left basal ganglia intracerebral hemorrhage. He has passed a 30-minute spontaneous breathing trial on pressure support with a rapid shallow breathing index of 62 breaths per minute per liter. He is on PEEP of 5 cmH2O and FiO2 of 0.3, he has a strong cough with suctioning, minimal secretions, and an audible cuff leak. He opens his eyes spontaneously and tracks the examiner across the room, but he is aphasic and does not follow commands. Which of the following is the most appropriate next step?

- **A.** Perform percutaneous dilatational tracheostomy within 24 hours
- **B.** Continue mechanical ventilation and repeat the spontaneous breathing trial daily until he follows commands
- **C.** Give intravenous methylprednisolone for 24 hours and then reassess for extubation
- **D.** Extubate to supplemental oxygen with close monitoring
- **E.** Extubate directly to bilevel noninvasive positive pressure ventilation

Answer: D. Extubate to supplemental oxygen with close monitoring

Inability to follow commands should not preclude a trial of extubation, and he has only one of the three predictors of extubation failure, which are weak cough, secretions, and inability to follow commands; two of those three confer a likelihood ratio of only 3.8 for failure. Patients with Glasgow Coma Scale scores as low as 3 and 4 have been extubated safely, and spontaneous eye opening and visual pursuit have been associated with extubation success in multivariate analysis. Delaying extubation is not benign, because it exposes him to ventilator-associated pneumonia and increased intensive care unit length of stay, and extubation is the single most effective intervention to prevent ventilator-associated pneumonia, so answer B is wrong. Tracheostomy is indicated only in patients who cannot be safely extubated or who have failed extubation, and he has not yet had a trial, so answer A is wrong. Prophylactic corticosteroids given 4 hours before extubation are for patients at high risk of post-extubation laryngeal edema, typically those without a cuff leak, and he has a cuff leak, so answer C is wrong. Noninvasive ventilation should be used with caution in patients with a decreased or altered level of consciousness, so answer E is wrong.

Key point:
not following commands is not a reason to withhold a trial of extubation, and only the combination of two of weak cough, secretions, and inability to follow commands raises the risk of failure meaningfully.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 11

Question 8

Airway, Mechanical Ventilation, and Respiratory Disease · Tracheostomy timing and technique · moderate

A 55-year-old woman with a large right middle cerebral artery infarct has been intubated for airway protection for three days and remains unable to protect her airway. The family asks whether performing a tracheostomy now rather than in the second week would improve her chances of recovery. Which statement best reflects the evidence described for tracheostomy in this population?

- **A.** Early tracheostomy reduced mortality and improved functional outcome compared with late tracheostomy in ventilated stroke patients
- **B.** Early tracheostomy in ventilated stroke patients shortened intensive care unit stay but significantly increased bleeding complications
- **C.** A randomized trial of early tracheostomy up to day 3 versus late tracheostomy on days 7 to 14 in ventilated stroke patients did not improve outcomes, but early tracheostomy was safe and feasible and may decrease sedation needs
- **D.** Surgical tracheostomy has a lower overall complication rate, about 3 percent, than percutaneous dilatational tracheostomy, about 12 percent
- **E.** Neurocritically ill patients wean from the ventilator more slowly after tracheostomy than general intensive care unit patients

Answer: C. A randomized trial of early tracheostomy up to day 3 versus late tracheostomy on days 7 to 14 in ventilated stroke patients did not improve outcomes, but early tracheostomy was safe and feasible and may decrease sedation needs

The randomized trial of early tracheostomy, performed up to day 3 from intubation, versus late tracheostomy on days 7 to 14 in patients ventilated for ischemic stroke, intracerebral hemorrhage, or subarachnoid hemorrhage did not demonstrate improved outcomes with early tracheostomy, but it did show that early tracheostomy is safe and feasible and may decrease the need for sedation, which makes answers A and B wrong. The complication rates are the reverse of answer D: percutaneous dilatational tracheostomy has an overall complication rate of about 3 percent versus about 12 percent for surgical tracheostomy, with lower rates of bleeding and infection when performed with bronchoscopy, and it can be done at the bedside. Neurocritically ill patients actually wean faster after tracheostomy than general intensive care unit patients, because they were tracheostomized for airway protection rather than primary lung pathology, so answer E is wrong. A separate meta-analysis in severe traumatic brain injury found early tracheostomy associated with shorter mechanical ventilation, shorter intensive care unit and hospital stay, and less ventilator-associated pneumonia, but no mortality benefit.

Key point:
early tracheostomy in ventilated stroke is safe and feasible and may reduce sedation, but it does not improve outcomes, and the percutaneous technique has the lower complication rate.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 11

Question 9

Airway, Mechanical Ventilation, and Respiratory Disease · Post-extubation laryngeal edema · easy

A 63-year-old woman is extubated on day 8 after a traumatic subdural hematoma evacuation. No cuff leak had been documented before extubation. Ten minutes after extubation she develops a high-pitched inspiratory sound audible across the room, suprasternal retractions, and rising work of breathing. Blood pressure is 156/88 mmHg, heart rate 118, respiratory rate 30, SpO_2 91 percent on a nonrebreather mask. Lung fields are clear without wheezing. Which of the following is the most appropriate next step?

- **A.** Administer nebulized epinephrine and intravenous methylprednisolone
- **B.** Administer nebulized albuterol 2.5 mg every 20 minutes
- **C.** Proceed directly to percutaneous dilatational tracheostomy
- **D.** Continue supplemental oxygen by nonrebreather mask alone and observe
- **E.** Administer magnesium 2 g intravenously over 20 minutes

Answer: A. Administer nebulized epinephrine and intravenous methylprednisolone

Stridor with retractions minutes after extubation in a patient who had no cuff leak is post-extubation laryngeal edema, which causes 15 percent of all reintubations; most patients develop symptoms within 30 minutes and almost half within 5 minutes of extubation. Treatment is intravenous methylprednisolone or dexamethasone for 24 to 48 hours after extubation to decrease the inflammatory response, combined with nebulized epinephrine, which helps through vasoconstriction, although optimal dosing of both is unclear. Because this is upper airway obstruction, simply adding supplemental oxygen will not effectively manage the patient and answer D is wrong; the cause must be treated or the airway secured. Albuterol and intravenous magnesium treat lower airway bronchospasm in acute severe asthma and do nothing for glottic edema, and her lungs are clear without wheezing, so answers B and E are wrong. Tracheostomy is not the immediate maneuver for acute post-extubation obstruction, since reintubation is the rescue if medical therapy fails, so answer C is wrong. Note that the presence of a cuff leak does not guarantee that a patient will not develop obstruction from post-extubation laryngeal edema.

Key point:
post-extubation stridor is treated with corticosteroids plus nebulized epinephrine, and steroids given 4 hours before planned extubation prevent it in high-risk patients.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 12

Question 10

Airway, Mechanical Ventilation, and Respiratory Disease · COPD exacerbation management · easy

A 70-year-old man with severe chronic obstructive pulmonary disease is admitted to the neurocritical care unit after a small thalamic intracerebral hemorrhage. On day 2 he becomes dyspneic and wheezy with increased purulent sputum. He is alert and protecting his airway. Blood pressure is 142/84 mmHg, heart rate 104, respiratory rate 26, SpO_2 84 percent on room air. He is started on nebulized albuterol and ipratropium, prednisone 40 mg orally daily, and antibiotics. To what oxygen saturation range should supplemental oxygen be titrated?

- **A.** 80 to 84 percent
- **B.** 84 to 88 percent
- **C.** 88 to 92 percent
- **D.** 92 to 96 percent
- **E.** 96 to 100 percent

Answer: C. 88 to 92 percent

In hypoxic patients with a chronic obstructive pulmonary disease exacerbation, supplemental oxygen should be given to maintain oxygen saturations between 88 and 92 percent, so answer C is correct and the lower ranges in answers A and B leave him hypoxemic. Targeting 92 to 96 or 96 to 100 percent, as in answers D and E, delivers more oxygen than is needed and runs counter to the general ventilatory principle of oxygenating with the least amount of oxygen required while avoiding oxygen toxicity. The rest of his regimen is appropriate: short-acting bronchodilators with an added short-acting anticholinergic, and systemic glucocorticoids and antibiotics, which shorten the exacerbation, decrease the risk of relapse, and improve lung function and oxygen saturation. Prednisone 40 mg daily for 5 days is commonly recommended, and a course of 7 days or fewer is as effective as a longer course without increasing treatment failure or relapse. Antibiotics are appropriate for patients requiring intubation or with worsening dyspnea and increased or purulent sputum, which he has.

Key point:
in a chronic obstructive pulmonary disease exacerbation, target an oxygen saturation of 88 to 92 percent and give prednisone 40 mg daily for 5 days.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 13-14

Question 11

Airway, Mechanical Ventilation, and Respiratory Disease · ARDS in the brain-injured patient · hard

A 44-year-old man, 70 kg ideal body weight, with severe traumatic brain injury and diffuse cerebral edema has a parenchymal intracranial pressure monitor reading 16 mmHg. Three days after admission he develops fever and hypoxemia. Chest radiograph shows new bilateral opacities, echocardiography shows normal biventricular function without volume overload, and on FiO2 of 0.8 with PEEP of 10 cmH2O his PaO2 is 68 mmHg with a PaCO2 of 41 mmHg. He is currently ventilated with tidal volumes of 550 mL and a plateau pressure of 32 cmH2O. Which of the following is the most appropriate ventilator strategy?

- **A.** Increase the tidal volume to 10 mL/kg of ideal body weight to prevent any rise in PaCO2
- **B.** Start high-frequency oscillatory ventilation, which has been shown to reduce mortality in this setting
- **C.** Initiate venovenous extracorporeal membrane oxygenation as first-line rescue therapy
- **D.** Reduce the tidal volume to 6 mL/kg of ideal body weight and accept permissive hypercapnia with a PaCO2 of 60 to 70 mmHg
- **E.** Reduce the tidal volume to 6 mL/kg of ideal body weight with a plateau pressure of 30 cmH2O or less, modifying the protocol by increasing the respiratory rate to avoid hypercarbia, while monitoring intracranial pressure

Answer: E. Reduce the tidal volume to 6 mL/kg of ideal body weight with a plateau pressure of 30 cmH2O or less, modifying the protocol by increasing the respiratory rate to avoid hypercarbia, while monitoring intracranial pressure

He meets the Berlin definition of acute respiratory distress syndrome, with onset within one week, bilateral opacities not explained by effusion, collapse, or nodules, no heart failure or fluid overload, and a PaO2 to FiO2 ratio of 85 mmHg on PEEP of at least 5 cmH2O, which is the severe category of 100 mmHg or less. The mainstay of treatment is lung-protective ventilation with tidal volumes of 6 mL/kg of ideal body weight and plateau pressures of 30 cmH2O or less, with PEEP titrated to an oxyhemoglobin saturation of 88 to 95 percent or a PaO2 of 55 to 80 mmHg, so his current 550 mL and plateau pressure of 32 cmH2O must come down. Because lung-protective ventilation can produce hypercarbia that is detrimental in a patient with cerebral edema, a modified acute respiratory distress syndrome protocol should be used to avoid hypercarbia, and an intracranial pressure monitor can be used when hypercarbia is unavoidable, which makes answer E correct and answer D wrong. Answer A abandons lung protection entirely. High-frequency oscillatory ventilation improves oxygenation when implemented early but has not been shown to improve mortality, so answer B overstates the evidence. Extracorporeal membrane oxygenation is a rescue therapy for patients who have failed or are not candidates for proning and high PEEP strategies, and severe neurologic injury is an absolute contraindication, so answer C is wrong.

Key point:

keep lung-protective ventilation at 6 mL/kg with plateau pressure 30 cmH2O or less in acute respiratory distress syndrome, but modify the protocol to avoid hypercarbia when the brain is injured.

Source: Practice of Neurocritical Care, 2nd ed., Airway, Mechanical Ventilation, and Respiratory Disease, pp. 15-16

Question 12

Acid-Base and Electrolyte Disorders · Hyperglycemic crises: HHS versus DKA · easy

A 58-year-old man with type 2 diabetes is brought to the emergency department after several days of increasing confusion and polyuria. He is obtunded and does not follow commands. Temperature 38.1 C, heart rate 118/min, blood pressure 96/58 mmHg. Fingerstick glucose is 950 mg/dL. Laboratory studies show sodium 130 mEq/L, potassium 5.3 mEq/L, chloride 89 mEq/L, bicarbonate 17 mmol/L, serum osmolality 355 mOsm/kg, and beta-hydroxybutyrate 50.7 mg/dL. Arterial blood gas shows pH 7.41, PCO2 27 mmHg, base excess -6.6 mmol/L. Noncontrast head CT is normal. Which of the following is the most likely diagnosis?

- **A.** Diabetic ketoacidosis
- **B.** Alcoholic ketoacidosis
- **C.** Hyperosmolar hyperglycemic syndrome with minor ketoacidotic features
- **D.** Lactic acidosis from septic shock
- **E.** Salicylate intoxication

Answer: C. Hyperosmolar hyperglycemic syndrome with minor ketoacidotic features

The defining features of hyperosmolar hyperglycemic syndrome (HHS) are a normal pH, severe hyperglycemia usually above 600 mg/dL and often near 1,000 mg/dL, and an osmolality typically greater than 320 mOsm/kg, all of which this patient has (pH 7.41, glucose 950 mg/dL, osmolality 355 mOsm/kg). Coma and severe encephalopathy are very common in HHS, and although HHS occurs less frequently than diabetic ketoacidosis it is more morbid and carries a higher mortality rate. Over 30% of patients have features of both DKA and HHS, and by the American Diabetes Association Consensus Statement this patient falls on that spectrum as HHS with a minor DKA feature (the mildly positive ketones and low bicarbonate). Pure DKA is wrong because DKA is defined by an acidemia, and this patient's pH is normal at 7.41. Alcoholic and starvation ketosis produce ketosis without this degree of hyperglycemia or hyperosmolality. Lactic acidosis and salicylism are other causes of metabolic acidosis on the differential but do not explain a glucose of 950 mg/dL with an osmolality of 355 mOsm/kg and a normal pH.

Key point:
a normal pH with glucose near 1,000 mg/dL and osmolality above 320 mOsm/kg is hyperosmolar hyperglycemic syndrome, not DKA.

Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 28-29

Question 13

Acid-Base and Electrolyte Disorders · Anion gap and delta/delta interpretation · hard

A 42-year-old man is on hospital day 3 after severe traumatic brain injury from a motor vehicle collision. He remains on mechanical ventilation, a continuous hypertonic saline infusion, pentobarbital, and two vasopressors. Today's arterial blood gas shows pH 7.20 and PCO2 37 mmHg with bicarbonate 14 mEq/L. The basic metabolic profile shows sodium 152 mEq/L, chloride 122 mEq/L, and total CO2 14 mEq/L. Which of the following best characterizes his acid-base disorder?

- **A.** Isolated hyperchloremic (normal anion gap) metabolic acidosis
- **B.** Anion gap metabolic acidosis with a coexisting hyperchloremic metabolic acidosis
- **C.** Anion gap metabolic acidosis with a superimposed metabolic alkalosis
- **D.** Respiratory acidosis with an appropriate metabolic compensation
- **E.** Primary respiratory alkalosis with a compensatory metabolic acidosis

Answer: B. Anion gap metabolic acidosis with a coexisting hyperchloremic metabolic acidosis

The pH of 7.20 indicates acidemia, and because the PCO2 is within the normal range of 35 to 45 mmHg the primary disorder must be a metabolic acidosis rather than a respiratory one. The anion gap is calculated as sodium minus the sum of chloride and bicarbonate, which is 152 - (122 + 14) = 16, an elevated gap. The delta/delta ratio is the change in anion gap divided by the change in bicarbonate, here (16 - 12) divided by (24 - 14), which equals 0.4. A delta/delta less than 1 indicates an underlying normal anion gap (hyperchloremic) metabolic acidosis in addition to the primary anion gap acidosis, which fits a patient receiving large volumes of chloride-rich hypertonic saline. Option C is wrong because a superimposed metabolic alkalosis would require a delta/delta greater than 2. Options D and E are excluded because the PCO2 is normal, so there is no primary respiratory process.

Key point:
an elevated anion gap with a delta/delta below 1 means a second, hyperchloremic non-gap acidosis is also present.

Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 31, 41

Question 14

Acid-Base and Electrolyte Disorders · Sodium bicarbonate in metabolic and lactic acidosis · moderate

A 66-year-old woman with pneumococcal meningitis and septic shock is intubated in the neurocritical care unit. She is on norepinephrine at an escalating dose. Arterial blood gas shows pH 7.15, PCO2 28 mmHg, bicarbonate 10 mEq/L, and serum lactate is 7.2 mmol/L. Creatinine is 1.2 mg/dL and there is no hyperkalemia. The bedside nurse asks whether bicarbonate should be given to improve her pressor responsiveness. Which of the following is the most appropriate next step?

- **A.** Continue targeted antimicrobial and hemodynamic therapy while optimizing brain oxygenation and tissue perfusion
- **B.** Give sodium bicarbonate 50 mEq intravenous push
- **C.** Start a sodium bicarbonate infusion titrated to a pH above 7.30
- **D.** Arrange emergent hemodialysis for acid clearance
- **E.** Increase the ventilator rate to drive the PCO2 below 25 mmHg

> **Answer: A.** Continue targeted antimicrobial and hemodynamic therapy while optimizing brain oxygenation and tissue perfusion
>
> In metabolic acid-base disorders the top priorities are reestablishing normal brain oxygenation and perfusion and treating the underlying cause promptly, which here means antimicrobials for sepsis and hemodynamic support. Infusion of sodium bicarbonate is not associated with an improved response to adrenergic agonists or with clinical improvement, most evidence shows it inconsistently produces only a modest rise in pH, and it has no role in lactic acidosis, so options B and C are wrong. Empiric bicarbonate has also been shown to have no impact on 28-day mortality in critically ill patients presenting with metabolic acidosis, and it carries the risks of volume overload, hypernatremia, impaired cardiac contractility, hyperosmolarity, and CO2 generation that can worsen respiratory acidosis. Dialysis is the treatment for acidosis from renal failure, and her creatinine is essentially normal, so option D is not indicated. Manipulating ventilator settings is not the appropriate initial step in managing a primary metabolic disorder, and driving the PCO2 below 25 mmHg risks ischemic hypoxia in a brain-injured patient.
>
> **Key point:**
> treat the cause and the perfusion, not the number, because bicarbonate does not improve hemodynamics, lactic acidosis, or 28-day mortality.
>
> Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 32, 41

Question 15

Acid-Base and Electrolyte Disorders · Drug-induced normal anion gap metabolic acidosis · easy

Which of the following antiseizure medications is most likely to produce a normal anion gap (hyperchloremic) metabolic acidosis through inhibition of carbonic anhydrase?

- **A.** Lamotrigine
- **B.** Carbamazepine
- **C.** Phenytoin
- **D.** Topiramate
- **E.** Levetiracetam

Answer: D. Topiramate

Topiramate is a carbonic anhydrase inhibitor that increases renal elimination of bicarbonate, and the resulting bicarbonate loss produces a metabolic acidosis without an elevated anion gap. The chapter lists carbonic anhydrase inhibitors, specifically acetazolamide, topiramate, and zonisamide, among the medication causes of non-anion gap metabolic acidosis, alongside amphotericin B. Lamotrigine, carbamazepine, phenytoin, and levetiracetam are not carbonic anhydrase inhibitors and are not listed as causes of a hyperchloremic acidosis. The other listed non-gap causes are excessive gastrointestinal losses from pancreatic or enteric fistula and diarrhea, and distal (type 1) and proximal (type 2) renal tubular acidosis. Anion gap causes, by contrast, include methanol, uremic renal failure, diabetic ketoacidosis, paraldehyde, isoniazid, lactic acidosis, ethylene glycol, and salicylates.

Key point:
topiramate, zonisamide, and acetazolamide inhibit carbonic anhydrase and cause a normal anion gap metabolic acidosis.

Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 32, 41

Question 16

Acid-Base and Electrolyte Disorders · Severe metabolic alkalemia management · moderate

A 71-year-old woman with a cerebellar hemorrhage has had high-output nasogastric suction and recurrent emesis for 4 days. She is euvolemic after 48 hours of volume repletion with normal saline plus aggressive potassium chloride and has no heart or liver failure. She becomes lethargic with circumoral paresthesias and carpopedal spasms. Arterial blood gas shows pH 7.64, PCO_2 52 mmHg, bicarbonate 54 mEq/L; potassium is 3.1 mEq/L and ionized calcium is low. Telemetry shows frequent premature ventricular complexes. Which of the following is the most appropriate next step?

- **A.** Continue normal saline with additional potassium chloride repletion alone
- **B.** Start acetazolamide to induce bicarbonate diuresis
- **C.** Deepen sedation to reduce minute ventilation and allow the PCO_2 to rise further
- **D.** Administer sodium bicarbonate and begin enteral free water
- **E.** Infuse dilute hydrochloric acid 0.10 to 0.15 N through a central venous catheter until the pH falls below 7.6

Answer: E. Infuse dilute hydrochloric acid 0.10 to 0.15 N through a central venous catheter until the pH falls below 7.6

Severe alkalemia is defined as a pH greater than 7.6 and produces exactly this picture: intracellular shift and renal wasting of potassium with low serum potassium, ionized calcium, and phosphate, causing circumoral paresthesias, carpopedal spasms, muscle cramps, obtundation, and, in the most severe cases, respiratory arrest, together with negative inotropism and an increased risk of supraventricular and ventricular arrhythmias. For severe alkalemia the chapter states that a diluted hydrochloric acid solution of 0.10 to 0.15 N can be given through a central line until a pH below 7.6 is achieved. Options A is insufficient because she has already received volume, chloride, and potassium repletion and remains at a pH of 7.64 with cardiac irritability. Acetazolamide (or potassium chloride) is the option reserved for patients who are hypervolemic or who have congestive heart failure or liver failure, and she is euvolemic with neither. Deepening sedation to permit hypoventilation does not address the primary metabolic alkalosis, and giving more base as in option D would worsen it.

Key point:
a pH above 7.6 is severe alkalemia, and when saline, potassium, and chloride repletion have failed, dilute hydrochloric acid 0.10 to 0.15 N through a central line is the rescue therapy.

Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 33

Question 17

Acid-Base and Electrolyte Disorders · Therapeutic hyperventilation and cerebral CO2 reactivity · moderate

A 29-year-old man is 8 hours out from a severe traumatic brain injury with diffuse cerebral edema. An external ventricular drain shows an ICP of 24 mmHg. He is normotensive with a mean arterial pressure of 85 mmHg. Over the past hour the resident has increased the respiratory rate, and the arterial blood gas now shows pH 7.55 and PCO2 22 mmHg. There is no brain tissue oxygen or jugular venous oximetry monitoring in place. Which of the following is the most appropriate next step?

- **A.** Maintain the PCO2 at 20 to 25 mmHg until the ICP is consistently below 20 mmHg
- **B.** Lower the PCO2 below 20 mmHg to obtain maximal cerebral vasoconstriction
- **C.** Adjust ventilation back to a PCO2 of 35 to 40 mmHg, reserving mild brief hypocapnia of 30 to 34 mmHg as a bridge to other therapies with brain tissue or jugular venous oxygen monitoring
- **D.** Maintain the PCO2 at 25 to 30 mmHg for the first 24 hours after injury
- **E.** Allow the PCO2 to rise above 45 mmHg to augment cerebral blood flow

Answer: C. Adjust ventilation back to a PCO2 of 35 to 40 mmHg, reserving mild brief hypocapnia of 30 to 34 mmHg as a bridge to other therapies with brain tissue or jugular venous oxygen monitoring

Cerebral resistance arterioles begin to constrict below a PCO2 of 35 mmHg and are maximally constricted below 20 mmHg, and at levels of 20 to 25 mmHg cerebral blood flow is cut by approximately half, producing ischemic hypoxia while also inducing excitatory amino acids and increasing glucose and oxygen consumption, impairing pulmonary gas exchange, and increasing hemoglobin oxygen affinity. Strict maintenance of normocapnia with a PCO2 of 35 to 40 mmHg is the mainstay of management; mild brief hypocapnia of 30 to 34 mmHg may still be used as a bridge to other therapies, but monitoring of brain tissue or jugular venous oxygen content is recommended at that level or beyond. Severe hypocapnia with a PCO2 below 25 mmHg is not recommended, and hypocapnia should be avoided for at least the first 24 hours after brain injury, which excludes options A, B, and D. Option E is wrong because hypercapnia is a potent cerebral vasodilator that increases cerebral blood flow, disrupts autoregulation, and would raise the ICP further. Any ICP benefit would also be short-lived, since renal compensation typically restores serum and therefore CSF pH after about 4 hours, and because roughly 70% of cerebral blood volume is venous.

Key point:
keep the PCO2 at 35 to 40 mmHg, use 30 to 34 mmHg only briefly and with brain oxygen monitoring, and never go below 25 mmHg or hyperventilate in the first 24 hours after injury.

Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 33-34

Question 18

Acid-Base and Electrolyte Disorders · Hyponatremia in acute intracranial pathology: SIADH versus CSWS · moderate

A 54-year-old woman is on day 5 after an aneurysmal subarachnoid hemorrhage that has been secured by coiling. She is arousable but has a new mild left pronator drift, and transcranial Doppler velocities are rising. Serum sodium has fallen from 138 to 126 mEq/L over 36 hours. Plasma osmolality is 262 mOsm/kg, urine osmolality is 520 mOsm/kg, and urine sodium is 65 mEq/L. She is on no diuretics and thyroid, adrenal, pituitary, and renal function are normal. Clinical volume status is difficult to assess at the bedside. Which of the following is the most appropriate treatment?

- **A.** Begin a hypertonic sodium chloride solution
- **B.** Restrict free water intake to 1 liter per day
- **C.** Start a vasopressin receptor antagonist
- **D.** Start demeclocycline
- **E.** Give a 1 liter bolus of 5% dextrose in water

Answer: A. Begin a hypertonic sodium chloride solution

Her laboratory profile satisfies the diagnostic criteria for SIADH, namely a plasma osmolarity below 270 mOsm/kg with a urine osmolarity above 100 mOsm/kg, a urine sodium above 20 mEq/L, and absence of diuretics and of thyroid, pituitary, adrenal, and renal insufficiency, but cerebral salt wasting cannot be excluded because the only real diagnostic difference is volume status, which is clinically difficult to determine with certainty. Because of that ambiguity, hypertonic sodium solutions are recommended over vasopressin antagonists or free water restriction in patients with acute intracranial pathology, so option A is correct. Free water restriction is the preferred treatment in the general population, but hypertonic sodium is preferred in patients with CNS pathology in whom the risk of hypovolemia is significant, such as those with elevated ICP or vasospasm, which makes option B dangerous here given her rising Doppler velocities. Vasopressin receptor antagonists act at V2 receptors to block ADH-mediated free water reabsorption and are reserved for refractory cases, and demeclocycline along with lithium carbonate is an alternative for chronic resistant cases. Dextrose in water would add free water and worsen the hyponatremia. If fluid balance remains negative despite hypertonic sodium, a short course of fludrocortisone may be effective, and evidence of chronic hyponatremia should be sought before treatment so the correction rate avoids osmotic demyelination syndrome.

Key point:
in acute intracranial pathology with hyponatremia, give hypertonic sodium rather than fluid restriction or a vaptan, because SIADH and cerebral salt wasting cannot be reliably separated at the bedside.

Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 34-35

Question 19

Acid-Base and Electrolyte Disorders · Central diabetes insipidus: criteria for ADH analog therapy · moderate

A 34-year-old man with a severe traumatic brain injury and basilar skull fracture is intubated and sedated on hospital day 2. Over the past 3 consecutive hours his urine output has been 380, 420, and 400 mL/hr. Serum sodium has risen from 142 to 149 mEq/L, urine osmolality is 180 mOsm/kg, and urine specific gravity is 1.003. He is not receiving mannitol, hypertonic saline, or diuretics, and his glucose is 130 mg/dL. Which of the following is the most appropriate next step?

- **A.** Continue free water via the feeding tube alone and recheck the sodium in 12 hours
- **B.** Start a 0.45% sodium chloride infusion
- **C.** Start fludrocortisone
- **D.** Initiate antidiuretic hormone analog therapy along with free water replacement
- **E.** Start a 3% sodium chloride infusion

Answer: D. Initiate antidiuretic hormone analog therapy along with free water replacement

He meets all three criteria for initiating ADH analog therapy in central diabetes insipidus: a urine osmolality below 300 mOsm/kg or a specific gravity below 1.005 (his are 180 mOsm/kg and 1.003), polyuria of more than 300 mL/hr for 2 or more consecutive hours (three hours near 400 mL/hr), and a sodium above 145 mEq/L (149 mEq/L). Options for therapy include a continuous aqueous vasopressin infusion, bolus dosing, or 1-desamino-8-D-arginine vasopressin, and fluid replacement should be almost solely dextrose-water solutions intravenously or free water by mouth or feeding tube with strict monitoring of fluid balance. Option A is inadequate because a sedated, intubated patient cannot respond to thirst, and this population is more likely to suffer complications from impaired thirst, inability to communicate thirst, inability to access water, and increased insensible losses. Half normal saline and 3% saline both add sodium and would worsen the hypernatremia, and fludrocortisone is used for persistently negative fluid balance in cerebral salt wasting, not for central DI. Prompt recognition matters because DI is associated with an overall mortality rate of 72.4% and with impending brain death in most patients with subarachnoid hemorrhage and traumatic brain injury, though the majority of cases are self-limited and last an average of 3 to 5 days.

Key point:
start an ADH analog when urine osmolality is below 300 mOsm/kg or specific gravity is below 1.005, urine output exceeds 300 mL/hr for at least 2 consecutive hours, and the sodium is above 145 mEq/L.

Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 35

Question 20

Acid-Base and Electrolyte Disorders · Correcting hypernatremia in a patient with cerebral edema · hard

A 65-year-old man weighing 100 kg is on hospital day 6 after a 45 mL left basal ganglia hemorrhage with perilesional edema. His hypertonic saline infusion was weaned and stopped over the past 24 hours, and his sodium goal is 145 to 155 mEq/L. Overnight he developed pulmonary edema and received two doses of a loop diuretic. This morning the sodium has risen from 157 to 160 mEq/L with bicarbonate 20 mEq/L and a critical chloride of 126 mEq/L. He has a functioning feeding tube, and follow-up imaging still shows perilesional edema. Which of the following is the safest way to bring his sodium back into the goal range?

- **A.** 5% dextrose in water at 145 mL/hr
- **B.** Enteral free water flushes of approximately 285 mL every 4 hours
- **C.** Enteral free water flushes of approximately 585 mL every 4 hours
- **D.** 0.45% sodium chloride at 70 mL/hr
- **E.** Replace half of the calculated free water deficit over the first 24 hours at a rate of up to 2 mEq/L per hour

Answer: B. Enteral free water flushes of approximately 285 mL every 4 hours

For patients with ongoing cerebral edema or in its early resolution phase whose sodium is above the desired goal, correction should be much slower than usual, at no more than 2 to 4 mEq/L per day, to prevent rebound edema; correcting from 160 to 157 mEq/L over the next 24 hours requires giving on the order of 1.7 L of free water over that 24 hours, which is approximately 285 mL of enteral free water every 4 hours. Option C would be expected to drop the sodium by about 6 mEq/L within 24 hours and could cause rebound cerebral edema. Hypotonic and dextrose-containing intravenous solutions should be avoided both because of the risk of rebound cerebral edema and because of the risk of hyperglycemia in brain-injured patients, which excludes option A, and half normal saline provides additional sodium and chloride that are not wanted in a patient whose chloride is already critically elevated at 126 mEq/L, excluding option D. Option E states the general critical-care rule, in which half the free water deficit is corrected in the first 24 hours and the remainder over the subsequent 48 hours with a maximum rate of 2 mEq/L per hour, but that rule does not apply to a patient with active cerebral edema. Note also that some patients develop high gastric residuals with large-volume boluses, so more frequent smaller-volume flushes may be necessary.

Key point:
with active cerebral edema, lower an elevated sodium by no more than 2 to 4 mEq/L per day using enteral free water, not by the standard 2 mEq/L per hour rule and not with hypotonic or dextrose intravenous fluids.

Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 35-36, 41

Question 21

Acid-Base and Electrolyte Disorders · Magnesium: evidence base and therapeutic uses · easy

Adjunctive magnesium therapy has demonstrated a clear outcome benefit in which of the following conditions?

- **A.** Acute ischemic stroke when magnesium is given in the prehospital setting
- **B.** Aneurysmal subarachnoid hemorrhage
- **C.** Severe traumatic brain injury
- **D.** Spontaneous intracerebral hemorrhage
- **E.** Eclamptic seizures

Answer: E. Eclamptic seizures

Magnesium therapy is established in severe preeclampsia and eclampsia, and in treating eclamptic seizures magnesium was found to be superior to both diazepam and phenytoin, although its mechanism in this disease process is not clear. The FAST-MAG trial in 2015 demonstrated only the safety of early prehospital magnesium administration in stroke patients and failed to show any improvement in outcomes at 90 days, which excludes options A and D. Evidence from the IMASH and MASH-2 trials does not support adjuvant magnesium therapy in patients with acute subarachnoid hemorrhage, excluding option B. Although neuroprotective and glioprotective effects have been reliably reproduced in diverse animal models of stroke and traumatic brain injury, these have not translated into a demonstrated clinical benefit, excluding option C. The other accepted critical care use is therapeutic hypermagnesemia with serum levels of 3 to 4 mg/dL during targeted temperature management to prevent shivering and possibly to shorten the time to target temperature, achieved with 4 g intravenous boluses of magnesium sulfate every 4 hours or an infusion of 0.5 to 1 g per hour.

Key point:
FAST-MAG, IMASH, and MASH-2 were all negative, so the proven neurologic indication for magnesium remains eclamptic seizures, where it beats both diazepam and phenytoin.

Source: Practice of Neurocritical Care, 2nd ed., Acid-Base and Electrolyte Disorders, pp. 36-37, 41

Cardiovascular Monitoring and Complications

Question 22

Cardiovascular Monitoring and Complications · Invasive arterial pressure monitoring: damping artifact · moderate

A 62-year-old woman with a Hunt-Hess 3 subarachnoid hemorrhage is in the neurocritical care unit on a nicardipine infusion. Her radial arterial line displays a systolic blood pressure of 92 mmHg and a mean arterial pressure of 78 mmHg, while a simultaneous oscillometric cuff on the same arm reads 126/70 mmHg. She is awake, warm, and well perfused with a urine output of 60 mL/h. The arterial waveform appears blunted and rounded with loss of the dicrotic notch. The transducer is leveled at the fourth intercostal space in the midaxillary line and has been zeroed. Which of the following is the most appropriate next step?

- **A.** Start a norepinephrine infusion to raise the systolic blood pressure above 120 mmHg
- **B.** Replace the fluid-filled tubing with stiffer, longer pressure tubing to improve signal fidelity
- **C.** Flush the catheter and tubing, check for air bubbles, clots, or kinks, and confirm the flush bag is adequately pressurized
- **D.** Remove the arterial catheter and manage blood pressure with oscillometric cuff readings every 15 minutes
- **E.** Re-zero the transducer at the level of the external auditory meatus to reflect cerebral perfusion pressure

Answer: C. Flush the catheter and tubing, check for air bubbles, clots, or kinks, and confirm the flush bag is adequately pressurized

A blunted, rounded waveform with loss of the dicrotic notch is the classic appearance of an overdamped system, which causes underestimation of the systolic blood pressure and explains why the arterial line reads lower than the cuff in a well-perfused patient. Causes of overdamping include compliant tubing, air bubbles, clots or fibrin in the tubing, catheter kinks, absent fluid, or a low flush bag pressure, and it is usually corrected by flushing the tubing, repositioning the patient or catheter, and ensuring adequate saline volume and pressure in the system. Option A treats an artifact rather than the patient and risks overtreatment, the exact error the text warns against. Option B would produce the opposite artifact: stiff tubing and increased vascular resistance cause underdamping, in which the waveform is narrow and peaked and the systolic pressure is overestimated; underdamping artifacts are seen in roughly a third of critically ill patients. Option D discards the gold standard for blood pressure monitoring, which is essential when antihypertensive, inotropic, or vasopressor drugs are being titrated, and oscillometric readings themselves diverge from invasive values at very high and very low pressures. Option E is wrong because cardiovascular pressures are referenced to the right atrium at the phlebostatic axis, the fourth intercostal space at the midaxillary line. Note also that the mean arterial pressure is less affected by damping artifact than the systolic pressure.

Key point:

A blunted, rounded arterial waveform means overdamping and a falsely low systolic pressure, so flush and troubleshoot the system before treating the number.

Source: Practice of Neurocritical Care, 2nd ed., Cardiovascular Monitoring and Complications, pp. 48

Question 23

Cardiovascular Monitoring and Complications · Pulse pressure variation and fluid responsiveness · easy

A 58-year-old man with a large left middle cerebral artery infarct is sedated and mechanically ventilated in volume-control mode with a tidal volume of 8 mL/kg. He is in sinus rhythm at 96 beats per minute. His blood pressure is 88/52 mmHg and his lactate is rising. The arterial waveform is well damped, and the monitor reports a pulse pressure variation of 18% over the respiratory cycle. Which of the following is the most appropriate interpretation and next step?

- **A.** The stroke volume is sensitive to preload, and a fluid bolus is likely to increase cardiac output
- **B.** The stroke volume is insensitive to preload, and further fluid is unlikely to increase cardiac output
- **C.** The value indicates volume overload, and intravenous furosemide should be given
- **D.** The value indicates vasoplegia, and norepinephrine should be started before any fluid is given
- **E.** The value is uninterpretable, and a pulmonary artery catheter should be placed to measure occlusion pressure

Answer: A. The stroke volume is sensitive to preload, and a fluid bolus is likely to increase cardiac output

Pulse pressure variation is calculated as the maximum pulse pressure minus the minimum pulse pressure divided by the mean pulse pressure over a respiratory cycle, and a value greater than 10% suggests fluid responsiveness, meaning the stroke volume is sensitive to fluctuations in preload. This patient's value of 18% is well above that threshold, so a fluid challenge is the reasonable next step. Option B states the opposite of what a value above 10% means. Option C is wrong because pulse pressure variation above the threshold reflects a preload-dependent, underfilled ventricle rather than volume overload. Option D confuses a dynamic preload index with a measure of systemic vascular resistance. Option E is not supported: neither central venous pressure nor pulmonary artery occlusion pressure reliably predicts ventricular preload or the response to volume, which is why dynamic indices such as pulse pressure variation are preferred. The analogous dynamic index from pulse contour devices is stroke volume variation, where a value of 15% or more predicts a response to fluid resuscitation.

Key point:
Pulse pressure variation greater than 10% in a ventilated patient predicts fluid responsiveness.

Source: Practice of Neurocritical Care, 2nd ed., Cardiovascular Monitoring and Complications, pp. 47-48

Question 24

Cardiovascular Monitoring and Complications · Bedside ultrasound of the inferior vena cava · moderate

A 71-year-old woman admitted with a cerebellar hemorrhage is spontaneously breathing on nasal cannula and becomes hypotensive with a blood pressure of 84/48 mmHg after a period of poor oral intake and vomiting. Bedside ultrasound in the sagittal plane just proximal to the hepatic vein shows an inferior vena cava diameter of 1.6 cm that collapses by approximately 70% with inspiration. Which of the following best describes the estimated right atrial pressure and the expected response to volume?

- **A.** Right atrial pressure of 10 to 20 mmHg, and the patient is unlikely to respond to fluid
- **B.** Right atrial pressure of 10 to 20 mmHg, and the patient is likely to respond to fluid
- **C.** Right atrial pressure of 6 to 12 mmHg, and the patient is unlikely to respond to fluid
- **D.** Right atrial pressure of 0 to 5 mmHg, and the patient is unlikely to respond to fluid
- **E.** Right atrial pressure of 0 to 5 mmHg, and the patient is likely to respond to fluid

Answer: E. Right atrial pressure of 0 to 5 mmHg, and the patient is likely to respond to fluid

In a spontaneously breathing patient, an inferior vena cava diameter of 2.1 cm or less that collapses more than 50% on inspiration suggests a normal right atrial pressure of 0 to 5 mmHg, and in a hypotensive patient that degree of collapse suggests volume depletion and possible fluid responsiveness. This patient's 1.6 cm vessel collapsing 70% meets both criteria, making options A, B, and D incorrect. The pattern described in options A and B, a right atrial pressure of 10 to 20 mmHg, corresponds instead to an inferior vena cava greater than 2.1 cm that collapses less than 50%, and those patients are less likely to benefit from fluid resuscitation. Option C quotes 6 to 12 mmHg, which is the normal range for pulmonary artery occlusion pressure, not a right atrial pressure category derived from inferior vena cava sonography. Note that in mechanically ventilated patients the inferior vena cava distends rather than collapses, so a distensibility index greater than 18% is used to predict fluid responsiveness. Inferior vena cava measurements should never be interpreted in isolation and should be combined with other physiologic data and the physical exam.

Key point:
A small, highly collapsible inferior vena cava in a spontaneously breathing hypotensive patient predicts a low right atrial pressure and possible fluid responsiveness.

Source: Practice of Neurocritical Care, 2nd ed., Cardiovascular Monitoring and Complications, pp. 51

Question 25

Cardiovascular Monitoring and Complications · Pulmonary artery catheter: evidence and preload estimation · hard

A 66-year-old man with a poor-grade subarachnoid hemorrhage develops shock of unclear cause on hospital day 5. He has a central venous catheter in place reading a central venous pressure of 14 mmHg. A consultant proposes floating a pulmonary artery catheter, arguing that a wedge pressure will settle whether the patient needs volume. Which of the following responses is best supported by the evidence cited in the chapter?

- **A.** Place the catheter, because the pulmonary artery occlusion pressure reliably reflects left ventricular end-diastolic volume in critically ill patients
- **B.** Defer the catheter, because randomized trials and a meta-analysis showed no benefit in mortality or hospital days and neither central venous pressure nor occlusion pressure reliably predicts preload
- **C.** Defer the catheter, because a central venous pressure of 14 mmHg establishes that the left ventricle is adequately filled and excludes fluid responsiveness
- **D.** Place the catheter, because thermodilution cardiac output is directly proportional to the magnitude of the blood temperature decrease it measures
- **E.** Place the catheter, because ventricular arrhythmia during insertion is the only recognized complication and it is self-limited

Answer: B. Defer the catheter, because randomized trials and a meta-analysis showed no benefit in mortality or hospital days and neither central venous pressure nor occlusion pressure reliably predicts preload

Multiple randomized controlled trials have reported no evidence of benefit or harm from pulmonary artery catheter use, and a meta-analysis in critically ill patients showed the catheter neither increased overall mortality or days in hospital nor conferred benefit, while some literature suggests an increase in complications; studies also show that neither central venous pressure nor pulmonary artery occlusion pressure is a useful predictor of ventricular preload or the response to volume infusion. Option A overstates the occlusion pressure, whose correlation with left ventricular end-diastolic volume is explicitly described as dubious. Option C is wrong in the opposite direction: a normal central venous pressure is variable and anywhere between 6 and 20 mmHg may be consistent with normovolemia, and a patient with a high central venous pressure may still have an underfilled, fluid-responsive left ventricle, as in acute right ventricular infarction. Option D reverses the physics, since cardiac output is inversely proportional to the temperature decrease and the duration of transit of the cooled blood, that is, the area under the thermodilution curve. Option E is false: although triggering ventricular arrhythmias while traversing the right ventricle is the most common complication, others include pulmonary infarction from balloon overinflation, pulmonary artery rupture, intracardiac perforation, embolism, valvular injury, and catheter knotting.

Key point:
Pulmonary artery catheter use has not improved outcomes in randomized trials, and neither central venous pressure nor wedge pressure reliably predicts preload.

Source: Practice of Neurocritical Care, 2nd ed., Cardiovascular Monitoring and Complications, pp. 50, 52-53

Question 26

Cardiovascular Monitoring and Complications · QT interval monitoring · easy

A neurocritical care unit is establishing a protocol for monitoring the corrected QT interval in patients receiving drugs that may cause torsades de pointes. Which of the following correctly pairs the upper limit of a normal corrected QT interval with the electrocardiographic lead most commonly used to measure it?

- **A.** 0.40 seconds in men and 0.42 seconds in women, measured in lead V1
- **B.** 0.42 seconds in men and 0.44 seconds in women, measured in lead I
- **C.** 0.45 seconds in men and 0.46 seconds in women, measured in lead aVF
- **D.** 0.45 seconds in men and 0.46 seconds in women, measured in lead II
- **E.** 0.50 seconds in men and 0.50 seconds in women, measured in lead V5

Answer: D. 0.45 seconds in men and 0.46 seconds in women, measured in lead II

The QT interval is measured from the beginning of the QRS complex to the end of the T wave and reflects ventricular repolarization; because repolarization time lengthens at slow rates and shortens at fast rates, it is corrected for heart rate using Bazett's formula. A normal corrected QT interval is less than 0.45 seconds in men and less than 0.46 seconds in women, and although any lead can be used, lead II is the lead most commonly used for the measurement, making options A, B, C, and E incorrect on one or both elements. Prolongation of the corrected QT interval increases the risk of torsades de pointes ventricular tachycardia. Class I indications for QT interval monitoring include patients given an antiarrhythmic drug known to cause torsades de pointes, patients who overdose on a potentially proarrhythmic agent, patients with new-onset bradyarrhythmias, and patients with severe hypokalemia or hypomagnesemia; patients with acute neurological events fall into class II. For reference, the normal PR interval is 120 to 200 ms and the normal QRS interval is less than 110 to 120 ms.

Key point:
Normal corrected QT is under 0.45 seconds in men and under 0.46 seconds in women by Bazett's formula, usually measured in lead II.

Source: Practice of Neurocritical Care, 2nd ed., Cardiovascular Monitoring and Complications, pp. 60, 62

Question 27

Cardiovascular Monitoring and Complications · End-tidal carbon dioxide during cardiopulmonary resuscitation · easy

A 70-year-old intubated man in the neurocritical care unit suffers a pulseless arrest. Continuous waveform capnography during chest compressions shows an end-tidal carbon dioxide of 8 mmHg. After a change of compressor and two more minutes of high-quality compressions, the end-tidal carbon dioxide abruptly rises to 40 mmHg. Which of the following is the most appropriate interpretation and next step?

- **A.** Return of spontaneous circulation has likely occurred; pause compressions to check the rhythm and pulse
- **B.** The endotracheal tube has migrated into the esophagus; remove it and reintubate
- **C.** Compressions have become inadequate; deepen compressions and reassess in two minutes
- **D.** A tension pneumothorax has developed; perform immediate needle decompression
- **E.** Iatrogenic hypercarbia has developed; increase the ventilator respiratory rate

Answer: A. Return of spontaneous circulation has likely occurred; pause compressions to check the rhythm and pulse

Because end-tidal carbon dioxide is determined in part by the amount of blood flow returning to the lungs, it is used to verify the effectiveness of cardiopulmonary resuscitation, and an abrupt increase during resuscitation indicates return of spontaneous circulation. Adequate chest compressions are associated with end-tidal carbon dioxide levels above 10 mmHg, so the initial value of 8 mmHg signaled inadequate perfusion, and the rise to 40 mmHg after improved compressions is the classic marker of restored circulation rather than a ventilation or airway problem. Option B is wrong because esophageal intubation produces an absent or very low carbon dioxide signal, not an abrupt rise; on a colorimetric detector, a color change from purple to yellow indicates correct airway placement while persistent purple indicates esophageal placement. Option C is contradicted by the data, since the value rose above the 10 mmHg adequacy threshold. Options D and E describe conditions that do not cause a sudden isolated rise in end-tidal carbon dioxide during compressions. Remember that even with correct tube placement, end-tidal carbon dioxide may remain low in cardiogenic shock or during ongoing resuscitation.

Key point:
During cardiopulmonary resuscitation, end-tidal carbon dioxide above 10 mmHg indicates adequate compressions and an abrupt rise indicates return of spontaneous circulation.

Source: Practice of Neurocritical Care, 2nd ed., Cardiovascular Monitoring and Complications, pp. 63

Question 28

Cardiovascular Monitoring and Complications · Carbon dioxide targets in traumatic brain injury · moderate

A 34-year-old man with severe traumatic brain injury is sedated and mechanically ventilated with an intracranial pressure monitor reading 14 mmHg and a cerebral perfusion pressure of 68 mmHg. His pupils are equal and reactive and there are no signs of herniation. Capnography shows an end-tidal carbon dioxide of 24 mmHg, and an arterial blood gas confirms an arterial partial pressure of carbon dioxide of 27 mmHg. Which of the following is the most appropriate ventilator adjustment?

- **A.** Continue current settings, since an arterial carbon dioxide of 25 to 30 mmHg is the standard target in traumatic brain injury
- **B.** Increase the respiratory rate to bring the arterial carbon dioxide to 20 to 25 mmHg
- **C.** Reduce minute ventilation to bring the arterial carbon dioxide to 35 to 45 mmHg
- **D.** Reduce minute ventilation to allow the arterial carbon dioxide to rise above 45 mmHg to augment cerebral blood flow
- **E.** Leave minute ventilation unchanged and increase positive end-expiratory pressure to improve carbon dioxide clearance

Answer: C. Reduce minute ventilation to bring the arterial carbon dioxide to 35 to 45 mmHg

The usual arterial carbon dioxide goal for patients in the neurocritical care unit is 35 to 45 mmHg, and patients with traumatic brain injury are highly sensitive to small changes in this value. An arterial carbon dioxide below 35 mmHg causes cerebral vasoconstriction, decreased cerebral blood flow, and ultimately cerebral ischemia, so this patient's value of 27 mmHg should be corrected upward by reducing minute ventilation, making options A and B wrong and potentially harmful, particularly since his intracranial pressure and cerebral perfusion pressure are currently acceptable and there is no herniation to treat. Option D overshoots in the other direction: an arterial carbon dioxide above 45 mmHg causes vasodilation, increased cerebral blood volume, and increased intracranial pressure. Option E does not address the hyperventilation and misattributes carbon dioxide clearance to positive end-expiratory pressure. Continuous end-tidal carbon dioxide monitoring at the bedside is specifically recommended to prevent episodes of hypocarbia and hypercarbia in these patients.

Key point:
Target an arterial carbon dioxide of 35 to 45 mmHg in brain-injured patients, because below 35 mmHg causes ischemic vasoconstriction and above 45 mmHg raises intracranial pressure.

Source: Practice of Neurocritical Care, 2nd ed., Cardiovascular Monitoring and Complications, pp. 63

Question 29

Cardiovascular Monitoring and Complications · Neurogenic stress cardiomyopathy after subarachnoid hemorrhage · moderate

A 24-year-old woman presents with a Hunt-Hess 4, Fisher 4 subarachnoid hemorrhage. On hospital day 1 she develops chest discomfort and dyspnea while remaining hemodynamically stable, with a blood pressure of 126/74 mmHg. Her electrocardiogram shows a prolonged QT interval and inverted T waves in the precordial leads, and her troponin is elevated. Bedside echocardiography demonstrates apical ballooning with hyperdynamic basal segments. Urgent coronary angiography shows coronary vessels free of obstruction or spasm. Which of the following is the most likely diagnosis and an appropriate therapy?

- **A.** Non-ST-elevation myocardial infarction, treated with aspirin, statin, and urgent percutaneous coronary intervention
- **B.** Cardiac tamponade, treated with emergent ultrasound-guided pericardiocentesis
- **C.** Stanford type A aortic dissection, treated with emergent surgical consultation
- **D.** Saddle pulmonary embolism, treated with systemic alteplase
- **E.** Takotsubo cardiomyopathy, treated with beta blockade

Answer: E. Takotsubo cardiomyopathy, treated with beta blockade

A young woman with subarachnoid hemorrhage, a prolonged QT interval, precordial T-wave inversions, elevated troponin, echocardiographic apical ballooning, and angiographically normal coronary arteries has Takotsubo, or stress, cardiomyopathy, and some evidence suggests beta blockade may be effective. Option A is excluded by the angiogram showing no coronary obstruction or spasm, which is the finding that separates stress cardiomyopathy from an acute coronary syndrome. Option B is wrong because tamponade presents with a narrowed pulse pressure, jugular venous distention, and pericardial fluid on echocardiography, none of which are described here, and the echocardiogram instead shows a regional wall motion abnormality. Options C and D do not explain apical ballooning with clean coronaries. The chapter's case study illustrates the same syndrome in a different form: a patient who became severely hypertensive with a mean arterial pressure above 140 mmHg after hemicraniectomy, then developed diffuse T-wave inversions, a prolonged QT interval, diffuse ventricular hypokinesis, and troponin above 80 ng/mL with normal coronary vessels, and whose wall motion abnormalities resolved over the following days with supportive care.

Key point:
Apical ballooning with a prolonged QT interval and clean coronary arteries after an acute brain injury is Takotsubo cardiomyopathy, and it is largely reversible.

Source: Practice of Neurocritical Care, 2nd ed., Cardiovascular Monitoring and Complications, pp. 46, 65, 67

Question 30

Cardiovascular Monitoring and Complications · ST-elevation myocardial infarction with intracranial injury · hard

A 55-year-old man is brought in after an all-terrain vehicle crash. His Glasgow Coma Scale score is 13, and head computed tomography shows a basilar skull fracture and a mild left frontal contusion. On telemetry he develops ST-segment elevations, and a 12-lead electrocardiogram confirms 2 mm of ST elevation in leads I, aVL, and V5 through V6. He is hemodynamically stable. The nearest hospital capable of percutaneous coronary intervention is 100 minutes away by air. Which of the following is the most appropriate management?

- **A.** Administer streptokinase and then transport by air to the interventional center
- **B.** Give oxygen and metoprolol and transport by air to the percutaneous coronary intervention center
- **C.** Administer tenecteplase and then transport by air to the interventional center
- **D.** Give aspirin with a 2,000 unit intravenous heparin bolus and observe on telemetry locally
- **E.** Give sublingual nitroglycerin, digoxin, and clopidogrel and observe on telemetry locally

Answer: B. Give oxygen and metoprolol and transport by air to the percutaneous coronary intervention center

This patient has an acute ST-elevation myocardial infarction in the setting of an acute traumatic intracranial injury, so fibrinolysis is not an option and he should receive oxygen and metoprolol and be flown to the percutaneous coronary intervention center. Options A and C are therefore wrong despite being reperfusion strategies: when fibrinolysis genuinely is the best option, tenecteplase is the typically recommended agent, but a basilar skull fracture with a fresh cerebral contusion precludes lytic therapy. The relevant time targets are a door-to-intervention window of 90 minutes or less for patients transported directly to a hospital capable of percutaneous coronary intervention, and 120 minutes or less for patients who are transported to a non-capable hospital first, so a 100-minute flight remains a reasonable path to catheterization. Options D and E withhold definitive reperfusion in an evolving infarct and, in the case of option E, combine agents with no role in this presentation.

Key point:
In ST-elevation myocardial infarction with acute intracranial injury, fibrinolysis is off the table and transfer for percutaneous coronary intervention is the correct reperfusion strategy, with target windows of 90 minutes direct and 120 minutes after initial presentation to a non-capable hospital.

Source: Practice of Neurocritical Care, 2nd ed., Cardiovascular Monitoring and Complications, pp. 65, 67

Endocrine Issues in Neurocritical Care

Question 31

Endocrine Issues in Neurocritical Care · Glycemic target in acute ischemic stroke · easy

A 68-year-old man with type 2 diabetes mellitus presents 2 hours after sudden onset of right hemiparesis and global aphasia. NIHSS is 14. Blood pressure is 168/88 mm Hg, heart rate 82/min. Non-contrast head CT shows no hemorrhage and an early left MCA hypodensity; CTA shows a left M2 occlusion. Fingerstick glucose is 268 mg/dL. He receives intravenous thrombolysis and is admitted to the neuro ICU on an insulin protocol. Which of the following blood glucose targets is recommended for this patient by current American Heart Association/American Stroke Association guidelines?

- **A.** 80-110 mg/dL
- **B.** 80-130 mg/dL
- **C.** 140-180 mg/dL
- **D.** Greater than 80 mg/dL and less than 200 mg/dL
- **E.** Less than 250 mg/dL

Answer: C. 140-180 mg/dL

The 2019 AHA/ASA acute ischemic stroke guideline recommends a target glucose of 140-180 mg/dL (Class IIa, LOE C), which is also the general ICU target endorsed across guidelines. Option A reproduces classic intensive insulin therapy (IIT); in NICE-SUGAR, IIT targeting 81-108 mg/dL produced a 2.6% absolute increase in 90-day death and severe hypoglycemia in 6.8% versus 0.5% with conventional control, so it is not recommended. Option B is the intensive arm of the SHINE trial (80-130 mg/dL), which in 1,151 stroke patients showed no difference in any measure of 90-day functional outcome versus subcutaneous insulin targeting 80-179 mg/dL, while hypoglycemia was far more common (3.6% versus 0.35%). Option D is the Neurocritical Care Society consensus target for aneurysmal subarachnoid hemorrhage (greater than 80 and less than 200 mg/dL), not for ischemic stroke. Option E is too permissive: relaxed control above 180 mg/dL also appears harmful and should be avoided, and hyperglycemia independently predicts hemorrhagic conversion after thrombolysis.

Key point:
Target 140-180 mg/dL after acute ischemic stroke; intensive insulin therapy adds hypoglycemia without improving outcome.

Source: Practice of Neurocritical Care, 2nd ed., Endocrine Issues in Neurocritical Care, pp. 74-75

Question 32

Endocrine Issues in Neurocritical Care · Diabetic ketoacidosis - potassium before insulin · moderate

A 34-year-old woman with type 1 diabetes mellitus is admitted to the neuro ICU after a generalized convulsive seizure. She has vomited for 2 days and stopped taking her insulin. Blood pressure is 96/58 mm Hg, heart rate 122/min, respirations 28/min and deep. Laboratory studies: glucose 480 mg/dL, arterial pH 7.12, bicarbonate 9 mEq/L, anion gap 24, serum and urine ketones positive, sodium 132 mEq/L, potassium 3.0 mEq/L, creatinine 1.4 mg/dL. She has received 1 L of 0.9% normal saline over the past hour. Which of the following is the most appropriate next step?

- **A.** Hold insulin and infuse potassium chloride 20 mEq per hour until the potassium exceeds 3.3 mEq/L
- **B.** Give regular insulin 0.1 U/kg as a bolus followed by an infusion at 0.1 U/kg/hr
- **C.** Begin a regular insulin infusion at 0.14 U/kg/hr without a bolus
- **D.** Administer sodium bicarbonate 100 mEq in 400 mL infused over 2 hours
- **E.** Replace phosphate and begin a regular insulin infusion at 0.1 U/kg/hr

Answer: A. Hold insulin and infuse potassium chloride 20 mEq per hour until the potassium exceeds 3.3 mEq/L

She meets criteria for diabetic ketoacidosis (glucose greater than 250 mg/dL, pH less than 7.30, bicarbonate less than 18 mEq/L, anion gap greater than 10, and ketonemia/ketonuria), but her potassium is below 3.3 mEq/L. Potassium replacement is necessary and required prior to starting insulin when the potassium level is low, because insulin drives potassium intracellularly and can precipitate life-threatening hypokalemia; insulin is held and potassium chloride 20 mEq/hr is given until the potassium exceeds 3.3 mEq/L. Options B and C are the correct insulin regimens (0.1 U/kg bolus then 0.1 U/kg/hr, or 0.14 U/kg/hr with no bolus) but are premature at this potassium. Option D is wrong because bicarbonate is not recommended routinely and is considered only when the pH is less than 6.9. Option E is wrong because phosphate is replaced only when the level is low, not routinely, and it does not address the hypokalemia before insulin. Once insulin is running, expect glucose to fall 50-75 mg/dL per hour, and when glucose reaches 250-300 mg/dL halve the infusion and add dextrose-containing fluid.

Key point:
In DKA, check potassium first, and if it is below 3.3 mEq/L, hold insulin and replete potassium before starting the infusion.

Source: Practice of Neurocritical Care, 2nd ed., Endocrine Issues in Neurocritical Care, pp. 77-78

Question 33

Endocrine Issues in Neurocritical Care · Emergency treatment of hypoglycemia · easy

A 57-year-old man with alcohol use disorder, malnutrition, and type 2 diabetes mellitus is brought to the emergency department unresponsive. Temperature is 35.8 C, blood pressure 110/70 mm Hg, heart rate 96/min. He is comatose with a GCS of 6, has no focal deficits, and is diaphoretic. Fingerstick glucose is 28 mg/dL. A peripheral intravenous line is in place. Which of the following is the most appropriate next step?

- **A.** Administer 25 grams of 50% dextrose intravenously, then intravenous thiamine
- **B.** Administer oral glucose gel delivering 15-20 grams of simple carbohydrate
- **C.** Administer glucagon 1 mg intramuscularly
- **D.** Administer 25 grams of 50% dextrose intravenously alone
- **E.** Administer intravenous thiamine, then 25 grams of 50% dextrose intravenously

Answer: E. Administer intravenous thiamine, then 25 grams of 50% dextrose intravenously

Intravenous thiamine is recommended before dextrose in comatose hypoglycemic patients with a history of alcohol abuse or malnutrition, in order to lessen the risk of precipitating Wernicke-Korsakoff syndrome; 25 grams of 50% dextrose intravenously is the correct glucose dose when the patient cannot take glucose enterally. Options A and D give dextrose first or alone, which is exactly the sequence that risks Wernicke-Korsakoff syndrome in this patient. Option B is the correct initial treatment only for patients who are awake and can swallow without difficulty or who have a gastric tube, and this patient is comatose. Option C, glucagon 1 mg subcutaneously or intramuscularly, is reserved for hypoglycemic patients without intravenous access and commonly causes nausea and vomiting, which is hazardous in an obtunded patient. Emergency correction should be followed by additional nutrition containing carbohydrate, protein, and fat to sustain the blood glucose, and clinicians should remember that hypoglycemia is a recognized stroke mimic.

Key point:
Give thiamine before dextrose in the comatose hypoglycemic patient with alcoholism or malnutrition.

Source: Practice of Neurocritical Care, 2nd ed., Endocrine Issues in Neurocritical Care, pp. 79

Question 34

Endocrine Issues in Neurocritical Care · Cerebral salt wasting versus SIADH after SAH · hard

A 52-year-old woman is on postoperative day 6 after clipping of a ruptured anterior communicating artery aneurysm (Hunt-Hess grade 3). Over 24 hours she becomes lethargic. Serum sodium has fallen from 138 to 126 mEq/L. Serum osmolality is 268 mOsm/kg, urine sodium 68 mmol/L, urine osmolality 480 mOsm/kg. Urine output has been 300-350 mL/hr, net fluid balance is negative 2.6 L, weight is down 2 kg, and BUN and hematocrit have risen. Heart rate is 108/min with orthostatic hypotension and low central venous pressure. Morning cortisol and TSH are normal. Which of the following is the most appropriate management?

- **A.** Restrict free water intake to 1 L per day
- **B.** Give 0.9% normal saline plus 3% hypertonic saline with enteral sodium chloride, and add fludrocortisone
- **C.** Start a vasopressin receptor antagonist
- **D.** Start furosemide together with sodium chloride tablets
- **E.** Administer desmopressin 1 mcg intravenously

Answer: B. Give 0.9% normal saline plus 3% hypertonic saline with enteral sodium chloride, and add fludrocortisone

Hyponatremia with natriuresis, a contracted intravascular volume, rising BUN and hematocrit, weight loss, and negative fluid balance after an acute neurologic insult defines cerebral salt wasting, in which natriuretic peptides such as BNP and reduced sympathetic tone drive renal sodium and water loss; SIADH by contrast is a euvolemic dilutional hyponatremia. Management targets restoration of intravascular volume and sodium: hypertonic saline and/or oral sodium chloride are the most commonly used initial therapies, with an initial bolus of 150-250 mL of 3% hypertonic saline followed by an infusion, and AHA/ASA guidelines recommend fludrocortisone plus hypertonic saline for hyponatremia related to SAH. Option A is dangerous here because fluid restriction in cerebral salt wasting produces hypovolemia and potentiates delayed cerebral ischemia and cerebral infarction during vasospasm. Options C and D are wrong because vasopressin receptor antagonists and diuretics are avoided in cerebral salt wasting, since both worsen intravascular volume depletion. Option E, desmopressin, would worsen free water retention and hyponatremia; a single 1 mcg intravenous dose is instead reserved for overly rapid rises in serum sodium. Sodium should be corrected no faster than 8-10 mEq/L per day, or roughly 0.5 mEq/L per hour, to avoid osmotic demyelination syndrome.

Key point:
Hyponatremia with hypovolemia after SAH is cerebral salt wasting; give salt and volume with fludrocortisone, and never fluid restrict.

Source: Practice of Neurocritical Care, 2nd ed., Endocrine Issues in Neurocritical Care, pp. 86-88

Question 35

Endocrine Issues in Neurocritical Care · Central diabetes insipidus - vasopressin versus desmopressin · moderate

A 24-year-old man is in the neuro ICU on hospital day 2 after a motorcycle crash causing severe traumatic brain injury (admission GCS 5) with diffuse cerebral edema and an external ventricular drain. Urine output rises to 400 mL/hr for 3 consecutive hours. Urine specific gravity is 1.002, urine osmolality 120 mOsm/kg, serum sodium has risen from 146 to 156 mEq/L, and serum osmolality is 312 mOsm/kg. He remains hypotensive on norepinephrine 0.3 mcg/kg/min, and his intracranial pressure is labile. Which of the following is the most appropriate therapy?

- **A.** Subcutaneous desmopressin 1-2 mcg every 6 to 8 hours
- **B.** Intranasal desmopressin 10-40 mcg daily
- **C.** Oral desmopressin 0.1-0.6 mcg daily
- **D.** Continuous intravenous vasopressin titrated to urine output, adjusted as often as every 30 minutes
- **E.** Enteral fludrocortisone 100 mcg daily

Answer: D. Continuous intravenous vasopressin titrated to urine output, adjusted as often as every 30 minutes

This is central diabetes insipidus (hypernatremia with serum osmolality above 295 mOsm/kg, large-volume dilute urine with osmolality below 200 mOsm/kg and specific gravity below 1.005) in a hemodynamically unstable patient with catastrophic TBI. Continuous intravenous vasopressin may be considered for patients with central DI who are hemodynamically unstable or in whom tight control of serum sodium and volume status is necessary; it is a V1 and V2 agonist with an intravenous onset of 5-20 minutes and duration of 10-20 minutes, and the rate can be adjusted as rapidly as every 30 minutes until urine output declines and specific gravity normalizes. Option A is reasonable in a stable patient, but desmopressin is a pure V2 agonist with an intravenous onset of 1-2 hours and duration of 6-12 hours, so it cannot be titrated quickly and provides no vasopressor support. Options B and C are avoided in the ICU: oral desmopressin has poor bioavailability, and intranasal delivery is unreliable in an intubated patient who cannot synchronize breaths or breathe through the nose. Option E treats cerebral salt wasting, not diabetes insipidus. Free water deficit replacement is given alongside, with the goal of normalizing serum sodium gradually at 0.5 mEq/L/hr; a multicenter cohort of 262 severe TBI patients found desmopressin doses of 0.125-10 mcg were not associated with worsening ICP, with sodium falling faster than 0.5 mEq/L/hr in fewer than 8% of patients.

Key point:
Use titratable intravenous vasopressin for central DI when the patient is hemodynamically unstable or needs tight sodium and volume control; desmopressin is preferred otherwise.

Source: Practice of Neurocritical Care, 2nd ed., Endocrine Issues in Neurocritical Care, pp. 83-85

Question 36

Endocrine Issues in Neurocritical Care · Myxedema coma · moderate

A 74-year-old woman is admitted in winter after being found unresponsive at home. Temperature is 33.4 C, heart rate 44/min, blood pressure 92/60 mm Hg, respirations 8/min. Examination shows dry coarse skin, non-pitting periorbital edema, macroglossia, and no focal deficits; head CT is unremarkable. Laboratory studies: sodium 126 mEq/L, glucose 54 mg/dL, creatinine 1.6 mg/dL, with leukocytosis and anemia. TSH is markedly elevated and free T4 is undetectable. Her family reports she stopped taking levothyroxine several months ago. Which of the following is the most appropriate next step in management?

- **A.** Administer intravenous levothyroxine alone
- **B.** Perform active external rewarming and defer hormone replacement pending repeat thyroid studies
- **C.** Administer intravenous hydrocortisone 50-100 mg every 8 hours before starting intravenous levothyroxine
- **D.** Administer oral liothyronine alone
- **E.** Administer intravenous levothyroxine together with fludrocortisone

Answer: C. Administer intravenous hydrocortisone 50-100 mg every 8 hours before starting intravenous levothyroxine

She has myxedema coma, defined by the triad of altered mentation, dysfunctional thermoregulation causing hypothermia, and a precipitating factor, here noncompliance with thyroid supplementation, one of the most common precipitants along with surgery and infection. Severe hypothyroidism is often accompanied by adrenal insufficiency, and thyroid hormone replacement increases the metabolism of cortisol and can precipitate adrenal crisis, so glucocorticoids such as hydrocortisone 50-100 mg every 8 hours are recommended before initiation of T4. Options A and D give thyroid hormone without glucocorticoid cover and risk precipitating adrenal crisis; the correct repletion strategy once steroids are on board is intravenous T4 with oral T3 started simultaneously. Option B is unsafe, since myxedema coma is potentially fatal in up to 60% of untreated patients and the elevated TSH with undetectable free T4 already establishes primary hypothyroidism. Option E adds a mineralocorticoid rather than the needed glucocorticoid and still gives T4 first. Her hypoglycemia, hyponatremia, anemia, leukocytosis, and elevated creatinine are all characteristic accompanying laboratory abnormalities, and depressed ventilatory drive may require mechanical ventilation and prolong weaning.

Key point:
In myxedema coma, give hydrocortisone before thyroid hormone, because T4 accelerates cortisol metabolism and can trigger adrenal crisis.

Source: Practice of Neurocritical Care, 2nd ed., Endocrine Issues in Neurocritical Care, pp. 89-90

Question 37

Endocrine Issues in Neurocritical Care · Pituitary apoplexy with acute adrenal insufficiency · moderate

A 58-year-old man taking warfarin for atrial fibrillation presents with the abrupt onset of severe headache, nausea, and double vision that began 4 hours ago. Examination shows a partial left third nerve palsy, a bitemporal field cut, and reduced visual acuity in the left eye; he is drowsy but arousable. Blood pressure is 82/48 mm Hg and heart rate 118/min despite 2 L of crystalloid. Sodium is 128 mEq/L and glucose is 52 mg/dL. MRI demonstrates hemorrhage into a pituitary macroadenoma with suprasellar extension. Which of the following is the most appropriate immediate treatment?

- **A.** Intravenous hydrocortisone 100-500 mg as a bolus, followed by a 2-4 mg per hour infusion
- **B.** Immediate transsphenoidal decompression before any hormonal therapy
- **C.** Intravenous levothyroxine
- **D.** Dexamethasone 4 mg every 6 hours started only after cosyntropin stimulation testing returns
- **E.** Oral fludrocortisone 50-200 mcg daily

Answer: A. Intravenous hydrocortisone 100-500 mg as a bolus, followed by a 2-4 mg per hour infusion

This is pituitary apoplexy with acute adrenal insufficiency, signaled by hypotension refractory to fluids, altered consciousness, reduced visual acuity and field deficits, hyponatremia, and hypoglycemia; anticoagulation is a recognized triggering factor, as are hypertension, major surgery, and intracranial hypertension. Acute adrenal insufficiency from apoplexy should be treated emergently with prompt corticosteroid replacement: hydrocortisone 100-500 mg as an intravenous bolus followed by a 2-4 mg per hour continuous infusion or 50-100 mg every 6 hours, later converted to oral hydrocortisone at a 1:1 ratio and tapered on clinical and hemodynamic stability. Option B is wrong in sequencing; although surgery is common (about 70% of patients in one series, at a median of 5 days from apoplexy) and improves recovery of ocular palsy and visual fields, steroid resuscitation must not wait for the operating room. Option C is not the emergency priority, and giving thyroid hormone before glucocorticoid can precipitate adrenal crisis. Option D delays lifesaving therapy for a confirmatory test; dexamethasone 4 mg every 6 hours is a dosing pattern used for low cortisol after transsphenoidal resection, not for an unstable apoplexy patient. Option E is an adjunct, used at 50-200 mcg orally daily if hyponatremia persists, not the initial resuscitative therapy; long-term steroid replacement is ultimately needed in 58-83% of these patients.

Key point:
Pituitary apoplexy with hypotension, obtundation, or visual loss demands immediate intravenous hydrocortisone, given before thyroid hormone replacement, confirmatory adrenal testing, or surgery.

Source: Practice of Neurocritical Care, 2nd ed., Endocrine Issues in Neurocritical Care, pp. 88-89

Question 38

Non-Neurological Trauma, Burns, and Thermal Injury · Primary survey - tension pneumothorax decompression · easy

A 22-year-old man is brought to the trauma bay after a high-speed motor vehicle collision. He is agitated and dyspneic. Blood pressure is 84/48 mmHg, heart rate 132 beats/min, respiratory rate 40 breaths/min, and SpO2 85% on a non-rebreather mask. Examination shows subcutaneous emphysema over the left chest, absent breath sounds on the left, hyperresonance to percussion on the left, distended neck veins, and tracheal deviation to the right. Immediate needle decompression is planned. At which site should the needle be placed?

- **A.** Second intercostal space at the midclavicular line on the left
- **B.** Fifth intercostal space at the midaxillary line on the left
- **C.** Second intercostal space at the midclavicular line on the right
- **D.** Fifth intercostal space at the midaxillary line on the right
- **E.** Seventh intercostal space at the posterior axillary line on the left

Answer: B. Fifth intercostal space at the midaxillary line on the left

This is a classic tension pneumothorax: absent breath sounds, hyperresonance, jugular venous distention, tracheal deviation away from the affected side, and obstructive shock from impaired venous return. Cadaver studies demonstrated improved success in reaching the thoracic cavity when the fourth or fifth intercostal space at the midaxillary line is used instead of the second intercostal space at the midclavicular line in adults, and the ATLS course now recommends this location for needle decompression in adults, making option B correct. Options A and C reflect the older, less reliable landmark, and C is also on the wrong side, as is D; the physical findings localize the pneumothorax to the left. Option E is too low and posterior for emergent decompression. If needle decompression fails to improve the patient, consider hemothorax or a kinked angiocatheter and perform a finger thoracostomy; a chest tube should follow immediate decompression in all cases.

Key point:
In adults, needle decompression for tension pneumothorax is performed at the fourth or fifth intercostal space in the midaxillary line, followed by tube thoracostomy.

Source: Practice of Neurocritical Care, 2nd ed., Non-Neurological Trauma, Burns, and Thermal Injury, pp. 107-108

Question 39

Non-Neurological Trauma, Burns, and Thermal Injury · Damage control resuscitation and tranexamic acid · moderate

A 34-year-old man arrives 50 minutes after a motorcycle crash. Blood pressure is 86/50 mmHg, heart rate 128 beats/min, and respiratory rate 30 breaths/min. He is pale and diaphoretic. FAST examination shows hemoperitoneum and the pelvic radiograph is normal. Two large-bore intravenous lines are in place, the massive transfusion protocol has been activated, and packed red blood cells, plasma, and platelets are being given in equal ratios while the operating room is prepared. Which of the following is the most appropriate additional pharmacologic intervention?

- **A.** Infuse 2 L of normal saline before further blood products
- **B.** Start a norepinephrine infusion titrated to a systolic blood pressure above 120 mmHg
- **C.** Administer hydroxyethyl starch to expand intravascular volume
- **D.** Administer tranexamic acid now
- **E.** Withhold tranexamic acid until at least 6 hours after injury

Answer: D. Administer tranexamic acid now

Tranexamic acid is the only agent demonstrated in large randomized controlled trials to be efficacious for attenuating the acute coagulopathy of trauma-shock, and it must be given within 3 hours of injury; this patient is only 50 minutes out, so option D is correct. Option E is wrong and potentially harmful, because there is potential for harm when tranexamic acid is given beyond the first 3 hours. Option A reflects the outdated 3:1 crystalloid rule and the ATLS 2 L crystalloid bolus; more recent data show that crystalloid resuscitation worsens outcomes in hemorrhage by exacerbating the lethal triad of acidemia, hypothermia, and coagulopathy and by diluting clotting factors. Option C substitutes colloid for definitive hemostatic resuscitation and does not address coagulopathy. Option B is inappropriate because vasoactive infusions have no role in the acute management of uncontrolled hemorrhagic shock, where the priority is blood products in balanced 1:1:1 ratios and surgical hemostasis.

Key point:
Give tranexamic acid within 3 hours of injury as an adjunct to hemostatic resuscitation; beyond 3 hours it may cause harm.

Source: Practice of Neurocritical Care, 2nd ed., Non-Neurological Trauma, Burns, and Thermal Injury, pp. 108-109

Question 40

Non-Neurological Trauma, Burns, and Thermal Injury · Massive hemothorax · easy

A 41-year-old man sustains a stab wound to the left lateral chest. Blood pressure is 88/54 mmHg and heart rate is 124 beats/min. Breath sounds are decreased on the left with a dull percussion note. A 36 French thoracostomy tube is placed in the left sixth intercostal space at the midaxillary line, with immediate return of 1,700 mL of blood. Which of the following is the most appropriate next step?

- **A.** Proceed to operative thoracotomy
- **B.** Place a second thoracostomy tube in the left chest
- **C.** Obtain computed tomographic angiography of the chest
- **D.** Observe and repeat a chest radiograph in 4 hours
- **E.** Perform pericardiocentesis

Answer: A. Proceed to operative thoracotomy

Traditional indications for surgical intervention in acute traumatic hemothorax include initial drainage of more than 1,500 mL following tube thoracostomy or ongoing drainage of more than 200 mL per hour for 4 hours, and this patient drained 1,700 mL immediately while remaining hypotensive, so thoracotomy is indicated. Option B does not address the ongoing source of great vessel or intercostal bleeding. Option C is inappropriate in a hemodynamically unstable patient; imaging should not delay operative hemorrhage control. Option D ignores both the volume threshold and the hemodynamic instability, and the chapter emphasizes that the patient's physiology, not the absolute chest tube output alone, should be the primary driver for operative intervention. Option E treats cardiac tamponade, which presents with Beck's triad of elevated venous pressure, hypotension, and muffled heart sounds rather than a dull percussion note with absent breath sounds.

Key point:
More than 1,500 mL of initial chest tube output, or more than 200 mL per hour for 4 hours, is a traditional indication for thoracotomy in traumatic hemothorax.

Source: Practice of Neurocritical Care, 2nd ed., Non-Neurological Trauma, Burns, and Thermal Injury, pp. 108-109

Question 41

Non-Neurological Trauma, Burns, and Thermal Injury · Burn resuscitation - rule of nines and Parkland formula · hard

An 80-kg man is rescued from a house fire. He has deep partial-thickness and full-thickness burns involving the entire anterior torso and circumferentially involving the right upper extremity. He is intubated for deep facial burns and a hoarse voice. Using the rule of nines and the Parkland formula, which of the following is the most appropriate initial fluid regimen for the first 24 hours?

- **A.** Normal saline at approximately 270 mL/hr for the first 8 hours, then approximately 540 mL/hr for the next 16 hours
- **B.** 5% albumin bolus followed by lactated Ringer's at approximately 300 mL/hr for 24 hours
- **C.** Lactated Ringer's at approximately 360 mL/hr for the full 24 hours
- **D.** Lactated Ringer's at approximately 270 mL/hr for the first 8 hours, then approximately 540 mL/hr for the next 16 hours
- **E.** Lactated Ringer's at approximately 540 mL/hr for the first 8 hours, then approximately 270 mL/hr for the next 16 hours

Answer: E. Lactated Ringer's at approximately 540 mL/hr for the first 8 hours, then approximately 270 mL/hr for the next 16 hours

By the rule of nines, the anterior torso is 18% total body surface area and a circumferential upper extremity is 9%, for a total of 27%. The Parkland formula gives 4 mL/kg per %TBSA, so 4 x 80 x 27 equals 8,640 mL over the first 24 hours, with half given in the first 8 hours (4,320 mL, or about 540 mL/hr) and the remaining half over the next 16 hours (4,320 mL, or about 270 mL/hr), making option E correct. Options A and D invert the sequence, giving the smaller volume first; the modified Brooke formula uses 2 mL/kg per %TBSA but still gives half in the first 8 hours. Option C spreads the volume evenly and ignores the front-loaded schedule. Option B is wrong because colloid should be reserved for patients unresponsive to crystalloid resuscitation and is avoided in the first 24 hours; first-line therapy is isotonic crystalloid, preferably lactated Ringer's. Resuscitation is then titrated to a urine output of 0.5 mL/kg/hr in adults and 1 mL/kg/hr in children.

Key point:
Parkland gives 4 mL/kg per %TBSA of lactated Ringer's over 24 hours, half in the first 8 hours, titrated to urine output of 0.5 mL/kg/hr in adults.

Source: Practice of Neurocritical Care, 2nd ed., Non-Neurological Trauma, Burns, and Thermal Injury, pp. 119-120

Question 42

Non-Neurological Trauma, Burns, and Thermal Injury · Abdominal compartment syndrome after burns · moderate

A 25-year-old man with 45% total body surface area burns, including circumferential burns of the torso, has received large-volume crystalloid resuscitation over the first 18 hours. He is intubated and sedated. Peak inspiratory pressures have risen from 26 to 42 cm H_2O, urine output has fallen to 8 mL/hr, and blood pressure is 84/50 mmHg. The abdomen is tense. Bladder pressure is 34 mmHg. Which of the following is the most appropriate next step?

- **A.** Increase the crystalloid infusion rate
- **B.** Begin a colloid infusion
- **C.** Perform escharotomy of the chest and abdomen
- **D.** Perform immediate decompressive laparotomy
- **E.** Initiate continuous renal replacement therapy

Answer: C. Perform escharotomy of the chest and abdomen

This patient has abdominal compartment syndrome, defined as a sustained intra-abdominal pressure above 20 mmHg associated with new organ dysfunction, here manifest as elevated peak inspiratory pressure, oliguria, and hypotension; in burns a bladder pressure above 30 mmHg requires immediate treatment. As opposed to trauma-induced abdominal compartment syndrome, first-line treatment in the setting of circumferential truncal burns is escharotomy first, performed through the midaxillary line and horizontally across the chest and abdominal junction. Option D is not wrong in principle but is the second step, reserved for when escharotomy fails to relieve symptoms. Options A and B worsen the problem, since compartment syndrome after burns is classically a complication of over-resuscitation, and colloid is reserved for patients unresponsive to crystalloid after the first 24 hours. Option E treats a consequence rather than the cause and does not lower intra-abdominal pressure.

Key point:
In circumferential truncal burns with abdominal compartment syndrome, escharotomy comes first and decompressive laparotomy follows only if escharotomy fails.

Source: Practice of Neurocritical Care, 2nd ed., Non-Neurological Trauma, Burns, and Thermal Injury, pp. 121-122

Question 43

Non-Neurological Trauma, Burns, and Thermal Injury · Accidental hypothermia and rewarming · moderate

A 57-year-old man is found unresponsive outdoors after a night of subfreezing temperatures. On arrival he is pulseless with ventricular fibrillation on the monitor. Esophageal core temperature is 27 degrees Celsius. Three defibrillation attempts have been unsuccessful. Which of the following is the most appropriate management?

- **A.** Withhold cardiopulmonary resuscitation until the core temperature reaches 32 degrees Celsius
- **B.** Passive external rewarming with warm dry blankets alone
- **C.** Aggressive diuresis to manage cold-induced volume shifts
- **D.** Continue cardiopulmonary resuscitation while initiating active core rewarming
- **E.** Administer empiric potassium to prevent hypokalemia during rewarming

Answer: D. Continue cardiopulmonary resuscitation while initiating active core rewarming

Accidental hypothermia is an unintentional core temperature below 35 degrees Celsius, and this patient is in the severe range of 20 to 28 degrees Celsius, where refractory ventricular fibrillation is expected; below 32 degrees Celsius ventricular fibrillation is often refractory to defibrillation. Cardiopulmonary resuscitation should be initiated for any patient who is pulseless or has a nonperfusing rhythm while rewarming is initiated, so option A is wrong. Active core rewarming is reserved for and indicated in patients with severe hypothermia below 30 degrees Celsius, using techniques such as airway rewarming, warm gastrointestinal irrigation, peritoneal or pleural lavage, hemodialysis, and extracorporeal support, making option D correct. Option B applies to mild hypothermia in young healthy patients with intact shivering. Option C is wrong because these patients are volume depleted from cold diuresis and hypotension is treated first with intravenous fluid. Option E is wrong because hyperkalemia, not hypokalemia, frequently occurs during rewarming, along with rewarming shock and profound hypoglycemia.

Key point:
In severe hypothermia below 30 degrees Celsius with cardiac arrest, continue cardiopulmonary resuscitation while performing active core rewarming.

Source: Practice of Neurocritical Care, 2nd ed., Non-Neurological Trauma, Burns, and Thermal Injury, pp. 122-123

Question 44

Nutrition and Metabolism · Energy and protein requirements in obesity · hard

A 58-year-old woman is admitted to the neurocritical care unit after aneurysmal subarachnoid hemorrhage and coiling. She is intubated and hemodynamically stable. Height is 157 cm and weight is 82 kg, giving a body mass index of 33 kg/m2. Renal function is normal and she is not receiving dialysis. Indirect calorimetry is unavailable. Which of the following is the most appropriate nutrition prescription?

- **A.** 25 to 30 kcal/kg actual body weight with 1.2 to 2 g/kg actual body weight of protein
- **B.** 11 to 14 kcal/kg actual body weight with 2 g/kg ideal body weight of protein
- **C.** 22 to 25 kcal/kg ideal body weight with 2 to 2.5 g/kg ideal body weight of protein
- **D.** 25 to 30 kcal/kg ideal body weight with 1.2 g/kg ideal body weight of protein
- **E.** 11 to 14 kcal/kg ideal body weight with 1.2 to 2 g/kg actual body weight of protein

Answer: B. 11 to 14 kcal/kg actual body weight with 2 g/kg ideal body weight of protein

Energy requirements are stratified by body mass index: 25 to 30 kcal/kg actual body weight for a body mass index of 29.9 or less, 11 to 14 kcal/kg actual body weight for a body mass index of 30 to 50, and 22 to 25 kcal/kg ideal body weight for a body mass index above 50. This patient's body mass index of 33 places her in the 11 to 14 kcal/kg actual body weight band. Protein for obese patients is based on ideal body weight calculated with the Hamwi equation, which for women is 100 lb at 60 inches plus 5 lb for each additional inch, giving roughly 50 kg here; the requirement is 2 g/kg ideal body weight for a body mass index of 30 to 40 and 2 to 2.5 g/kg ideal body weight above 40, so option B is correct. Option A applies to a body mass index of 29.9 or less. Option C applies only to a body mass index above 50. Options D and E mix the wrong weight basis for calories or protein, and D also underdoses protein. For obese patients receiving enteral nutrition, calories provided should not exceed 65% to 70% of the target energy requirement.

Key point:
For a body mass index of 30 to 40, feed 11 to 14 kcal/kg actual body weight with 2 g/kg ideal body weight of protein.

Source: Practice of Neurocritical Care, 2nd ed., Nutrition and Metabolism, pp. 135-136

Question 45

Nutrition and Metabolism · Nutrition assessment and biochemical markers · easy

A 70-year-old man is in the neurocritical care unit on hospital day 5 after a large intracerebral hemorrhage. He is receiving continuous enteral nutrition through a nasojejunal tube. Serum albumin is 2.3 g/dL and prealbumin is low; C-reactive protein is elevated. Renal function is normal and urine collection is reliable. Which of the following best assesses the adequacy of his protein provision?

- **A.** A 24-hour urine urea nitrogen measurement to calculate nitrogen balance
- **B.** Serial serum albumin levels twice weekly
- **C.** Serial serum prealbumin levels twice weekly
- **D.** Serial C-reactive protein levels
- **E.** A serum transferrin level

Answer: A. A 24-hour urine urea nitrogen measurement to calculate nitrogen balance

Nitrogen balance, calculated as protein intake in grams divided by 6.25 minus urine urea nitrogen excretion plus 3 to 5 g, is the measure recommended for assessing the adequacy of protein supplementation, and the 24-hour urine urea nitrogen also guides clinicians on the degree of catabolism and approximate protein needs. It requires normal renal function, accurate urine collection, and knowledge of 24-hour protein intake, all present here. Laboratory testing of serum albumin, prealbumin, and transferrin is no longer recommended to assess nutritional status because these acute phase proteins are influenced by the inflammatory process and acute phase response of illness, which excludes options B, C, and E; this patient's elevated C-reactive protein confirms an ongoing inflammatory state. Option D is wrong because research is insufficient regarding C-reactive protein, interleukin-1, interleukin-6, tumor necrosis factor, calcitonin, and citrulline as indicators of nutritional status, and their routine clinical use is not recommended. Note also that positive nitrogen balance is difficult to achieve in critically ill patients, so the value is used to gauge catabolism rather than as a target to be normalized.

Key point:
Albumin, prealbumin, and transferrin no longer assess nutritional status; use the 24-hour urine urea nitrogen and nitrogen balance to judge protein adequacy.

Source: Practice of Neurocritical Care, 2nd ed., Nutrition and Metabolism, pp. 134

Question 46

Nutrition and Metabolism · Timing and route of enteral nutrition · moderate

A 65-year-old woman is admitted after a motor vehicle collision with a Glasgow Coma Scale score of 8 and a right subdural hematoma with 12 mm of midline shift. She undergoes decompressive craniectomy and arrives in the intensive care unit sedated with propofol and fentanyl. She is hemodynamically stable with stable intracranial pressure and normal respiratory parameters. A large-bore nasogastric tube is on free drainage and a nasojejunal tube is in place. Computed tomography showed no significant intra-abdominal injury. Which of the following is the most appropriate nutrition plan?

- **A.** Start parenteral nutrition now through the central venous catheter
- **B.** Withhold nutrition until hospital day 7 or until sedation is discontinued
- **C.** Start a standard polymeric formula through the nasojejunal tube within 24 to 48 hours, counting the 1.1 calories per milliliter delivered by propofol
- **D.** Start enteral nutrition and check gastric residual volumes every 4 hours to guide advancement
- **E.** Start an elemental formula through the nasogastric tube and place it on free drainage between feeds

Answer: C. Start a standard polymeric formula through the nasojejunal tube within 24 to 48 hours, counting the 1.1 calories per milliliter delivered by propofol

Enteral nutrition should be initiated within 24 to 48 hours of intensive care unit admission in intubated patients or those unable to safely take an oral diet, because early enteral nutrition down-regulates systemic immune responses, reduces oxidative stress, and improves mortality and infectious complications. A standard polymeric formula is appropriate for most neurocritical care patients, and specialty or elemental formulas should be avoided in the absence of significant abdominal injury, which makes option E wrong; feeding is delivered through the nasojejunal tube while the nasogastric tube remains on free drainage. Propofol delivers 1.1 calories of fat per milliliter, similar to a 10% parenteral lipid formulation, and clevidipine delivers 2 calories per milliliter, so failure to account for these calories causes overfeeding and hypertriglyceridemia with a risk of pancreatitis; triglycerides should be followed every 3 days during the infusion. Option A is wrong because enteral nutrition is favored over parenteral, which is reserved for a nonfunctioning gastrointestinal tract or intolerance of enteral trials. Option B causes iatrogenic underfeeding. Option D is wrong because checking gastric residuals is no longer recommended to monitor tolerance.

Key point:
Start enteral nutrition within 24 to 48 hours with a standard polymeric formula, and subtract propofol calories at 1.1 calories per milliliter from the energy target.

Source: Practice of Neurocritical Care, 2nd ed., Nutrition and Metabolism, pp. 133, 137-138

Question 47

Nutrition and Metabolism · Drug and enteral formula interactions · moderate

A 47-year-old man with a traumatic brain injury is receiving continuous enteral nutrition through a nasogastric tube and phenytoin for seizure prophylaxis. Despite appropriate weight-based dosing and good compliance with administration, his total phenytoin levels remain persistently subtherapeutic. Albumin and renal function are normal, and no interacting intravenous medications have been added. Which of the following is the most appropriate management?

- **A.** Switch the patient to parenteral nutrition
- **B.** Change to an elemental enteral formula
- **C.** Discontinue enteral nutrition for 24 hours and recheck the level
- **D.** Hold enteral nutrition 1 to 2 hours before and after each phenytoin dose and adjust the feeding rate over the rest of the day
- **E.** Crush an extended-release formulation and give it with the feeds

Answer: D. Hold enteral nutrition 1 to 2 hours before and after each phenytoin dose and adjust the feeding rate over the rest of the day

Phenytoin binds both to components of enteral nutrition and to the feeding tube itself, which lowers absorption and produces subtherapeutic levels. The recommended approach is to hold enteral nutrition 1 to 2 hours before and after each phenytoin dose while adjusting the rate of enteral nutrition for the remainder of the day to meet similar caloric goals; when continuous feeding cannot be held or dosing frequency is too high, the phenytoin dose can instead be increased and given without holding feeds to overcome binding. Option A abandons a functioning gastrointestinal tract and is not indicated, since parenteral nutrition is reserved for a nonfunctioning gut or intolerance of enteral feeding. Option B does not address the binding interaction, and specialty formulas are discouraged. Option C causes unnecessary iatrogenic underfeeding, one of the commonest quality problems in these units. Option E is wrong because enteric-coated, delayed-release, and extended-release medications should never be crushed and should be changed to another formulation. Fluoroquinolones bind similarly, and the analogous rule is to hold feeds 1 hour before and 2 hours after the dose or increase the dose by 50%.

Key point:
Phenytoin binds to enteral feeds and tubing, so hold feeds 1 to 2 hours before and after each dose or increase the dose to overcome binding.

Source: Practice of Neurocritical Care, 2nd ed., Nutrition and Metabolism, pp. 138-139

Question 48

Nutrition and Metabolism · Glycemic control in neurocritical care · moderate

A 71-year-old woman with type 2 diabetes mellitus is admitted 6 hours after an acute middle cerebral artery ischemic stroke. She is receiving enteral nutrition. Fingerstick glucose values over the past 12 hours have ranged from 190 to 240 mg/dL. Which of the following glucose targets is most appropriate for this patient?

- **A.** 80 to 110 mg/dL
- **B.** 100 to 140 mg/dL
- **C.** 110 to 140 mg/dL
- **D.** 140 to 160 mg/dL
- **E.** Less than 180 mg/dL

Answer: E. Less than 180 mg/dL

The general recommendation in the neurocritical care population is a moderate approach to glucose control with a target of less than 180 mg/dL, making option E correct. The NICE-SUGAR investigators published a trial in 2009 that challenged prior aggressive glucose control strategies and found lower mortality in patients receiving the less intensive insulin regimen. The SHINE trial specifically addressed ischemic stroke and found that patients treated in the first 72 hours to an aggressive lower glucose goal did not do better than those allowed to reach a higher glucose level of less than 180 mg/dL, which is exactly this patient's situation. Option A is the range shown by cerebral microdialysis studies to carry a high incidence of central nervous system glucopenia at low normal serum glucose levels of 80 to 110 mg/dL. Options B and C are similarly tight targets without demonstrated benefit and with a risk of hypoglycemia and brain metabolic distress. Option D is arbitrary and not supported by the cited trials. Cerebral microdialysis can be used in appropriate patients to individualize control.

Key point:
Target a serum glucose of less than 180 mg/dL in neurocritical care patients; tighter control offered no benefit in NICE-SUGAR or SHINE and risks central nervous system glucopenia.

Source: Practice of Neurocritical Care, 2nd ed., Nutrition and Metabolism, pp. 140

Part II: Neurocritical Care

Acute Ischemic Stroke

Question 49

Acute Ischemic Stroke · Pathophysiology: core, penumbra, and ischemic thresholds · easy

A 68-year-old man presents 2 hours after sudden onset of left hemiparesis and neglect. CT angiography shows a right M1 occlusion. CT perfusion demonstrates a small region of severely reduced blood flow surrounded by a much larger region of moderately reduced flow. The stroke team explains that reperfusion therapy is directed at a specific tissue compartment. Which of the following best characterizes the ischemic penumbra?

- **A.** Tissue below the infarct-necrosis threshold that is already irreversibly injured within minutes of occlusion
- **B.** Tissue with normal cerebral blood flow that is transiently dysfunctional from remote deafferentation
- **C.** Collaterally perfused tissue near the ischemic threshold of 18 to 20 mL/100 g/min that will infarct over 12 to 24 hours if flow is not restored
- **D.** Tissue with mildly reduced flow that remains above the ischemic threshold and survives regardless of reperfusion
- **E.** Tissue that restricts on diffusion-weighted imaging at the time of first presentation

Answer: C. Collaterally perfused tissue near the ischemic threshold of 18 to 20 mL/100 g/min that will infarct over 12 to 24 hours if flow is not restored

The ischemia threshold for cerebral blood flow is 18 to 20 mL/100 g tissue/min; below this, adenosine triphosphate cannot be produced aerobically and irreversible injury becomes likely. In focal ischemia, collateral flow through the circle of Willis and pial-pial anastomoses means perfusion is rarely completely absent, and moving centripetally away from the core these collaterals supply progressively more flow. The tissue perfused just around this threshold is the penumbra, and it will progressively join the core over hours, a process that is likely complete in 12 to 24 hours if flow is not restored, which is exactly why it is the target of all acute reperfusion therapy. Option A describes the core, which begins to infarct within minutes and is usually already dead when the patient reaches medical attention. Option D describes benign oligemia, tissue with lower than normal perfusion that is still above the ischemic threshold. Option E describes the core as seen on diffusion-weighted imaging, not the penumbra. Option B is not a described mechanism in acute stroke triage.

Key point:
the penumbra is threshold-level, collaterally perfused, time-dependent tissue, and salvaging it, not the core, is the goal of thrombolysis and thrombectomy.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 150-151

Question 50

Acute Ischemic Stroke · IV alteplase dosing · easy

A 72-year-old woman weighing 80 kg presents 90 minutes after onset of aphasia and right hemiparesis, NIHSS 14. Noncontrast CT shows no hemorrhage, glucose is 118 mg/dL, INR is 1.0, and blood pressure is 168/92 mmHg. She is eligible for intravenous thrombolysis. Which of the following is the correct alteplase regimen?

- **A.** 0.9 mg/kg total, maximum 90 mg, given entirely as a bolus over 5 minutes
- **B.** 0.9 mg/kg total, maximum 90 mg, with 10% as a bolus over 1 minute and the remainder infused over 1 hour
- **C.** 0.9 mg/kg total, maximum 100 mg, with 20% as a bolus over 2 minutes and the remainder infused over 2 hours
- **D.** 0.6 mg/kg total, maximum 60 mg, with 15% as a bolus and the remainder infused over 1 hour
- **E.** 0.25 mg/kg as a single intravenous bolus, maximum 25 mg

Answer: B. 0.9 mg/kg total, maximum 90 mg, with 10% as a bolus over 1 minute and the remainder infused over 1 hour

Once a patient is eligible for intravenous thrombolysis, alteplase should be initiated without delay at a total dose of 0.9 mg/kg to a maximum of 90 mg, with 10% given as an intravenous bolus over 1 minute and the remainder infused over 1 hour. For this 80 kg patient that is 72 mg total, with roughly 7 mg as the bolus. Option E is the tenecteplase regimen studied in thrombectomy-eligible patients with large vessel occlusion, 0.25 mg/kg to a maximum of 25 mg, not alteplase. Option D reflects a lower-dose regimen not endorsed in this chapter. Option A gives the entire dose as a single bolus rather than a 10% bolus followed by a 1-hour infusion, and option C misstates both the maximum dose and the bolus-to-infusion split.

Key point:
alteplase is 0.9 mg/kg, maximum 90 mg, 10% bolus over 1 minute and the rest over 1 hour.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 157

Question 51

Acute Ischemic Stroke · Contraindications to IV alteplase · moderate

A 66-year-old man arrives 2 hours after sudden onset of right hemiparesis and dysarthria, NIHSS 12. Noncontrast CT shows no hemorrhage and no significant edema. Which of the following additional findings, if present, would be an absolute contraindication to intravenous alteplase?

- **A.** Blood pressure of 178/104 mmHg after two doses of intravenous labetalol
- **B.** Serum glucose of 210 mg/dL
- **C.** International normalized ratio of 1.4 in a patient taking warfarin
- **D.** Elective total knee arthroplasty 10 days ago
- **E.** Age of 84 years

Answer: D. Elective total knee arthroplasty 10 days ago

Major surgery in the preceding 14 days is listed as a contraindication to intravenous alteplase, and a knee arthroplasty 10 days ago falls inside that window. The glucose contraindications are a serum glucose above 400 mg/dL or below 50 mg/dL, so 210 mg/dL is acceptable. The anticoagulation contraindication is oral anticoagulant use with an INR above 1.7, so an INR of 1.4 does not disqualify him. Sustained blood pressure above 185/110 mmHg is the contraindication, and 178/104 mmHg after labetalol is below that threshold and therefore permissible. The listed age criterion is age 18 years or older, with no upper age limit in the guidelines or the United States label, although the European label specifies an upper limit of 80 years while the European Stroke Organisation recommends no upper limit. Other bleeding-related exclusions include gastrointestinal bleeding within the preceding 21 days and ischemic stroke or severe head trauma within 3 months.

Key point:
memorize the alteplase exclusion numbers, glucose above 400 or below 50, INR above 1.7, major surgery within 14 days, gastrointestinal bleeding within 21 days, and stroke or severe head trauma within 3 months.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 157-158

Question 52

Acute Ischemic Stroke · Blood pressure management after thrombolysis · easy

A 70-year-old man with a left MCA syndrome required two doses of intravenous labetalol to bring his blood pressure below 185/110 mmHg before alteplase was started. The infusion is now complete and his blood pressure has drifted up to 186/104 mmHg in the neurointensive care unit. Which of the following is the correct blood pressure goal for this patient?

- **A.** Below 220/120 mmHg for the next 24 hours
- **B.** Below 185/110 mmHg for the duration of the infusion only, with no goal thereafter
- **C.** Below 180/105 mmHg for 24 hours after the alteplase infusion
- **D.** Below 160/90 mmHg for 24 hours after the alteplase infusion
- **E.** Below 140/90 mmHg for 72 hours after the alteplase infusion

Answer: C. Below 180/105 mmHg for 24 hours after the alteplase infusion

Blood pressure must be lowered below 185/110 mmHg before alteplase is given, and then maintained below 180/105 mmHg during the infusion and for 24 hours afterward, because elevated blood pressure increases the risk of hemorrhagic complications of thrombolysis. Option A is the threshold that applies to patients who did not receive thrombolytics, in whom blood pressure is generally lowered only above 220/120 mmHg. Option B is wrong because the requirement extends well past the infusion. Options D and E impose targets lower and longer than any guideline recommendation and risk hypotension, which threatens the penumbra and can extend the stroke. Options for control include labetalol 10 to 20 mg intravenously over 1 to 2 minutes, or infusions of nicardipine or clevidipine.

Key point:
185/110 mmHg before alteplase, 180/105 mmHg for the 24 hours after it.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 157, 163

Question 53

Acute Ischemic Stroke · Blood pressure management without thrombolysis · moderate

A 74-year-old woman presents 14 hours after last known well with a right MCA syndrome. She is not a candidate for thrombolysis or thrombectomy. Blood pressure is 208/112 mmHg. ECG shows no ST elevation, troponin is normal, and there is no hematuria. Her home medications are metoprolol, clonidine, lisinopril, and amlodipine. Which of the following is the most appropriate blood pressure management?

- **A.** Withhold the lisinopril and amlodipine, continue metoprolol and clonidine at half the home dose, and treat only if blood pressure exceeds 220/120 mmHg
- **B.** Start a nicardipine infusion targeting a systolic blood pressure below 140 mmHg
- **C.** Resume all four home antihypertensives at full dose immediately
- **D.** Begin an intravenous sodium nitroprusside infusion now
- **E.** Lower the blood pressure below 185/110 mmHg with intravenous labetalol

Answer: A. Withhold the lisinopril and amlodipine, continue metoprolol and clonidine at half the home dose, and treat only if blood pressure exceeds 220/120 mmHg

In a patient who has not received thrombolysis, current American Stroke Association and European Stroke Organisation guidance is to lower blood pressure only above 220/120 mmHg, provided there is no end-organ damage such as cardiac ischemia with ST elevation or a troponin rise above 1 mcg/L, or acute renal failure with hematuria. It is common practice to discontinue prestroke antihypertensives, with two exceptions: beta-blockers are continued, usually at lower doses, because abrupt withdrawal risks myocardial infarction, and clonidine is continued at half the home dose because abrupt discontinuation causes rebound hypertension. Options B and E impose thresholds that belong to thrombolysed patients or are lower still, and aggressive lowering risks extending the infarct by threatening penumbral perfusion. Option C ignores the standard practice of holding most agents. Sodium nitroprusside is reserved for blood pressure that is not controlled by first-line agents or for a diastolic blood pressure above 140 mmHg, so option D is premature.

Key point:
without thrombolysis, treat only above 220/120 mmHg, but keep the beta-blocker and continue clonidine at half dose.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 163-164

Question 54

Acute Ischemic Stroke · Wake-up stroke and DWI-FLAIR mismatch · moderate

A 59-year-old woman awoke at 6:00 am with aphasia and right hemiparesis. She was last seen normal at 11:00 pm. She arrives at 7:30 am with an NIHSS of 9. Noncontrast CT shows no hemorrhage and no established infarct. CT angiography shows no large vessel occlusion. MRI shows a left MCA territory lesion that is hyperintense on diffusion-weighted imaging with no corresponding signal change on fluid attenuated inversion recovery. Which of the following is the most appropriate next step?

- **A.** Give aspirin 325 mg and admit for observation
- **B.** Repeat the MRI in 6 hours to look for FLAIR evolution
- **C.** Proceed to mechanical thrombectomy
- **D.** Start therapeutic intravenous heparin
- **E.** Administer intravenous alteplase at 0.9 mg/kg

Answer: E. Administer intravenous alteplase at 0.9 mg/kg

In patients with unknown onset time, a lesion that is positive on diffusion-weighted imaging but not yet visible on fluid attenuated inversion recovery suggests the stroke is hyperacute and likely within the treatment window. The WAKE-UP trial randomized more than 500 such patients and found a favorable outcome at 90 days in 53% of alteplase-treated patients versus 42% of placebo patients, with an odds ratio of 1.6 and a 95% confidence interval of 1.09 to 2.36. As with all thrombolysis trials there was an increased risk of symptomatic intracerebral hemorrhage, but the overall signal of efficacy was not negated by it. Option C is not applicable because CT angiography shows no large vessel occlusion, and thrombectomy trials required a proven occlusion. Option D is wrong because intravenous heparin has not been shown to prevent early recurrent stroke or improve outcome. Options A and B forfeit a time-sensitive opportunity, and because outcome is so time-dependent it is better to treat than to delay.

Key point:
diffusion-positive, FLAIR-negative unknown-onset stroke is a WAKE-UP indication for intravenous alteplase.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 155

Question 55

Acute Ischemic Stroke · Extended-window thrombolysis with perfusion selection · hard

A 68-year-old man presents 7 hours after last known well with mild right-sided weakness and dysarthria, NIHSS 8. Noncontrast CT shows no hemorrhage. CT angiography shows no large vessel occlusion. CT perfusion shows an ischemic core of 12 mL and a hypoperfused volume of 60 mL. Which of the following statements best supports the management decision here?

- **A.** Alteplase is contraindicated beyond 4.5 hours from onset regardless of the perfusion imaging findings
- **B.** Intravenous alteplase may be offered, since the EXTEND trial showed better functional outcome with perfusion-mismatch selection between 4.5 and 9 hours from onset
- **C.** Intra-arterial alteplase should be given as salvage therapy in place of the intravenous route
- **D.** He should proceed to thrombectomy on the basis of the DAWN trial criteria
- **E.** Tenecteplase is the only thrombolytic that has been studied beyond the 4.5-hour window

Answer: B. Intravenous alteplase may be offered, since the EXTEND trial showed better functional outcome with perfusion-mismatch selection between 4.5 and 9 hours from onset

Two imaging-based extended-window trials of thrombolysis versus placebo have been published, ECASS 4 using magnetic resonance mismatch and EXTEND using CT or MR mismatch, and both enrolled patients between 4.5 and 9 hours from onset or with wake-up stroke within 9 hours of the midpoint of sleep. ECASS 4 did not show a significant benefit over placebo, but EXTEND produced better functional outcome among patients with ischemic stroke and salvageable penumbra, which is exactly this patient's profile with a small core and a large mismatch. Notably, only about half of the patients in these trials had large vessel occlusions, so imaging-selected late thrombolysis fills a gap for patients like this one who are not thrombectomy candidates. Option A ignores the imaging-selected extended window noted in the alteplase indication table. Option D is wrong because DAWN required an internal carotid artery terminus or M1 occlusion, which he does not have. Option C is wrong because intra-arterial alteplase alone has not been shown to be beneficial as salvage therapy and should not substitute for the intravenous route in eligible patients.

Key point:
EXTEND supports intravenous alteplase for perfusion-mismatch patients 4.5 to 9 hours out, including wake-up stroke within 9 hours of the sleep midpoint.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 156

Question 56

Acute Ischemic Stroke · Tenecteplase · easy

A comprehensive stroke center uses tenecteplase rather than alteplase for patients with large vessel occlusion who are eligible for thrombectomy. Which of the following is the tenecteplase regimen studied in this population?

- **A.** 0.25 mg/kg as a single intravenous bolus, maximum 25 mg
- **B.** 0.9 mg/kg total, maximum 90 mg, with 10% as a bolus and the remainder over 1 hour
- **C.** 0.5 mg/kg as a single intravenous bolus, maximum 50 mg
- **D.** 0.4 mg/kg infused over 1 hour with no bolus
- **E.** 1.0 mg/kg as a single intravenous bolus, maximum 100 mg

Answer: A. 0.25 mg/kg as a single intravenous bolus, maximum 25 mg

Tenecteplase is a tissue plasminogen activator with greater fibrin specificity and a longer half-life than alteplase, allowing single-bolus administration at 0.25 mg/kg to a maximum of 25 mg. In patients with internal carotid, basilar, or middle cerebral artery occlusion who were eligible for thrombectomy, tenecteplase was noninferior to alteplase, with reperfusion of more than 50% of the ischemic territory or absence of thrombus on subsequent angiography in 22% versus 10%, P equal to 0.002. It was also associated with a better modified Rankin Scale at 90 days as a secondary outcome, although not with more patients left with minimal or no deficit, and a large Norwegian study failed to show noninferiority, limited in part by a high number of stroke mimics. Option B is the alteplase regimen. Options C, D, and E are not studied stroke regimens. Tenecteplase is not approved for use in ischemic stroke, but updated guidelines support it as an alternative to alteplase in patients with minor stroke symptoms or those eligible for thrombectomy.

Key point:
tenecteplase is a single 0.25 mg/kg bolus, maximum 25 mg, and is a guideline-supported alternative in minor stroke and thrombectomy-eligible patients.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 158

Question 57

Acute Ischemic Stroke · Orolingual angioedema after alteplase · moderate

A 63-year-old woman taking lisinopril is 20 minutes into an intravenous alteplase infusion for an acute left MCA stroke when the nurse notes asymmetric swelling of her tongue and lower lip. She is phonating normally with no stridor and oxygen saturation is 98%. Which of the following is the most appropriate immediate management?

- **A.** Continue the alteplase infusion and give intravenous diphenhydramine alone
- **B.** Continue the alteplase infusion and observe with an airway cart at the bedside
- **C.** Stop the alteplase and proceed directly to surgical cricothyrotomy
- **D.** Stop the alteplase, hold the angiotensin-converting enzyme inhibitor, and give intravenous methylprednisolone, diphenhydramine, and famotidine
- **E.** Stop the alteplase and administer intravenous protamine sulfate and cryoprecipitate

Answer: D. Stop the alteplase, hold the angiotensin-converting enzyme inhibitor, and give intravenous methylprednisolone, diphenhydramine, and famotidine

Acute orolingual angioedema can occur immediately after initiation of intravenous alteplase and is treated by stopping the alteplase, holding angiotensin-converting enzyme inhibitors, and giving intravenous methylprednisolone 125 mg, diphenhydramine 50 mg, and famotidine 20 mg. If the angioedema continues to progress, epinephrine 0.1% may be given as 0.3 mL subcutaneously or 0.5 mL by nebulizer. Options A and B are unsafe because continuing the drug allows progression, and angiotensin-converting enzyme inhibitors are a recognized potentiating factor that must be held. Option C skips the appropriate airway ladder; endotracheal intubation may be needed for edema of the oropharynx, larynx, or floor of the mouth, and awake fiberoptic intubation is the optimal technique, with surgical airway reserved as a last resort. Option E is not indicated, since these agents do not reverse alteplase-associated angioedema. Airway compromise, not the swelling itself, is what mandates urgent tracheal intubation.

Key point:
for alteplase-associated orolingual edema, stop the drug, hold the ACE inhibitor, give methylprednisolone 125 mg, diphenhydramine 50 mg and famotidine 20 mg, and escalate to epinephrine and awake fiberoptic intubation if it progresses.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 157

Question 58

Acute Ischemic Stroke · Endovascular thrombectomy in the early window · moderate

A 70-year-old man presents 3 hours after onset of a left hemispheric syndrome with an NIHSS of 18. Noncontrast CT shows an ASPECTS of 8. CT angiography shows a left internal carotid artery terminus occlusion. Intravenous alteplase has been started and a neurointerventional team is immediately available. Which of the following statements best supports proceeding directly to thrombectomy?

- **A.** A mandatory observation period after alteplase should first be completed to see whether the deficit improves
- **B.** Thrombectomy benefit is limited to patients who are ineligible for or did not receive intravenous alteplase
- **C.** Thrombectomy benefit is restricted to patients younger than 80 years with an NIHSS below 20
- **D.** Thrombectomy improves recanalization rates but has not been shown to improve functional outcome
- **E.** Stent-retriever thrombectomy within 6 hours improves functional outcome with a number needed to treat of 2 to 3, across a broad range of ages and NIHSS scores

Answer: E. Stent-retriever thrombectomy within 6 hours improves functional outcome with a number needed to treat of 2 to 3, across a broad range of ages and NIHSS scores

A meta-analysis of 1,287 patients from the five 2015 stent-retriever trials, MR CLEAN, ESCAPE, REVASCAT, SWIFT PRIME, and EXTEND IA, showed conclusively that embolectomy improves clinical outcome when performed within 6 hours of symptom onset for anterior circulation ischemia, that it is safe to combine with intravenous alteplase, that the results are robust across a broad range of NIHSS scores and ages, and that outcomes are better the faster the patient is treated. The number needed to treat to improve clinical outcome is 2 to 3, and each minute of delay costs the patient another day of disability. The American Stroke Association and European Stroke Organisation both recommend anterior circulation embolectomy if groin puncture can occur within 6 hours. Option A refers to the obligatory waiting period used in MR CLEAN to see whether alteplase produced improvement, which is explicitly not recommended because outcomes are time sensitive. Option B is contradicted by the meta-analysis, in which benefit was essentially identical in patients who did and did not receive alteplase. Options C and D are contradicted by the same pooled data, which showed benefit including in patients 80 years and older and with NIHSS 21 or greater.

Key point:
within 6 hours, stent-retriever thrombectomy for anterior circulation large vessel occlusion has a number needed to treat of 2 to 3, and no post-alteplase waiting period is justified.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 158-159

Question 59

Acute Ischemic Stroke · DAWN trial selection criteria · hard

An 83-year-old woman is brought in 20 hours after last known well with a right hemispheric syndrome and an NIHSS of 14. CT angiography shows a right internal carotid artery terminus occlusion, and automated perfusion software reports an infarct core of 18 mL. Which of the following best describes her candidacy for mechanical thrombectomy?

- **A.** She is ineligible because thrombectomy trials excluded patients older than 80 years
- **B.** She is eligible under DAWN group A criteria: age 80 years or older, NIHSS of 10 or more, and infarct core below 21 mL
- **C.** She is eligible only if the infarct core is below 31 mL and the NIHSS is greater than 20
- **D.** She is ineligible because the 6-to-24-hour window applies only to basilar artery occlusion
- **E.** She is eligible under DEFUSE 3, which enrolled patients presenting 6 to 16 hours after last known well

Answer: B. She is eligible under DAWN group A criteria: age 80 years or older, NIHSS of 10 or more, and infarct core below 21 mL

DAWN randomized patients with internal carotid artery terminus or M1 occlusion, or both, in the 6-to-24-hour window using CT or MR perfusion imaging, and stratified eligibility into three groups. Group A is age 80 years or older with an NIHSS of 10 or more and an infarct core below 21 mL, which this patient meets exactly at age 83, NIHSS 14, and core 18 mL. Group B is age below 80 with an NIHSS of 10 or more and a core below 31 mL, and group C is age below 80 with an NIHSS greater than 20 and a core of 31 to 50 mL, so options C misapplies the younger-patient criteria. The trial enrolled 206 subjects, met both coprimary endpoints, and was stopped early for efficacy, with functional independence in 49% of the thrombectomy group versus 13% of the medical group. Option A is wrong because DAWN specifically created a group for patients 80 and older, and the pooled early-window data also showed benefit in that age band. Option D is wrong because basilar embolectomy has not actually been studied in randomized trials, and the late-window trials were anterior circulation studies. Option E is wrong on timing, since DEFUSE 3 enrolled 6 to 16 hours and she is at 20 hours.

Key point:
in the 6-to-24-hour window, DAWN group A allows thrombectomy in patients 80 or older with NIHSS 10 or more and a core under 21 mL.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 162

Question 60

Acute Ischemic Stroke · DEFUSE 3 trial selection criteria · moderate

A 61-year-old man presents 9 hours after last known well with an NIHSS of 12. CT angiography shows a left M1 occlusion. CT perfusion shows an ischemic core of 45 mL and a total hypoperfused volume of 110 mL. Which of the following sets of criteria establishes his eligibility for thrombectomy?

- **A.** Infarct core below 21 mL with an NIHSS of 10 or more
- **B.** Infarct core below 31 mL with age below 80 years
- **C.** Infarct core below 70 mL with a ratio of tissue at risk to core greater than 1.8, presenting 6 to 16 hours after last known well
- **D.** Infarct core of 31 to 50 mL with an NIHSS greater than 20
- **E.** Any infarct core volume, provided the patient presents within 24 hours

Answer: C. Infarct core below 70 mL with a ratio of tissue at risk to core greater than 1.8, presenting 6 to 16 hours after last known well

DEFUSE 3 studied late-window anterior circulation patients between 6 and 16 hours from last known well who had evidence of salvageable tissue on diffusion-weighted MR perfusion or CT perfusion mismatch, requiring a core volume below 70 mL and a ratio of tissue at risk to core volume greater than 1.8. This patient at 9 hours with a 45 mL core and a mismatch ratio of about 2.4 satisfies all three requirements. The odds ratio for a favorable outcome of modified Rankin Scale 0 to 2 was 2.77, P less than 0.001, favoring endovascular therapy, with a borderline-significant drop in mortality from 26% to 14%, P equal to 0.05. Options A, B, and D are the DAWN group A, B, and C criteria respectively, which are narrower and would exclude this man because his core exceeds 31 mL while his NIHSS is not above 20. Option E is wrong because core volume ceilings are the central selection principle in both late-window trials.

Key point:
DEFUSE 3 is the broader late-window rule, 6 to 16 hours, core below 70 mL, and mismatch ratio greater than 1.8.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 162

Question 61

Acute Ischemic Stroke · Antithrombotic timing after reperfusion therapy · moderate

A 66-year-old man presents 8 hours after last known well with an M1 occlusion and is taken directly for mechanical thrombectomy without intravenous thrombolysis, achieving complete recanalization. He returns to the neurointensive care unit 6 hours after the procedure, neurologically improved, with a stable examination. The 24-hour follow-up CT has not yet been obtained. Which of the following is the most appropriate antithrombotic management?

- **A.** Aspirin may be started within the first 24 hours, because antithrombotics are not contraindicated after endovascular therapy alone
- **B.** Withhold all antithrombotics for 24 hours, because the post-thrombolysis rule applies to any reperfusion therapy
- **C.** Start a therapeutic intravenous heparin infusion to prevent reocclusion
- **D.** Load with clopidogrel in addition to aspirin as routine post-stroke therapy
- **E.** Withhold all antithrombotics for 7 days given the size of the territory at risk

Answer: A. Aspirin may be started within the first 24 hours, because antithrombotics are not contraindicated after endovascular therapy alone

The 24-hour antithrombotic moratorium applies specifically to patients who received intravenous thrombolytics; the use of antithrombotics following endovascular therapy alone is not contraindicated in the first 24 hours. Because this patient never received alteplase, option B misapplies the rule. Aspirin has the most data in the acute stroke setting when given within 48 hours, and unless alteplase has been administered, aspirin 81 to 325 mg should be given as soon as feasible, within 24 hours, with the rectal route available if the patient cannot swallow and has no enteral access. Option C is wrong because several trials found little to no advantage to routine heparin, and most favor antiplatelet agents; heparin is reserved for selected situations such as stroke in progression with large vessel intracranial occlusion or basilar thrombosis where aspirin has failed. Option D is wrong because there are no recommendations for clopidogrel or aspirin-dipyridamole in acute ischemic stroke outside of initiation for secondary prevention, and dual antiplatelet loading in this chapter is tied to acute carotid or intracranial stenting, given after a control CT at 6 to 12 hours excludes hemorrhage. Option E has no basis and needlessly exposes the patient to early recurrence.

Key point:
hold antithrombotics for 24 hours after intravenous thrombolysis, but not after thrombectomy alone.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 165

Question 62

Acute Ischemic Stroke · Malignant cerebral edema and decompressive hemicraniectomy · hard

A 48-year-old right-handed man had a left internal carotid artery terminus occlusion with failed recanalization. Thirty hours after onset he becomes progressively drowsy, opening his eyes only to noxious stimulation. Repeat CT shows hypodensity involving more than two-thirds of the left MCA territory including the basal ganglia, with 4 mm of midline shift. Which of the following is the most appropriate management?

- **A.** Intravenous dexamethasone 10 mg followed by 4 mg every 6 hours
- **B.** Induced hypothermia to 33 degrees Celsius
- **C.** Placement of an intracranial pressure monitor, treating only if the pressure exceeds 22 mmHg
- **D.** Start osmotherapy and proceed urgently to decompressive hemicraniectomy
- **E.** Defer surgery because the infarct involves the dominant hemisphere

Answer: D. Start osmotherapy and proceed urgently to decompressive hemicraniectomy

Malignant swelling is most likely in younger patients with proximal internal carotid artery or M1 occlusion and poor collaterals, with extensive early infarction signs in more than two-thirds of the hemisphere including the basal ganglia, which describes this patient exactly. The most specific sign of significant cerebral edema after stroke is an interval decline in the level of consciousness, and there is no reason to wait for overt herniation or a large midline shift; hemicraniectomy should as a rule be performed within 48 hours of onset and preferably within the first 24 hours, with earlier treatment yielding better outcomes in the randomized trials. Where craniectomy is the chosen first-line therapy, osmotherapy is started while surgery is being prepared, using mannitol 0.5 to 1.5 g/kg or hypertonic saline. Option A is wrong because the edema is cytotoxic, and corticosteroids in conventional or large doses lack efficacy and increase infectious complications. Option B is wrong because there are insufficient data on hypothermia and barbiturates in ischemic swelling and they are not recommended. Option C is misleading because clinical deterioration results from displacement of midline structures rather than globally raised pressure, and intracranial pressure is in fact not elevated in the early days after hemispheric infarct. Option E is wrong because subgroup analysis showed no difference in eventual disability between dominant and nondominant hemicraniectomy.

Key point:
declining consciousness with a greater-than-two-thirds MCA infarct calls for osmotherapy plus early hemicraniectomy, within 48 hours and ideally within 24, regardless of hemispheric dominance.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 167

Question 63

Acute Ischemic Stroke · Cerebellar infarction with mass effect and hydrocephalus · moderate

A 57-year-old woman with a right posterior inferior cerebellar artery territory infarct becomes increasingly somnolent on hospital day 2. CT shows cerebellar swelling with effacement of the fourth ventricle and enlarged lateral and third ventricles. Which of the following is the most appropriate management?

- **A.** Placement of an external ventricular drain as definitive therapy
- **B.** Mannitol alone with hourly neurological examinations
- **C.** Dexamethasone with continued observation
- **D.** Induction of barbiturate coma
- **E.** Suboccipital decompressive craniectomy, with ventriculostomy added as needed

Answer: E. Suboccipital decompressive craniectomy, with ventriculostomy added as needed

All cerebellar infarct patients should be closely monitored and referred for suboccipital decompression if significant mass effect from edema is present, and an infarct that is large for the posterior fossa is one of the situations in which decompressive surgery is the recommended first-line therapy rather than osmotherapy. Ventriculostomy is recommended for obstructive hydrocephalus after a cerebellar infarct, but it should be followed or accompanied by decompressive craniectomy, so option A alone is inadequate and risks upward displacement without relieving the posterior fossa mass effect. Option B is inadequate as sole therapy, since osmotherapy works quickly but only for a short time and should be used while surgery is being prepared. Option C is wrong because the edema is cytotoxic and corticosteroids are ineffective and increase infectious complications. Option D is wrong because there are insufficient data on barbiturates in ischemic cerebellar swelling and they are not recommended. Removal of infarcted cerebellar tissue is an option, particularly with hemorrhagic transformation, and close monitoring is essential for the first several days because sudden respiratory arrest and death can occur.

Key point:
for a swelling cerebellar infarct, suboccipital decompression is the definitive treatment, and a ventriculostomy must never stand alone.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 167-168

Question 64

Acute Ischemic Stroke · Glucose management after acute ischemic stroke · easy

A 70-year-old man with type 2 diabetes received intravenous alteplase 6 hours ago for a right MCA stroke. His serum glucose is 265 mg/dL on two consecutive checks in the neurointensive care unit. Which of the following is the appropriate serum glucose target?

- **A.** 80 to 110 mg/dL using a continuous insulin infusion
- **B.** 110 to 140 mg/dL using a continuous insulin infusion
- **C.** 140 to 180 mg/dL
- **D.** 180 to 220 mg/dL
- **E.** No treatment unless the glucose exceeds 300 mg/dL

Answer: C. 140 to 180 mg/dL

Current recommendations for acute ischemic stroke patients are to keep the serum glucose in the 140 to 180 mg/dL range, and this matters particularly after alteplase because hyperglycemia portends a higher rate of intracerebral hemorrhage. Option A is the tight-control target from the earlier surgical intensive care unit trial that reduced all-cause mortality from 8% to 4.6% among 1,548 patients and drove widespread use of insulin drips, but the later NICE-SUGAR trial found increased mortality and a significantly higher risk of hypoglycemia with intensive insulin infusion therapy, so this target is no longer used. Option B is likewise below the recommended range and carries hypoglycemia risk. Option D is above the recommended range. Option E is wrong: although serum glucose above 300 mg/dL has been shown to exacerbate stroke in animal models, likely through local acidosis in ischemic tissue, treatment should begin well below that level.

Key point:
target 140 to 180 mg/dL after ischemic stroke, because NICE-SUGAR showed that tighter control causes harm.

Source: Practice of Neurocritical Care, 2nd ed., Acute Ischemic Stroke, pp. 168

Acute Neuromuscular Disorders

Question 65

Acute Neuromuscular Disorders · Neuromuscular respiratory failure: PFT thresholds and airway decision · moderate

A 37-year-old woman presents with one week of ascending paresthesias and weakness, mild slurring of speech, and dyspnea that is worse when supine. Vital signs: temperature 37.0 C, heart rate 118/min, blood pressure 152/88 mm Hg, respiratory rate 28/min, oxygen saturation 96% on room air. She is anxious and diaphoretic and uses accessory muscles with each breath. Examination shows proximal greater than distal weakness in all four limbs, diffuse areflexia, and a stocking-glove sensory deficit. Chest radiograph is clear. Arterial blood gas: pH 7.42, $PaCO_2$ 44 mm Hg, PaO_2 78 mm Hg. Bedside pulmonary function testing performed by a respiratory therapist with an adequate seal shows forced vital capacity 14 mL/kg, maximal inspiratory pressure -23 cm H_2O, and maximal expiratory pressure 32 cm H_2O. Which of the following is the most appropriate next step in management?

- **A.** Continue close observation with pulmonary function tests every 4 hours
- **B.** Initiate noninvasive ventilation with bilevel positive airway pressure
- **C.** Apply a nonrebreather mask at 15 L/min
- **D.** Perform elective endotracheal intubation and initiate mechanical ventilation
- **E.** Start high-flow nasal cannula oxygen with incentive spirometry

Answer: D. Perform elective endotracheal intubation and initiate mechanical ventilation

This presentation is characteristic of Guillain-Barre syndrome, and her pulmonary function tests violate the "20/30/40 rule" (forced vital capacity < 20 mL/kg, maximal inspiratory pressure weaker than -30 cm H_2O, maximal expiratory pressure < 40 cm H_2O), and her maximal inspiratory pressure of -23 cm H_2O is weaker than -30, values validated principally in Guillain-Barre syndrome and associated with progression to respiratory failure. In Guillain-Barre syndrome specifically, a forced vital capacity below 15 mL/kg, or a greater than 30% decline in any of these variables, necessitates intubation and mechanical ventilation, and her forced vital capacity of 14 mL/kg is below that threshold. Continued monitoring alone (A) ignores objective evidence of established respiratory muscle failure and risks emergent rather than elective intubation, which the chapter explicitly advises against. Bilevel positive airway pressure (B) does not appear to be a safe or effective option in Guillain-Barre syndrome. Supplemental oxygen by nonrebreather (C) or high-flow nasal cannula (E) treats hypoxemia from microatelectasis but does nothing for the failing respiratory pump, and normal blood gases do not exclude significant neuromuscular respiratory involvement because a nonsustainable tachypnea can hold the $PaCO_2$ down; hypercarbia is a late finding.

Key point:
In Guillain-Barre syndrome, intubate for a forced vital capacity below 15 to 20 mL/kg, maximal inspiratory pressure weaker than -30 cm H_2O, maximal expiratory pressure below 40 cm H_2O, or a greater than 30% drop in any of these, and do not rely on bilevel positive airway pressure.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 181-182, 185

Question 66

Acute Neuromuscular Disorders · Bedside signs of diaphragmatic failure · easy

A 67-year-old man presents with progressive ascending paresthesias, weakness, and shortness of breath. He is anxious and pauses between words to catch his breath. Heart rate is 110/min and respiratory rate is 26/min with an oxygen saturation of 95%. Contraction of the sternocleidomastoid, pectoral, and latissimus dorsi muscles is noted with each inspiration. He can count only to 12 on a single breath. With one hand on the chest and one on the abdomen in the supine position, the abdomen is seen to move inward during inspiration. Examination also reveals facial diplegia, proximal greater than distal weakness, and areflexia. Which of the following findings is the most specific indicator of diaphragmatic failure?

- **A.** Facial diplegia
- **B.** Single-breath count of 12
- **C.** Sternocleidomastoid contraction with inspiration
- **D.** Heart rate of 110/min
- **E.** Inward movement of the abdomen during inspiration

Answer: E. Inward movement of the abdomen during inspiration

Paradoxical breathing, the inward movement of the abdomen during inspiration, reflects failure of diaphragmatic descent and the resulting loss of the normal rise in intra-abdominal pressure, so that unopposed negative intrathoracic pressure draws abdominal contents into the thorax. It is the cardinal clinical expression of diaphragmatic failure in Guillain-Barre syndrome and is a late sign that often signals the need for immediate intubation. A reduced single-breath count (B) is a useful bedside surrogate for vital capacity, since counting from 1 to 20 in one-second increments approximates preserved vital capacity, but it is nonspecific and reflects overall respiratory reserve rather than diaphragmatic mechanics. Accessory muscle recruitment (C), detectable by inspection or by palpating the "respiratory pulse," and tachycardia (D) indicate increased work of breathing and are nearly always present but are not specific to the diaphragm. Facial diplegia (A) is a common cranial nerve finding in Guillain-Barre syndrome and points to oropharyngeal rather than diaphragmatic weakness.

Key point:
Abdominal paradox on inspiration is the bedside hallmark of diaphragmatic failure and portends imminent need for ventilatory support.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 179, 183-184

Question 67

Acute Neuromuscular Disorders · GBS subtypes and pathophysiology (AMAN) · moderate

A 42-year-old man is admitted with generalized weakness and dyspnea. Two weeks earlier he returned from Mexico with a diarrheal illness that resolved with supportive care. He developed leg weakness one week later that ascended to involve the arms, followed by dysarthria and choking with meals. Examination shows facial diplegia, a weak cough and gag, proximal and distal weakness in all limbs, diffuse areflexia, and a normal sensory examination. Cerebrospinal fluid shows elevated protein with a normal white cell count. Serum GM1 antibodies are present and stool culture grows Campylobacter jejuni. Nerve conduction studies show diffuse reduction in the excitability of motor nerves. Which of the following best describes the mechanism of this patient's neuropathy?

- **A.** CD4 T cell-mediated attack on myelin proteins P2, P0, and PMP22 with macrophage-mediated demyelination
- **B.** Antibodies against gangliosides on the axolemma that activate macrophages attacking the nodes of Ranvier
- **C.** Autoantibodies against presynaptic voltage-gated calcium channels impairing acetylcholine vesicle release
- **D.** Autoantibody attack on postsynaptic acetylcholine receptors of the neuromuscular junction
- **E.** Thick-filament myosin loss with muscle fiber inexcitability from an acquired sodium channelopathy

Answer: B. Antibodies against gangliosides on the axolemma that activate macrophages attacking the nodes of Ranvier

The antecedent Campylobacter jejuni diarrhea, GM1 antibodies, pure motor deficit with normal sensation, and early motor nerve inexcitability identify acute motor axonal neuropathy, the axonal form of Guillain-Barre syndrome. In this variant, antibodies directed against antigens on the axolemma activate macrophages that attack the nodes of Ranvier, producing axonal damage with relative preservation of the myelin sheath; molecular mimicry between the Campylobacter jejuni bacterial wall and gangliosides in peripheral axonal membranes is thought to be responsible. Option A describes acute inflammatory demyelinating polyradiculoneuropathy, in which a CD4 T cell-mediated response against myelin proteins P2, P0, or PMP22 drives demyelination and no antibody has a proven pathogenic role. Option C describes Lambert-Eaton myasthenic syndrome, which is paraneoplastic in 50% to 60% of cases, usually with small cell lung cancer, and rarely causes respiratory failure. Option D describes myasthenia gravis, and option E describes critical illness myopathy. Note also that early motor nerve inexcitability is a reliable marker of an axonal polyradiculoneuropathy rather than of severe demyelination.

Key point:
Campylobacter jejuni plus GM1 antibodies plus a pure motor, axonal picture equals acute motor axonal neuropathy caused by antiganglioside antibodies attacking the nodes of Ranvier through molecular mimicry.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 182-183, 185

Question 68

Acute Neuromuscular Disorders · GBS immunotherapy: IVIG regimen · easy

A patient with severe Guillain-Barre syndrome is to be treated with intravenous immunoglobulin. Which of the following is the standard dosing regimen for this indication?

- **A.** 0.4 g/kg per day for 5 days, for a total dose of 2 g/kg
- **B.** 0.4 g/kg as a single dose
- **C.** 1 g/kg per day for 5 days, for a total dose of 5 g/kg
- **D.** 2 g/kg per day for 2 days, for a total dose of 4 g/kg
- **E.** 0.1 g/kg per day for 10 days, for a total dose of 1 g/kg

Answer: A. 0.4 g/kg per day for 5 days, for a total dose of 2 g/kg

The standard intravenous immunoglobulin regimen for Guillain-Barre syndrome is a total dose of 2 g/kg given in divided doses of 0.4 g/kg per day for 5 days; an accepted alternative is 1 g/kg per day for 2 days, which delivers the same 2 g/kg total. Options B and E deliver far less than the required 2 g/kg total, and options C and D deliver two to two and a half times the standard total dose. Intravenous immunoglobulin and plasma exchange are considered equivalent in efficacy, with seven randomized trials analyzed in a Cochrane review showing no significant difference in outcomes, although adverse event rates are lower with intravenous immunoglobulin. The most common adverse event of intravenous immunoglobulin is headache, possibly related to aseptic meningitis, which can also raise the cerebrospinal fluid nucleated cell count and confound the albuminocytologic dissociation; rare serious events include renal failure, myocardial infarction, and stroke. Patients with a smaller rise in serum immunoglobulin G may have more severe, prolonged courses, and those with low immunoglobulin G levels 2 weeks after treatment may benefit from a second course.

Key point:
Intravenous immunoglobulin for Guillain-Barre syndrome is 2 g/kg total, given as 0.4 g/kg per day for 5 days or 1 g/kg per day for 2 days.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 187

Question 69

Acute Neuromuscular Disorders · GBS dysautonomia: adynamic ileus · hard

A 54-year-old man with severe Guillain-Barre syndrome is on day 8 of mechanical ventilation. His course has been complicated by dysautonomia, including two episodes of symptomatic bradycardia to the 30s requiring atropine. He receives fentanyl for severe neuropathic pain. Over 24 hours he develops abdominal distention, absent bowel sounds, and rising gastric residuals with intolerance of enteral feeds. Abdominal radiograph shows diffuse bowel dilation consistent with adynamic ileus, without evidence of mechanical obstruction or perforation. Which of the following is the most appropriate next step in management?

- **A.** Administer intravenous neostigmine
- **B.** Administer intravenous metoclopramide
- **C.** Place a nasogastric tube for intermittent suction and insert a rectal tube
- **D.** Administer oral pyridostigmine
- **E.** Begin parenteral nutrition without gastric or rectal decompression

Answer: C. Place a nasogastric tube for intermittent suction and insert a rectal tube

Adynamic ileus occurs in up to 15% of patients with severe Guillain-Barre syndrome and may be triggered by opiates used for pain; the combined use of intermittent gastric suctioning and a rectal tube usually suffices to allow recovery of intestinal motility without complications. Intravenous neostigmine (A) has been used successfully for adynamic ileus in other populations, but in Guillain-Barre syndrome it is riskier because it can be complicated by severe bradycardia, a particular hazard in this patient who has already had symptomatic bradyarrhythmias. Intravenous metoclopramide (B) stimulates gastric peristalsis but must be used with extreme caution because it has been reported to induce sinus arrest in this population. Oral pyridostigmine (D) is a cholinesterase inhibitor used in myasthenia gravis, carries the same bradycardia risk, and cannot be absorbed reliably in a patient with gastroparesis and ileus. Parenteral nutrition (E) is appropriate only for refractory cases with protracted intolerance of gastric feeding and does not address decompression.

Key point:
In Guillain-Barre syndrome with dysautonomia, treat ileus with gastric suction and a rectal tube, and avoid neostigmine and metoclopramide because of bradycardia and sinus arrest.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 184, 186

Question 70

Acute Neuromuscular Disorders · GBS dysautonomia: bradyarrhythmia management · hard

A 61-year-old woman with severe Guillain-Barre syndrome is quadriplegic and mechanically ventilated. On hospital day 6, during tracheal suctioning, she develops asystole lasting 9 seconds with loss of pulse, treated with atropine and brief chest compressions. The following day she has a second episode of profound bradycardia with hypotension that occurs at rest, with no identifiable trigger. Telemetry shows markedly reduced R-R interval variability, and she has had recurrent severe hypertensive spells alternating with hypotension. Electrolytes are normal and she is on no nodal blocking agents. Which of the following is the most appropriate next step in management?

- **A.** Order scheduled intravenous atropine every 6 hours
- **B.** Start oral propranolol to blunt the autonomic swings
- **C.** Start a nicardipine infusion to control the hypertensive spells
- **D.** Administer a repeat course of intravenous immunoglobulin
- **E.** Arrange elective pacemaker placement

Answer: E. Arrange elective pacemaker placement

Sudden bradycardia and asystole in Guillain-Barre syndrome result from exaggerated vagal activity, and the risk of subsequent severe bradycardia and asystole is increased after an initial episode of symptomatic bradyarrhythmia; such patients should be considered candidates for elective pacemaker placement unless a trigger for the bradycardia can be identified and avoided. Here the second episode had no identifiable trigger, so trigger avoidance alone is insufficient. Reduced R-R interval variation and severe hypertension are themselves predictors of increased risk for serious arrhythmias, reinforcing the need for definitive protection while telemetry monitoring continues and atropine remains at the bedside. Scheduled atropine (A) is not a substitute for pacing, and nodal blocking agents such as propranolol (B) should be avoided in Guillain-Barre syndrome because beta blockers may precipitate hypotension and bradycardia; if a beta blocker is truly necessary, esmolol is preferred for its very short half-life. Nicardipine (C) is a reasonable short-acting agent for sustained hypertension requiring treatment but does not address the life-threatening bradyarrhythmia. Repeat intravenous immunoglobulin (D) is considered for treatment-related fluctuations of weakness, not for autonomic arrhythmias.

Key point:
After a symptomatic bradyarrhythmia in Guillain-Barre syndrome without an avoidable trigger, arrange elective pacemaker placement and avoid nodal blocking agents.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 184, 186

Question 71

Acute Neuromuscular Disorders · Myasthenic crisis: noninvasive ventilation · moderate

A 27-year-old woman with acetylcholine receptor antibody-positive myasthenia gravis presents with 3 days of worsening diplopia, nasal speech, dysphagia, proximal weakness, and dyspnea. Respiratory rate is 30/min, heart rate 113/min, oxygen saturation 94% on room air. She has prominent bulbar weakness with pooling of oral secretions and uses accessory respiratory muscles, but no paradoxical abdominal movement is seen and she is alert and protecting her airway with a weak but effective cough. Bedside pulmonary function testing is unreliable because facial weakness prevents an adequate mouthpiece seal and effort is poor. Arterial blood gas: pH 7.41, PaCO2 40 mm Hg, PaO2 82 mm Hg. Which of the following is the most appropriate next step in management?

- **A.** Perform rapid sequence intubation
- **B.** Initiate noninvasive ventilation with bilevel positive airway pressure
- **C.** Apply a nonrebreather mask at 15 L/min
- **D.** Observe with frequent pulmonary function tests alone
- **E.** Give high-dose intravenous methylprednisolone without ventilatory support

Answer: B. Initiate noninvasive ventilation with bilevel positive airway pressure

In myasthenic exacerbation without hypercapnia, early bilevel positive airway pressure can avert intubation in a substantial proportion of patients, reduces the duration of ventilatory assistance by more than half, shortens intensive care unit stay, and is associated with lower rates of atelectasis and pneumonia than invasive ventilation; the benefit persists despite bulbar weakness or pooled secretions. The effectiveness of noninvasive ventilation declines once hypercapnia has developed, so its early use, while the PaCO2 remains below about 45 mm Hg as here, is crucial. Immediate intubation (A) is reserved for hypercapnia, aspiration, or severe respiratory distress with severe bulbar dysfunction. Supplemental oxygen alone (C) does not unload fatiguing respiratory muscles. Watchful waiting with serial pulmonary function tests (D) is inadequate because the 20/30/40 cutoffs were developed in Guillain-Barre syndrome, are not validated in myasthenia gravis, and are unreliable here given poor seal and the fluctuating nature of the disease. Immunotherapy (E) is essential but works over days and is not a substitute for ventilatory support.

Key point:
In myasthenic crisis with a normal PaCO2, start bilevel positive airway pressure early, even with bulbar weakness and secretions.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 188-190

Question 72

Acute Neuromuscular Disorders · Myasthenia gravis: intubation and paralytic dosing · moderate

A 43-year-old man with known myasthenia gravis is in crisis. Despite 12 hours of bilevel positive airway pressure he becomes somnolent with paradoxical abdominal breathing, and arterial blood gas shows pH 7.28 with $PaCO_2$ 58 mm Hg. The decision is made to perform rapid sequence intubation. Which of the following neuromuscular blocking agents and doses is most appropriate?

- **A.** Succinylcholine 0.5 mg/kg
- **B.** Succinylcholine 1 mg/kg
- **C.** Rocuronium 0.5 mg/kg
- **D.** Rocuronium 1 mg/kg
- **E.** Rocuronium 1.5 mg/kg

Answer: C. Rocuronium 0.5 mg/kg

Because antibody-mediated loss of functional acetylcholine receptors alters the response to agents acting at the neuromuscular junction, nondepolarizing blockers must be given in reduced doses in myasthenia gravis to prevent prolonged blockade; the chapter recommends about 0.5 mg/kg of rocuronium instead of the usual 1 mg/kg. Options D and E give the usual or a higher dose and risk prolonged paralysis that confounds the postintubation neurologic examination and delays weaning. Succinylcholine at any dose (A and B) should generally be avoided in patients with progressive neuromuscular disease because of the risk of inducing severe hyperkalemia, and higher rather than lower doses of depolarizing agents would be needed to achieve the same effect in myasthenia gravis. The clinician should also anticipate that intubation can amplify a sympathetic surge in patients with dysautonomia, and that agents used to blunt it can precipitate hypotension and bradycardia.

Key point:
For intubation in myasthenia gravis, use roughly half the usual nondepolarizing dose, such as rocuronium 0.5 mg/kg, and avoid succinylcholine.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 182, 190

Question 73

Acute Neuromuscular Disorders · Corticosteroids in myasthenic exacerbation · easy

A 58-year-old woman with generalized myasthenia gravis is admitted with moderate bulbar weakness and dyspnea. She is not intubated, has a $PaCO_2$ of 38 mm Hg, and is being managed with close monitoring and plasma exchange. The team plans to add high-dose intravenous methylprednisolone. Which of the following adverse effects of this specific therapy should be anticipated?

- **A.** Clinically meaningful worsening of weakness within days of starting treatment in one-third to one-half of patients
- **B.** Aseptic meningitis presenting as headache
- **C.** Acute renal failure
- **D.** Symptomatic hypocalcemia
- **E.** Line-related bacteremia

Answer: A. Clinically meaningful worsening of weakness within days of starting treatment in one-third to one-half of patients

Corticosteroids are standard of care in myasthenic exacerbation and crisis and should be given to all such patients, but caution is warranted with high-dose intravenous methylprednisolone in nonintubated patients because one-third to one-half will experience clinically meaningful exacerbation of weakness within days of starting treatment, which can precipitate the need for intubation. Headache from aseptic meningitis (B) and rare renal failure, myocardial infarction, and stroke (C) are adverse effects of intravenous immunoglobulin, not corticosteroids. Hypocalcemia (D) and line-related bacteremia (E), along with hypotension, are complications of plasma exchange, where most problems relate to venous access. Daily prednisone at an initial dose of about 1 mg/kg of ideal body weight is used first, with a switch to alternate-day dosing when feasible to limit side effects, and a steroid-sparing agent such as azathioprine or mycophenolate mofetil is reasonably started during the acute hospitalization to allow earlier steroid reduction and reduce recurrent crises. All patients on high-dose corticosteroids should receive gastric protection against peptic ulcers.

Key point:
High-dose intravenous steroids can transiently worsen myasthenic weakness in one-third to one-half of nonintubated patients, so anticipate deterioration and monitor closely.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 187, 191

Question 74

Acute Neuromuscular Disorders · ALS in the ICU: secretion and cough management · moderate

A 63-year-old man with bulbar-onset amyotrophic lateral sclerosis diagnosed 2 years ago is admitted to the neurocritical care unit with aspiration pneumonia. He is on antibiotics and nocturnal bilevel positive airway pressure. He is alert, oriented, and states clearly that he does not want a tracheostomy. Examination shows a weak, ineffective cough with copious retained secretions; measured peak cough expiratory flow is 210 L/min. Which of the following interventions is most appropriate for his secretion clearance?

- **A.** Proceed with tracheostomy and long-term mechanical ventilation
- **B.** Start riluzole to improve cough strength
- **C.** Start edaravone to improve respiratory muscle function
- **D.** Provide a mechanical cough assist device with chest physiotherapy and nebulized mucolytics
- **E.** Perform daily therapeutic bronchoscopy for secretion removal

Answer: D. Provide a mechanical cough assist device with chest physiotherapy and nebulized mucolytics

Secretion management with chest physiotherapy, nebulized mucolytics, and anticholinergics to reduce secretion production is a mainstay of treatment for nearly all amyotrophic lateral sclerosis patients admitted to the neurocritical care unit, and cough assist devices are most beneficial for patients with an impaired cough, defined as a peak cough expiratory flow below 270 L/min. His value of 210 L/min falls under that threshold. Tracheostomy with mechanical ventilation (A) can prolong survival by about 10 months but must align with the patient's goals of care given the poor prognosis, and he has explicitly declined it. Riluzole (B) and edaravone (C) are the only current disease-directed treatments, but they only slow progression, with a modest survival benefit on the order of months, and neither improves cough or acute secretion burden. Daily bronchoscopy (E) is invasive, not a maintenance strategy, and is not recommended in the chapter. Prompt treatment of pulmonary infection and deep vein thrombosis prophylaxis are also key, since most patients with amyotrophic lateral sclerosis die of respiratory complications after 3 to 5 years.

Key point:
In amyotrophic lateral sclerosis, a peak cough expiratory flow below 270 L/min identifies the patient who benefits from a cough assist device.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 191

Question 75

Acute Neuromuscular Disorders · Critical illness neuromyopathy: diagnosis · moderate

A 56-year-old man was admitted with septic shock and multiorgan failure requiring vasopressors, mechanical ventilation, and 5 days of continuous neuromuscular blockade, which was stopped 10 days ago. His infection has resolved, but he now repeatedly fails spontaneous breathing trials. Examination shows flaccid quadriparesis that is symmetric, depressed deep tendon reflexes, and preserved facial and extraocular movements. Serum creatine kinase is 180 U/L, electrolytes are normal, and brain and spine imaging and cerebrospinal fluid studies are unremarkable. Which of the following electrodiagnostic findings would most strongly support the diagnosis of critical illness neuromyopathy?

- **A.** A greater than 10% decremental compound muscle action potential response with repetitive nerve stimulation at 2 to 5 Hz
- **B.** Elevated cerebrospinal fluid protein with a normal cell count
- **C.** Diffuse nerve root enhancement on magnetic resonance imaging of the spine
- **D.** Markedly elevated creatine kinase with myoglobinuria
- **E.** Severe reduction in compound muscle action potential amplitude with prolonged compound muscle action potential duration and muscle inexcitability on direct muscle stimulation

Answer: E. Severe reduction in compound muscle action potential amplitude with prolonged compound muscle action potential duration and muscle inexcitability on direct muscle stimulation

Severe reduction in compound muscle action potential amplitude together with prolongation of compound muscle action potential duration are the most useful electrodiagnostic findings in critical illness neuromyopathy, and direct muscle stimulation demonstrating muscle inexcitability supports critical illness myopathy and helps distinguish it from critical illness polyneuropathy; the inexcitability reflects an acquired sodium channelopathy. A decremental response on repetitive nerve stimulation (A) is not seen in critical illness neuromyopathy, where repetitive stimulation is normal, and would instead suggest a neuromuscular junction disorder such as residual neuromuscular blockade or myasthenia gravis. Albuminocytologic dissociation (B) and nerve root enhancement (C) point toward Guillain-Barre syndrome, whereas neuroimaging and lumbar puncture are unremarkable in critical illness neuromyopathy. Marked creatine kinase elevation with myoglobinuria (D) suggests rhabdomyolysis or the rare acute necrotizing myopathy of intensive care; in ordinary critical illness neuromyopathy creatine kinase is usually normal or only mildly elevated. His risk factors of sepsis, multiorgan failure, prolonged ventilation, and neuromuscular blockade are typical, and there is no specific treatment, so prevention with intensive insulin therapy, minimized sedation, and early rehabilitation starting as soon as 48 hours after admission is the priority.

Key point:
In critical illness neuromyopathy, look for low-amplitude, prolonged compound muscle action potentials with muscle inexcitability on direct stimulation, and normal repetitive nerve stimulation.

Source: Practice of Neurocritical Care, 2nd ed., Acute Neuromuscular Disorders, pp. 191-192

Clinical Evaluation of Coma and Brain Death

Question 76

Clinical Evaluation of Coma and Brain Death · Metabolic versus structural coma · easy

A 58-year-old woman with alcoholic cirrhosis is brought to the emergency department after two days of worsening confusion and is now unresponsive. Temperature is 37.1 C, blood pressure 118/70 mmHg, heart rate 96/min, and respirations are 26/min with a smooth crescendo-decrescendo pattern interrupted by brief apneic pauses. Pupils are 3 mm and briskly reactive bilaterally. Oculocephalic reflexes are intact. She withdraws symmetrically in all four limbs to nail bed pressure without posturing. Noncontrast head CT is unremarkable. Ammonia is elevated, and EEG shows diffuse slowing with triphasic waves. Which of the following examination findings most strongly supports a metabolic rather than a structural cause of her coma?

- **A.** The crescendo-decrescendo respiratory pattern with apneic pauses
- **B.** Symmetric withdrawal to noxious stimulation in all four limbs
- **C.** Preserved pupillary light reflexes
- **D.** The unremarkable noncontrast head CT
- **E.** Intact oculocephalic responses

Answer: C. Preserved pupillary light reflexes

Pupillary responsiveness may be the single most important part of the neurological examination for distinguishing metabolic from structural causes of coma; preservation of the pupillary light reflex despite respiratory depression, extensor rigidity, or motor flaccidity points to a metabolic etiology. Cheyne-Stokes respiration is nonspecific, and although it is described in diffuse bilateral brain toxicity it is also produced by uremia, hepatic failure, heart failure, and bilateral hemispheric infarcts or forebrain masses, so it does not discriminate. Symmetric withdrawal indicates the absence of a lateralized motor deficit but does not itself separate metabolic from structural disease. A normal noncontrast CT does not exclude a structural cause: MRI is more sensitive for anoxic-ischemic injury, small strokes, and encephalitis, and basilar occlusion is a frequently overlooked structural cause of coma that requires vascular imaging. Intact oculocephalic responses indicate an intact brainstem but are less reliable than oculovestibular testing and again do not localize the process to a metabolic cause. The diffuse slowing with triphasic waves in the vignette is the classic metabolic EEG pattern, whereas structural disease generally produces a focal EEG abnormality.

Key point:
reactive pupils in a deeply comatose patient are the strongest bedside clue that the coma is metabolic rather than structural.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 205-207

Question 77

Clinical Evaluation of Coma and Brain Death · Intracranial pressure waveforms and the Monro-Kellie doctrine · moderate

A 24-year-old man with severe traumatic brain injury has an external ventricular drain in place. His baseline intracranial pressure is 15 to 18 mmHg. Over the course of the night, the nurse documents several discrete episodes in which the intracranial pressure abruptly rises to 55 to 60 mmHg and remains sustained at that level for roughly 20 minutes before falling back to baseline. During the episodes his blood pressure rises and he becomes less responsive, and between episodes his examination returns to baseline. The drain waveform is dampened only during the episodes and tracings are otherwise normal. Which of the following best explains these findings?

- **A.** Cushing reflex from established medullary compression
- **B.** Plateau waves, also known as Lundberg A waves
- **C.** Overdamped waveform artifact from a malpositioned external ventricular drain
- **D.** Nonconvulsive status epilepticus with autonomic accompaniments
- **E.** Progressive refractory intracranial hypertension from an expanding hematoma

Answer: B. Plateau waves, also known as Lundberg A waves

Critical increases in intracranial pressure that occur once intracranial compliance is exhausted produce plateau waves, also called Lundberg A waves, which are large sustained elevations that can recur every 15 to 30 minutes. Sudden rises of this kind reduce cerebral perfusion pressure and produce changes in consciousness ranging from confusion to lethargy and ultimately coma, exactly as described here, with recovery between waves. The Cushing reflex or triad, described by Cushing and Kocher, is the combination of depressed mental status, bradycardia, hypertension, and slowed or changed respiratory rate arising from loss of accommodation and anoxemia of the bulbar centers; it is a sustained clinical syndrome, not a recurrent self-terminating pressure wave. Waveform artifact would not be accompanied by reproducible clinical deterioration and recovery. Refractory intracranial hypertension from an expanding hematoma would show a progressive rather than an episodic pattern with return to a normal baseline. The underlying substrate is the Monro-Kellie doctrine: the combined volume of brain, cerebrospinal fluid, and intracranial blood must be constant, and once cerebrospinal fluid displacement can no longer compensate, small volume changes produce large pressure swings.

Key point:
recurrent sustained intracranial pressure elevations lasting 15 to 30 minutes with transient depression of consciousness are plateau or Lundberg A waves, a sign of exhausted intracranial compliance.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 203

Question 78

Clinical Evaluation of Coma and Brain Death · Locked-in syndrome · moderate

A 63-year-old man with atrial fibrillation who is not anticoagulated develops sudden quadriplegia and anarthria. On examination his eyes are open and he appears awake. He cannot move his face, limbs, or tongue and cannot phonate. He does not move his eyes horizontally, but he reliably blinks once for yes and twice for no and looks up and down on command, answering orientation questions correctly. Pupils are 2 mm and reactive. Which of the following is the most likely site of his lesion?

- **A.** Bilateral medial thalami
- **B.** Midbrain tectum and pretectum
- **C.** Bilateral frontal lobes
- **D.** Ventral pons
- **E.** Bilateral ventrolateral medulla

Answer: D. Ventral pons

This is locked-in syndrome, in which a fully alert patient loses all voluntary motor activity except blinking and vertical eye movements because of a ventral pontine lesion, most often a pontine hemorrhage or top-of-the-basilar syndrome. Mesencephalic structures are intact, which is why arousal and vertical gaze are preserved, and this is the key reason locked-in syndrome must not be mistaken for coma or unawareness. If the thalamus is also involved in top-of-the-basilar syndrome, the patient will instead have depressed alertness, so bilateral medial thalamic injury does not fit an alert patient. A tectal or pretectal midbrain lesion causes unreactive fixed pupils, which is not the case here. Bilateral frontal lobe injury produces akinetic mutism, an abulic emotionless state with preserved tracking eye movements, not quadriplegia with intact cognition. Bilateral ventrolateral medullary lesions affect the ventral respiratory centers and cause apnea. Neuromuscular causes such as botulism, severe Guillain-Barre syndrome, and neuromuscular blocking agents can produce a locked-in type syndrome and should also be considered.

Key point:
preserved vertical eye movements and blinking in an otherwise motionless but alert patient localize to the ventral pons.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 209

Question 79

Clinical Evaluation of Coma and Brain Death · Respiratory patterns in coma · moderate

A 70-year-old man with a large cerebellar hemorrhage and progressive brainstem compression is being observed off sedation before intubation. His breathing is quick and shallow, with irregular breaths of varying amplitude separated by random periods of apnea. There is no crescendo-decrescendo cycling, and no sustained inspiratory pauses are seen. Which of the following is the most likely level of the lesion producing this respiratory pattern?

- **A.** Pontomedullary junction
- **B.** Bilateral diencephalon
- **C.** Rostral pons and midbrain
- **D.** Bilateral ventrolateral medulla
- **E.** Bilateral cerebral hemispheres

Answer: A. Pontomedullary junction

The pattern described, quick shallow irregular breaths with periods of apnea and no crescendo-decrescendo cycling, is Biot's respiration, also called cluster or ataxic breathing, and it is associated with lesions at the pontomedullary junction. Cheyne-Stokes respiration is the smooth crescendo-decrescendo pattern with apneic periods lasting up to 10 to 20 seconds and reflects bilateral hemispheric or diencephalic dysfunction, metabolic processes such as uremia or hepatic failure, or heart failure, so choices B and E do not fit an irregular noncycling pattern. Central neurogenic hyperventilation, described by Plum and Swanson, arises from structural lesions in the brainstem, typically the pons, but is a sustained rapid deep pattern rather than an irregular one, and it can also be seen with primary pulmonary processes such as pneumonia. Apneustic breathing, with inspiratory pauses of 2 to 3 seconds followed by end-expiratory pauses, is seen with bilateral pontine lesions and is explicitly absent here. Bilateral lesions of the ventral respiratory centers in the ventrolateral medulla produce frank apnea rather than an irregular pattern.

Key point:
irregular, quick, shallow breaths with random apneas and no crescendo-decrescendo cycling are ataxic or Biot's breathing and localize to the pontomedullary junction.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 211

Question 80

Clinical Evaluation of Coma and Brain Death · Oculovestibular reflex testing · easy

When performing cold caloric (oculovestibular) testing in a comatose patient with an intact tympanic membrane, to what angle should the head of the bed be elevated in order to obtain maximal stimulation?

- **A.** Flat, at 0 degrees
- **B.** 30 degrees
- **C.** 45 degrees
- **D.** 60 degrees
- **E.** 90 degrees

Answer: B. 30 degrees

The oculovestibular reflex is elicited by injecting ice water into the ear canal with the head raised to 30 degrees, because at 30 degrees the horizontal semicircular canals are properly oriented, giving maximal stimulation and the highest probability of eliciting a response. The injection stimulates the semicircular canals and the eighth cranial nerve, which projects to the vestibular nuclei and then to the third, fourth, and sixth nerve nuclei, producing gradual tonic eye movement toward the side of the cold ear in a patient with an intact brainstem. Flat, 45, 60, and 90 degree positions all malalign the horizontal canals and reduce or abolish the response, so they risk a falsely absent reflex. The oculocephalic or doll's eyes maneuver, performed by turning the head 90 degrees to each side, is less reliable than oculovestibular testing, although if present it does indicate an intact brainstem. Repetitive oculovestibular testing that produces disconjugate eye movements suggests a structural brainstem lesion, and in drug-induced coma the eyes may conjugately deviate downward after lateral movement. Absence of the oculocephalic or oculovestibular reflex correlates with poor outcome.

Key point:
cold caloric testing is performed with the head of bed at 30 degrees so the horizontal semicircular canals are maximally stimulated.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 210, 223

Question 81

Clinical Evaluation of Coma and Brain Death · Emergency management of raised intracranial pressure · moderate

A 55-year-old woman, 80 kg, is on day 3 after a large right middle cerebral artery infarction. She is intubated and sedated. She acutely develops a dilated nonreactive right pupil and extensor posturing on the left. Repeat noncontrast head CT shows 10 mm of midline shift with effacement of the right ambient cistern. Blood pressure is 160/85 mmHg, and the current arterial $PaCO_2$ is 42 mmHg. Neurosurgery has been called for consideration of decompressive craniectomy. Which of the following is the most appropriate immediate medical therapy?

- **A.** Dexamethasone 10 mg intravenously followed by 4 mg every 6 hours
- **B.** Sodium nitroprusside infusion titrated to a systolic pressure below 120 mmHg
- **C.** Place the bed flat to augment cerebral perfusion pressure
- **D.** Adjust the ventilator to a target $PaCO_2$ of 50 to 55 mmHg
- **E.** Mannitol 20 percent, 0.75 to 1.25 g/kg intravenously to a maximum of 100 g, with the head of bed at 30 degrees

Answer: E. Mannitol 20 percent, 0.75 to 1.25 g/kg intravenously to a maximum of 100 g, with the head of bed at 30 degrees

Hyperosmolar therapy with mannitol 20 percent at 0.75 to 1.25 g/kg, with a maximum dose of 100 g, targeting a serum osmolality at or below 320 mOsm/kg and an osmolar gap below 12, is the appropriate immediate hyperosmolar therapy, and the head of bed should be elevated to 30 degrees or more to promote venous drainage. Hypertonic saline given as a bolus such as 23.4 percent or as an infusion of 2, 3, or 7 percent is an accepted alternative, and available studies are inconclusive as to which delivery method is superior. Steroids have no place in the treatment of traumatic brain injury, ischemic stroke, or intracerebral hemorrhage; dexamethasone 10 mg followed by 4 to 6 mg every 6 hours is reserved for edema surrounding a neoplasm, so choice A is wrong for this infarct. Vasodilatory agents such as nitrates should be avoided because they increase cerebral blood flow and can raise intracranial pressure; labetalol, nicardipine, or clevidipine are preferred when blood pressure must be lowered. Placing the bed flat opposes venous drainage and worsens intracranial pressure. Hypercapnia causes reflexive cerebral vasodilation and raises intracranial pressure, so a $PaCO_2$ target of 50 to 55 mmHg is harmful; the emergency maneuver is transient hyperventilation to a $PaCO_2$ of 25 to 35 mmHg, with a normal target otherwise of 35 to 45 mmHg.

Key point:
for acute herniation, elevate the head to 30 degrees, give mannitol 0.75 to 1.25 g/kg up to 100 g keeping osmolality at or below 320, and hyperventilate transiently to a $PaCO_2$ of 25 to 35 mmHg.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 213-214

Question 82

Clinical Evaluation of Coma and Brain Death · Glycemic target and the NICE-SUGAR trial · easy

Based on the NICE-SUGAR study, what is the recommended target blood glucose range for a critically ill comatose patient in the neurointensive care unit?

- **A.** 80 to 110 mg/dL
- **B.** 110 to 140 mg/dL
- **C.** 140 to 180 mg/dL
- **D.** 180 to 220 mg/dL
- **E.** 220 to 260 mg/dL

Answer: C. 140 to 180 mg/dL

The NICE-SUGAR study demonstrated benefit to targeting glucose levels of 140 to 180 mg/dL rather than tighter control, and this is the goal blood glucose range used in comatose critically ill patients. Tight targets of 80 to 110 and 110 to 140 mg/dL expose the patient to hypoglycemia, which is itself a metabolic cause of coma and encephalopathy and was the harm signal that drove the trial result. Targets of 180 to 220 and 220 to 260 mg/dL permit sustained hyperglycemia, which is common in comatose patients because of pre-existing diabetes, physiological stress, and medications, and which the chapter identifies as requiring active management. Nursing-led insulin protocols are beneficial for maintaining the target range while preventing hypoglycemia. Comatose patients should also be started on enteral nutrition within 24 to 48 hours of admission, which interacts directly with glycemic control.

Key point:
target blood glucose 140 to 180 mg/dL in comatose critically ill patients, per NICE-SUGAR.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 214, 216

Question 83

Clinical Evaluation of Coma and Brain Death · Prerequisites for brain death determination · hard

A 44-year-old man had a witnessed cardiac arrest and underwent targeted temperature management. Fentanyl and midazolam infusions were discontinued 8 hours ago; his hepatic and renal function are normal. Core bladder temperature is now 34.8 C during passive rewarming. He is on norepinephrine with a mean arterial pressure of 55 mmHg. On examination he is comatose with no pupillary, corneal, oculocephalic, cough, or gag reflexes and no motor response to noxious stimulation. A urine drug screen is negative. The intensivist asks you to proceed with brain death determination. Which of the following is the most appropriate next step?

- **A.** Proceed directly to apnea testing, since the brainstem reflexes are already absent
- **B.** Obtain a cerebral angiogram and declare brain death if there is no intracranial flow
- **C.** Obtain an EEG to document electrocerebral silence before completing the examination
- **D.** Rewarm the patient to a core temperature of at least 36 C, correct the hypotension, and wait at least five elimination half-lives of the sedatives before performing the determination
- **E.** Arrange for two separate physicians to perform the clinical examination and then declare

Answer: D. Rewarm the patient to a core temperature of at least 36 C, correct the hypotension, and wait at least five elimination half-lives of the sedatives before performing the determination

An evaluation for brain death requires a clear cause of the neurological catastrophe, evidence of irreversibility, and the exclusion of confounders: hypothermia, hypotension, drug intoxication, neuromuscular blockade assessed by train-of-four testing, and electrolyte, acid-base, or endocrinological derangement. The lowest core temperature allowable for clinical determination of brain death is 36 C, so at 34.8 C the examination cannot be used. In addition, a negative drug screen is not sufficient; excluding drug effect also requires waiting at least five elimination half-lives for any sedatives or narcotics, assuming normal hepatic and renal function, which is not satisfied 8 hours after stopping infusions. Apnea testing before confounders are corrected is invalid and the patient is hypotensive, with a mean arterial pressure below the 60 mmHg floor, which is itself both a prerequisite violation and a reason for the test to be aborted. Ancillary tests such as cerebral angiography, nuclear medicine studies, or transcranial Doppler are not required under the American Academy of Neurology or Society of Critical Care Medicine protocols and are reserved for situations in which part of the examination cannot be completed, so they cannot be used here to bypass uncorrected confounders. Requiring two examiners does not correct hypothermia, hypotension, or residual sedation, and in the United States a single physician examination is generally sufficient in adults, although institutional protocols vary and two examinations are required in pediatric patients.

Key point:
brain death determination requires a core temperature of at least 36 C, normal blood pressure, and at least five elimination half-lives after sedatives before the examination counts.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 217, 223

Question 84

Clinical Evaluation of Coma and Brain Death · Movements compatible with brain death · moderate

A 31-year-old woman with a massive intracerebral hemorrhage and intraventricular extension is being evaluated for brain death. All prerequisites have been met. She is comatose with absent pupillary, corneal, oculocephalic, oculovestibular, cough, and gag reflexes. During nail bed pressure applied to the right foot, the examiner observes brief flexion at the hip, knee, and ankle on that side. Deep tendon reflexes are present in the legs and an extensor plantar response is noted bilaterally. Which of the following is the most appropriate interpretation?

- **A.** These are spinally mediated responses that are compatible with brain death, and apnea testing should proceed
- **B.** The motor response indicates preserved brainstem function, and brain death cannot be determined
- **C.** The findings represent decorticate posturing and preclude a diagnosis of brain death
- **D.** An ancillary blood flow study is now mandatory before the determination can continue
- **E.** The examination should be repeated in 24 hours before any further testing

Answer: A. These are spinally mediated responses that are compatible with brain death, and apnea testing should proceed

A clear distinction must be made between decerebrate or decorticate posturing, which is brainstem mediated and is not compatible with brain death, and spinal reflexes or responses such as triple flexion, which are compatible with brain death. Deep tendon reflexes, the Babinski sign, triple flexion, and even the dramatic Lazarus sign, in which the patient sits up with the arms raised, are all spinally mediated and permissible, so the examination may proceed to assessment of respiratory drive with an apnea test. Choice B is incorrect because the stereotyped brief triple flexion described is not a brainstem-mediated motor response. Choice C is incorrect because decorticate posturing is pathological flexion in the upper limbs with leg extension and would in fact exclude brain death, unlike triple flexion of one leg. Ancillary testing is not required by the American Academy of Neurology or Society of Critical Care Medicine protocols and becomes relevant only when a portion of the examination cannot be completed, such as with significant facial trauma, or when the patient cannot tolerate apnea testing. Waiting 24 hours is not required in adults, in whom a single physician examination is generally sufficient in the United States, although institutional protocols vary. Families should be prepared for these movements, since they may appear purposeful and can create false hope.

Key point:
triple flexion, deep tendon reflexes, the Babinski sign, and the Lazarus sign are spinal reflexes compatible with brain death, whereas decorticate or decerebrate posturing is not.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 217, 219, 222-223

Question 85

Clinical Evaluation of Coma and Brain Death · Apnea testing · moderate

A 52-year-old man with a devastating brainstem hemorrhage is comatose with absent brainstem reflexes, and all prerequisites for brain death determination have been met. He is hemodynamically stable and is not a chronic carbon dioxide retainer. Observation of the ventilator confirms he is not triggering breaths above the set rate. Which of the following describes the appropriate performance of the apnea test and a result consistent with nonfunctional medullary chemoreceptors?

- **A.** Preoxygenate with 100 percent FiO_2, adjust the ventilator to a baseline $PaCO_2$ of 25 to 30 mmHg, deliver oxygen at 10 L/min, observe for 3 minutes, and accept a post-test pCO_2 above 45 mmHg
- **B.** Preoxygenate with 40 percent FiO_2, adjust the ventilator to a baseline $PaCO_2$ of 30 to 35 mmHg, deliver oxygen at 2 L/min, observe for 15 minutes, and accept a post-test pCO_2 above 50 mmHg
- **C.** Preoxygenate with 100 percent FiO_2, leave the patient on the ventilator with the rate set to zero, observe for 5 minutes, and accept a post-test pCO_2 rise of 5 mmHg
- **D.** Preoxygenate with 100 percent FiO_2, adjust the ventilator to a baseline $PaCO_2$ of 55 to 60 mmHg, deliver oxygen at 4 to 6 L/min, observe for 8 to 10 minutes, and accept any post-test pCO_2 above the baseline value
- **E.** Preoxygenate with 100 percent FiO_2, adjust the ventilator to a baseline $PaCO_2$ of 40 to 45 mmHg, disconnect and deliver oxygen at 4 to 6 L/min through a catheter in the endotracheal tube, observe for 8 to 10 minutes, and accept a post-test pCO_2 above 60 mmHg or a rise of 20 mmHg

Answer: E. Preoxygenate with 100 percent FiO_2, adjust the ventilator to a baseline $PaCO_2$ of 40 to 45 mmHg, disconnect and deliver oxygen at 4 to 6 L/min through a catheter in the endotracheal tube, observe for 8 to 10 minutes, and accept a post-test pCO_2 above 60 mmHg or a rise of 20 mmHg

The apnea test is performed after preoxygenation with 100 percent FiO_2 to reduce the risk of desaturation, with recruitment maneuvers using pressure control and incremental PEEP as an option, and the ventilator adjusted to achieve a baseline $PaCO_2$ of 40 to 45 mmHg. The patient is then disconnected and given oxygen at 4 to 6 L/min, for example through a catheter placed in the endotracheal tube, and observed for 8 to 10 minutes for any spontaneous respiratory effort before the ventilator is reconnected and a repeat arterial blood gas is drawn. A post-test pCO_2 above 60 mmHg in adults, or a rise of 20 mmHg which is the criterion specified for carbon dioxide retainers, is consistent with nonfunctional medullary chemoreceptors. Choices A and B use inadequate baseline carbon dioxide levels, wrong observation times, and thresholds too low to establish absent respiratory drive. Choice C is invalid because the patient must be disconnected from the ventilator, and a 5 mmHg rise does not meet any threshold. Choice D starts from a hypercapnic baseline outside the specified 40 to 45 mmHg window and accepts any rise at all, which would not demonstrate absent chemoreceptor function. If the patient becomes unstable during the test, with desaturation, hypotension, or arrhythmia, the ventilator is reconnected and the test aborted, and an ancillary test should then be performed if the completed portions of the examination are consistent with brain death.

Key point:
apnea testing uses a baseline $PaCO_2$ of 40 to 45 mmHg, oxygen at 4 to 6 L/min, 8 to 10 minutes of observation, and a post-test pCO_2 above 60 mmHg or a rise of 20 mmHg.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 219

Question 86

Clinical Evaluation of Coma and Brain Death · Prognosis after cardiac arrest · hard

A 62-year-old man with hypertension and diabetes suffers a ventricular tachycardia cardiac arrest with return of spontaneous circulation after 25 minutes. He undergoes targeted temperature management at 32 to 34 C for 24 hours and is rewarmed to normothermia over 12 hours. Off all sedation after complete rewarming, he is comatose with no eye opening to noxious stimulation. Pupillary, corneal, and oculocephalic reflexes are present. He has extensor posturing to noxious stimuli and intermittent myoclonic jerking of the face, eyes, and upper extremities. The family asks about his prognosis. Which of the following findings, if present, would most strongly indicate a poor neurological outcome?

- **A.** Burst suppression on EEG
- **B.** Myoclonic jerking of the face and upper extremities
- **C.** Bilaterally absent N20 responses on median nerve somatosensory evoked potentials
- **D.** Diffuse slowing with triphasic waves on EEG
- **E.** Preserved pupillary and corneal reflexes off sedation

Answer: C. Bilaterally absent N20 responses on median nerve somatosensory evoked potentials

Absent bilateral N20 responses on somatosensory evoked potentials are associated with poor outcome in patients who remain comatose after cardiac arrest, and somatosensory evoked potentials are specifically recommended in this chapter to help determine the degree of brain injury and to assist prognostication. Burst suppression on EEG was previously thought to be a marker of poor outcome, but recent data suggest it may actually represent an intrinsic mechanism to rescue neurons from cell death, so it should not be used as a poor prognostic sign. Diffuse slowing with triphasic waves indicates a metabolic encephalopathy rather than conveying prognosis after arrest. Preserved pupillary and corneal reflexes reflect intact brainstem function, and since assessment for brainstem involvement, meaning abnormal brainstem reflexes or neuroimaging evidence of brainstem injury, is one of the most important components of accurate outcome prediction, their preservation argues against the worst outcomes. Myoclonus is described as a movement disorder seen in coma, consisting of sudden nonrhythmic gross jerking of the face and proximal limbs, and is not by itself the strongest prognostic marker in this chapter. Note also that this patient's brainstem reflexes are present and he has extensor posturing, so he is not brain dead and brain death testing is not appropriate.

Key point:
after cardiac arrest, bilaterally absent N20 responses on somatosensory evoked potentials predict poor outcome, while burst suppression is no longer a reliable marker of futility.

Source: Practice of Neurocritical Care, 2nd ed., Clinical Evaluation of Coma and Brain Death, pp. 213, 216

Hypoxic-Ischemic Brain Injury After Cardiac Arrest

Question 87

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Pathophysiology: primary vs secondary injury · easy

A 35-year-old woman suffers a witnessed pulseless electrical activity arrest in the emergency department from a saddle pulmonary embolism and achieves return of spontaneous circulation after alteplase, with an estimated total downtime of 5 minutes. Immediately after resuscitation she has preserved brainstem reflexes and localizes with both upper extremities. Over the next 4 days, despite hemodynamic stability off vasopressors, she becomes comatose with absent corneal reflexes and no motor response to noxious stimulation. A trainee asks what caused the brain injury at the moment circulation ceased. Which of the following best describes the mechanism of the primary ischemic insult?

- **A.** Loss of inhibitory interneurons with increased glutamatergic transmission
- **B.** Microvascular dysfunction producing the no-reflow phenomenon
- **C.** Adenosine triphosphate depletion with failure of transmembrane ionic gradients and anoxic depolarization
- **D.** Free radical formation and activation of nitric oxide synthase
- **E.** Systemic inflammatory response with microthrombotic events after reperfusion

Answer: C. Adenosine triphosphate depletion with failure of transmembrane ionic gradients and anoxic depolarization

During cardiac arrest, oxygen and glucose delivery fails, adenosine triphosphate is depleted, and ion pumps can no longer maintain the transmembrane gradients required for cell signaling; the resulting massive cation influx produces anoxic depolarization, which is the main mechanism of primary ischemic brain injury. Excitotoxicity from loss of inhibitory interneurons and increased glutamatergic transmission is a delayed, secondary injury mechanism, as are free radical formation and nitric oxide synthase activation. The no-reflow phenomenon is by definition a post-reperfusion event, reflecting suboptimal restoration of forward flow due to microvascular dysfunction. The systemic inflammatory response and microthrombosis of postcardiac arrest syndrome likewise occur after return of spontaneous circulation. This is the two-hit model: a primary insult during circulatory cessation followed by secondary injury from the imbalance between cerebral metabolic demand and substrate supply, which explains why an initially reassuring exam can deteriorate over days.

Key point:
primary hypoxic-ischemic injury is adenosine triphosphate failure leading to anoxic depolarization; excitotoxicity, free radicals, and no-reflow drive secondary injury.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 231

Question 88

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Post-arrest coronary angiography and the COACT trial · moderate

A 58-year-old man has a witnessed out-of-hospital ventricular fibrillation arrest with bystander cardiopulmonary resuscitation and return of spontaneous circulation after two shocks. He is comatose and intubated. Blood pressure is 104/62 mmHg on low-dose norepinephrine, and the postresuscitation 12-lead electrocardiogram shows sinus rhythm with no ST-segment elevation. Troponin is mildly elevated and there is no shock or ongoing arrhythmia. Which of the following is the most appropriate next step?

- **A.** Admit to the intensive care unit for postarrest care and defer coronary angiography
- **B.** Proceed immediately to the catheterization laboratory for coronary angiography
- **C.** Administer systemic thrombolysis before intensive care unit admission
- **D.** Start a prophylactic amiodarone infusion to prevent recurrent ventricular arrhythmia
- **E.** Cannulate for extracorporeal cardiopulmonary resuscitation as a rescue strategy

Answer: A. Admit to the intensive care unit for postarrest care and defer coronary angiography

Coronary angiography is standard of care for arrest patients with ST-segment elevation, but the COACT trial showed that delaying angiography in patients without ST elevation did not affect survival, so deferring the catheterization laboratory in a stable patient is acceptable. Immediate angiography is not wrong in every case, since between 25 percent and 58 percent of patients without ST changes have an offending lesion and a Parisian registry found a culprit lesion in close to one-third of such patients, but the randomized evidence does not support urgency here; note also that fewer than 5 percent of COACT subjects had an acute thrombotic occlusion, so the cohort may not represent populations with heavy vascular risk. Systemic thrombolysis is recommended for arrest from pulmonary embolism but its widespread use in cardiac arrest has not improved outcomes. Prophylactic amiodarone to prevent recurrent ventricular arrhythmia after out-of-hospital arrest is not supported by current evidence, and decisions should involve cardiac electrophysiology. Extracorporeal cardiopulmonary resuscitation is a rescue strategy for selected patients with a reversible etiology who fail conventional resuscitation, not for a patient already resuscitated and perfusing.

Key point:
without ST elevation and without instability, coronary angiography can be deferred per COACT, while postarrest brain-directed care begins immediately.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 232

Question 89

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · TTM target selection with spontaneous hypothermia · hard

A 47-year-old man is resuscitated from an out-of-hospital ventricular fibrillation arrest after 12 minutes of downtime. On arrival to the emergency department he is intubated, does not follow commands, and withdraws to noxious stimulation. Esophageal core temperature is 32.4 degrees C, mean arterial pressure is 72 mmHg on low-dose norepinephrine, and there is no active bleeding or coagulopathy. Which of the following is the most appropriate temperature management strategy?

- **A.** Actively rewarm to 37.5 degrees C at 1 degree C per hour, then maintain controlled normothermia
- **B.** Rewarm to 36 degrees C over 2 hours and maintain that target for 24 hours
- **C.** Withhold temperature control because the patient is already below all recommended targets
- **D.** Maintain a target of 33 degrees C for 24 hours, then rewarm no faster than 0.25 to 0.5 degrees C per hour
- **E.** Deepen cooling to 30 degrees C for 24 hours to maximize metabolic suppression

Answer: D. Maintain a target of 33 degrees C for 24 hours, then rewarm no faster than 0.25 to 0.5 degrees C per hour

All patients who are unconscious or unable to follow commands after return of spontaneous circulation, meaning a Glasgow Coma Scale motor subscore less than 6, should be considered for temperature modulation regardless of etiology, location of arrest, or rhythm, and this patient has no contraindication. When a patient presents with spontaneous hypothermia, so that the intended target is above the current core temperature, the induction phase must either use a controlled rewarming pace no faster than 0.25 degrees C per hour or use a lower target, because rapid rewarming and temperature fluctuation are deleterious in acute brain injury; choosing 33 degrees C avoids rewarming altogether. Options A and B both rewarm this patient faster than the recommended 0.25 to 0.5 degrees C per hour ceiling. Withholding temperature control abandons the only intervention shown to have a neuroprotective effect after arrest and also abandons fever prevention during the first 72 hours. Targets below 32 degrees C are not supported and carry prolonged PR and QT intervals and a higher risk of malignant arrhythmias, with refractoriness to defibrillation at or below 28 degrees C.

Key point:
in a spontaneously hypothermic postarrest patient, pick the lower target rather than rewarming, and never rewarm faster than 0.25 to 0.5 degrees C per hour.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 233, 235-236

Question 90

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · HYPERION trial and nonshockable rhythms · moderate

A 68-year-old woman has an unwitnessed in-hospital cardiac arrest from asphyxia with an initial rhythm of asystole. Return of spontaneous circulation is achieved after 18 minutes. She remains comatose with a Glasgow Coma Scale score of 3 and requires norepinephrine for circulatory shock. Core temperature is 36.8 degrees C. Which temperature strategy is supported by the best available randomized evidence for this patient population?

- **A.** Controlled normothermia at 37.5 degrees C for 40 hours
- **B.** Targeted temperature management at 33 degrees C for 24 hours
- **C.** No active temperature control, with antipyretics as needed
- **D.** Targeted temperature management at 36 degrees C for 24 hours
- **E.** Targeted temperature management at 32 degrees C for 48 hours

Answer: B. Targeted temperature management at 33 degrees C for 24 hours

The HYPERION trial randomized 581 subjects with nonshockable rhythms, including both in-hospital and out-of-hospital arrest and both cardiac and noncardiac etiologies, to 33 degrees C versus targeted normothermia at 37 degrees C, and found more favorable neurologic outcomes with 33 degrees C, at 10.2 percent versus 5.7 percent achieving Cerebral Performance Category 1 to 2 at 3 months. This trial is the best evidence for exactly the patients excluded from earlier trials: prolonged unwitnessed asystole, in-hospital arrest, and noncardiac etiology, and roughly 80 percent of its cohort was asystolic with half arresting from asphyxia with circulatory shock. Normothermia at 37.5 degrees C was studied in TTM2, which enrolled only out-of-hospital arrest of cardiac or presumed cardiac etiology, so it does not extend to this patient; 37 degrees C was inferior to 33 degrees C in HYPERION. Withholding temperature control is not supported, as targeted temperature management is the only neuroprotective therapy demonstrated after arrest. A target of 36 degrees C has been adequately studied only in out-of-hospital arrest of cardiac etiology, and 32 degrees C for 48 hours is not a validated regimen; guidelines recommend cooling for at least 24 hours.

Key point:
for nonshockable rhythms, including in-hospital and asphyxial arrest, HYPERION supports 33 degrees C over 37 degrees C.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 233-235

Question 91

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · TTM2 trial results · moderate

A cardiology colleague asks you to justify your unit's temperature protocol in light of the 2021 Targeted Hypothermia versus Targeted Normothermia after Out-of-Hospital Cardiac Arrest trial, known as TTM2. Which of the following statements most accurately reflects the findings of that trial?

- **A.** Hypothermia at 33 degrees C reduced 6-month mortality compared with targeted normothermia
- **B.** Hypothermia at 33 degrees C improved functional outcome but increased major bleeding
- **C.** The trial enrolled only nonshockable rhythms and demonstrated a benefit of 33 degrees C
- **D.** The normothermia group achieved its target without any active temperature control devices
- **E.** Mortality and functional outcome were similar, but arrhythmia causing hemodynamic compromise was more frequent with 33 degrees C, at 24 percent versus 16 percent

Answer: E. Mortality and functional outcome were similar, but arrhythmia causing hemodynamic compromise was more frequent with 33 degrees C, at 24 percent versus 16 percent

TTM2 randomized 1,861 patients with out-of-hospital arrest of presumed cardiac etiology, excluding unwitnessed asystole, to 33 degrees C or targeted normothermia of 37.5 degrees C or less, and found no difference in mortality, 50 percent versus 48 percent, or in favorable functional neurologic outcome, 44.6 percent in each arm at 6 months. The single significant safety difference was arrhythmia resulting in hemodynamic compromise, which occurred in 24 percent of the hypothermia group versus 16 percent of the normothermia group, with a P value less than 0.001, a signal not seen in TTM1. Mortality was not reduced, and functional outcome was not improved, so options A and B are incorrect; bleeding was not the differentiating adverse event. The trial enrolled both shockable and nonshockable rhythms, with 72 percent shockable, so option C is wrong, and it was HYPERION that studied nonshockable rhythms. Option D is wrong because 46 percent of the normothermia group still required a surface or intravascular device to keep temperature at or below 37.5 degrees C, which is why normothermia in TTM2 is active temperature control rather than no temperature control.

Key point:
TTM2 found 33 degrees C no better than controlled normothermia for out-of-hospital cardiac-etiology arrest and raised a signal of hemodynamically significant arrhythmia at 33 degrees C.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 234

Question 92

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Shivering during targeted temperature management · moderate

A 52-year-old man is cooled to a target of 33 degrees C with external surface pads after an anterior ST-elevation myocardial infarction arrest and successful percutaneous coronary intervention. He is receiving scheduled acetaminophen, fentanyl at 200 mcg per hour, and propofol at 50 mcg per kg per minute. Four hours into the induction phase his esophageal temperature is 37 degrees C, the device water temperature reads 3 degrees C, and water flow through the pads is documented as good. Which of the following is the most appropriate next step?

- **A.** Exchange the surface pads for an endovascular cooling catheter because of device malfunction
- **B.** Add scheduled antipyretics for presumed central fever
- **C.** Administer a bolus of a neuromuscular blocking agent, such as vecuronium 0.1 mg per kg
- **D.** Lower the device set point to 32 degrees C to increase the temperature gradient
- **E.** Infuse 2 liters of 4 degrees C normal saline to accelerate induction

Answer: C. Administer a bolus of a neuromuscular blocking agent, such as vecuronium 0.1 mg per kg

A very low water temperature with documented good flow means the device is working maximally, so the failure to reach target reflects high heat generation by the patient, and shivering is by far the most likely cause. Shivering must be treated promptly because it causes substantial energy expenditure and oxygen consumption and negates the effects of hypothermia; this patient is already on high-dose sedation, so shivering is refractory to sedation alone and neuromuscular blockade is the most effective way to abort it, with short-term neuromuscular blockade associated with improved survival during targeted temperature management after arrest. Device malfunction is unlikely given the water temperature and flow readings. Central fever is possible but far less likely than shivering in this setting, and additional antipyretics will not overcome ongoing muscular thermogenesis. Lowering the set point or infusing chilled saline does not address the source of the heat load, and prehospital chilled saline in particular was associated with more rearrests, 26 percent versus 21 percent, and more pulmonary edema, 41 percent versus 30 percent, without outcome benefit.

Key point:
failing to reach target with a maximally cold, well-flowing device means shivering, and refractory shivering on high-dose sedation is treated with neuromuscular blockade.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 237, 252

Question 93

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Ventilation and oxygenation targets after ROSC · easy

You are writing the ventilator order set for comatose patients admitted after return of spontaneous circulation. Which of the following sets of targets is most consistent with guideline recommendations and the brain-centered goals for postcardiac arrest syndrome?

A. Oxygen saturation greater than 94 percent, partial pressure of carbon dioxide 35 to 45 mmHg, avoiding partial pressure of oxygen below 60 mmHg or above 300 mmHg

B. Oxygen saturation greater than 88 percent, partial pressure of carbon dioxide 25 to 30 mmHg, partial pressure of oxygen above 300 mmHg

C. Oxygen saturation of 100 percent on a fraction of inspired oxygen of 1.0, partial pressure of carbon dioxide 30 to 35 mmHg

D. Oxygen saturation greater than 94 percent, partial pressure of carbon dioxide 20 to 30 mmHg, partial pressure of oxygen 60 to 80 mmHg

E. Oxygen saturation greater than 90 percent, with routine targeted hypercapnia of 50 to 60 mmHg

Answer: A. Oxygen saturation greater than 94 percent, partial pressure of carbon dioxide 35 to 45 mmHg, avoiding partial pressure of oxygen below 60 mmHg or above 300 mmHg

The 2015 American Heart Association guidelines recommend maintaining the partial pressure of carbon dioxide within 35 to 45 mmHg and titrating the fraction of inspired oxygen to keep oxyhemoglobin saturation at 94 percent or greater, and the brain-centered postarrest goals add avoiding a partial pressure of oxygen below 60 mmHg or above 300 mmHg and avoiding a partial pressure of carbon dioxide below 35 mmHg. Both carbon dioxide and oxygen tensions have an independent U-shaped relationship with hospital mortality, demonstrated in a retrospective cohort of over 5,000 arrest patients and replicated in a meta-analysis of 23,434 patients. Hypocapnia causes cerebral vasoconstriction and reduced cerebral blood flow, while hyperoxia increases free radical formation and can worsen secondary injury, which is why options B, C, and D are harmful. Option E is premature: permissive or mild hypercapnia is a promising signal from small trials and is being tested prospectively in the Targeted Therapeutic Mild Hypercapnia after Resuscitated Cardiac Arrest trial, but it is not yet a standard target.

Key point:
after return of spontaneous circulation, target saturation above 94 percent and carbon dioxide 35 to 45 mmHg, and avoid both hypoxemia below 60 mmHg and hyperoxia above 300 mmHg.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 233, 239

Question 94

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Hemodynamic targets after cardiac arrest · easy

A 61-year-old woman is 8 hours out from an out-of-hospital cardiac arrest and is comatose on targeted temperature management. Her mean arterial pressure drifts between 55 and 60 mmHg with a systolic blood pressure in the 80s despite adequate volume resuscitation. Which of the following is the most appropriate blood pressure goal?

- **A.** Mean arterial pressure 85 to 100 mmHg with mixed venous oxygen saturation 65 to 75 percent
- **B.** Mean arterial pressure greater than or equal to 50 mmHg
- **C.** Systolic blood pressure greater than or equal to 140 mmHg
- **D.** Mean arterial pressure greater than or equal to 65 mmHg or systolic blood pressure greater than or equal to 100 mmHg
- **E.** Permissive hypotension with no specific target while sedated and cooled

Answer: D. Mean arterial pressure greater than or equal to 65 mmHg or systolic blood pressure greater than or equal to 100 mmHg

Current guidelines recommend maintaining a mean arterial pressure of at least 65 mmHg or a systolic blood pressure of at least 100 mmHg after cardiac arrest. Hypotension occurs in up to two-thirds of postarrest patients and is associated with impaired functional status among survivors and increased risk of death, and autoregulatory failure occurs in 35 to 78 percent of these patients, so allowing permissive hypotension is not acceptable. Higher targets have been tested and failed: the COMACARE trial compared a mean arterial pressure of 65 to 75 mmHg with 80 to 100 mmHg and found no difference in neuron-specific enolase levels, electroencephalogram findings, cerebral oxygenation, or 6-month neurologic outcome, and the Neuroprotect trial found that early goal-directed augmentation to a mean arterial pressure of 85 to 100 mmHg with mixed venous saturation of 65 to 75 percent during the first 36 hours was safe and improved cerebral oxygenation but did not change hypoxic-ischemic injury burden on imaging or neurologic outcome. A systolic target of 140 mmHg has no evidentiary basis in this population. The ideal target remains unknown, and individualized goals are reasonable given the heterogeneity of postcardiac arrest syndrome.

Key point:

after arrest, keep the mean arterial pressure at or above 65 mmHg or the systolic pressure at or above 100 mmHg; routinely chasing higher pressures has not improved outcomes.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 238-239

Question 95

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Postanoxic myoclonus and EEG phenotypes · hard

A 44-year-old man is 36 hours out from a witnessed ventricular fibrillation arrest and remains comatose after rewarming. The bedside nurse reports generalized jerks of the limbs and trunk. Continuous electroencephalography shows a continuous background with narrow, midline vertex-predominant spike-and-wave discharges occurring in lockstep with the jerks. Somatosensory evoked potentials show preserved bilateral N20 peaks. The family asks whether this finding means recovery is impossible. Which of the following is the most accurate statement?

- **A.** This pattern is pathognomonic of a poor outcome and supports withdrawal of life-sustaining therapy
- **B.** Roughly half of patients with this electroclinical phenotype achieve functional independence, so aggressive treatment is warranted
- **C.** The finding is consistent with brain death and confirmatory testing should be arranged
- **D.** A burst-suppression background with high-amplitude polyspikes in lockstep with the jerks carries the same favorable prognosis
- **E.** Myoclonus occurring within the first 24 to 48 hours is always subcortical and therefore benign

Answer: B. Roughly half of patients with this electroclinical phenotype achieve functional independence, so aggressive treatment is warranted

Postanoxic myoclonic status epilepticus was historically treated as pathognomonic of a poor outcome, but electroclinical phenotype separates prognosis: a continuous background with narrow, vertex spike-wave discharges in lockstep with the jerks carried a 50 percent chance of achieving functional independence, whereas a suppression-burst background with high-amplitude polyspikes in lockstep with the jerks portended a poor prognosis in 100 percent of patients, which makes option D wrong. Because of the unacceptably high false positive rates of myoclonus as a poor-outcome predictor, modern guidelines have tempered its use, so option A is incorrect. Status epilepticus is incompatible with brain death, since dead neurons cannot seize, so option C is wrong. Myoclonus may be cortical, meaning it has a time-locked electroencephalographic correlate, or subcortical, meaning there is no cerebral correlate underlying the muscle artifact, and timing does not determine the generator, so option E is wrong; this patient's jerks have a clear cortical correlate. Other features favoring potential for independence with aggressive treatment include emergence of status epilepticus from a continuous background, preserved N20 peaks on somatosensory evoked potentials, neuron-specific enolase below 37 mcg per liter, and a low injury burden on neuroimaging.

Key point:
myoclonic status epilepticus after arrest is not uniformly fatal, and the electroencephalographic background, continuous versus burst-suppressed, drives the prognosis.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 238, 242

Question 96

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Timing of multimodal neuroprognostication · moderate

A 66-year-old man treated with targeted temperature management at 33 degrees C completed rewarming and reached normothermia 12 hours ago. Sedation with midazolam and fentanyl was discontinued at the same time. He is now 48 hours from return of spontaneous circulation. His examination shows extensor posturing to noxious stimulation, giving a Glasgow Coma Scale motor subscore of 2, with preserved pupillary and corneal reflexes. The intensive care team asks whether they can tell the family that the prognosis is hopeless. Which of the following is the most appropriate next step?

- **A.** Declare a poor prognosis on the basis of the Glasgow Coma Scale motor subscore of 2
- **B.** Obtain somatosensory evoked potentials now and act on absent N20 responses
- **C.** Send a neuron-specific enolase level and use a value above the cutoff to declare a poor outcome
- **D.** Perform formal brain death determination
- **E.** Defer a definitive prognostic statement until at least 72 hours after return to normothermia and discontinuation of sedatives, using multimodal testing

Answer: E. Defer a definitive prognostic statement until at least 72 hours after return to normothermia and discontinuation of sedatives, using multimodal testing

Modern guidelines recommend that the clinical examination used for prognostication be performed 72 hours after return to normothermia and discontinuation of sedatives in patients treated with targeted temperature management, and this patient is only 12 hours past both milestones. A Glasgow Coma Scale motor subscore of 2 or less has an unacceptably high false positive rate of 10 to 35 percent, so American Heart Association and European guidelines suggest using it only in combination with other tools. Somatosensory evoked potentials are a robust tool, but bilateral absence of N20 peaks should be applied at 24 hours or more after arrest only in patients not treated with targeted temperature management and at 72 hours or more in those who were treated with it, since hypothermia prolongs latencies. Neuron-specific enolase should never be used alone, and most modern guidelines have omitted a specific cutoff because hemolysis, seizures, age, comorbidities, and assay type all alter the value. Brain death determination is inappropriate with preserved pupillary and corneal reflexes, and delayed awakening is expected here given the low temperature target and longer-acting sedatives.

Key point:
in targeted temperature management patients, prognosticate no earlier than 72 hours after normothermia and off sedation, and always multimodally, never on a single test.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 233, 241, 243-245

Question 97

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Neuron-specific enolase as a prognostic biomarker · moderate

In a substudy of the first Targeted Temperature Management trial, neuron-specific enolase measured at 48 hours after return of spontaneous circulation was assessed using the cutoff value of 33 ng per mL recommended by the 2006 American Academy of Neurology guidelines. Which of the following best describes the performance of that cutoff for predicting a poor neurologic outcome?

- **A.** Sensitivity 65 percent and specificity 91 percent
- **B.** Sensitivity 91 percent and specificity 65 percent
- **C.** Sensitivity 100 percent and specificity 100 percent
- **D.** Sensitivity 32 percent and specificity 100 percent
- **E.** Sensitivity 61 percent and specificity 100 percent

Answer: A. Sensitivity 65 percent and specificity 91 percent

In the TTM1 substudy, a neuron-specific enolase cutoff of 33 ng per mL at 48 hours after return of spontaneous circulation yielded a sensitivity of 65 percent and a specificity of 91 percent for a poor outcome, which is why it is inadequate as a standalone predictor. Option C is implausible, since no clinical, electrophysiologic, radiographic, or laboratory test after arrest is infallible. Option D describes the Neurological Pupil index below 2, which has 100 percent specificity and positive predictive value for Cerebral Performance Category 3 to 5 but only 32 percent sensitivity, and option E describes quantitative pupillary light reflex below 13 percent at 48 hours, which had 100 percent positive predictive value for mortality with 61 percent sensitivity and 100 percent specificity. The trajectory of neuron-specific enolase over the first 72 hours may predict outcome better than any single absolute value, and a decline over the first 3 days has been associated with good outcomes. Most modern guidelines have omitted a specific neuron-specific enolase cutoff, and the biomarker should be used only within multimodal prognostication because hemolysis, seizures, age, comorbid conditions, and assay type distort it.

Key point:
a neuron-specific enolase of 33 ng per mL at 48 hours is only 65 percent sensitive and 91 percent specific for poor outcome, so it can support but never replace multimodal prognostication.

Source: Practice of Neurocritical Care, 2nd ed., Hypoxic-Ischemic Brain Injury After Cardiac Arrest, pp. 243, 247

Intracerebral Hemorrhage

Question 98

Intracerebral Hemorrhage · ICH volume by ABC/2 and the ICH Score · moderate

A 72-year-old man with long-standing hypertension is brought to the emergency department 1 hour after sudden left hemiparesis and vomiting. Blood pressure is 208/114 mmHg, heart rate 92 beats/min. His Glasgow Coma Scale score after resuscitation is 8 (E2, V2, M4). Noncontrast head CT shows a right putaminal hemorrhage with extension into the right lateral ventricle. On the axial slice containing the largest hematoma area, the longest diameter is 6 cm and the longest perpendicular diameter is 4 cm; hemorrhage is visible on six 0.5-cm slices, each containing more than 75% of the area of the largest slice. Which of the following is his ICH Score?

- **A.** 1
- **B.** 2
- **C.** 3
- **D.** 4
- **E.** 5

Answer: C. 3

Using the ABC/2 method, A is the longest diameter on the largest axial slice (6 cm), B is the longest perpendicular diameter (4 cm), and C is the number of slices with hemorrhage multiplied by slice thickness; because all six slices contain more than 75% of the largest slice area, each counts as a full slice, so C is 6 x 0.5 = 3 cm, and the volume is (6 x 4 x 3)/2 = 36 mL. The ICH Score assigns 1 point for GCS 5-12, 1 point for ICH volume of 30 mL or greater, 1 point for any intraventricular hemorrhage, 1 point for infratentorial origin, and 1 point for age 80 years or older, with 2 points for GCS 3-4. This patient earns 1 point for GCS 8, 1 point for a 36-mL volume, and 1 point for IVH; he is 72 years old and the hemorrhage is supratentorial, so he earns no points for those items, giving a total of 3. Answers A and B underestimate by omitting the volume or IVH point, and D and E would require an additional point such as age 80 or older, infratentorial location, or a GCS of 3-4. Note that slices containing an estimated less than 25% of the largest slice area are not counted at all and those with 25%-75% are counted as half slices, which is where most calculation errors arise.

Key point:

ICH volume by ABC/2 with correct slice weighting drives the volume point in the ICH Score, which awards one point each for volume 30 mL or greater, IVH, infratentorial location, and age 80 or older, plus one or two points for GCS.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 269-270

Question 99

Intracerebral Hemorrhage · Acute blood pressure control: choice of agent · easy

A 58-year-old man with untreated hypertension presents 90 minutes after sudden onset of right hemiparesis and aphasia. Blood pressure is 212/112 mmHg and heart rate is 88 beats/min. Noncontrast head CT shows a 20-mL left basal ganglia hemorrhage without intraventricular extension. He is awake and protecting his airway. Which of the following is the most appropriate agent for acute blood pressure lowering?

- **A.** Intravenous hydralazine boluses
- **B.** Sodium nitroprusside infusion
- **C.** Sublingual nifedipine
- **D.** Nicardipine infusion
- **E.** Intravenous enalaprilat boluses

Answer: D. Nicardipine infusion

The preferred approach to acute blood pressure lowering in ICH is intermittent intravenous labetalol push or a continuous infusion of an agent such as nicardipine, urapidil, or clevidipine, titrated to the target while avoiding abrupt overshoot. Nicardipine is the agent used in the ATACH trials and is the prototypical continuous infusion choice, allowing minute-to-minute titration. Sodium nitroprusside and hydralazine may cause significant venodilation, disturb cerebral autoregulation, and increase intracranial pressure, so they are not first-line in ICH. Sublingual nifedipine and intravenous enalaprilat provide unpredictable, poorly titratable reductions and are not among the recommended agents. The final choice should still be individualized on the basis of heart rate and comorbidities such as renal or cardiac failure, and these patients should be managed in a neurological intensive care unit or specialized stroke unit to ensure optimal acute control.

Key point:
Use titratable intravenous labetalol, nicardipine, urapidil, or clevidipine for acute BP control in ICH, and avoid nitroprusside and hydralazine because they can raise intracranial pressure.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 273

Question 100

Intracerebral Hemorrhage · Vitamin K antagonist-related ICH reversal · moderate

A 66-year-old woman on warfarin for atrial fibrillation presents 2 hours after acute headache and left-sided weakness. Blood pressure is 186/98 mmHg, INR is 3.4, and platelet count is 220,000/microL. Noncontrast head CT shows a 25-mL right frontal hemorrhage. Which of the following is the most appropriate hemostatic management?

- **A.** Fresh frozen plasma 10-15 mL/kg intravenously alone
- **B.** Four-factor prothrombin complex concentrate plus vitamin K 10 mg intravenously
- **C.** Vitamin K 10 mg intravenously alone
- **D.** Andexanet alfa bolus followed by a 2-hour infusion
- **E.** Recombinant activated factor VII 80 mcg/kg intravenously

Answer: B. Four-factor prothrombin complex concentrate plus vitamin K 10 mg intravenously

Current guidelines for warfarin reversal in life-threatening hemorrhage emphasize a rapid reversal agent, prothrombin complex concentrate, given together with vitamin K; the Neurocritical Care Society recommends that patients with ICH on warfarin with an INR of 1.4 or higher receive 4-factor PCC intravenously plus vitamin K 10 mg intravenously, and the European Stroke Organisation recommends an initial PCC dose of 30 units/kg. In the INCH trial, patients with vitamin K antagonist-related ICH randomized to PCC versus FFP within 3 hours of bleeding onset had significantly faster INR normalization with PCC, with less hematoma expansion at 3 and 24 hours and a lower (though not statistically significant) mortality. FFP 10-15 mL/kg should be considered only if PCC is unavailable, so answer A is inferior. Vitamin K alone is far too slow to prevent early hematoma expansion, which occurs in up to 35%-50% of anticoagulation-related ICH. Andexanet alfa is a factor Xa inhibitor antidote for rivaroxaban and apixaban, not for warfarin, and recombinant activated factor VII is not recommended for ICH, including in coagulopathic patients.

Key point:
Warfarin-related ICH is reversed with 4-factor PCC plus vitamin K 10 mg intravenously, with FFP reserved for when PCC is unavailable.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 273, 275

Question 101

Intracerebral Hemorrhage · Dabigatran-related ICH reversal and dosing · hard

A 74-year-old man taking dabigatran for atrial fibrillation is brought in 2 hours after his last dose with acute aphasia and right hemiparesis. Noncontrast head CT shows a 30-mL left temporal hemorrhage with intraventricular extension. Which of the following is the most appropriate reversal regimen?

- **A.** Andexanet alfa low-dose bolus followed by a 2-hour infusion
- **B.** Four-factor prothrombin complex concentrate 25 units/kg intravenously
- **C.** Protamine sulfate 50 mg intravenously
- **D.** Fresh frozen plasma 15 mL/kg intravenously
- **E.** Idarucizumab 5 g intravenously given as two 2.5-g doses no more than 15 minutes apart

Answer: E. Idarucizumab 5 g intravenously given as two 2.5-g doses no more than 15 minutes apart

Dabigatran is a direct thrombin inhibitor and was the first direct oral anticoagulant with a specific antidote, idarucizumab, which is given as 5 g intravenously in two separate 2.5-g doses administered no more than 15 minutes apart, supplied as two 2.5 g/50 mL vials. Neurocritical Care Society guidance also suggests considering activated charcoal 50 g if it can be given within 2 hours of the last dabigatran dose, which applies to this patient; if idarucizumab is unavailable, alternatives are activated PCC 50 units/kg or 4-factor PCC 50 units/kg, so answer B is both the wrong agent choice and an incorrect dose. In RE-VERSE AD, idarucizumab produced rapid normalization of the dilute thrombin time and ecarin clotting time within minutes, and 30-day mortality among ICH patients was 14%, although the absence of a control group limits interpretation. Andexanet alfa targets factor Xa inhibitors such as rivaroxaban and apixaban, not dabigatran. Protamine reverses unfractionated heparin and partially reverses low-molecular-weight heparin, and FFP has no specific role in DOAC reversal.

Key point:
For dabigatran-related ICH, give idarucizumab 5 g intravenously as two 2.5-g doses no more than 15 minutes apart, with activated charcoal 50 g if within 2 hours of the last dose.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 274-275

Question 102

Intracerebral Hemorrhage · Antiplatelet-associated ICH and platelet transfusion · moderate

A 77-year-old woman taking aspirin 81 mg daily for primary prevention presents with sudden dysarthria and left arm weakness. Blood pressure is 178/92 mmHg and platelet count is 240,000/microL. Noncontrast head CT shows an 18-mL right thalamic hemorrhage without intraventricular extension. She is neurologically stable, and no surgical intervention is planned. Which of the following is the most appropriate management of her antiplatelet exposure?

- **A.** Transfuse one pool of platelets immediately
- **B.** Administer desmopressin 0.4 mcg/kg intravenously
- **C.** Continue standard medical care without platelet transfusion
- **D.** Transfuse platelets only after a platelet function assay confirms aspirin effect
- **E.** Administer recombinant activated factor VII

Answer: C. Continue standard medical care without platelet transfusion

The PATCH trial randomized 97 ICH patients on antiplatelet therapy to platelet transfusion and 93 to standard care and found that death or dependence at 3 months was higher in the transfusion group. Serious in-hospital adverse events occurred in 42% of transfused patients versus 29% of those receiving standard care, and in-hospital mortality was 24% versus 17%, a statistically significant difference. Platelet transfusion therefore cannot be recommended for patients with acute ICH taking antiplatelet drugs, making answers A and D inappropriate; obtaining a platelet function assay does not change this, because PATCH tested the intervention itself rather than the history of aspirin use. Most PATCH patients were on aspirin and none underwent surgery, so it remains unclear whether transfusion has a role in patients on clopidogrel or dual antiplatelet therapy or those going to hematoma evacuation, but this patient matches the trial population exactly. Desmopressin was not evaluated in this setting, and recombinant activated factor VII is not recommended as a treatment for ICH.

Key point:
Platelet transfusion for antiplatelet-associated ICH increased death or dependence in PATCH and should not be given to nonsurgical aspirin-treated patients.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 276

Question 103

Intracerebral Hemorrhage · Blood pressure targets and the INTERACT/ATACH trials · easy

A 62-year-old man presents 2 hours after onset of a 15-mL left thalamic hemorrhage. His initial systolic blood pressure is 190 mmHg and he is awake with a mild right hemiparesis. According to the 2015 American Heart Association and American Stroke Association guidelines for spontaneous ICH, which of the following is the most appropriate acute blood pressure target?

- **A.** Systolic below 220 mmHg
- **B.** Systolic below 140 mmHg
- **C.** Systolic below 180 mmHg
- **D.** Systolic below 200 mmHg
- **E.** Systolic below 110 mmHg

Answer: B. Systolic below 140 mmHg

The 2015 AHA/ASA guidelines state that early rapid reduction of systolic blood pressure to below 140 mmHg is probably safe and reasonable, especially when the presenting systolic pressure is between 150 and 220 mmHg, which describes this patient. This recommendation is based on the INTERACT trials: INTERACT-1 randomized 400 patients and showed that targeting systolic below 140 mmHg rather than below 180 mmHg reduced the absolute risk of significant hematoma growth, defined as 33% or more of baseline volume, by 8% without increasing adverse events, and INTERACT-2 randomized 2,794 patients treated within 6 hours with no difference in the primary death and disability endpoint but a significant benefit on the secondary ordinal modified Rankin Scale analysis. Answers A, C, and D reflect the less aggressive comparator arms or no meaningful treatment at all. Answer E is dangerous and unstudied as a target; ATACH-2 showed no benefit from intensive lowering and a 2.3-fold higher incidence of renal adverse events, with concern that pressure in the intensive group may have been lowered too far. No trial has compared blood pressure lowering with allowing pressures to remain above 180 mmHg.

Key point:
For acute ICH with a presenting systolic between 150 and 220 mmHg, early lowering to below 140 mmHg is considered safe and reasonable by the 2015 AHA/ASA guidelines.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 272-273

Question 104

Intracerebral Hemorrhage · Hemostatic agents: tranexamic acid and TICH-2 · hard

A 59-year-old man with hypertension presents 5 hours after onset of a 22-mL right frontal lobar hemorrhage. He takes no antithrombotic drugs, and his INR and platelet count are normal. A resident asks whether tranexamic acid should be given to limit hematoma expansion. Which of the following best describes the evidence?

- **A.** Tranexamic acid given as a 1-g bolus followed by 1 g over 8 hours improves 90-day modified Rankin Scale outcomes
- **B.** Tranexamic acid reduces 90-day mortality by approximately 10% in absolute terms
- **C.** Tranexamic acid is recommended specifically for patients with a positive spot sign
- **D.** Tranexamic acid modestly reduces hematoma expansion but does not improve clinical outcome and is not recommended
- **E.** Tranexamic acid is recommended only for ICH related to oral anticoagulants

Answer: D. Tranexamic acid modestly reduces hematoma expansion but does not improve clinical outcome and is not recommended

TICH-2 randomized 2,325 ICH patients not on anticoagulation within 8 hours of onset to placebo or tranexamic acid given as a 1-g bolus followed by 1 g infused over 8 hours. Functional outcome at 90 days on the modified Rankin Scale was no different between groups and mortality was essentially identical at 21.6% with placebo versus 21.7% with tranexamic acid, refuting answers A and B. There was a small but statistically significant reduction in hematoma expansion by day 2, 25% versus 29% with P equal to 0.03, so the drug has a measurable biological effect without a clinical one. The trial specifically tested whether spot-sign-positive patients benefited more and did not support that hypothesis, so answer C is wrong. Hemostatic therapy with either tranexamic acid or recombinant activated factor VII is not currently recommended for ICH patients, including those with coagulopathy, which also excludes answer E; recombinant activated factor VII reduced hematoma expansion in its phase III trial but produced no significant change in death or severe disability.

Key point:
TICH-2 showed tranexamic acid slightly reduces hematoma expansion without improving outcome, and hemostatic agents are not recommended in ICH.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 275

Question 105

Intracerebral Hemorrhage · Cerebellar hemorrhage and surgical indications · moderate

A 68-year-old woman with hypertension develops sudden vertigo, vomiting, and truncal ataxia. Noncontrast head CT shows a right cerebellar hemorrhage measuring 3.5 cm in greatest diameter with effacement of the fourth ventricle and early hydrocephalus. Over 45 minutes her Glasgow Coma Scale score falls from 14 to 10. Which of the following is the most appropriate next step?

- **A.** Surgical evacuation of the cerebellar hematoma as soon as possible
- **B.** Placement of an external ventricular drain alone with continued observation
- **C.** Mannitol bolus and repeat CT in 6 hours
- **D.** Hyperventilation to a $PaCO_2$ of 25 mmHg maintained for 48 hours
- **E.** Medical management with a systolic blood pressure target below 140 mmHg alone

Answer: A. Surgical evacuation of the cerebellar hematoma as soon as possible

Case series report that patients with large cerebellar hematomas, generally considered greater than 3 cm in diameter, or with brainstem compression or hydrocephalus may have a favorable outcome with surgical intervention, and the 2015 AHA/ASA guidelines recommend surgical removal of the hematoma as soon as possible in cerebellar hemorrhage patients who are deteriorating neurologically or who have brainstem compression or hydrocephalus from ventricular obstruction. The same guidelines explicitly advise against attempting to manage these patients solely with an external ventricular drain, which makes answer B incorrect despite the hydrocephalus. Osmotic therapy and blood pressure control are appropriate adjuncts but are not definitive in a deteriorating patient with posterior fossa mass effect, so answers C and E delay necessary surgery. Hyperventilation reduces ICP only transiently, over a few hours, and is a temporizing measure in preparation for definitive treatment rather than a sustained therapy. It should be noted that no randomized trial of cerebellar ICH surgery analogous to STICH exists, and a recent large meta-analysis found no improvement in functional outcome with evacuation, yet guidelines and practice still favor surgery in this exact clinical scenario.

Key point:
A cerebellar hemorrhage larger than about 3 cm with brainstem compression, hydrocephalus, or neurologic deterioration warrants prompt surgical evacuation, not an EVD alone.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 278

Question 106

Intracerebral Hemorrhage · Venous thromboembolism prophylaxis · moderate

A 64-year-old man is admitted with a 25-mL left putaminal hemorrhage. Intermittent pneumatic compression devices were placed on the day of admission. On hospital day 2, a repeat noncontrast head CT shows a stable hematoma, he has no coagulopathy, and his blood pressure is well controlled. Which of the following is the most appropriate next step for venous thromboembolism prevention?

- **A.** Add graduated compression stockings to the pneumatic compression devices
- **B.** Continue pneumatic compression alone until hospital day 10
- **C.** Begin therapeutic-dose low-molecular-weight heparin
- **D.** Obtain lower extremity duplex ultrasonography before any pharmacologic prophylaxis
- **E.** Begin subcutaneous unfractionated heparin 5,000 units every 8 hours

Answer: E. Begin subcutaneous unfractionated heparin 5,000 units every 8 hours

A study comparing initiation of low-dose subcutaneous heparin on the second, fourth, or tenth day after ICH found a statistically significant decrease in pulmonary embolism when unfractionated heparin 5,000 units every 8 hours was started on the second day, with no increase in intracranial rebleeding. Neurocritical Care Society guidelines affirm intermittent pneumatic compression on the day of admission and suggest starting pharmacologic prophylaxis within 48 hours of admission in patients with stable hematomas and no ongoing coagulopathy, which this patient meets, so continuing mechanical prophylaxis alone until day 10 forgoes a proven benefit. Graduated compression stockings should not be used because they are not beneficial overall and may reduce circulation to the affected limb if poorly sized or kinked; the 2015 AHA/ASA emphasis on pneumatic compression on day 1 derives from CLOTS 3, where hemorrhagic stroke patients had a DVT rate of 6.7% versus 17.0%, odds ratio 0.36 with a 95% confidence interval of 0.17-0.75. Therapeutic anticoagulation is not prophylaxis and would risk hematoma expansion. Screening duplex is not required before starting prophylactic dosing.

Key point:
In ICH, use pneumatic compression from day 1 and start prophylactic subcutaneous heparin 5,000 units every 8 hours as early as day 2 once the hematoma is stable.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 277

Question 107

Intracerebral Hemorrhage · Seizures and antiseizure prophylaxis · easy

A 61-year-old woman is admitted with a 20-mL right parietal lobar hemorrhage. She is alert with a mild left visual field deficit and has had no clinical seizure. Which of the following is the most appropriate approach to antiseizure medication?

- **A.** Start levetiracetam prophylactically for 7 days
- **B.** Start phenytoin prophylactically for 30 days
- **C.** Do not start an antiseizure medication and treat only if a seizure occurs
- **D.** Start prophylaxis because lobar location alone is an indication
- **E.** Start prophylaxis and continue indefinitely given the risk of late epilepsy

Answer: C. Do not start an antiseizure medication and treat only if a seizure occurs

Current AHA/ASA and European Stroke Organisation guidelines recommend against routine anticonvulsant prophylaxis after ICH but do recommend treating seizures that occur. Clinically apparent seizures occur in 4%-8% of ICH patients within the first 30 days and lobar location does increase that likelihood, but increased risk has not translated into a demonstrated benefit from prophylaxis, so answers A, B, D, and E all overtreat. It is unknown whether prophylaxis reduces the incidence of either overt or nonconvulsive seizures, and analysis of the placebo group of a neuroprotective ICH trial found that patients who received prophylactic antiepileptic treatment had worse outcomes even after adjustment for other factors. Nonconvulsive seizures are found on continuous EEG in about 20% of ICH patients and have been associated with worse midline shift, higher NIH Stroke Scale scores, and less favorable outcomes, which supports monitoring unexplained depressed consciousness rather than blanket prophylaxis.

Key point:
Do not give routine antiseizure prophylaxis after ICH; treat only clinical or electrographic seizures that actually occur.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 277

Question 108

Intracerebral Hemorrhage · Cerebral amyloid angiopathy · easy

An 83-year-old woman with 3 years of progressive memory decline and no history of hypertension presents with transient word-finding difficulty. She takes no antiplatelet or anticoagulant drugs. Blood pressure is 128/74 mmHg. Imaging shows an acute right parietal hemorrhage at the gray-white junction, and susceptibility-weighted MRI demonstrates numerous microbleeds with a lobar predominance. Which of the following is the most likely etiology?

- **A.** Chronic hypertensive small vessel disease
- **B.** Cerebral amyloid angiopathy
- **C.** Hemorrhagic metastasis
- **D.** Ruptured arteriovenous malformation
- **E.** Cerebral venous sinus thrombosis with hemorrhagic transformation

Answer: B. Cerebral amyloid angiopathy

Cerebral amyloid angiopathy is an increasingly recognized cause of ICH in elderly patients and is associated with dementia; its hemorrhages are typically located in the peripheral lobar white matter near the gray-white matter junction, and susceptibility-weighted or gradient echo MRI typically shows numerous microbleeds with a lobar predominance, exactly as described here. Hypertensive ICH accounts for about 60%-70% of cases and typically occurs in the basal ganglia, thalamus, cerebellum, medulla, and pons, and this normotensive patient has neither the risk factor nor the location, so answer A is unlikely. Hemorrhagic brain tumors are usually metastatic and would be expected to show an underlying mass or history of a primary cancer, and vascular malformations are the higher-yield consideration in young patients, those without hypertension, those with atypical locations, or those with primary intraventricular hemorrhage, generally investigated by angiography. Venous sinus thrombosis produces hemorrhage in a venous distribution, often with headache and a hypercoagulable context. Practically, recurrent hemorrhage risk from amyloid deposition is tripled by the apolipoprotein E epsilon 2 and epsilon 4 alleles, and there is no treatment for CAA, so management focuses on avoiding anticoagulation.

Key point:
Lobar hemorrhage at the gray-white junction with multiple lobar microbleeds in an elderly, normotensive, demented patient is cerebral amyloid angiopathy.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 266, 284

Question 109

Intracerebral Hemorrhage · CT angiography spot sign · hard

A 57-year-old man undergoes CT angiography 100 minutes after the onset of a right thalamic hemorrhage. The noncontrast study shows a 14-mL hematoma, and the CTA source images demonstrate a focus of contrast extravasation within the center of the clot. No aneurysm or arteriovenous malformation is identified. Which of the following does this finding best predict?

- **A.** An occult arteriovenous malformation requiring conventional angiography
- **B.** A high likelihood of benefit from recombinant activated factor VII
- **C.** An increased risk of nonconvulsive seizures on continuous EEG
- **D.** An increased risk of hematoma expansion and higher 3-month mortality
- **E.** An underlying hemorrhagic neoplasm

Answer: D. An increased risk of hematoma expansion and higher 3-month mortality

The described focus of active contrast extravasation is the spot sign, which is a predictor of hematoma expansion and worsened outcome. In the PREDICT prospective observational study, a spot sign was seen in about 30% of patients with acute ICH scanned within about 160 minutes of ictus, with a sensitivity of only 63% but a specificity of 90% for hematoma expansion, and its presence was significantly associated with higher mortality at 3 months. The likelihood of seeing a spot sign is time-dependent and higher the earlier the contrast study is obtained, which is why this patient scanned at 100 minutes is in the highest-yield window. The spot sign is not a marker of an underlying vascular malformation or tumor, which are suspected instead on the basis of young age, atypical location, absence of hypertension, or primary intraventricular hemorrhage. It has been used as a selection criterion for hemostatic therapy, but SPOTLIGHT and STOP-IT did not demonstrate an effect of recombinant activated factor VII in spot-sign-positive patients and TICH-2 did not confirm greater tranexamic acid benefit in this subgroup, so answer B is not supported.

Key point:
The CTA spot sign has 63% sensitivity and 90% specificity for hematoma expansion and predicts higher 3-month mortality, but it does not currently identify patients who benefit from hemostatic therapy.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 271

Question 110

Intracerebral Hemorrhage · Stepwise medical management of intracranial pressure · moderate

A 49-year-old woman with a large right hemispheric ICH and intraventricular hemorrhage has an external ventricular drain in place. Despite head of bed elevation to 30 degrees, a neutral neck position, adequate analgesia and sedation, and CSF drainage, her intracranial pressure remains 32 mmHg with a stable blood pressure. Which of the following is the most appropriate next step?

- **A.** Osmotic therapy with mannitol or hypertonic saline
- **B.** Sustained hyperventilation to a $PaCO_2$ of 25 mmHg for 48 hours
- **C.** Pentobarbital coma with continuous EEG monitoring
- **D.** Induced hypothermia to 33 degrees Celsius
- **E.** Neuromuscular blockade with continued sedation

Answer: A. Osmotic therapy with mannitol or hypertonic saline

Guidelines advocate a graded stepwise approach that begins with less invasive measures such as head of bed elevation to 30 degrees, neutral neck positioning to facilitate jugular venous drainage, and adequate analgesia and sedation, followed by CSF drainage through an EVD; when these fail, osmotic agents such as mannitol or hypertonic saline are the appropriate next tier. Hyperventilation does rapidly lower ICP through cerebral arterial vasoconstriction, but the effect is generally transient over a few hours, it reduces cerebral blood flow, and it shifts the oxyhemoglobin dissociation curve so that less oxygen is released to tissue, so it is used only as a temporizing measure while more definitive therapy is arranged, not for 48 hours. Neuromuscular blockade may be considered in refractory intracranial hypertension but carries an increased risk of infection and critical illness neuromuscular disease and comes after osmotic therapy. Barbiturate coma and induced hypothermia have not been systematically studied in ICH and are considered third-tier salvage therapies, with barbiturates causing significant hypotension and requiring continuous EEG to titrate. Note that prophylactic mannitol before intracranial hypertension develops has not been shown to be beneficial, and although an ICP threshold above 20 mmHg is commonly used, no single validated trigger exists; in one analysis of 100 patients, 90% of readings were below 20 mmHg and the percentage of readings above 30 mmHg independently predicted mortality and poor 30-day modified Rankin score.

Key point:
Escalate ICP therapy in tiers, positioning and sedation, then CSF drainage, then osmotic therapy, reserving hyperventilation as a bridge and barbiturates or hypothermia as salvage.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 276-277

Question 111

Intracerebral Hemorrhage · Secondary prevention and resumption of anticoagulation · moderate

A 71-year-old man with nonvalvular atrial fibrillation had a warfarin-associated 20-mL left basal ganglia hemorrhage that was reversed with 4-factor prothrombin complex concentrate and vitamin K. He has no mechanical heart valve. On hospital day 5 he is neurologically stable with a stable hematoma, and the family asks when anticoagulation for stroke prevention can be restarted. Which of the following is the most appropriate recommendation?

- **A.** Resume oral anticoagulation within 24 hours
- **B.** Resume oral anticoagulation after 7 to 10 days
- **C.** Resume oral anticoagulation after 72 hours
- **D.** Never resume oral anticoagulation after any ICH
- **E.** Resume oral anticoagulation after 4 weeks

Answer: E. Resume oral anticoagulation after 4 weeks

For oral anticoagulant-related ICH, reinitiation of anticoagulation after 4 weeks is currently recommended, and this patient with a deep hypertensive hemorrhage and no mechanical valve fits that default. The 7 to 10 day option applies specifically to patients with a mechanical heart valve, in whom earlier reinitiation might be considered because of the very high thromboembolic risk, so answer B is a distractor tied to a different clinical situation. Resumption within 24 to 72 hours is far earlier than recommended and risks rebleeding while the hematoma is still evolving. Anticoagulation is not permanently contraindicated after every ICH, but it may be prudent to avoid anticoagulation entirely in patients with lobar ICH and likely cerebral amyloid angiopathy, which is not this patient. Additional secondary prevention includes immediate and sustained blood pressure control with a long-term goal of 130/80 mmHg or below, based on the dramatically lowered ICH risk seen in the SPS3 trial, and continuation of statin therapy in patients who were taking a statin at the time of hemorrhage.

Key point:
After anticoagulant-related ICH, restart oral anticoagulation at about 4 weeks, at 7 to 10 days if a mechanical valve is present, and generally not at all in lobar ICH from suspected amyloid angiopathy.

Source: Practice of Neurocritical Care, 2nd ed., Intracerebral Hemorrhage, pp. 280

Metabolic Encephalopathies and Delirium

Question 112

Metabolic Encephalopathies and Delirium · Pharmacologic management of ICU delirium · easy

A 68-year-old man is on hospital day 3 in the neurocritical care unit after clipping of a ruptured anterior communicating artery aneurysm. Overnight he becomes agitated, pulls at his lines, and is disoriented and inattentive, with fluctuating symptoms that worsen at night. He has no history of alcohol use. Temperature is 37.0 C, heart rate 96/min, blood pressure 138/78 mm Hg. Serum sodium, glucose, ammonia, and renal function are normal; urinalysis is unremarkable and head CT shows no new lesion. Electrocardiography shows a QTc of 430 msec and the serum magnesium is 2.0 mg/dL. Nonpharmacologic measures have failed to control the agitation. Which of the following is the most appropriate pharmacologic management?

- **A.** Haloperidol 1 to 5 mg intravenously every 4 to 6 hours with serial QTc monitoring
- **B.** Lorazepam 4 mg intravenously every 4 hours on a scheduled basis
- **C.** Olanzapine 20 mg orally daily
- **D.** Diphenhydramine 50 mg intravenously every 6 hours
- **E.** Bilateral wrist restraints without pharmacologic therapy

Answer: A. Haloperidol 1 to 5 mg intravenously every 4 to 6 hours with serial QTc monitoring

Delirium in the ICU is treated as a medical emergency because it is associated with longer ICU stay and higher mortality, and after the underlying cause is sought, antipsychotics are the usual pharmacologic option. The chapter specifies haloperidol in low, frequent doses of 1 to 5 mg intravenously every 4 to 6 hours with monitoring of the electrocardiographic QTc interval and normal magnesium levels, both of which this patient has. Benzodiazepines such as lorazepam are reserved for alcohol or cocaine withdrawal, and randomized trials of dexmedetomidine versus midazolam or lorazepam showed either no difference or less delirium with dexmedetomidine, so scheduled lorazepam is a poor choice here. The alternative atypical agents listed in the chapter are quetiapine 25 to 100 mg per day orally and olanzapine 2.5 to 7.5 mg per day orally, so olanzapine 20 mg per day is far above the cited range. Diphenhydramine is an antihistamine with anticholinergic activity, a listed cause of metabolic encephalopathy, and would worsen delirium. Physical restraints are themselves a listed risk factor for delirium and should be minimized.

Key point:
haloperidol 1 to 5 mg IV every 4 to 6 hours, with QTc and magnesium monitoring, is the chapter's cited antipsychotic regimen for ICU delirium.

Source: Practice of Neurocritical Care, 2nd ed., Metabolic Encephalopathies and Delirium, pp. 293-295

Question 113

Metabolic Encephalopathies and Delirium · Hepatic encephalopathy - first-line ammonia-lowering therapy · moderate

A 52-year-old man with alcohol-related cirrhosis is admitted with 2 days of increasing somnolence. He is lethargic but arousable to voice, disoriented to time, has asterixis, and performs subtraction poorly. He is jaundiced. Serum ammonia is 118 mcmol/L, international normalized ratio is 1.6, and head CT shows no focal lesion or hydrocephalus. Electroencephalography shows triphasic waves. Which of the following is the most appropriate first-line therapy for his encephalopathy?

- **A.** Rifaximin 550 mg orally twice daily as monotherapy
- **B.** Intravenous flumazenil to reverse endozepine activity
- **C.** L-ornithine-L-aspartate infusion
- **D.** Lactulose 20 to 30 g orally two to four times daily
- **E.** Placement of an intraparenchymal intracranial pressure monitor

Answer: D. Lactulose 20 to 30 g orally two to four times daily

This patient has grade 2 hepatic encephalopathy by the West Haven criteria, with lethargy, disorientation to time, and impaired subtraction. Lactulose is a nonabsorbable disaccharide and is the first-line agent for overt hepatic encephalopathy, given orally 2 to 4 times daily at 20 to 30 g per dose and titrated to stool output. Rifaximin, at 550 mg orally twice daily, is described as an add-on to lactulose for secondary prophylaxis rather than as initial monotherapy. Flumazenil is listed only as an adjunct in the intensive management of intracranial hypertension from acute liver failure, not as first-line therapy for overt encephalopathy. L-ornithine-L-aspartate was studied and did not affect ammonia or survival, unlike glycerol phenylbutyrate, which reduced hepatic encephalopathy events and ammonia. Intracranial pressure monitoring is reserved for acute liver failure with advanced encephalopathy after correction of coagulopathy, and even then its benefit has been questioned.

Key point:
lactulose 20 to 30 g orally two to four times daily, titrated to stool output, is first-line for overt hepatic encephalopathy, with rifaximin 550 mg twice daily added for secondary prophylaxis.

Source: Practice of Neurocritical Care, 2nd ed., Metabolic Encephalopathies and Delirium, pp. 296-297

Question 114

Metabolic Encephalopathies and Delirium · Hyponatremic encephalopathy - correction limits · hard

A 47-year-old malnourished woman with a long history of heavy alcohol use is brought in with 4 days of progressive confusion. She is somnolent and disoriented but has had no seizures. Serum sodium is 104 mEq/L, potassium is 2.9 mEq/L, and a value of 106 mEq/L is documented from an outside clinic 5 days earlier. She is started on careful sodium correction. Which of the following is the maximum increase in serum sodium that should be permitted over the first 24 hours?

- **A.** 2 mEq/L
- **B.** 4 mEq/L
- **C.** 8 mEq/L
- **D.** 12 mEq/L
- **E.** 18 mEq/L

Answer: C. 8 mEq/L

This is chronic hyponatremia, since the sodium has been at or below 120 mEq/L for more than 48 hours, and the brain has already adapted by shedding solute, so rapid correction risks osmotic demyelination. The chapter identifies patients at high risk for osmotic demyelination as those with sodium at or below 105 mEq/L, hypokalemia, alcohol addiction, malnutrition, or advanced liver disease, and in these patients correction should not exceed 8 mEq/L in 24 hours. This patient meets several of those criteria at once. Correction of 4 to 6 mEq/L in 24 hours is the stated goal for gradual correction of chronic hyponatremia and is a reasonable target, but the question asks for the maximum permitted ceiling, which is 8 mEq/L. Increases of 10 to 12 mEq/L over 24 hours, or up to 18 mEq/L over 48 hours, apply only to patients at normal risk for demyelination, and 10 mEq/L in 24 hours is the general upper limit even in acute symptomatic cases treated with hypertonic saline. Osmotic demyelination, formerly called central pontine myelinolysis, can range from spasticity to locked-in syndrome to coma.

Key point:
in patients at high risk for osmotic demyelination, sodium correction must not exceed 8 mEq/L in any 24-hour period.

Source: Practice of Neurocritical Care, 2nd ed., Metabolic Encephalopathies and Delirium, pp. 300

Question 115

Metabolic Encephalopathies and Delirium · Wernicke's encephalopathy - thiamine dosing · easy

A 44-year-old woman with metastatic gastric cancer has been receiving chemotherapy and has eaten almost nothing for 3 weeks. She develops inattention, indifference, and impaired memory. Examination shows horizontal nystagmus, bilateral lateral rectus palsies, and truncal ataxia. MRI FLAIR sequences show high signal in the medial thalami, mammillary bodies, periaqueductal gray, and midbrain tectum. Which of the following is the most appropriate initial treatment regimen?

- **A.** Thiamine 100 mg intravenously once daily
- **B.** Thiamine 200 to 500 mg intravenously every 8 hours
- **C.** Thiamine 50 mg orally once daily
- **D.** Intravenous dextrose infusion followed by oral multivitamins
- **E.** Methylprednisolone 1 g intravenously daily for 5 days

Answer: B. Thiamine 200 to 500 mg intravenously every 8 hours

The ocular signs, truncal ataxia, and the medial thalamic, mammillary body, periaqueductal, and tectal FLAIR changes are characteristic of Wernicke's encephalopathy, which can follow malnutrition, cancer, chemotherapy, or liver or kidney disease and is not confined to alcoholism. Clinical practice guidelines cited in the chapter recommend high-dose thiamine, 200 to 500 mg intravenously every 8 hours, as the initial regimen, with the frequency based on pharmacokinetics rather than the once-daily dosing used in earlier studies. Lower doses of 50 to 100 mg per day will improve the ocular symptoms within hours to days, but they are not the recommended initial regimen and the ataxia improves less often. Oral thiamine is inadequate in a malnourished patient with acute encephalopathy. Giving dextrose without high-dose thiamine risks precipitating or worsening the syndrome. Corticosteroids treat Hashimoto's or steroid-responsive encephalopathy, not thiamine deficiency.

Key point:
initial treatment of Wernicke's encephalopathy is thiamine 200 to 500 mg intravenously every 8 hours.

Source: Practice of Neurocritical Care, 2nd ed., Metabolic Encephalopathies and Delirium, pp. 299-300

Question 116

Metabolic Encephalopathies and Delirium · Alcohol withdrawal - timing of syndromes · moderate

A 55-year-old man who drinks a liter of vodka daily was admitted 3 days ago after a fall. His last drink was approximately 72 hours before the current evaluation. He now has a temperature of 38.4 C, heart rate 132/min, blood pressure 176/98 mm Hg, profuse diaphoresis, coarse tremor, and vivid visual and tactile hallucinations, and he is disoriented to place and time. He has received no antipsychotics. Serum sodium, glucose, and ammonia are normal, and head CT is unremarkable. Which of the following is the most likely diagnosis?

- **A.** Alcohol withdrawal seizures
- **B.** Wernicke's encephalopathy
- **C.** Neuroleptic malignant syndrome
- **D.** Sepsis-associated encephalopathy
- **E.** Delirium tremens

Answer: E. Delirium tremens

The chapter gives an explicit timeline for alcohol withdrawal encephalopathy: tremulousness and hyperactivity begin within hours of cessation and peak at 10 to 30 hours, withdrawal seizures have an onset of 6 to 48 hours and peak at 13 to 24 hours, hallucinations begin at 8 to 48 hours and last 1 to 6 days, and delirium tremens has an onset of 60 to 96 hours with severe autonomic hyperactivity. This patient at 72 hours with fever, marked tachycardia, hypertension, diaphoresis, hallucinations, and confusion fits delirium tremens, which carries 15 percent mortality if left untreated. Withdrawal seizures would have occurred earlier and this patient has had none. Wernicke's encephalopathy is characterized by ocular signs such as nystagmus and lateral rectus palsy plus ataxia, which are absent here. Neuroleptic malignant syndrome is a complication of antipsychotics, which he has not received. Sepsis-associated encephalopathy would require an infectious source, and the workup is otherwise unrevealing.

Key point:
delirium tremens characteristically begins 60 to 96 hours after the last drink, whereas withdrawal seizures peak at 13 to 24 hours.

Source: Practice of Neurocritical Care, 2nd ed., Metabolic Encephalopathies and Delirium, pp. 299

Question 117

Metabolic Encephalopathies and Delirium · NMDA receptor encephalitis - initial management · moderate

A 22-year-old woman presents after a week of headache, fever, and vomiting, followed by paranoia, delusions, and echolalia. Over the next several days she develops orofacial dyskinesias, choreoathetoid limb movements, sialorrhea, tachycardia, and periods of hyperventilation. CSF shows lymphocytic pleocytosis with mildly elevated protein and oligoclonal bands, brain MRI is unremarkable, and continuous EEG shows rhythmic delta activity with superimposed bursts of 20 to 30 Hz beta activity riding on each delta wave. Serum and CSF N-methyl-D-aspartate receptor antibodies are positive. Which of the following is the most appropriate first step in her treatment?

- **A.** Imaging of the abdomen and pelvis to detect and remove an ovarian teratoma
- **B.** Immediate initiation of chronic azathioprine
- **C.** Rituximab combined with cyclophosphamide as the initial regimen
- **D.** Continuous propofol and sevoflurane sedation to suppress the dyskinesias
- **E.** Empiric high-dose acyclovir alone pending repeat CSF testing

Answer: A. Imaging of the abdomen and pelvis to detect and remove an ovarian teratoma

The prodromal, early psychiatric, and late movement disorder and autonomic phases, together with the extreme delta brush pattern on EEG and positive antibodies, establish NMDA receptor encephalitis. The chapter states that the first step in treating these patients is tumor detection and removal if a tumor is present, and ovarian teratomas are found in 26 to 59 percent of cases, with 80 percent of patients being female. Immunotherapy with IVIG or plasma exchange plus methylprednisolone follows, with rituximab alone or with cyclophosphamide reserved for the most refractory cases, so option C is not the initial regimen. Azathioprine or mycophenolate mofetil are for chronic immunosuppression in the 20 to 25 percent of patients who recur, not for initial therapy. Propofol and sevoflurane should specifically be avoided during anesthesia in these patients because they can exacerbate dyskinesias and seizures; benzodiazepines, opioids, and curares are used instead. Acyclovir alone does not treat an established antibody-mediated encephalitis.

Key point:
in NMDA receptor encephalitis the first therapeutic step is to find and remove an underlying tumor, most often an ovarian teratoma, before or alongside immunotherapy.

Source: Practice of Neurocritical Care, 2nd ed., Metabolic Encephalopathies and Delirium, pp. 302

Question 118

Metabolic Encephalopathies and Delirium · Posterior reversible leukoencephalopathy syndrome · moderate

A 34-year-old woman is 6 weeks post allogeneic stem cell transplantation and is maintained on tacrolimus. She develops severe headache, vomiting, visual disturbance, and a generalized tonic-clonic seizure. Blood pressure is 196/112 mm Hg. MRI shows bilateral parieto-occipital cortical and subcortical FLAIR hyperintensities without restricted diffusion, and additional patchy frontal lobe involvement. Which of the following is the most appropriate management?

- **A.** Intravenous thrombolysis for presumed posterior circulation ischemia
- **B.** Therapeutic anticoagulation for cerebral venous sinus thrombosis
- **C.** Induced hypertension to augment posterior circulation perfusion
- **D.** Emergent decompressive suboccipital craniectomy
- **E.** Blood pressure control and withdrawal of the tacrolimus

Answer: E. Blood pressure control and withdrawal of the tacrolimus

The clinical picture of headache, vomiting, seizure, visual disturbance, severe hypertension, and parieto-occipital vasogenic edema in a transplant patient on a calcineurin inhibitor is posterior reversible leukoencephalopathy syndrome, classically associated with eclampsia, severe hypertension, and immunosuppression with cyclosporine or tacrolimus. The mainstay of treatment is control of hypertension, removal of the offending agent, and treatment of any predisposing condition. Seizures occur in up to 90 percent of cases and hypertension is present in only 50 to 70 percent, so a normal blood pressure would not exclude the diagnosis. The frontal lobe involvement here does not argue against the diagnosis, since PRES can also affect frontal, watershed, or subcortical areas. Thrombolysis, anticoagulation, induced hypertension, and craniectomy all either treat the wrong disease or would worsen the hyperperfusion and edema. The syndrome is not uniformly reversible: in up to 25 percent of cases the lesions are irreversible and deficits persist, 44 percent of survivors of severe PRES had severe functional impairment at day 90, and hyperglycemia on day 1 and time to control of the causative factor were strong early predictors of outcome.

Key point:
treat PRES by lowering the blood pressure and removing the offending agent, and remember that it is neither always posterior nor always reversible.

Source: Practice of Neurocritical Care, 2nd ed., Metabolic Encephalopathies and Delirium, pp. 298-299

Multimodality Neuromonitoring

Question 119

Multimodality Neuromonitoring · Rationale and pathophysiology of secondary brain injury · moderate

A 32-year-old man is admitted to the neurocritical care unit after a high-speed motor vehicle collision. After intubation and adequate resuscitation his Glasgow Coma Scale score is 6 off sedation, with no eye opening and extensor posturing on the left. Blood pressure is 132/74 mmHg, pulse 88 beats per minute, and oxygen saturation 100%. A repeat noncontrast head CT at 6 hours shows stable traumatic subarachnoid hemorrhage and diffuse cerebral edema without herniation or mass effect requiring surgery. The attending proposes placing an intracranial multimodality monitoring bolt, and a trainee asks why serial neurological examinations and repeat CT imaging are not sufficient. Which of the following best explains the pathophysiologic rationale for invasive multimodality monitoring in this patient?

- **A.** True ischemic burden after severe traumatic brain injury involves the majority of the brain volume, so global imaging reliably identifies tissue at risk
- **B.** Secondary injury develops in patchy islands of metabolically vulnerable tissue in which blood flow is uncoupled from metabolism, and global imaging and the motor examination detect this only after injury is established
- **C.** Neuroworsening after traumatic brain injury occurs in fewer than 5% of patients, so monitoring serves mainly to enroll patients in research protocols
- **D.** Ischemia after traumatic brain injury is confined to the first 24 hours, after which normal flow-metabolism coupling is restored and monitoring adds little
- **E.** Pupillary asymmetry and a declining motor Glasgow Coma Scale subscore are sensitive early markers of regional metabolic distress in comatose patients

Answer: B. Secondary injury develops in patchy islands of metabolically vulnerable tissue in which blood flow is uncoupled from metabolism, and global imaging and the motor examination detect this only after injury is established

Positron emission tomography and combined PET-microdialysis studies in severe traumatic brain injury show a surprisingly low true ischemic burden of only 1.5% to 6% of total brain volume, yet more than three-quarters of patients harbor metabolically vulnerable tissue, so option A is wrong. These diffuse injuries create patchy islands of tissue with capillary transit time heterogeneity, diffusion hypoxia, and mitochondrial failure, all of which uncouple blood flow and substrate delivery from metabolism, which is exactly why measurements that infer regional metabolic status add information beyond the exam and CT. Option D is incorrect because the largest PET study of 68 patients with traumatic brain injury showed ischemia occurring up to 10 days after trauma, uncoupled from the normal flow-metabolism relationship. Option C is wrong because neurodeterioration occurs in 26% of patients with traumatic brain injury and raises mortality from 10% to 56%; delayed decline also occurs in 30% of subarachnoid hemorrhage and 13% to 29% of intracerebral hemorrhage. Option E is wrong because clinical changes involving critical structures such as pupillary asymmetry or a worsening motor subscore reflect pathology that already involves large tissue volumes and therefore detect neurodeterioration too late.

Key point:
multimodality monitoring exists because secondary brain injury is regional and metabolic, and global measures such as CT and the coma examination miss it until it is irreversible.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 317

Question 120

Multimodality Neuromonitoring · ICP monitoring: treatment threshold and device selection · moderate

A 45-year-old woman is admitted with severe traumatic brain injury and a post-resuscitation Glasgow Coma Scale score of 5. Head CT shows diffuse cerebral edema with slit-like lateral ventricles and no hydrocephalus. Her international normalized ratio is 1.0, platelet count is 210 x 10^9/L, and she takes no antiplatelet agents. The team plans intracranial pressure monitoring. Which of the following statements is most consistent with current guidelines and the comparative data on monitor type?

A. Treatment should be initiated once intracranial pressure exceeds 15 mmHg, and an external ventricular drain is preferred because it carries a lower infection risk than a parenchymal monitor

B. Intracranial pressure should be documented hourly alongside other vital signs, a practice that captures essentially all clinically important elevations

C. A ventriculostomy is required in this patient because parenchymal monitors cannot provide reliable measurements when the ventricles are small

D. Treatment should be initiated once intracranial pressure exceeds 22 mmHg, and a parenchymal monitor is reasonable here and carries lower hemorrhage and infection risk than an external ventricular drain

E. Intracranial pressure monitoring should be omitted because a randomized trial showed that monitored patients received more therapies and had worse functional outcomes

Answer: D. Treatment should be initiated once intracranial pressure exceeds 22 mmHg, and a parenchymal monitor is reasonable here and carries lower hemorrhage and infection risk than an external ventricular drain

Current severe traumatic brain injury guidelines recommend intracranial pressure monitoring to reduce in-hospital and two-week mortality and recommend treating intracranial pressure above 22 mmHg, both at level IIb evidence, so option A uses the wrong threshold. A parenchymal monitor is appropriate when there is no need or ability to drain cerebrospinal fluid, and it is the safer device: parenchymal sensor hemorrhage risk is roughly 1% and infection 0.6% to 2.1%, versus external ventricular drain hemorrhage of 6% to 7% and infection of about 8%, with a meta-analysis confirming higher infection and hemorrhage risk with external ventricular drainage and a post hoc randomized trial analysis linking external ventricular drains to worse functional and cognitive outcomes. Option B is wrong because low-frequency hourly documentation misses about one-quarter of intracranial pressure elevations above 20 mmHg lasting 15 minutes or longer, whereas minute-to-minute data show that 89% of patients with traumatic brain injury have elevations and that intracranial pressure burden correlates with functional outcome. Option C is wrong because parenchymal monitors are placed in subcortical tissue and do not require patent ventricles; slit ventricles in fact favor a parenchymal device. Option E misstates the randomized trial of 324 patients from low-income countries, in which patients managed without monitors received more therapies and there was no difference in functional outcome.

Key point:
treat intracranial pressure above 22 mmHg, and choose a parenchymal monitor when cerebrospinal fluid diversion is not needed, since it is safer than an external ventricular drain.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 321

Question 121

Multimodality Neuromonitoring · Brain tissue oxygen: probe validation and the oxygen challenge · easy

A 28-year-old man is admitted after an unhelmeted motorcycle crash with a Glasgow Coma Scale score of 6 after intubation. A right frontal bolt with combined intracranial pressure and brain tissue oxygen catheters was placed 2 hours ago, and a confirmatory noncontrast head CT shows the catheters within subcortical white matter with no periprocedural hemorrhage. Current values are heart rate 80 beats per minute, mean arterial pressure 85 mmHg, intracranial pressure 18 mmHg, cerebral perfusion pressure 67 mmHg, and brain tissue oxygen 7 mmHg. Oxygen saturation is 99% on a fraction of inspired oxygen of 0.35, and hemoglobin is 10.2 g/dL. Which of the following is the most appropriate next step?

- **A.** Increase the fraction of inspired oxygen to 1.0 for 10 to 15 minutes and observe the brain tissue oxygen value for a rise
- **B.** Administer mannitol 1 g/kg intravenously over 15 minutes to lower the intracranial pressure
- **C.** Start norepinephrine to raise the cerebral perfusion pressure above 70 mmHg
- **D.** Transfuse packed red blood cells to a hemoglobin above 10 g/dL to increase oxygen delivery
- **E.** Remove the bolt and replace the brain tissue oxygen catheter at a contralateral site

Answer: A. Increase the fraction of inspired oxygen to 1.0 for 10 to 15 minutes and observe the brain tissue oxygen value for a rise

A brain tissue oxygen value of 7 mmHg is well below the abnormal threshold of 20 mmHg and below the 15 mmHg level associated with hypoxic-ischemic injury, but before acting on it the probe's validity must be established, because invasive devices require time to stabilize and brain tissue oxygen equilibration can take 2 or more hours after placement. The recommended maneuver is an oxygen challenge: raise the fraction of inspired oxygen, thereby raising arterial oxygen tension, and watch for a proportional rise in dissolved brain tissue oxygen over 10 to 15 minutes; failure to rise indicates a faulty or malpositioned probe. Option B is not indicated because the intracranial pressure of 18 mmHg is below the 22 mmHg treatment threshold. Option C is premature and not benign, since a randomized trial of driving cerebral perfusion pressure above 70 mmHg in severe traumatic brain injury produced significantly more acute lung injury from the fluids and pressors required. Option D is unhelpful because the hemoglobin is already above 10 g/dL, and option E is wrong because the CT has already confirmed satisfactory catheter position and the probe has not yet been challenged.

Key point:
before treating a low brain tissue oxygen value, confirm the probe works by raising the fraction of inspired oxygen to 1.0 and watching for a rise over 10 to 15 minutes.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 316, 322, 325

Question 122

Multimodality Neuromonitoring · Autoregulation, pressure reactivity index, and optimum CPP · moderate

A 34-year-old woman with Hunt-Hess grade 4 aneurysmal subarachnoid hemorrhage is on postbleed day 4 and postoperative day 3 after coiling. She remains intubated with a Glasgow Coma Scale score of 7. Multimodality monitoring shows intracranial pressure 24 mmHg, mean arterial pressure 80 mmHg, and cerebral perfusion pressure 56 mmHg. The bedside nurse reports that during the neurological examination the mean arterial pressure rose and the intracranial pressure fell precipitously, and that the intracranial pressure rose again when sedation was restarted and the mean arterial pressure fell. She has received a hypertonic saline infusion and her serum sodium is 158 mEq/L; she is euvolemic. Transcranial Doppler shows only very mild proximal vasospasm. Which of the following is the most appropriate next step in managing her intracranial pressure?

- **A.** Administer mannitol 1 g/kg intravenously over 15 minutes
- **B.** Start a neuromuscular blocking agent to reduce intrathoracic pressure and improve cerebral venous return
- **C.** Start a vasopressor to raise the mean arterial pressure by approximately 10 mmHg and reassess the intracranial pressure
- **D.** Take the patient to angiography for intra-arterial verapamil
- **E.** Hyperventilate to a partial pressure of carbon dioxide of 25 to 30 mmHg

Answer: C. Start a vasopressor to raise the mean arterial pressure by approximately 10 mmHg and reassess the intracranial pressure

The observed inverse relationship, in which a rise in mean arterial pressure lowers the intracranial pressure, indicates that autoregulation is at least partly preserved and that the cerebral perfusion pressure of 56 mmHg sits below the lower limit of autoregulation, where compensatory arteriolar vasodilation increases cerebral blood volume and therefore intracranial pressure. The correct maneuver is a trial of raising blood pressure toward the optimum cerebral perfusion pressure, which is identified as the nadir of the U-shaped curve relating the pressure reactivity index to cerebral perfusion pressure; it averages about 70 mmHg after severe traumatic brain injury and rises from a mean of 78 to a mean of 98 mmHg in high-grade subarachnoid hemorrhage complicated by delayed cerebral ischemia. Options A and E are second-line here and osmotherapy is additionally constrained by a serum sodium already at 158 mEq/L, while neuromuscular blockade in option B is a later tier of therapy that would eliminate the exam without addressing the pressure-dependent physiology. Option D is wrong because transcranial Doppler shows minimal spasm and intra-arterial calcium channel blockers such as verapamil can raise intracranial pressure by causing relative vasodilation and increasing cerebral blood volume within a closed cranial vault. Note that a positive pressure reactivity index, particularly above 0.25, would instead indicate dysfunctional autoregulation in which raising mean arterial pressure passively raises intracranial pressure.

Key point:
when raising the mean arterial pressure lowers the intracranial pressure, the perfusion pressure is below the lower limit of autoregulation, and the first move is to raise blood pressure toward the optimum cerebral perfusion pressure rather than escalate osmotherapy.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 315-316, 322

Question 123

Multimodality Neuromonitoring · Cerebral microdialysis: ischemia versus mitochondrial dysfunction · hard

A 40-year-old man is on hospital day 3 after severe traumatic brain injury with right frontal multimodality monitoring in place, including intracranial pressure, brain tissue oxygen, thermal diffusion flowmetry, and cerebral microdialysis catheters. Current values are intracranial pressure 14 mmHg, cerebral perfusion pressure 72 mmHg, brain tissue oxygen 28 mmHg, and regional cerebral blood flow 22 mL/100 g per minute. Serum glucose is 145 mg/dL, hemoglobin is 11.5 g/dL, and temperature is 37.2 degrees Celsius. Microdialysis shows a lactate-pyruvate ratio of 48, a pyruvate of 145 micromol/L, and a brain glucose of 1.6 mmol/L, or 29 mg/dL. Which of the following best explains this pattern?

- **A.** Ischemia from inadequate delivery of substrate to the monitored tissue
- **B.** Systemic hypoglycemia causing a brain substrate deficit
- **C.** Large-vessel vasospasm within the anterior circulation
- **D.** Artifact from failure to discard the first hour of collected dialysate
- **E.** Mitochondrial dysfunction, also called energy failure, in the monitored tissue

Answer: E. Mitochondrial dysfunction, also called energy failure, in the monitored tissue

Interpretation of an elevated lactate-pyruvate ratio depends entirely on the pyruvate concentration: pyruvate falls in ischemic tissue, whereas pyruvate remains normal or elevated in tissue that cannot use it efficiently, which defines energy failure or mitochondrial dysfunction. Here the ratio of 48 exceeds the abnormal threshold of 25 to 40 but the pyruvate of 145 micromol/L is well above 70 micromol/L, and mitochondrial dysfunction defined by a ratio above 25 with pyruvate above 70 is roughly 5 times as common as ischemia after traumatic brain injury; in subarachnoid hemorrhage, mitochondrial dysfunction defined by a ratio above 30 with pyruvate above 70 occurred in 64% of patients versus 18% for ischemia defined by pyruvate below 70. Option A is excluded because ischemia requires a low pyruvate and would be expected alongside a low brain tissue oxygen, low regional blood flow, and low brain glucose, none of which are present. Option B is excluded because the serum glucose is 145 mg/dL and the brain glucose of 29 mg/dL is well above the low-brain-glucose threshold of 0.8 mmol/L, or 14.4 mg/dL. Option C is unlikely with a preserved perfusion pressure, normal white matter blood flow of 22 mL/100 g per minute, and normal brain tissue oxygen, and option D is wrong because the requirement to discard the first hour of dialysate applies only at the start of monitoring, not on day 3.

Key point:
an elevated lactate-pyruvate ratio with pyruvate above 70 micromol/L means mitochondrial dysfunction, whereas the same ratio with pyruvate below 70 micromol/L means ischemia.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 324-325

Question 124

Multimodality Neuromonitoring · Brain glucose and glycemic targets · moderate

A 52-year-old woman with severe traumatic brain injury is on hospital day 4 with a right frontal multimodality monitoring bolt. She is on an insulin infusion per a unit protocol targeting a serum glucose of 90 to 110 mg/dL, and her current serum glucose is 100 mg/dL. Intracranial pressure is 14 mmHg, cerebral perfusion pressure is 70 mmHg, brain tissue oxygen is 26 mmHg, and temperature is 37.0 degrees Celsius. Cerebral microdialysis shows a brain glucose of 0.6 mmol/L, or 10.8 mg/dL, a lactate-pyruvate ratio of 32, and a pyruvate of 110 micromol/L. Which of the following is the most appropriate next step?

- **A.** Intensify the insulin infusion to keep the serum glucose below 100 mg/dL
- **B.** Administer hypertonic saline to lower the intracranial pressure
- **C.** Start a vasopressor to raise the cerebral perfusion pressure above 80 mmHg
- **D.** Liberalize the serum glucose target to 120 to 180 mg/dL
- **E.** Replace the microdialysis catheter because the values are internally inconsistent

Answer: D. Liberalize the serum glucose target to 120 to 180 mg/dL

Consensus recommendations define low brain glucose as less than 0.8 mmol/L, or 14.4 mg/dL, and this patient's brain glucose of 0.6 mmol/L is below that threshold despite a normal serum glucose, illustrating that the usual correlation between serum and brain glucose is lost in injured brain. The brain-to-serum glucose ratio here is 10.8 divided by 100, or 0.108, and a ratio below 0.12 has been associated with metabolic crisis and worse outcome, including in-hospital mortality. Tight serum glucose goals such as less than 120 mg/dL have been associated with lower brain glucose and higher lactate-pyruvate ratios compared with targets of 120 to 180 mg/dL, and tight glycemic control in traumatic brain injury is associated with more frequent episodes of metabolic crisis, so option A would worsen the problem; mortality occurs more often in patients whose average brain glucose is 0.5 mmol/L, or 9 mg/dL. Option B is not indicated with an intracranial pressure of 14 mmHg, and option C is not indicated because the cerebral perfusion pressure and brain tissue oxygen are already adequate. Option E is wrong because the pattern is internally coherent: a ratio above 25 with a pyruvate above 70 micromol/L reflects mitochondrial dysfunction rather than probe malfunction.

Key point:
brain glucose below 0.8 mmol/L or a brain-to-serum glucose ratio below 0.12 signals metabolic crisis, and the fix is to abandon tight glycemic control in favor of a serum glucose of 120 to 180 mg/dL.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 325, 327

Question 125

Multimodality Neuromonitoring · Cortical EEG, invasive seizure detection, and spreading depolarizations · moderate

A 60-year-old man with severe traumatic brain injury and a post-intubation Glasgow Coma Scale score of 6 undergoes emergency craniotomy for evacuation of an acute subdural hematoma. At the end of the case, a six-contact subdural strip electrode is placed over the cortex and connected to a full-band amplifier as part of an integrated multimodality monitoring strategy, and continuous scalp electroencephalography is also running. Which of the following statements about cortical electroencephalography in this setting is correct?

- **A.** Continuous scalp electroencephalography is more sensitive than a cortical strip electrode for detecting electrographic seizures
- **B.** The strip electrode must be removed in the operating room at the conclusion of monitoring
- **C.** Focal seizures are recorded from the cortex in roughly half of monitored patients, and approximately 40% of these are missed by scalp electroencephalography
- **D.** Spreading depolarizations propagate at approximately 3 centimeters per second and appear as bursts of fast activity on the cortical recording
- **E.** Spreading depolarizations are rare after traumatic brain injury, occurring in fewer than 10% of monitored patients

Answer: C. Focal seizures are recorded from the cortex in roughly half of monitored patients, and approximately 40% of these are missed by scalp electroencephalography

When electroencephalography is recorded directly from the cortex, focal seizures are found in 49% to 61% of patients, and approximately 40% of these are missed by scalp recording, because a seizure must involve roughly 6 to 10 centimeters of cortex to be visible on the scalp; this makes option A the reverse of the truth. Option B is incorrect because strip electrodes are removed at the bedside with gentle traction, complications of bedside removal are rare at about 1%, and patients with brain injury have not shown higher rates of bleeding or infection than the epilepsy surgery population. Option D misstates the physiology: spreading depolarizations propagate to contiguous tissue at approximately 3 millimeters per minute and manifest on full-band electroencephalography as a very slow shift in the direct current potential, which is precisely why a full-band amplifier is required. Option E is wrong because spreading depolarizations occur in 55% to 100% of patients with acute brain injury, and the incidence was 60% in the largest prospective series of surgical traumatic brain injury. These events matter clinically because both seizures and spreading depolarizations produce brain tissue hypoxia and metabolic crisis, and cerebral blood flow paradoxically falls in the wake of a spreading depolarization when neurovascular coupling is lost.

Key point:
cortical electrodes detect focal seizures in about half of monitored patients with roughly 40% invisible to scalp electroencephalography, and full-band recording is needed to see spreading depolarizations, which travel at about 3 millimeters per minute.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 323-324

Question 126

Multimodality Neuromonitoring · BOOST-II and brain oxygen-directed therapy · easy

In the phase II Brain Oxygen Optimization in Severe Traumatic Brain Injury (BOOST-II) randomized trial, which compared management directed at intracranial pressure alone with management directed at both intracranial pressure and brain tissue oxygen, which of the following describes the brain tissue oxygen target used and the principal result?

- **A.** A target above 15 mmHg, with a halving of mortality in the combined-management arm
- **B.** A target above 20 mmHg, with the proportion of patients experiencing brain tissue hypoxia falling from 44% to 15% and a trend toward improved functional outcome
- **C.** A target above 35 mmHg, with no reduction in the burden of brain tissue hypoxia in either arm
- **D.** A jugular venous oxygen saturation target above 50%, with a significant improvement in 6-month functional outcome
- **E.** A target above 20 mmHg, with a significant mortality reduction offset by a higher rate of acute lung injury

Answer: B. A target above 20 mmHg, with the proportion of patients experiencing brain tissue hypoxia falling from 44% to 15% and a trend toward improved functional outcome

BOOST-II used a brain tissue oxygen target above 20 mmHg and showed that active management reduced the proportion of patients with brain tissue hypoxia from 44% to 15%, a 66% decrease in the duration of brain tissue hypoxia, with only a trend toward improvement in functional outcome, which is why the definitive phase III BOOST-III trial with a target enrollment of 1,094 patients is underway. Option A uses the wrong threshold, since 15 mmHg is the level associated with hypoxic-ischemic injury while 20 mmHg is the abnormal threshold and treatment target; the trial was not powered for and did not show a mortality halving. Option C is wrong on both counts, since 25 to 35 mmHg is the normal range rather than a treatment target, and hypoxia burden fell substantially. Option D describes jugular venous oximetry, a separate modality that carries level III support in severe traumatic brain injury guidelines with a threshold saturation below 50%. Option E misattributes the acute lung injury finding, which came from the randomized trial of driving cerebral perfusion pressure above 70 mmHg, not from BOOST-II. Supporting evidence includes a meta-analysis in which brain tissue oxygen monitoring in severe traumatic brain injury was associated with double the odds of favorable outcome.

Key point:
BOOST-II targeted brain tissue oxygen above 20 mmHg and cut brain tissue hypoxia from 44% to 15% with only a trend toward better outcome, leaving the definitive answer to BOOST-III.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 322, 326-327

Question 127

Multimodality Neuromonitoring · Relative contraindications to invasive monitoring · easy

A 74-year-old man is admitted after falling down a flight of stairs. His Glasgow Coma Scale score after intubation and resuscitation is 6, and he does not follow commands. His alcohol screen is negative and he has received no long-acting sedative or paralytic agents. Head CT shows bifrontal contusions and diffuse cerebral edema with no surgical lesion. The team plans to place an intracranial multimodality monitoring bolt. Which of the following findings represents a relative contraindication to this procedure?

- **A.** An arterial lactate of 4.5 mmol/L
- **B.** A platelet count of 140 x 10^9/L
- **C.** An age of 74 years
- **D.** A shortened thromboelastography R time
- **E.** An international normalized ratio of 1.8

Answer: E. An international normalized ratio of 1.8

Published relative contraindications to invasive multimodality monitoring include coagulopathy defined as an international normalized ratio above 1.5, current use of non-vitamin K antagonist anticoagulation, a partial thromboplastin time greater than 1.5 times normal, or dual antiplatelet therapy, so an international normalized ratio of 1.8 qualifies; because it is relative rather than absolute, a compelling indication may still permit placement after appropriate reversal. The other listed relative contraindications are age above 80 years, expected mortality within 24 to 48 hours or preexisting advance directives, a rapidly improving exam within 6 to 8 hours or reliable command following, and lack of clinical or technical support, so option C is wrong because this patient is 74. Lactic acidosis is not a contraindication, making option A incorrect, and a platelet count of 140 x 10^9/L is well above the 100 x 10^9/L generally considered adequate for intracranial procedures, making option B incorrect. A shortened thromboelastography R time reflects faster clot initiation and does not indicate coagulopathy, whereas a prolonged R time would suggest decreased coagulation factor activity, so option D is incorrect. This risk assessment matters because significant hemorrhage after bolt placement occurs in roughly 2% of patients, and a confirmatory noncontrast head CT is recommended after placement to assess for periprocedural hemorrhage and catheter position.

Key point:
an international normalized ratio above 1.5, a partial thromboplastin time above 1.5 times normal, dual antiplatelet therapy, and age above 80 years are the classic relative contraindications to invasive intracranial monitoring.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 318

Question 128

Multimodality Neuromonitoring · Detecting delayed cerebral ischemia with regional monitoring · hard

A 45-year-old woman with Hunt-Hess grade 4 aneurysmal subarachnoid hemorrhage underwent clipping and remains intubated with a Glasgow Coma Scale score of 7. On postbleed day 3 a multimodality bolt was placed in the left frontal subcortical white matter, contralateral to the craniotomy, at the border zone between the anterior and middle cerebral artery territories, with intracranial pressure, brain tissue oxygen, thermal diffusion flowmetry, and microdialysis catheters. On postbleed day 7 she develops flaccid right leg weakness and abulia. Which of the following sets of monitoring values is most consistent with delayed cerebral ischemia within the monitored region?

- **A.** Intracranial pressure 17 mmHg, cerebral perfusion pressure 95 mmHg, brain tissue oxygen 7 mmHg, brain glucose 6.3 mg/dL, lactate-pyruvate ratio 42, regional cerebral blood flow 6 mL/100 g per minute
- **B.** Intracranial pressure 14 mmHg, cerebral perfusion pressure 90 mmHg, brain tissue oxygen 25 mmHg, brain glucose 25 mg/dL, lactate-pyruvate ratio 23, regional cerebral blood flow 25 mL/100 g per minute
- **C.** Intracranial pressure 18 mmHg, cerebral perfusion pressure 95 mmHg, brain tissue oxygen 18 mmHg, brain glucose 20 mg/dL, lactate-pyruvate ratio 28, regional cerebral blood flow 45 mL/100 g per minute
- **D.** Intracranial pressure 32 mmHg, cerebral perfusion pressure 70 mmHg, brain tissue oxygen 15 mmHg, brain glucose 23 mg/dL, lactate-pyruvate ratio 38, regional cerebral blood flow 15 mL/100 g per minute
- **E.** Intracranial pressure 8 mmHg, cerebral perfusion pressure 75 mmHg, brain tissue oxygen 30 mmHg, brain glucose 30 mg/dL, lactate-pyruvate ratio 15, regional cerebral blood flow 22 mL/100 g per minute

Answer: A. Intracranial pressure 17 mmHg, cerebral perfusion pressure 95 mmHg, brain tissue oxygen 7 mmHg, brain glucose 6.3 mg/dL, lactate-pyruvate ratio 42, regional cerebral blood flow 6 mL/100 g per minute

Right leg weakness with abulia localizes to the left medial frontal lobe in the anterior cerebral artery distribution, and because bolts are placed at the anterior-middle cerebral artery border zone, the probes should register the event. Option A shows the complete ischemic signature: brain tissue oxygen below the 15 mmHg ischemic threshold, brain glucose of 6.3 mg/dL far below the 14.4 mg/dL low-brain-glucose threshold, a lactate-pyruvate ratio above 40, and a regional blood flow of 6 mL/100 g per minute, which is below the subcortical white matter ischemic threshold of 7.4 mL/100 g per minute; a ratio above 40 may rise up to 16 hours before delayed cerebral ischemia becomes clinically apparent, and continuously measured flow falls below 15 mL/100 g per minute in patients who develop delayed cerebral ischemia. Option B is essentially normal subcortical physiology in the setting of an elevated perfusion pressure. Option C shows borderline oxygen and glucose with a hyperemic flow of 45 mL/100 g per minute, above the 35 mL/100 g per minute hyperemia threshold, suggesting dysregulated flow or capillary transit time heterogeneity rather than ischemia. Option D reflects intracranial hypertension above 22 mmHg with a perfusion pressure still within the recommended 60 to 70 mmHg range and only borderline oxygen and oligemic flow, and option E is normal across every parameter.

Key point:
regional ischemia is defined by the convergence of brain tissue oxygen below 15 mmHg, brain glucose below 14.4 mg/dL, a lactate-pyruvate ratio above 40, and white matter blood flow below about 7.4 mL/100 g per minute.

Source: Practice of Neurocritical Care, 2nd ed., Multimodality Neuromonitoring, pp. 323, 327, 331-332

Neurocritical Care Nursing Special Considerations

Question 129

Neurocritical Care Nursing Special Considerations · Nursing interventions and intracranial pressure - endotracheal suctioning · moderate

A 58-year-old man with severe traumatic brain injury is intubated and has an intraparenchymal intracranial pressure monitor in place, with pressures running between 12 and 18 mm Hg. His oxygen saturation drifts to 91 percent and coarse secretions are audible in the endotracheal tube. The nurse orienting on the unit is reluctant to suction him for fear of raising the intracranial pressure. Which of the following is the most appropriate approach?

- **A.** Withhold endotracheal suctioning entirely while the intracranial pressure monitor is in place
- **B.** Hyperoxygenate with up to 100 percent oxygen for 30 to 60 seconds, then limit suctioning to one or two catheter passes of no more than 15 seconds each
- **C.** Suction with four to five catheter passes of 30 seconds each to clear the secretions in a single episode
- **D.** Suction only after a bolus of mannitol has been given to create intracranial pressure headroom
- **E.** Suspend repositioning and chest physiotherapy for 24 hours because both reliably raise intracranial pressure

Answer: B. Hyperoxygenate with up to 100 percent oxygen for 30 to 60 seconds, then limit suctioning to one or two catheter passes of no more than 15 seconds each

Endotracheal suctioning is a standard of care in mechanically ventilated patients and is part of bronchial hygiene therapy, preventing tube obstruction by secretions. Although intracranial pressure may rise transiently from coughing, hypoxia, or tracheal stimulation, suctioning itself has not been associated with immediate elevation of intracranial pressure. The recommended technique is hyperoxygenation by increasing delivered oxygen up to 100 percent for 30 to 60 seconds before suctioning, with safe suctioning limited to one or two passes of the catheter, each lasting no more than 15 seconds. Withholding suctioning entirely risks tube obstruction and hypoxia, which is itself a driver of intracranial hypertension. Prolonged multi-pass suctioning exceeds the cited safety limits. Osmotic therapy is not a prerequisite for routine suctioning. Repositioning every 2 hours had no effect on intracranial pressure after 1 minute in a multicenter observational study, and chest physiotherapy should be performed even in patients at risk for intracranial pressure elevation, so suspending both is wrong.

Key point:
hyperoxygenate to 100 percent for 30 to 60 seconds, then suction with one or two passes of no more than 15 seconds each.

Source: Practice of Neurocritical Care, 2nd ed., Neurocritical Care Nursing Special Considerations, pp. 343-344

Question 130

Neurocritical Care Nursing Special Considerations · Enteral nutrition - gastric residual volumes · hard

A 66-year-old woman with a large left middle cerebral artery infarction is receiving continuous nasogastric tube feedings. On a routine check the gastric residual volume is 280 mL, and a repeat check returns 300 mL. Her abdomen is soft and nontender with normal bowel sounds, she has no nausea, vomiting, or distention, and the head of the bed is at 30 degrees. Which of the following is the most appropriate next step?

- **A.** Stop the tube feedings and obtain an urgent abdominal radiograph
- **B.** Stop the tube feedings permanently and begin parenteral nutrition
- **C.** Hold the feedings for 12 hours and then restart at half the previous rate
- **D.** Continue the feedings and notify the provider, who may order a promotility agent
- **E.** Lower the head of the bed to flat and recheck the residual volume in 1 hour

Answer: D. Continue the feedings and notify the provider, who may order a promotility agent

The American Society for Parenteral and Enteral Nutrition and Society of Critical Care Medicine guidelines cited in the chapter advise against holding tube feedings for gastric residual volumes below 500 mL unless the patient has other signs or symptoms of intolerance, and this patient has none. The chapter also states that if the gastric residual volume is 250 mL or greater after a second residual check, the provider should be notified because a promotility agent may be indicated, which is exactly the situation here at 300 mL on the second check. Gastric residual volumes should be monitored every 4 hours, or per facility protocol, in critically ill patients on continuous feedings to reduce aspiration risk. Stopping feeds or converting to parenteral nutrition is unwarranted at this volume without other intolerance, and early nutrition may reduce disease severity, decrease ICU length of stay, and improve outcomes. Feedings should be stopped immediately only for acute abdominal pain, abdominal rigidity, or vomiting. Lowering the head of the bed is the opposite of what is advised, since it should be kept at 30 degrees or greater to prevent reflux and aspiration.

Key point:
do not hold feeds for a residual below 500 mL without other intolerance, but notify the provider about a promotility agent when the second check is 250 mL or greater.

Source: Practice of Neurocritical Care, 2nd ed., Neurocritical Care Nursing Special Considerations, pp. 347

Question 131

Neurocritical Care Nursing Special Considerations · Targeted temperature management - shivering · easy

A 61-year-old man undergoing targeted temperature management for controlled normothermia after a large hemispheric infarction begins to shiver, confirmed on the Bedside Shiver Assessment Scale. He has been extubated, is taking oral medications, and has a documented history of provoked seizures, so the team wishes to avoid agents that lower the seizure threshold. Counterwarming and adjusting the water-cooling device have not controlled the shivering. Which of the following antishivering medications is available in an oral formulation?

- **A.** Buspirone
- **B.** Meperidine
- **C.** Dexmedetomidine
- **D.** Magnesium sulfate
- **E.** Cisatracurium

Answer: A. Buspirone

Shivering increases metabolism, oxygen consumption, energy expenditure, and carbon dioxide production and can negate the neuroprotective benefit of targeted temperature management, so it must be recognized and treated, usually first by the nurse using a standardized tool such as the Bedside Shiver Assessment Scale. Buspirone is described as the only medication used to treat shivering that is available orally, which makes it the correct choice for this extubated patient. Meperidine given as intermittent intravenous doses is effective for shivering but must be used with caution in the neurologic population because it lowers the seizure threshold, a particular concern in this patient. Dexmedetomidine, magnesium sulfate, and propofol are alternatives given as continuous intravenous infusions, not orally. Cisatracurium and other neuromuscular blocking agents are a last resort because they eliminate the ability to obtain a neurologic examination. Nonpharmacologic measures such as counterwarming, titrating the water-cooling machine temperature, or layering a cooling blanket between sheets should be tried first, as they were here.

Key point:
buspirone is the only orally available antishivering agent, and neuromuscular blockade is reserved for last because it abolishes the neurologic exam.

Source: Practice of Neurocritical Care, 2nd ed., Neurocritical Care Nursing Special Considerations, pp. 345

Question 132

Neurocritical Care Nursing Special Considerations · Targeted temperature management - temperature measurement and devices · moderate

A 47-year-old woman with a subarachnoid hemorrhage is started on targeted temperature management for controlled normothermia after repeated fevers. The unit currently monitors her temperature with a tympanic thermometer and cools her with ice packs and an air-cooling blanket, but the recorded temperature swings widely and repeatedly overshoots below the target. Which of the following changes is most consistent with guideline recommendations?

- **A.** Continue tympanic monitoring but increase the frequency of readings to every 15 minutes
- **B.** Switch to axillary temperature monitoring and continue the air-cooling blanket
- **C.** Switch to oral temperature monitoring and add ice packs to the groin and axillae
- **D.** Change to an air-cooling blanket alone and monitor with a temporal artery scanner
- **E.** Switch to an esophageal or bladder temperature probe and use an intravascular catheter or gel-pad device

Answer: E. Switch to an esophageal or bladder temperature probe and use an intravascular catheter or gel-pad device

Accurate temperature measurement is essential to successful targeted temperature management. Although tympanic, axillary, and oral sites are the most convenient, they have been proven less accurate than esophageal and bladder probes, so options A, B, and C all retain an inferior measurement site. Guidelines cited in the chapter recommend using an intravascular catheter or gel pads to maintain a constant patient temperature, and intranasal, surface, or intravascular temperature-modulating devices are recommended over air-cooling blankets or ice packs because they achieve the target temperature faster, improve the likelihood of reaching the target, and decrease the likelihood of overshooting, which is precisely this patient's problem. Option D keeps both an inaccurate site and the less effective air-cooling method. The greatest benefit from targeted temperature management is obtained when therapy is started within hours of the initial insult, and hyperthermia is associated with increased length of stay, higher mortality, and larger infarct size.

Key point:
measure with an esophageal or bladder probe and cool with an intravascular catheter or gel pads rather than air-cooling blankets or ice packs.

Source: Practice of Neurocritical Care, 2nd ed., Neurocritical Care Nursing Special Considerations, pp. 344-345

Question 133

Neuroimmunology in the Neurocritical Care Unit · Anti-NMDAR encephalitis: timing of tumor resection · moderate

A 22-year-old woman is admitted to the neurocritical care unit after 3 weeks of new psychiatric symptoms and short-term memory loss culminating in generalized seizures and orofacial dyskinesias. She is intubated for airway protection and has episodes of tachycardia to 150 beats/min alternating with sinus pauses. MRI of the brain is unremarkable. Continuous EEG shows an extreme delta brush pattern. CSF shows 28 white blood cells/microliter (88% lymphocytes), protein 48 mg/dL, glucose 62 mg/dL, and negative HSV PCR. Transvaginal ultrasound demonstrates a 4-cm complex right ovarian mass with calcification. Serum and CSF autoantibody panels have been sent, and the laboratory reports a 2- to 3-week turnaround. She has been started on high-dose methylprednisolone. Which of the following is the most appropriate next step in management?

- **A.** Withhold further immunotherapy and repeat the lumbar puncture in 1 week to document a rising cell count
- **B.** Administer tocilizumab to suppress the interleukin-6 driven inflammatory response
- **C.** Proceed with resection of the ovarian mass now, without waiting for the autoantibody results
- **D.** Defer any surgical intervention until CSF anti-NMDA receptor antibodies return positive
- **E.** Obtain a stereotactic brain biopsy of the mesial temporal lobe to confirm inflammation before escalating therapy

Answer: C. Proceed with resection of the ovarian mass now, without waiting for the autoantibody results

This patient has a clear-cut clinical syndrome of anti-N-methyl-D-aspartate receptor (NMDAR) encephalitis with an ovarian teratoma: subacute psychiatric change, memory loss, seizures, dyskinesias, dysautonomia, an extreme delta brush on EEG (the EEG pattern the chapter links specifically to anti-NMDAR encephalitis), and an ovarian mass, which the chapter lists as the imaging finding associated with anti-NMDA encephalitis on transvaginal ultrasound. In paraneoplastic disease the single most important therapeutic intervention is treatment of the underlying malignancy to decrease the antigen load, and the chapter states explicitly that treatment of a presumed symptomatic tumor should not await confirmation of autoantibody status, because testing can take weeks while neurologic deterioration occurs over hours; the text uses resection of an ovarian teratoma in exactly this scenario as its example, making D wrong. A is wrong because a normal or static CSF profile never excludes an autoimmune process and delaying therapy risks irreversible injury. B is wrong because tocilizumab is used for non-neurologic cytokine release syndrome after CAR T cell therapy, is not effective against neurologic symptoms, and may even worsen neurotoxicity; it has no role here. E is wrong because the clinical syndrome is already well defined, biopsy carries a risk of permanent neurologic injury and is reserved for cases in which workup yields only nonspecific inflammation, and prior glucocorticoids reduce the diagnostic yield of tissue biopsy.

Key point:
when a symptomatic tumor is identified in a patient with a recognizable paraneoplastic syndrome, resect it immediately rather than waiting on antibody serology.

Source: Practice of Neurocritical Care, 2nd ed., Neuroimmunology in the Neurocritical Care Unit, pp. 365-367

Question 134

Neuroimmunology in the Neurocritical Care Unit · Intracellular versus surface antigens: T cell-mediated paraneoplastic disease · hard

A 50-year-old woman with newly diagnosed ovarian carcinoma develops 6 weeks of progressive gait ataxia, dysarthria, and limb dysmetria, and is admitted after a fall with aspiration. MRI of the brain shows no cerebellar atrophy or enhancement. CSF shows 12 white blood cells/microliter, protein 62 mg/dL, and two oligoclonal bands not present in serum. Serum anti-Purkinje cell cytoplasmic antibody type 1 (anti-Yo) is positive. Which of the following is the most appropriate therapeutic strategy?

- **A.** Observation with serial examinations, since her tumor burden is small and the neurologic deficit is mild
- **B.** Physical therapy and referral to gynecologic oncology for tumor management alone
- **C.** Intravenous immunoglobulin alone, because antibody removal is the definitive therapy for this syndrome
- **D.** Aggressive immunotherapy directed at T cell-mediated pathology combined with aggressive treatment of the malignancy
- **E.** Plasma exchange alone, because circulating pathogenic autoantibodies drive the cerebellar injury

Answer: D. Aggressive immunotherapy directed at T cell-mediated pathology combined with aggressive treatment of the malignancy

Anti-Yo (PCA-1) targets an intracellular cytoplasmic antigen, and the chapter emphasizes that autoantibodies against intracellular antigens such as ANNA-1 (Hu), ANNA-2 (Ri), CV2/CRMP5, and PCA-1 (Yo) are more likely biomarkers of T cell-mediated disease than directly pathogenic; T cell-mediated immunity is much more likely to cause cell death, so the injury is more likely to be irreversible and the response to delayed treatment unfavorable. That mechanism argues for prompt, aggressive immunotherapy aimed at T cells (the chapter lists glucocorticoids, cyclophosphamide, and anti-CD20 therapies for this category) together with aggressive control of the malignancy, since the most important intervention in paraneoplastic disease is reducing tumor antigen load. A is wrong because observation allows a destructive process to progress and the deficit is unlikely to improve once established. B is wrong because tumor treatment without immunomodulation leaves the T cell-mediated attack unopposed. C and E are wrong because antibody-directed strategies such as intravenous immunoglobulin and plasma exchange are the rational choice for pathogenic surface antigen antibodies such as AQP4, NMDAR, and LGI1, not for a biomarker antibody against an intracellular target.

Key point:
intracellular antigen antibodies signal destructive T cell-mediated disease that requires early aggressive T cell-directed immunotherapy plus tumor control, whereas surface antigen antibodies predict reversibility with antibody-directed therapy.

Source: Practice of Neurocritical Care, 2nd ed., Neuroimmunology in the Neurocritical Care Unit, pp. 358-360, 367

Question 135

Neuroimmunology in the Neurocritical Care Unit · CSF markers of CNS inflammation · moderate

A 45-year-old man is admitted with 8 weeks of progressive confusion, apathy, and two witnessed seizures. Toxicology, ammonia, thyroid studies, and vitamin B12 are normal. MRI of the brain shows subtle bilateral mesiotemporal T2/FLAIR hyperintensity without enhancement. Lumbar puncture shows 3 white blood cells/microliter, protein 38 mg/dL, and glucose 64 mg/dL with a serum glucose of 96 mg/dL. HSV-1 and HSV-2 PCR and metagenomic sequencing are negative, and cytology and flow cytometry are normal. Five oligoclonal bands are present in CSF and absent from a simultaneously drawn serum sample, and the IgG index is 0.85. Which of the following is the most appropriate interpretation of these results?

- **A.** The oligoclonal bands and elevated IgG index indicate intrathecal immunoglobulin production and support starting empiric immunotherapy
- **B.** The normal cell count and protein exclude an inflammatory process, so immunotherapy should be withheld
- **C.** The oligoclonal band pattern is specific for multiple sclerosis and establishes that diagnosis
- **D.** The IgG index is the more sensitive and specific test, so the oligoclonal band result can be disregarded
- **E.** A CSF glucose below 45 mg/dL would be required before an inflammatory cause could be entertained

Answer: A. The oligoclonal bands and elevated IgG index indicate intrathecal immunoglobulin production and support starting empiric immunotherapy

The chapter defines CSF inflammation as one or more of pleocytosis above 5 white blood cells, protein above 50 mg/dL, oligoclonal bands greater than 1 (laboratory dependent), and an IgG index above 0.66; this patient meets the criteria on the last two despite a bland cell count and protein. The text states explicitly that often the only markers of CNS inflammation are CSF autoantibodies or oligoclonal bands, and that even in the absence of elevated protein or pleocytosis these findings may be signs of pathology, which makes B wrong. C is wrong because oligoclonal bands are simply a marker of intrathecal immunoglobulin production and are not unique to multiple sclerosis; they are also seen in ADEM, neurosarcoidosis, Behcet disease, lupus, Sjogren syndrome, paraneoplastic and autoimmune antineuronal disorders, CNS infections, and lymphoma. D reverses the chapter's statement that quantitative IgG analysis is a complementary test and not a substitute for qualitative oligoclonal band assessment, which has higher sensitivity and specificity. E is wrong because hypoglycorrhachia, defined as glucose below 45 mg/dL, typically points toward bacterial or fungal infection or leptomeningeal carcinomatosis and is not required for an inflammatory diagnosis, although it can occur in neurosarcoidosis, primary angiitis of the CNS, and neuro-Behcet disease. Note also that the paired serum sample was correctly drawn, since comparison with serum is mandatory whenever CSF is sent for oligoclonal bands.

Key point:
normal CSF cell count and protein do not exclude CNS inflammation; oligoclonal bands and an IgG index above 0.66 can be the only evidence of intrathecal immune activation.

Source: Practice of Neurocritical Care, 2nd ed., Neuroimmunology in the Neurocritical Care Unit, pp. 363, 366

Question 136

Neuroimmunology in the Neurocritical Care Unit · Search for occult malignancy in paraneoplastic syndromes · easy

An 80-year-old woman with a 50-pack-year smoking history is admitted with 5 weeks of progressive truncal ataxia and opsoclonus. MRI of the brain with and without gadolinium is normal. CT of the chest, abdomen, and pelvis shows no mass. CSF shows 22 white blood cells/microliter with 90% lymphocytes and protein 68 mg/dL, with negative cytology and flow cytometry. Which of the following is the most appropriate next step in the workup?

- **A.** Observe clinically and repeat CT of the chest, abdomen, and pelvis in 1 year
- **B.** Obtain whole-body fluorodeoxyglucose PET/CT to search for an occult tumor
- **C.** Obtain MR spectroscopy of the cerebellum to characterize the cerebellar dysfunction
- **D.** Obtain continuous EEG to evaluate for nonconvulsive seizures as the cause of the ataxia
- **E.** Obtain a stereotactic biopsy of the cerebellum to establish the inflammatory diagnosis

Answer: B. Obtain whole-body fluorodeoxyglucose PET/CT to search for an occult tumor

Subacute cerebellar degeneration with opsoclonus and inflammatory CSF in an elderly smoker is a classic paraneoplastic presentation, and the chapter associates opsoclonus-myoclonus-ataxia with anti-ANNA-2 (Ri) and anti-ANNA-1 (Hu) antibodies. Paraneoplastic syndromes may be caused by small lesions that are missed by CT but detected on whole-body FDG-PET, which the chapter lists as the study that identifies areas of FDG avidity indicating malignancy or inflammation; the opening case of the chapter illustrates exactly this, with a normal CT of the chest, abdomen, and pelvis but an FDG-avid axillary node and breast lesion on PET. A is wrong because these disorders can progress irreversibly over weeks and a 1-year delay forfeits the chance to reduce antigen load. C is wrong because MR spectroscopy is used to look for a lactate peak suggesting metabolic disease and would not identify the driving tumor. D is wrong because ataxia and opsoclonus are not seizure manifestations and EEG is rarely diagnostic in these disorders. E is wrong because biopsy is reserved for cases in which the workup yields only nonspecific inflammation, targets must lie in radiographically active regions, and the MRI here is normal, so the yield would be low relative to the risk.

Key point:
when conventional CT is negative in a suspected paraneoplastic syndrome, whole-body FDG-PET is the next test to find the occult tumor.

Source: Practice of Neurocritical Care, 2nd ed., Neuroimmunology in the Neurocritical Care Unit, pp. 363, 365

Question 137

Neuroimmunology in the Neurocritical Care Unit · Neurologic immune-related adverse events of checkpoint inhibitors · moderate

A 50-year-old man with metastatic melanoma who began pembrolizumab 6 weeks ago presents with 10 days of progressive ptosis, binocular diplopia, and fatigue that worsens through the day. He is now using accessory muscles and his negative inspiratory force is declining. Laboratory studies show creatine phosphokinase 2,400 units/liter, TSH 0.02 microunits/milliliter with a low free T4, and sodium 126 milliequivalents/liter. MRI of the brain shows no metastatic lesions but demonstrates an enlarged, enhancing pituitary gland. Which of the following is the most appropriate management?

- **A.** Continue pembrolizumab and add low-dose steroids for symptomatic relief of perilesional inflammation
- **B.** Discontinue pembrolizumab and switch to an alternative cytotoxic regimen for progressive melanoma
- **C.** Continue pembrolizumab and obtain a chest CT with plans for thymectomy
- **D.** Continue pembrolizumab and treat with tocilizumab for the cytokine-mediated syndrome
- **E.** Discontinue pembrolizumab and begin high-dose glucocorticoids

Answer: E. Discontinue pembrolizumab and begin high-dose glucocorticoids

Pembrolizumab is a programmed cell death protein 1 antagonist, and the chapter states that checkpoint inhibitors trigger immune responses against the pituitary gland and against specific neuronal antigens, producing hypophysitis, autoimmune encephalitis, and myasthenia gravis, with neurologic immune-related adverse events reported in as many as 1% of treated patients. This patient has the characteristic heterogeneous checkpoint inhibitor syndrome of myositis (elevated creatine phosphokinase), myasthenia gravis (fatigable ptosis, diplopia, and respiratory decline), and hypophysitis (central hypothyroidism, hyponatremia, enhancing pituitary). Management of immune-related adverse events consists of interruption or permanent discontinuation of the checkpoint inhibitor plus corticosteroids, with additional immunosuppressants such as TNF-alpha inhibitors or mycophenolate mofetil if needed, and always in collaboration with the primary oncologist. A and C are wrong because continuing the offending drug allows the autoimmune attack to progress, and the syndrome is drug-induced rather than a thymoma-associated or perilesional problem. B is wrong because the MRI shows no new metastatic disease, so this represents toxicity rather than treatment failure. D is wrong because tocilizumab is not a treatment for checkpoint inhibitor toxicity and is specifically noted to be ineffective for and potentially worsening of neurologic toxicity in the immune effector cell setting.

Key point:
fatigable weakness, elevated creatine phosphokinase, and endocrinopathy in a patient on a checkpoint inhibitor means stop the drug and give high-dose steroids.

Source: Practice of Neurocritical Care, 2nd ed., Neuroimmunology in the Neurocritical Care Unit, pp. 360-361, 367

Question 138

Neuroimmunology in the Neurocritical Care Unit · CAR T cell-associated neurotoxicity: pharmacologic management · hard

A 61-year-old man with refractory diffuse large B-cell lymphoma received CD19-directed CAR T cell therapy 5 days ago. He was treated for cytokine release syndrome on day 2 with tocilizumab, and his fevers and hypotension resolved. On day 5 he becomes aphasic, then rapidly obtunded, and is intubated. Head CT shows diffuse sulcal effacement and loss of gray-white differentiation consistent with cerebral edema. He has received high-dose methylprednisolone. Which of the following is the most appropriate additional therapy for his neurotoxicity?

- **A.** Antithymocyte globulin
- **B.** Repeat dosing of tocilizumab
- **C.** Natalizumab
- **D.** Intravenous immunoglobulin
- **E.** Eculizumab

Answer: A. Antithymocyte globulin

This is immune effector cell-associated neurotoxicity syndrome, whose typical course includes encephalopathy as the most common symptom, aphasia as the most common focal deficit, and, rarely, fulminant cerebral edema progressing to coma and death. The chapter states that neurologic toxicity of CAR T cell therapy is typically treated with corticosteroids, and that in more severe cases the anti-interleukin-6 antibody siltuximab, the anti-interleukin-1 receptor antibody anakinra, or, particularly in cases of cerebral edema, antithymocyte globulin may be used, which makes A correct. B is wrong and is the key distractor: tocilizumab is an anti-interleukin-6 receptor antibody used for the non-neurologic toxicities of CAR T cell therapy, is explicitly not effective against neurologic symptoms, and may even worsen neurotoxicity; his cytokine release syndrome has already resolved, so there is no remaining indication. C, D, and E are not among the agents the chapter describes for this syndrome; natalizumab is listed for T cell-mediated demyelinating disease, intravenous immunoglobulin for antibody-mediated disease, and eculizumab as an anti-C5 agent for antibody-mediated disease. It should be recognized that antithymocyte globulin will by design abrogate the antitumor effect of the CAR T cells, so the decision must be made with the primary oncologist, whereas judicious corticosteroid use probably does not impair the CAR T therapeutic effect.

Key point:
escalate CAR T neurotoxicity with steroids and then siltuximab, anakinra, or antithymocyte globulin for cerebral edema, and do not use tocilizumab for the neurologic syndrome.

Source: Practice of Neurocritical Care, 2nd ed., Neuroimmunology in the Neurocritical Care Unit, pp. 361, 367-369

Question 139

Neuroimmunology in the Neurocritical Care Unit · Induction immunotherapy dosing · easy

Which of the following correctly states a standard induction regimen for acute immunomodulatory therapy in autoimmune neurologic disease?

- **A.** Methylprednisolone 125 mg intravenously daily for 14 days
- **B.** Plasma exchange of 0.25 plasma volumes daily for 10 consecutive days
- **C.** Rituximab 100 mg intravenously once monthly for 6 months
- **D.** Azathioprine 10 mg/kg/day started at full dose on day 1
- **E.** Intravenous immunoglobulin 2 g/kg divided over 3 to 5 days

Answer: E. Intravenous immunoglobulin 2 g/kg divided over 3 to 5 days

The chapter's dosing table gives intravenous immunoglobulin as 2 g/kg administered over 3 to 5 days, making E correct. A misstates glucocorticoid induction, which is methylprednisolone 1 g intravenously daily for 3 to 5 days, or prednisone 1 mg/kg/day in the range of 60 to 80 mg daily, or dexamethasone 1 to 40 mg every 6 hours. B misstates plasma exchange, which uses 1 to 1.5 plasma volumes, typically 5 exchanges given every other day to allow vascular compartment equilibration between treatments. C misstates rituximab, which is 1,000 mg every 2 weeks for 2 doses or 375 mg/m2 weekly for 4 doses, usually repeated every 6 months. D misstates azathioprine, which is started at 1 mg/kg/day and titrated up to 3 mg/kg/day, with thiopurine methyltransferase genotyping before treatment. Glucocorticoids, intravenous immunoglobulin, and plasma exchange are all reserved for induction rather than maintenance because of their rapid onset and nonspecific immune effects.

Key point:
induction dosing is methylprednisolone 1 g daily for 3 to 5 days, intravenous immunoglobulin 2 g/kg over 3 to 5 days, or plasma exchange of 1 to 1.5 plasma volumes for about 5 every-other-day exchanges.

Source: Practice of Neurocritical Care, 2nd ed., Neuroimmunology in the Neurocritical Care Unit, pp. 366, 370

Neuroinfectious Disease

Question 140

Neuroinfectious Disease · Bacterial meningitis - empiric antimicrobial therapy · moderate

A 68-year-old man with type 2 diabetes is brought to the emergency department with 12 hours of fever, headache, and progressive confusion. Temperature is 39.2 C, heart rate 112/min, blood pressure 104/62 mm Hg. He is oriented to person only and has marked nuchal rigidity without focal deficits. Noncontrast head CT is unremarkable. Lumbar puncture shows an opening pressure of 28 cm H_2O, CSF WBC 4,200/microliter with 92% neutrophils, glucose 18 mg/dL with a simultaneous serum glucose of 118 mg/dL, and protein 260 mg/dL. Gram stain and cultures are pending. Which of the following is the most appropriate empiric antimicrobial regimen?

- **A.** Ceftriaxone alone
- **B.** Vancomycin plus ceftriaxone
- **C.** Vancomycin plus ceftriaxone plus ampicillin
- **D.** Vancomycin plus ampicillin
- **E.** Ampicillin plus gentamicin

Answer: C. Vancomycin plus ceftriaxone plus ampicillin

The CSF profile - elevated opening pressure, neutrophilic pleocytosis, hypoglycorrhachia with a CSF-to-blood glucose ratio well below 0.6, and elevated protein - is classic for acute bacterial meningitis. IDSA guidelines recommend empiric vancomycin plus a third-generation cephalosporin, with ampicillin added when the patient is over 50 years old or immunocompromised, because Listeria monocytogenes joins Streptococcus pneumoniae, Neisseria meningitidis, and aerobic gram-negative bacilli as an expected pathogen in that age group. Option B is the correct regimen for an immunocompetent patient aged 2-50 years but leaves Listeria uncovered in a 68-year-old. Option A omits vancomycin, which is needed for resistant pneumococcus, and also omits Listeria coverage. Option D omits the third-generation cephalosporin needed for pneumococcus and meningococcus, and option E is a regimen for Enterococcus rather than empiric community-acquired meningitis. If this patient had a beta-lactam allergy, sulfamethoxazole-trimethoprim could substitute for ampicillin and aztreonam, meropenem, or moxifloxacin for the cephalosporin.

Key point:
add ampicillin for Listeria coverage to vancomycin and a third-generation cephalosporin whenever the patient is over 50 years old or immunocompromised.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 383-384

Question 141

Neuroinfectious Disease · Bacterial meningitis - adjunctive dexamethasone · easy

Adjunctive corticosteroid therapy is recommended in adults with suspected or confirmed pneumococcal meningitis. Which of the following describes the recommended dexamethasone regimen and its timing relative to antimicrobial therapy?

- **A.** 0.15 mg/kg every 6 hours for 2-4 days, with the first dose 10-20 minutes before or concomitant with the first antibiotic dose
- **B.** 0.15 mg/kg every 12 hours for 7 days, beginning once CSF cultures confirm the organism
- **C.** 0.30 mg/kg every 6 hours for 4-6 days, beginning 24 hours after the first antibiotic dose
- **D.** 0.50 mg/kg every 8 hours for 3-5 days, beginning with the second antibiotic dose
- **E.** 0.75 mg/kg every 8 hours for 1-3 days, given only if the CSF Gram stain shows gram-positive diplococci

Answer: A. 0.15 mg/kg every 6 hours for 2-4 days, with the first dose 10-20 minutes before or concomitant with the first antibiotic dose

Animal models showed that morbidity and mortality in bacterial meningitis track with subarachnoid space inflammation, and blunting the proinflammatory cytokine response is thought to reduce neuronal injury, cerebral edema, ICP elevation, and hearing loss. The recommended adult regimen is dexamethasone 0.15 mg/kg every 6 hours for 2-4 days, with the first dose given 10-20 minutes prior to or concomitant with the first dose of antimicrobial therapy. Options B, C, and D all delay steroids until after antibiotics have been started or cultures have returned, which forfeits the benefit, since the drug must be on board before or with the antibiotic-induced bacterial lysis. Option E is wrong on both dose and on waiting for the Gram stain: therapy is indicated for suspected as well as confirmed pneumococcal meningitis. The doses in options C, D, and E are all supratherapeutic relative to the guideline.

Key point:
dexamethasone 0.15 mg/kg every 6 hours for 2-4 days, started 10-20 minutes before or with the first antibiotic dose, for suspected or confirmed pneumococcal meningitis.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 384

Question 142

Neuroinfectious Disease · Bacterial meningitis - neuroimaging before lumbar puncture · moderate

Five patients present to the emergency department with fever and headache, and each is suspected of having bacterial meningitis. Empiric antibiotics are ordered for all of them. Which of the following findings most strongly indicates that head CT should be obtained before lumbar puncture?

- **A.** Severe nuchal rigidity with a positive Kernig sign
- **B.** A petechial rash over the trunk and extremities
- **C.** Photophobia with two episodes of emesis
- **D.** Temperature of 39.8 C with a peripheral WBC of 24,000/microliter
- **E.** A new left sixth cranial nerve palsy in a patient taking antiretroviral therapy for HIV

Answer: E. A new left sixth cranial nerve palsy in a patient taking antiretroviral therapy for HIV

Imaging should precede lumbar puncture when there is clinical suspicion of raised intracranial pressure such as a depressed level of consciousness, when there are focal neurologic signs suggestive of mass effect, or when the patient has a known immunocompromised state; a new abducens palsy plus HIV satisfies two of these. Performing a lumbar puncture in a patient with a mass lesion and cerebral edema can precipitate herniation. Options A, B, C, and D are all typical features of meningitis itself - nuchal rigidity, the petechial rash of meningococcal infection, photophobia, emesis, fever, and leukocytosis - and none of them implies mass effect or immunocompromise, so none mandates prior imaging. Importantly, empiric antimicrobial therapy may be initiated without obtaining CSF, so waiting for imaging must never delay antibiotics; blood cultures should be drawn first when possible. Note also that the absence of Kernig and Brudzinski signs does not exclude meningitis.

Key point:
image before lumbar puncture for depressed consciousness, focal signs suggesting mass effect, or immunocompromise - but give empiric antibiotics first.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 383

Question 143

Neuroinfectious Disease · Viral encephalitis - HSV treatment and dosing · moderate

A 44-year-old woman presents with 3 days of fever, headache, and malaise followed by confusion and an olfactory hallucination. Temperature is 38.9 C. Exam shows expressive aphasia and a right pronator drift. MRI shows T2 hyperintensity in the left medial temporal lobe and insula. EEG shows left temporal spike and slow wave activity. CSF shows 180 WBC/microliter with lymphocytic predominance, protein 92 mg/dL, glucose 62 mg/dL, and 90 RBC/microliter; HSV PCR is pending. Which of the following is the most appropriate therapy?

- **A.** Acyclovir 5 mg/kg IV every 8 hours for 7 days
- **B.** Acyclovir 10 mg/kg IV, dosed on ideal body weight, every 8 hours for 14-21 days
- **C.** Ganciclovir 5 mg/kg IV every 12 hours for 21 days
- **D.** Foscarnet 60 mg/kg IV every 8 hours for 14 days
- **E.** Ganciclovir plus foscarnet for 21 days

Answer: B. Acyclovir 10 mg/kg IV, dosed on ideal body weight, every 8 hours for 14-21 days

IDSA guidelines direct that all patients with suspected encephalitis receive intravenous acyclovir 10 mg/kg dosed on ideal body weight every 8 hours until HSV PCR returns negative, and for confirmed HSV encephalitis the course is 14-21 days at that dose. The vignette is classic HSV-1 encephalitis: influenza-like prodrome, temporal lobe involvement with aphasia and behavioral change, and the temporal spike and slow wave EEG pattern that carries roughly 60% sensitivity and 80% specificity. Option A underdoses acyclovir and truncates the course. Option C is the cytomegalovirus dose of ganciclovir, and options D and E reflect regimens for CMV, in which ganciclovir and foscarnet are combined; neither is first-line for HSV. Because acyclovir is an alkaline irritant that can cause phlebitis and cutaneous necrosis and can be nephrotoxic with rapid infusion or poor hydration, it should be infused over 1 hour with adequate hydration maintained. CSF PCR, not brain biopsy, is now the gold standard for diagnosis.

Key point:
intravenous acyclovir 10 mg/kg of ideal body weight every 8 hours, empirically in all suspected encephalitis and for 14-21 days when HSV is confirmed.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 386

Question 144

Neuroinfectious Disease · Recurrent aseptic meningitis - HSV-2 · easy

A 55-year-old immunocompetent man presents with headache, photophobia, and nuchal rigidity. He reports this is the third such episode in 2 years; he received an antiviral after the first episode and has been on no therapy since, recovering completely each time. He recalls genital lesions about a week before the first episode. Temperature is 37.9 C and the neurologic exam is nonfocal. CSF shows 220 WBC/microliter with lymphocytic predominance, normal glucose, and mildly elevated protein. Which of the following is the most likely causative organism?

- **A.** Herpes simplex virus type 1
- **B.** Varicella-zoster virus
- **C.** Human herpesvirus 6
- **D.** Herpes simplex virus type 2
- **E.** Epstein-Barr virus

Answer: D. Herpes simplex virus type 2

HSV-2 accounts for about 10% of HSV infections and classically causes a recurrent aseptic meningitis, with genital lesions typically appearing about a week before the onset of neurologic symptoms; the recurrent form is known as Mollaret meningitis. Recovery is usually full, and the recurrent episodes are not treated with antiviral therapy - acyclovir may be used for the initial infection but does not change the chance of recurrence, which fits this patient's course exactly. HSV-1 is the most common cause of fatal encephalitis in adults and produces temporal lobe encephalitis with focal deficits and seizures, not a benign relapsing meningitis. HHV-6 and CMV are seen chiefly in immunocompromised and transplant patients, and VZV reactivation more often causes encephalitis with focal deficits, seizures, and a large-vessel arteritis, again typically in the immunocompromised. EBV encephalitis is associated with saliva exposure and presents with seizures, coma, ataxia, and cranial nerve palsies.

Key point:
recurrent benign aseptic meningitis in an immunocompetent adult is HSV-2 Mollaret meningitis, and the recurrences are not treated with antivirals.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 386

Question 145

Neuroinfectious Disease · Arboviral encephalitis - West Nile virus · easy

An 85-year-old woman is brought in confused and unable to walk. She was last seen at her baseline 5 days earlier on a camping trip. Temperature is 38.6 C. She is oriented to person only, has a coarse tremor and multifocal myoclonus, and has flaccid, areflexic paraplegia with preserved sensation. CSF shows 90 WBC/microliter with lymphocytic predominance, protein 88 mg/dL, and normal glucose. MRI shows T2 and FLAIR hyperintensity in the substantia nigra, basal ganglia, and thalami. Which of the following is the most likely pathogen?

- **A.** Herpes simplex virus type 1
- **B.** Cytomegalovirus
- **C.** Epstein-Barr virus
- **D.** Human herpesvirus 6
- **E.** West Nile virus

Answer: E. West Nile virus

West Nile virus is mosquito-borne with birds as the reservoir, and the elderly and immunocompromised are at highest risk, so a recent camping exposure in an 85-year-old is the key epidemiologic clue. Its presentation includes acute fever, headache, nuchal rigidity, and vomiting along with tremor, myoclonus, parkinsonism, and a poliomyelitis-like flaccid paralysis, and MRI classically shows T2 and FLAIR hyperintense lesions in the substantia nigra, basal ganglia, and thalami. Diagnosis rests on CSF IgM, since CSF PCR is positive in fewer than 60% of cases. HSV-1 would produce temporal lobe T2 hyperintensity with language and behavioral changes rather than flaccid paralysis and basal ganglia lesions. CMV and HHV-6 occur in immunocompromised and transplant patients, with CMV showing subependymal enhancement and T2 white matter lesions and HHV-6 showing frontal, parietal, temporal, and limbic lesions after a recent exanthem. EBV is acquired through saliva exposure and causes seizures, coma, ataxia, and cranial nerve palsies.

Key point:
mosquito exposure plus a flaccid poliomyelitis-like paralysis with substantia nigra, basal ganglia, and thalamic lesions is West Nile virus, diagnosed by CSF IgM.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 387

Question 146

Neuroinfectious Disease · Brain abscess - surgical versus medical management · moderate

A 47-year-old man with long-standing chronic otitis media presents with 10 days of progressive right-sided headache, low-grade fever, and 2 days of left arm weakness. He has had no prior neurosurgery and no head trauma. Contrast head CT shows a 3.5 cm ring-enhancing lesion in the right temporal lobe with surrounding hypodensity and 6 mm of midline shift; MRI shows a hyperintense DWI signal with low ADC within the lesion. Which of the following is the most appropriate management?

- **A.** Surgical aspiration for organism identification and mass effect reduction, plus empiric vancomycin, ceftriaxone, and metronidazole
- **B.** Empiric vancomycin, ceftriaxone, and metronidazole alone, with repeat imaging in 2 weeks
- **C.** Lumbar puncture for CSF culture, with antibiotics withheld until an organism is identified
- **D.** Empiric vancomycin, cefepime, and metronidazole alone for 6-8 weeks
- **E.** Dexamethasone and observation pending blood culture results

Answer: A. Surgical aspiration for organism identification and mass effect reduction, plus empiric vancomycin, ceftriaxone, and metronidazole

The DWI-hyperintense, low-ADC ring-enhancing lesion with vasogenic edema is a mature brain abscess, and an abscess larger than 2.5 cm in diameter is the recommended decision point for surgical intervention. Medical therapy alone is usually ineffective and is appropriate only when the causative pathogen has already been identified and the abscess measures less than 2.5 cm, which excludes options B and D here; surgical aspiration also achieves organism identification and reduces mass effect. Option D additionally uses the wrong regimen: vancomycin plus a fourth-generation cephalosporin such as cefepime plus metronidazole for 6-8 weeks is the empiric choice for postneurosurgical patients or head trauma with skull fracture, whereas a non-neurosurgical patient at low risk for Pseudomonas receives vancomycin, a third-generation cephalosporin such as ceftriaxone, and metronidazole. Option C is wrong twice over - empiric antibiotic therapy should begin immediately, and lumbar puncture is performed only when meningitis or ventricular rupture is suspected and there is minimal concern for herniation, which is not the case with 6 mm of shift. Chronic otitis media as the contiguous focus predicts streptococci, Bacteroides, Pseudomonas aeruginosa, and Enterobacteriaceae.

Key point:
an abscess larger than 2.5 cm calls for surgical drainage plus immediate empiric antibiotics; medical therapy alone is reserved for smaller abscesses with a known pathogen.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 388-389

Question 147

Neuroinfectious Disease · Subdural empyema - diagnosis and imaging · hard

A 34-year-old man underwent a right frontal craniotomy for tumor resection 3 weeks ago. He returns with headache, fever to 38.7 C, meningismus, and a single generalized seizure. CT shows a crescentic hypodense extra-axial collection over the right convexity with peripheral enhancement. MRI shows the collection to be hypointense on T1 and hyperintense on T2, with reduced diffusion and an ADC value lower than that of normal cortical gray matter. Which of the following is the most likely diagnosis?

- **A.** Reactive subdural effusion
- **B.** Chronic subdural hematoma
- **C.** Subdural empyema
- **D.** Postoperative pseudomeningocele
- **E.** Epidural abscess with dural enhancement

Answer: C. Subdural empyema

Subdural empyema complicates about 4% of neurosurgical cases and usually appears 1-8 weeks after the procedure with headache, meningismus, fever, confusion, nausea, vomiting, seizures, and focal deficits, matching this presentation at 3 weeks. The single most useful imaging discriminator is diffusion: subdural empyema shows reduced diffusion with an ADC lower than that of normal cortical gray matter, whereas a reactive subdural effusion shows no DWI changes, which excludes option A. On CT the empyema is a hypodense collection with enhancement, and on MRI it is dark on T1 and bright on T2, as described. A chronic subdural hematoma and a pseudomeningocele would not restrict diffusion and would not typically produce fever and meningismus, excluding options B and D, and the collection here is described as subdural and crescentic over the convexity rather than epidural. Staphylococci and gram-negative bacilli are the usual pathogens; lumbar puncture is not recommended because of the increased risk of herniation. Management is surgical drainage, performed in about 96% of cases for source control, with empiric vancomycin plus a fourth-generation cephalosporin such as cefepime, tailored to intraoperative cultures and usually continued 3-6 weeks.

Key point:
a post-craniotomy extra-axial collection that restricts diffusion with an ADC below cortical gray matter is a subdural empyema requiring surgical drainage plus vancomycin and cefepime.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 390

Question 148

Neuroinfectious Disease · Healthcare-associated ventriculitis - empiric therapy · moderate

A 23-year-old man with a severe traumatic brain injury has had an external ventricular drain in place for 8 days. On ICU day 8 he becomes febrile to 38.8 C and his neurologic exam declines from following commands to localizing only. CSF from the drain shows 640 WBC/microliter with neutrophil predominance, glucose 32 mg/dL, and protein 180 mg/dL; Gram stain and culture have been sent. Which of the following is the most appropriate empiric antimicrobial regimen?

- **A.** Ceftriaxone plus vancomycin
- **B.** Vancomycin plus cefepime
- **C.** Meropenem plus linezolid
- **D.** Ceftazidime plus nafcillin
- **E.** Ampicillin plus gentamicin

Answer: B. Vancomycin plus cefepime

Empiric therapy for ventriculitis after neurosurgery, for CSF shunt infections, and after penetrating or blunt trauma is vancomycin plus cefepime, with vancomycin plus ceftazidime or vancomycin plus meropenem as accepted alternatives. The regimen must cover both the gram-positive skin flora that predominate in drain and shunt infections - Staphylococcus epidermidis, Staphylococcus aureus, Propionibacterium acnes, streptococci - and gram-negative organisms including Pseudomonas, which account for up to 25% of these infections and are especially common after trauma and brain surgery. Option A fails because ceftriaxone has no antipseudomonal activity. Option C provides both classes but is not the recommended empiric pairing, and option D pairs an antipseudomonal cephalosporin with nafcillin, which does not cover methicillin-resistant staphylococci. Option E is an Enterococcus regimen and covers neither Pseudomonas nor resistant staphylococci. Infected hardware should be removed or replaced, since the ventricles act as an immunologically isolated reservoir; treatment usually runs 10-14 days with CSF surveillance, 21 days for gram-negative bacilli, and 5-7 days when the drain is exchanged.

Key point:
empiric coverage for drain- or shunt-related ventriculitis is vancomycin plus cefepime, with removal or replacement of the infected hardware.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 391

Question 149

Neuroinfectious Disease · CSF lactate in the postoperative neurosurgical patient · hard

A 60-year-old woman is on postoperative day 4 from a posterior fossa tumor resection with an external ventricular drain in place. She develops a fever of 38.5 C and increased somnolence. CSF sampled from the drain shows 210 WBC/microliter, protein 96 mg/dL, glucose 46 mg/dL with a serum glucose of 128 mg/dL, and 3,400 RBC/microliter; Gram stain is negative and cultures are pending. The team notes that her blood-brain barrier injury and hemorrhagic CSF make the routine cell count, protein, and glucose difficult to interpret. Which of the following CSF findings would best support initiating empiric antimicrobial therapy while cultures are pending?

- **A.** A CSF red blood cell count above 1,000/microliter
- **B.** A CSF protein above 50 mg/dL
- **C.** A CSF opening pressure above 20 cm H_2O
- **D.** A CSF lactate concentration of 4.0 mmol/L or higher
- **E.** A single positive CSF culture with an otherwise normal CSF cell count, glucose, and protein

Answer: D. A CSF lactate concentration of 4.0 mmol/L or higher

In patients with prior neurologic injury and blood-brain barrier breakdown, the CSF cell count, protein, and glucose are already deranged, and CSF lactate is the measure that retains discriminative value. Current guidelines recommend that in the postoperative neurosurgical patient, empirical antimicrobial therapy should be considered if the CSF lactate concentration is 4.0 mmol/L or higher pending final cultures; in a cohort of patients with acute meningitis, a CSF lactate above 4.2 mmol/L had 96% sensitivity, 100% specificity, 100% positive predictive value, and 97% negative predictive value for bacterial meningitis, and CSF lactate appears superior to the CSF-to-blood glucose ratio for this diagnosis. Options A, B, and C are all expected nonspecific consequences of recent surgery and hemorrhagic CSF and do not distinguish infection. Option E describes the definition of external ventricular drain contamination - an isolated positive culture or Gram stain with an otherwise expected CSF profile - and a positive culture alone should never trigger a reflexive diagnosis of ventriculitis. Note also that CSF lactate is most useful when sampled before antimicrobial therapy is given.

Key point:
in the postoperative neurosurgical patient with uninterpretable routine CSF indices, a CSF lactate of 4.0 mmol/L or higher justifies starting empiric antimicrobials pending cultures.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 391

Question 150

Neuroinfectious Disease · Cryptococcal meningitis - hydrocephalus and CSF sampling · hard

A 45-year-old man with HIV who has been off antiretroviral therapy is found down by police and brought to the emergency department. Temperature is 38.9 C, blood pressure 128/74 mm Hg. He opens his eyes to voice, has a gait he cannot demonstrate because he will not stand, and has bilateral sixth nerve palsies without meningismus. CD4 count is 50 cells/microliter. Noncontrast head CT shows no ischemia or hemorrhage but demonstrates ventriculomegaly with transependymal flow. Which of the following is the most appropriate next step?

- **A.** Perform an immediate lumbar puncture for opening pressure and cryptococcal antigen
- **B.** Start fluconazole 800 mg IV daily as sole antifungal therapy
- **C.** Consult neurosurgery for placement of an external ventricular drain
- **D.** Send an HIV viral load and defer further intervention until it returns
- **E.** Begin high-dose dexamethasone for presumed inflammatory cerebral edema

Answer: C. Consult neurosurgery for placement of an external ventricular drain

The picture of advanced HIV with a CD4 count of 50, fever, cranial neuropathies, and obstructive hydrocephalus with transependymal flow is highly suggestive of CNS cryptococcal disease, and CSF should be sampled only if it is safe to do so from an intracranial pressure standpoint. With obstructive hydrocephalus on CT, an immediate lumbar puncture as in option A risks precipitating herniation, whereas an external ventricular drain both relieves the obstruction and provides CSF for cell count, protein, glucose, culture, and cryptococcal antigen. Option B is wrong because initial therapy for cryptococcal disease is a fungicidal agent - amphotericin B, with or without oral flucytosine - while fungistatic azoles are reserved for subsequent therapy when completing a total course of 12-18 months, and voriconazole is reserved for salvage. Option D delays treatment of a neurologic emergency for a test that will not change acute management. Option E is premature: corticosteroids can temporize inflammation-mediated edema, including the paradoxical immune response that follows treatment initiation, but they must be used cautiously because of their immunosuppressive effects, and they do not address obstructive hydrocephalus. Refractory obstruction may also require serial high-volume lumbar punctures once safe, or shunting.

Key point:
in suspected cryptococcal disease with obstructive hydrocephalus, divert CSF with an external ventricular drain rather than performing a lumbar puncture, and treat with fungicidal amphotericin B with or without flucytosine.

Source: Practice of Neurocritical Care, 2nd ed., Neuroinfectious Disease, pp. 394-395

Neurorehabilitation in the Neurocritical Care Unit

Question 151

Neurorehabilitation in the Neurocritical Care Unit · Timing of early mobilization after stroke (AVERT) · moderate

A 68-year-old woman is admitted to the neurocritical care unit 6 hours after onset of right hemiparesis and aphasia from a left middle cerebral artery infarct. She received intravenous thrombolysis. Her blood pressure is 148/82 mmHg, heart rate 78 and regular, and she is afebrile. NIHSS is 12. Repeat CT shows no hemorrhage. Eight hours after admission the nursing team asks whether the patient should be taken out of bed for frequent, high-dose mobilization sessions beginning now. Which of the following is the most appropriate approach to mobilization in this patient?

- **A.** Begin frequent high-dose out-of-bed mobilization immediately, because very early mobilization improved 3-month outcomes in the AVERT trial
- **B.** Use caution with out-of-bed mobilization during the first 24 hours while arranging a rehabilitation team assessment to determine candidacy for mobilization after the first day
- **C.** Withhold any rehabilitation assessment or therapy involvement until at least 7 days after stroke onset
- **D.** Mobilize aggressively only if her modified Rankin Scale score is greater than 2 at 24 hours
- **E.** Defer all mobilization decisions until a videofluoroscopic swallow study has been completed

Answer: B. Use caution with out-of-bed mobilization during the first 24 hours while arranging a rehabilitation team assessment to determine candidacy for mobilization after the first day

In A Very Early Rehabilitation Trial (AVERT), a single-blinded multicenter randomized trial of 2,104 patients with ischemic or hemorrhagic stroke, patients assigned to very early mobilization, defined as mobilization within 24 hours of stroke onset with more frequent sessions, were more likely to have an unfavorable outcome, meaning a modified Rankin Scale score greater than 2, at three months than patients receiving usual care; in a pre-specified subgroup analysis the effect was larger in patients with severe stroke and intracerebral hemorrhage. This makes option A incorrect, since AVERT showed potential harm rather than benefit. Option C is incorrect because the chapter emphasizes establishing a pattern of early rehabilitation assessment with the therapy team, and the Liu trial of 243 patients with intracerebral hemorrhage found that rehabilitation started within 48 hours produced shorter length of stay, better quality of life, greater independence in activities of daily living and lower six-month mortality compared with rehabilitation starting after seven days. Option D misuses the modified Rankin Scale, which was an outcome measure in AVERT rather than a trigger for therapy, and the cutoffs of 2 or less versus greater than 2 were themselves criticized as too coarse. Option E confuses dysphagia assessment with mobility; swallow evaluation is a separate pathway and does not gate physical therapy. Important critiques of AVERT include that ICU hospitalization was an exclusion criterion, so the findings may not generalize to neurocritical care patients, and that the difference in time to first mobilization between groups was only about 4 hours, with 92 percent of the early group versus 59 percent of the usual care group mobilized within 24 hours.

Key point:
be cautious with very early intensive mobilization in the first 24 hours after stroke, but still obtain early rehabilitation assessment so mobilization can begin after day one.

Source: Practice of Neurocritical Care, 2nd ed., Neurorehabilitation in the Neurocritical Care Unit, pp. 409

Question 152

Neurorehabilitation in the Neurocritical Care Unit · Posttraumatic agitation after TBI · hard

A 24-year-old man is in the neurocritical care unit on hospital day 6 after a severe traumatic brain injury with a right frontal contusion. He is arousable and follows simple commands intermittently but is restless, strikes at nursing staff, and pulls at his lines. Temperature is 37.1 C, white blood cell count is normal, urinalysis is negative, and there is no history of alcohol or benzodiazepine use. His agitation is tracked with the Agitated Behavior Scale. Scheduled acetaminophen has been started, opioids are timed before therapy sessions, the room has been made low-stimulus, and restraints have been minimized, but the agitated behavior persists. Which of the following is the most appropriate first-line medication?

- **A.** Scheduled lorazepam
- **B.** Haloperidol
- **C.** Propranolol
- **D.** Buspirone
- **E.** Diphenhydramine

Answer: C. Propranolol

When nonpharmacologic measures fail and other causes such as infection, withdrawal, psychiatric comorbidity, medication effects and uncontrolled pain have been excluded, the chapter identifies beta-blockers, specifically propranolol and pindolol, and mood-regulating antiseizure agents as first-line pharmacologic treatment for posttraumatic agitation, with the guiding principle of improving cognitive function rather than producing sedation. Options A, B, D and E are second-line or harmful choices: antipsychotics, antidepressants such as selective serotonin reuptake inhibitors, benzodiazepines and buspirone are all listed as second-line treatments. Buspirone additionally lowers the seizure threshold and must be used with caution. Benzodiazepines, typical antipsychotics and anticholinergic agents such as diphenhydramine appear on the list of medication classes that may delay cognitive recovery, increase sedation and worsen behavioral dysfunction after traumatic brain injury. Note also that posttraumatic agitation is an expected phase of recovery that resolves within 7 to 10 days in most patients and should be monitored with a validated instrument such as the Agitated Behavior Scale, so the temptation to pharmacologically suppress behavior should be resisted while protecting the patient from self-harm and educating family and staff.

Key point:
for persistent posttraumatic agitation, propranolol or another beta-blocker, or a mood-regulating antiseizure agent, is first line, while benzodiazepines, antipsychotics and anticholinergics risk delaying cognitive recovery.

Source: Practice of Neurocritical Care, 2nd ed., Neurorehabilitation in the Neurocritical Care Unit, pp. 414-415

Question 153

Neurorehabilitation in the Neurocritical Care Unit · Autonomic dysreflexia in spinal cord injury · moderate

A 42-year-old woman with a T4 ASIA A paraplegia sustained six months ago returns to the ICU after uneventful revision spinal surgery. She arrives from the postanesthesia care unit on 2 liters per minute of supplemental oxygen with a peripheral intravenous line but no intravenous fluids running; her indwelling urinary catheter was removed just before transfer. Thirty minutes after arrival she reports a pounding headache, and she is flushed and sweating profusely above the level of her lesion with piloerection and nasal congestion. Blood pressure is 230/110 mmHg and heart rate is 54. Which of the following is the most appropriate immediate action?

- **A.** Administer intravenous morphine for presumed postoperative pain
- **B.** Administer intravenous naloxone for presumed opioid withdrawal
- **C.** Administer intravenous labetalol to lower the blood pressure urgently
- **D.** Lay the patient flat and place her in Trendelenburg position
- **E.** Sit the patient upright and immediately drain the bladder by replacing the urinary catheter

Answer: E. Sit the patient upright and immediately drain the bladder by replacing the urinary catheter

This is autonomic dysreflexia, a life-threatening emergency seen in patients with a spinal cord injury at or above T6, which occurs in 48 to 90 percent of susceptible patients after spinal shock resolves. It is triggered by a noxious stimulus below the level of injury, most commonly bladder or bowel related, that produces unopposed sympathetic outflow with peripheral vasoconstriction and acute hypertension, while the parasympathetic response above the lesion causes headache, sweating, piloerection, flushing, pupillary constriction and sinus congestion. Here the catheter was removed just before transfer, so a distended bladder is the likely stimulus, and treatment is to remove the noxious stimulus by sitting the patient up and decompressing the bladder, which makes option E correct. Option A would worsen the situation and does not address the trigger, and there is no evidence of opioid withdrawal to justify option B. Option C is unnecessary once the distended bladder is decompressed, and lowering blood pressure pharmacologically without relieving the stimulus risks hypotension when the trigger is finally removed. Option D is the opposite of correct positioning, since sitting the patient upright helps lower the blood pressure by orthostatic pooling. Other recognized triggers include bowel impaction, pressure ulcers, fractures, ingrown toenails, infections, bladder stones, gastric ulcers, labor and abdominal emergencies such as appendicitis, cholecystitis and pancreatitis.

Key point:
in autonomic dysreflexia with a lesion at or above T6, sit the patient up and find and remove the noxious stimulus, starting with the bladder, before reaching for antihypertensives.

Source: Practice of Neurocritical Care, 2nd ed., Neurorehabilitation in the Neurocritical Care Unit, pp. 416

Question 154

Neurorehabilitation in the Neurocritical Care Unit · SSRIs and poststroke depression · easy

A 55-year-old woman with hypertension and diabetes is in the ICU after a large right middle cerebral artery infarct. She is awake and alert with a flat affect and dense left hemiparesis. Her family has read that fluoxetine helps stroke recovery and asks whether she should be started on it now. Which of the following is the most appropriate recommendation?

- **A.** Screen her with a validated depression instrument such as the 9-item Patient Health Questionnaire and start an antidepressant only if the screen is positive
- **B.** Start fluoxetine empirically in all stroke patients, based on the FLAME trial
- **C.** Avoid antidepressants entirely, since they are contraindicated in the immediate poststroke period
- **D.** Start fluoxetine for motor recovery based on the FOCUS trial, which demonstrated improved functional outcomes
- **E.** Defer any depression screening until at least 6 months after stroke, when poststroke depression typically appears

Answer: A. Screen her with a validated depression instrument such as the 9-item Patient Health Questionnaire and start an antidepressant only if the screen is positive

Approximately one third of stroke survivors develop poststroke depression at some point, and it is associated with both increased mortality and worse functional outcomes, so all stroke patients should be screened with a validated tool such as the Center of Epidemiological Studies Depression Scale, the Hamilton Depression Rating Scale, or the 9-item Patient Health Questionnaire, which is often the most feasible in a busy clinical setting. The evidence for empiric selective serotonin reuptake inhibitor treatment to promote recovery is conflicting: the FLAME trial in 2011 suggested a benefit for motor recovery in 118 patients treated with fluoxetine, but the much larger FOCUS trial in 2019, which enrolled 3,127 patients and included patients with intracerebral hemorrhage, showed no benefit. This makes options B and D incorrect, and option D also misstates the FOCUS result. Option C is wrong because antidepressants are not contraindicated and are advocated when depression is present. Option E is wrong because screening should occur during the acute admission, and risk factors for poststroke depression include physical disability, more severe stroke, preexisting depression and cognitive impairment, all of which apply here. Because optimal timing, threshold and drug choice remain unclear, the chapter recommends starting antidepressants only in the presence of depression rather than empirically in every stroke patient.

Key point:
screen every stroke patient for depression and treat when the screen is positive, rather than starting a selective serotonin reuptake inhibitor empirically for motor recovery.

Source: Practice of Neurocritical Care, 2nd ed., Neurorehabilitation in the Neurocritical Care Unit, pp. 417

Pediatric Neurocritical Care

Question 155

Pediatric Neurocritical Care · Pediatric arterial ischemic stroke - acute recanalization · moderate

A 9-year-old previously healthy girl is brought to the emergency department 3 hours after the acute onset of right face and arm weakness with difficulty speaking. She had been entirely well the week before. Temperature is 37.0 C, heart rate 92/min, blood pressure 106/62 mmHg (persistently below the 95th percentile for age), and fingerstick glucose is 96 mg/dL. Her Pediatric NIH Stroke Scale score is 12. Non-contrast head CT shows no hemorrhage, and CT angiography demonstrates occlusion of the left middle cerebral artery M1 segment with a corresponding early infarct. Platelet count is 240,000/microL, PT is 13 seconds (INR 1.0), and PTT is normal. She has had no recent surgery, trauma, intracranial hemorrhage, or bleeding. Which of the following is the most appropriate next step in management?

- **A.** Withhold thrombolysis because intravenous alteplase is contraindicated in all patients younger than 18 years
- **B.** Consider intravenous alteplase, as she meets the typical selection criteria used for thrombolysis in children
- **C.** Proceed directly to mechanical thrombectomy, which is the established standard of care for pediatric large vessel occlusion
- **D.** Begin aspirin 3 to 5 mg/kg/day and admit for observation without recanalization therapy
- **E.** Elevate the head of the bed to 45 degrees and induce hypertension to a systolic pressure 20 percent above the 95th percentile for age

Answer: B. Consider intravenous alteplase, as she meets the typical selection criteria used for thrombolysis in children

The typical pediatric selection criteria for intravenous alteplase are age 2 years or older, treatment within 4.5 hours of symptom onset, a persistent focal deficit with both a proven infarct and an arterial occlusion on MR or CT/CTA, systolic blood pressure persistently less than 15 percent above the 95th percentile for age, a PedNIHSS less than 25, glucose greater than 50 mg/dL and less than 400 mg/dL, platelets greater than 100,000, and PT less than 15 seconds (INR less than 1.4) - and this patient meets all of them. The chapter states that alteplase may be considered for older children with a confirmed large vessel thrombus and a PedNIHSS of 6 or greater, and that the risk of symptomatic intracranial hemorrhage when alteplase is given within 4.5 hours of onset is low. Option A is wrong because alteplase, while not adequately tested below age 18, can be considered case by case in patients 2 to 17 years; the absolute prohibition is only for children younger than 2 years. Option C is wrong because there is no consensus on mechanical thrombectomy in children: the Save ChildS Study included only 73 children from 42 centers over 19 years and does not establish thrombectomy as standard of care, so it is offered case by case by an experienced multidisciplinary team. Option D is wrong because aspirin 3 to 5 mg/kg/day is used for secondary prevention once hemorrhage is excluded, but choosing it here forgoes recanalization in an eligible child still inside the window. Option E is wrong because acute neuroprotection in suspected pediatric stroke means laying the patient flat with strict bedrest, treating fever to keep temperature below 37 C, and giving IV fluids; hypertension is not routinely treated and is never induced, and if blood pressure must be lowered it is reduced by 25 percent over 24 hours.

Key point:
intravenous alteplase can be considered in a child 2 years or older within 4.5 hours who has a proven infarct plus arterial occlusion and a PedNIHSS of at least 6, whereas thrombectomy remains case by case.

Source: Practice of Neurocritical Care, 2nd ed., Pediatric Neurocritical Care, pp. 430, 434

Question 156

Pediatric Neurocritical Care · Post-cardiac arrest syndrome - targeted temperature management · moderate

A 4-year-old boy is resuscitated after an unwitnessed pool submersion. The first documented rhythm was pulseless electrical activity, and return of spontaneous circulation was achieved after 12 minutes of CPR. On arrival in the pediatric ICU he is intubated and comatose, with a core temperature of 38.1 C, blood pressure 88/50 mmHg on a low-dose epinephrine infusion, $PaCO_2$ 40 mmHg, and SpO_2 97 percent. Continuous EEG has been started. Which of the following temperature strategies is most appropriate?

- **A.** Targeted temperature management to prevent fever, maintaining core temperature between 36.0 C and 37.5 C for the first 48 hours
- **B.** Therapeutic hypothermia to 33 C for 48 hours, which has been shown to reduce mortality compared with normothermia
- **C.** No active temperature control, treating fever with acetaminophen only if the temperature exceeds 39 C
- **D.** Deep hypothermia to 28 to 30 C for 24 hours followed by rewarming at 2 C per hour
- **E.** Continuous neuromuscular blockade with sustained hypothermia to 32 C for 7 days

Answer: A. Targeted temperature management to prevent fever, maintaining core temperature between 36.0 C and 37.5 C for the first 48 hours

Randomized trials of therapeutic hypothermia to 33 C for 48 hours after both in-hospital and out-of-hospital pediatric cardiac arrest did not reduce mortality or improve 1-year functional outcome compared with therapeutic normothermia, so hypothermia cannot be offered as an outcome-improving therapy (making B wrong). What the 2019 AHA guidelines do recommend is targeted temperature management to prevent fever in the first 48 hours after arrest, maintaining temperature between 36 C and 37.5 C; this is safe, feasible, and should be regarded as the standard of care. For a child who remains comatose after ROSC it is also reasonable to maintain TTM at 36 to 37.5 C for 5 days, or 2 days of continuous hypothermia at 32 to 34 C followed by 3 days of normothermia at 36 to 37.5 C - but not the regimens in D or E, which use temperatures and durations outside anything described. Option C is wrong because fever markedly increases cerebral metabolic demand and avoiding it is the central goal of post-arrest temperature control. Option E is additionally wrong because sedatives are preferred to neuromuscular blocking agents during temperature management so that seizures are not masked; when shivering does occur it is managed with a multifaceted approach of sedation, analgesia, magnesium, and only then neuromuscular blockade.

Key point:
after pediatric cardiac arrest, hypothermia to 33 C did not improve survival or 1-year outcome, and the standard of care is targeted temperature management at 36 to 37.5 C to prevent fever for the first 48 hours.

Source: Practice of Neurocritical Care, 2nd ed., Pediatric Neurocritical Care, pp. 436, 440

Question 157

Pediatric Neurocritical Care · Benzodiazepine-refractory convulsive status epilepticus · easy

A 5-year-old boy weighing 20 kg is brought to the emergency department with generalized tonic-clonic convulsions that began 20 minutes earlier. He receives two doses of intravenous lorazepam 0.1 mg/kg given 5 minutes apart. Ten minutes after the second dose he continues to convulse. His airway is patent, oxygen saturation is 96 percent on a non-rebreather mask, blood pressure is 96/58 mmHg, and glucose is 104 mg/dL. Which of the following is the most appropriate next step?

- **A.** Begin a propofol infusion titrated to seizure suppression
- **B.** Give a third dose of intravenous lorazepam 0.1 mg/kg
- **C.** Give pentobarbital 5 mg/kg by intravenous push
- **D.** Load with midazolam and begin a continuous midazolam infusion at 100 mcg/kg/hour
- **E.** Give levetiracetam 60 mg/kg intravenously, to a maximum of 4500 mg

Answer: E. Give levetiracetam 60 mg/kg intravenously, to a maximum of 4500 mg

Pediatric status epilepticus protocols call for 1 to 2 doses of a benzodiazepine as first tier, followed by a second-tier agent - fosphenytoin, levetiracetam, or phenobarbital - before any continuous anesthetic infusion. The Established Status Epilepticus Treatment Trial, conducted in 57 United States emergency departments in adults and children older than 2 years, found fosphenytoin 20 mg/kg, levetiracetam 60 mg/kg, and valproate 40 mg/kg to be equally effective, with 49 to 52 percent efficacy, and equally safe at those doses; two other randomized trials comparing phenytoin 20 mg/kg with levetiracetam 40 mg/kg likewise showed equal efficacy of 50 to 70 percent. Option A is wrong and dangerous: propofol is contraindicated for the treatment of seizures in children because of the increased risk of propofol related infusion syndrome. Option B is wrong because protocols specify only 1 to 2 benzodiazepine doses, and further doses add respiratory depression without added benefit. Options C and D are wrong because midazolam and pentobarbital infusions are reserved for refractory status epilepticus, defined as persistent seizures after first- and second-tier antiseizure medications have both been given; escalating to an anesthetic before a second-tier drug skips a proven step and commits the child to intubation and pressors.

Key point:
after 1 to 2 benzodiazepine doses fail, give a full second-tier antiseizure medication - levetiracetam 40 to 60 mg/kg or fosphenytoin 20 mg/kg - and never propofol in a child.

Source: Practice of Neurocritical Care, 2nd ed., Pediatric Neurocritical Care, pp. 447-448

Question 158

Pediatric Neurocritical Care · Severe TBI - tiered management of intracranial hypertension · hard

An 8-year-old girl with severe traumatic brain injury (post-resuscitation GCS 5) is in the pediatric ICU on day 2 with an external ventricular drain and an intraparenchymal ICP monitor. Her head is midline, the head of bed is at 30 degrees, the cervical collar has been loosened, and she is receiving fentanyl and midazolam infusions. Core temperature is 36.8 C, PaCO2 is 38 mmHg, SpO2 is 98 percent, and continuous EEG shows no seizures. Mean arterial pressure is 88 mmHg. Despite intermittent CSF drainage, her ICP has been 27 to 29 mmHg for the past 10 minutes. Serum sodium is 141 mEq/L and measured osmolarity is 295 mOsm/L. Which of the following is the most appropriate next step?

- **A.** Begin a pentobarbital infusion titrated to burst suppression with an interburst interval of 3 to 5 seconds
- **B.** Institute sustained hyperventilation to a PaCO2 of 25 to 30 mmHg
- **C.** Give a 3 percent hypertonic saline bolus of 2 to 5 mL/kg over 10 to 20 minutes, then start a continuous infusion at 0.1 to 1 mL/kg/hour
- **D.** Take her immediately for decompressive craniectomy
- **E.** Lower the head of bed to 0 degrees and give a 20 mL/kg normal saline bolus to raise cerebral perfusion pressure

Answer: C. Give a 3 percent hypertonic saline bolus of 2 to 5 mL/kg over 10 to 20 minutes, then start a continuous infusion at 0.1 to 1 mL/kg/hour

An ICP above 20 mmHg sustained for more than 5 minutes is the trigger to escalate therapy, and every first-tier item has already been addressed here - hypotension corrected, collar loosened, seizures excluded on cEEG, PaCO2 at the target of 37 plus or minus 2 mmHg, and adequate sedation and analgesia. The next step in the second tier, after intermittent CSF drainage, is hyperosmolar therapy: a 3 percent saline bolus of 2 to 5 mL/kg over 10 to 20 minutes followed by a continuous infusion of 0.1 to 1 mL/kg/hour titrated to keep ICP below 20 mmHg, targeting the lowest sodium that achieves the goal, with sodium and osmolarity checked every 2 to 4 hours, holding for osmolarity above 360 mOsm/L or sodium of 160 mEq/L, and avoiding sodium changes greater than 8 mEq per 24 hours. Options A and D are third-tier therapies and are premature before hyperosmolar therapy and neuromuscular blockade have been tried. Option B is wrong because hyperventilation to a PaCO2 of 25 to 30 mmHg is a third-tier maneuver that must be transient - less than 10 minutes - and is reserved for signs of herniation or an ICP above 30 mmHg for more than 5 minutes while other interventions take effect; prophylactic or prolonged hyperventilation has been associated with increased mortality. Option E is wrong because her cerebral perfusion pressure is already 88 minus 28, or about 60 mmHg, which meets both the minimum of 40 mmHg and the age-based target of 50 to 60 mmHg for a child older than 6 years, and lowering the head of bed to flat is not indicated when CPP is adequate.

Key point:
for ICP above 20 mmHg for more than 5 minutes with first-tier measures exhausted, the second tier is CSF drainage then 3 percent saline, 2 to 5 mL/kg bolus and 0.1 to 1 mL/kg/hour infusion, held for osmolarity above 360 or sodium of 160.

Source: Practice of Neurocritical Care, 2nd ed., Pediatric Neurocritical Care, pp. 442, 445-447

Question 159

Pediatric Neurocritical Care · Postoperative diabetes insipidus · moderate

A 7-year-old boy is in the pediatric ICU on the evening of a transsphenoidal resection of a craniopharyngioma. Over the past 3 hours his urine output has been 6 mL/kg/hour. Urine osmolality is 120 mOsm/kg with a specific gravity of 1.003. Whole blood sodium is 149 mEq/L and calculated serum osmolality is 305 mOsm/kg. He is drowsy and cannot reliably drink. Heart rate is 128/min and blood pressure is 96/54 mmHg. No mannitol has been given. Which of the following is the most appropriate next step?

- **A.** Restrict free water and begin a 3 percent saline infusion for the syndrome of inappropriate antidiuretic hormone secretion
- **B.** Give a 20 mL/kg bolus of 0.9 percent saline and begin fludrocortisone for cerebral salt wasting
- **C.** Observe with hourly sodium measurements and withhold treatment until serum osmolality exceeds 320 mOsm/kg
- **D.** Start a vasopressin infusion at 0.2 milliunits/kg/hour and change maintenance fluid to D5 one-quarter normal saline with 20 mEq/L potassium chloride
- **E.** Give oral desmopressin now and continue his current maintenance fluids unchanged

Answer: D. Start a vasopressin infusion at 0.2 milliunits/kg/hour and change maintenance fluid to D5 one-quarter normal saline with 20 mEq/L potassium chloride

He meets all of the diagnostic criteria for diabetes insipidus, which must all be present: polyuria with urine output above 3 mL/kg/hour for 2 hours, hypotonic urine with urine osmolality below 300 mOsm/kg or specific gravity below 1.010, and whole blood sodium above 145 mEq/L, supported here by a plasma osmolality above 300 mOsm/kg. Diabetes insipidus develops in roughly 10 to 44 percent of children after transsphenoidal pituitary surgery and usually within the first 12 hours, and treatment should begin as soon as criteria are met because delay causes dehydration, hypotension, and severe hypernatremia. The protocol is to start a vasopressin infusion at 0.2 milliunits/kg/hour, change IV fluids to D5 one-quarter normal saline with 20 mEq/L potassium chloride at 1 L/m2/day, replace urine output in excess of intake 1 to 1 with D5W, and titrate by 0.2 milliunits/kg/hour hourly to keep urine output between 1 and 3.5 mL/kg/hour, sodium between 140 and 155 mEq/L, and sodium shifts within 8 mEq/L. Options A and B are wrong because both SIADH and cerebral salt wasting produce hyponatremia, not the hypernatremia seen here; SIADH additionally requires a plasma osmolality below 275 mOsm/kg and urine osmolality above 100 mOsm/kg with euvolemia or hypervolemia, and cerebral salt wasting requires sodium below 135 mEq/L with hypovolemia and net negative fluid balance. Option C is wrong because increased plasma osmolality is not even required to initiate treatment once the other criteria are met. Option E is wrong because oral desmopressin is reserved for transition, generally about 24 hours after surgery, once the child's neurologic status allows safe oral intake and urine output and sodium are within goal.

Key point:
postoperative polyuria above 3 mL/kg/hour with dilute urine and a whole blood sodium above 145 mEq/L is diabetes insipidus - start vasopressin at 0.2 milliunits/kg/hour immediately; hyponatremia points instead to SIADH or cerebral salt wasting.

Source: Practice of Neurocritical Care, 2nd ed., Pediatric Neurocritical Care, pp. 469-471

Question 160

Pediatric Neurocritical Care · Bacterial meningitis - adjunctive dexamethasone · easy

A previously healthy 3-year-old girl presents with 12 hours of fever to 39.4 C, vomiting, photophobia, and neck stiffness. She is irritable but arousable with no focal deficits and no papilledema. Lumbar puncture yields cloudy CSF with 4,200 white cells/microL (92 percent neutrophils), glucose 18 mg/dL, and protein 180 mg/dL; Gram stain shows gram-positive diplococci. No antimicrobial agent has yet been given, and vancomycin plus ceftriaxone is being drawn up. Which of the following best describes the appropriate use of adjunctive corticosteroid therapy?

- **A.** Dexamethasone 0.15 mg/kg IV every 6 hours for 4 days, given 10 to 20 minutes before or together with the first dose of antimicrobial therapy
- **B.** Dexamethasone 0.15 mg/kg IV every 6 hours for 4 days, started 24 hours after antibiotics have been established
- **C.** Dexamethasone withheld unless Haemophilus influenzae is subsequently isolated from the CSF
- **D.** No corticosteroid, because adjunctive steroids have shown no benefit for any organism in high-income countries
- **E.** Methylprednisolone 30 mg/kg/day IV, to a maximum of 1 gram, for 3 to 5 days

Answer: A. Dexamethasone 0.15 mg/kg IV every 6 hours for 4 days, given 10 to 20 minutes before or together with the first dose of antimicrobial therapy

Dexamethasone should be given ideally 10 to 20 minutes before the first antimicrobial dose or at the same time as it, at 0.15 mg/kg every 6 hours for 4 days; doses used in pediatric studies vary around 0.4 to 0.6 mg/kg/day divided into 4 daily doses for 2 to 4 days, and a 4-day course is what is recommended. Option B is wrong because if antibiotic therapy has already been administered, adding dexamethasone is unlikely to improve outcomes - the timing relative to the first antibiotic dose is the whole point. Option C is wrong because dexamethasone improves outcome in bacterial meningitis due to both Haemophilus influenzae and Streptococcus pneumoniae, and in high-income countries steroid treatment reduced mortality specifically for Streptococcus pneumoniae, the organism suggested here by gram-positive diplococci. Option D is wrong because in high-income countries corticosteroids reduced severe hearing loss, any hearing loss, and short-term neurologic sequelae, and the most recent Cochrane review supports adjunctive steroids there for all organisms; it was in a developing-country trial that adjuvant dexamethasone failed to improve outcome. Option E is wrong because high-dose methylprednisolone at 30 mg/kg/day for 3 to 5 days is the regimen for autoimmune encephalitis, acute transverse myelitis, and ADEM, not for bacterial meningitis. One practical corollary: because steroids may decrease penetration of antibiotics such as vancomycin into the CNS, adding rifampin 20 mg/kg/day to a maximum of 600 mg may be considered when dexamethasone is used.

Key point:
adjunctive dexamethasone for pediatric bacterial meningitis is 0.15 mg/kg every 6 hours for 4 days, given 10 to 20 minutes before or with the first antibiotic dose, and is of little value once antibiotics are already in.

Source: Practice of Neurocritical Care, 2nd ed., Pediatric Neurocritical Care, pp. 461

Question 161

Perioperative Neurosurgical Critical Care · Posterior fossa craniotomy complications · moderate

A 61-year-old woman is 29 hours status post suboccipital craniectomy for resection of a cerebellar metastasis. She is awake and alert. Over the past hour the nurse has given two doses of intravenous labetalol and started a nicardipine infusion because the blood pressure keeps climbing; it is now 192/100 mmHg with a heart rate of 58. She appears uncomfortable and slightly more drowsy than earlier in the shift. Temperature is 37.2 C and oxygen saturation is 97 percent on 2 liters per minute by nasal cannula. Which of the following is the most appropriate next step?

- **A.** Give an additional as-needed dose of intravenous labetalol
- **B.** Titrate the nicardipine infusion upward until the blood pressure is controlled
- **C.** Give intravenous analgesia and reassess in one hour
- **D.** Perform a careful neurologic examination and obtain a stat head CT
- **E.** Place an external ventricular drain for presumed obstructive hydrocephalus

Answer: D. Perform a careful neurologic examination and obtain a stat head CT

Sudden changes in level of arousal, neurologic examination, or vital signs after a posterior fossa operation should immediately raise concern for a hematoma or severe swelling and trigger immediate imaging and evaluation, because unlike supratentorial hemorrhage a posterior fossa emergency leaves a much slimmer margin for error. Refractory hypertension in this setting is a vital sign change, and the accompanying drowsiness makes an expanding posterior fossa hematoma the leading concern. Options A and B are wrong because further escalation of antihypertensive therapy consumes valuable time while a hemorrhage expands. Option C is wrong for the same reason: sedation and analgesia may lower blood pressure in a patient who is in pain, but more serious pathology must be excluded first, and over-sedation also clouds the neurologic examination. Option E is wrong because although hydrocephalus is a recognized posterior fossa complication, an external ventricular drain should not be placed without radiographic evidence, which the head CT will provide. Recall that postoperative cerebral edema typically begins within hours and peaks at 48 to 72 hours, so this patient remains inside the window of maximal risk.

Key point:

refractory hypertension or any acute change after posterior fossa surgery means an urgent examination and stat head CT, not more antihypertensive.

Source: Practice of Neurocritical Care, 2nd ed., Perioperative Neurosurgical Critical Care, pp. 502

Question 162

Perioperative Neurosurgical Critical Care · Diabetes insipidus after trans-sphenoidal surgery · moderate

A 38-year-old woman is admitted to the ICU after endoscopic trans-sphenoidal resection of a large pituitary macroadenoma. Overnight her urine output rises to 400 mL per hour for three consecutive hours. Urine specific gravity is 1.003, serum sodium has risen from 140 to 149 mmol/L, and serum osmolality is elevated. She is intubated and sedated and cannot report thirst. Blood pressure is 104/62 mmHg and heart rate is 104. Which of the following is the most appropriate management?

- **A.** Restrict free water and repeat the serum sodium in 6 hours
- **B.** Replace urine output with hypotonic intravenous fluid and give desmopressin titrated to urine output, specific gravity, and serum sodium
- **C.** Give stress-dose hydrocortisone and observe, since the polyuria is most likely from cortisol insufficiency
- **D.** Give a normal saline bolus and attribute the polyuria to intraoperative fluid resuscitation
- **E.** Start a hypertonic saline infusion targeting a serum sodium of 155 mmol/L

Answer: B. Replace urine output with hypotonic intravenous fluid and give desmopressin titrated to urine output, specific gravity, and serum sodium

This patient has diabetes insipidus after trans-sphenoidal surgery, a common complication whose risk rises with resection of very large tumors and which is generally less frequent with an endoscopic approach. The diagnostic findings given in the chapter are high urine output greater than 300 mL per hour, low urine specific gravity less than 1.005, and elevated serum sodium and osmolality, with polydipsia when thirst is intact; all are present here. Management is hypotonic intravenous fluid given to match urine output plus desmopressin or vasopressin, intermittent or as a continuous infusion, titrated to normalize urine output, urine specific gravity, and serum sodium, making option B correct. Option A worsens the free water deficit. Option C is wrong because although hypothalamic or pituitary injury can impair adrenocorticotropic hormone secretion and precipitate cortisol insufficiency, treated with perioperative stress-dose steroids such as hydrocortisone, cortisol insufficiency does not produce dilute polyuria with rising sodium. Option D reflects the important caveat that most postoperative polyuria is not diabetes insipidus but rather aggressive intraoperative fluid resuscitation or mannitol-induced osmotic diuresis, yet the dilute urine with a rising sodium here distinguishes true diabetes insipidus. Option E is dangerous, since hypernatremia can become severe when thirst is impaired or cannot be expressed, exactly the situation in a sedated postoperative patient.

Key point:
urine output above 300 mL per hour with specific gravity below 1.005 and a rising sodium after pituitary surgery is diabetes insipidus, treated with hypotonic fluid matched to output plus desmopressin.

Source: Practice of Neurocritical Care, 2nd ed., Perioperative Neurosurgical Critical Care, pp. 503

Question 163

Perioperative Neurosurgical Critical Care · Neck hematoma after carotid endarterectomy · hard

A 68-year-old man with chronic obstructive pulmonary disease and obstructive sleep apnea is admitted to the neurocritical care unit after carotid endarterectomy under general anesthesia. Three hours after arrival he reports difficulty breathing. At the bedside his neck is visibly swollen and tense, he has inspiratory stridor, and oxygen saturation is 91 percent on 4 liters per minute by nasal cannula. He is anxious and using accessory muscles. Which of the following is the most appropriate next step?

- **A.** Call the surgeon and anesthesia for emergent opening of the wound at the bedside followed by intubation
- **B.** Call the surgeon and anesthesia for intubation first, followed by transfer to the operating room
- **C.** Give supplemental oxygen and a racemic epinephrine nebulizer
- **D.** Obtain a stat carotid duplex ultrasound
- **E.** Obtain a stat neck CT and CT angiogram

Answer: A. Call the surgeon and anesthesia for emergent opening of the wound at the bedside followed by intubation

Neck dissection for carotid exposure, like anterior cervical spine surgery, can be complicated by a postoperative hematoma with airway compromise, and identification of an enlarging neck hematoma must prompt swift action to evacuate the clot and secure the airway. Emergently opening the wound at the bedside followed by intubation can be a lifesaving intervention prior to return to the operating room for definitive hemostasis and closure, which is why option A is correct. Option B inverts the order: intubation is likely to be extremely difficult or impossible until the pressure from the hematoma is relieved, so the wound should be opened first. Options D and E are wrong because the clinical picture demands immediate intervention and leaves no time for diagnostic imaging. Option C is wrong because oxygen and racemic epinephrine do not treat mechanical compression from an expanding hematoma. Other recognized complications of carotid endarterectomy include transient bradycardia from stretch or manipulation of the carotid sinus baroreceptors, early thromboembolic stroke, hemorrhagic stroke from newly increased flow to previously underperfused brain, and cranial nerve deficits in about 5 percent of cases, most of which are transient and involve cranial nerves nine, ten, eleven, and twelve.

Key point:
an expanding neck hematoma with stridor after carotid or anterior cervical surgery is treated by opening the wound at the bedside first and then intubating.

Source: Practice of Neurocritical Care, 2nd ed., Perioperative Neurosurgical Critical Care, pp. 504-505

Question 164

Perioperative Neurosurgical Critical Care · Hemodynamic management of acute spinal cord injury · easy

A 48-year-old man is admitted after a motorcycle crash with a C6 fracture-dislocation and an incomplete cervical spinal cord injury. He has undergone decompression and posterior instrumented fusion. In the ICU his blood pressure is 88/52 mmHg with a heart rate of 52, his extremities are warm, and there is no evidence of ongoing hemorrhage. According to current neurosurgical guidelines cited in this chapter, which of the following best describes the hemodynamic target and initial treatment?

- **A.** Mean arterial pressure greater than 65 mmHg for 24 hours, treated with volume resuscitation alone
- **B.** Systolic blood pressure greater than 140 mmHg for 72 hours, treated with high-dose methylprednisolone
- **C.** Mean arterial pressure greater than 70 mmHg for 14 days, treated with albumin and hydrocortisone
- **D.** Systolic blood pressure greater than 100 mmHg for 7 days, with permissive bradycardia and no vasopressor
- **E.** Mean arterial pressure greater than 85 mmHg for 7 days, treated with volume resuscitation and, if needed, vasopressors such as dopamine, norepinephrine, or phenylephrine

Answer: E. Mean arterial pressure greater than 85 mmHg for 7 days, treated with volume resuscitation and, if needed, vasopressors such as dopamine, norepinephrine, or phenylephrine

Current neurosurgical guidelines recommend augmentation of circulation with a mean arterial pressure goal greater than 85 mmHg for seven days after spinal cord injury in order to minimize cord ischemia, although the chapter notes the supporting evidence is limited; at a minimum, hypotension should be strictly avoided because episodes of reduced perfusion place the cord at risk of ischemic injury. This patient's warm extremities, hypotension, and relative bradycardia are classic for neurogenic shock from sympathetic disruption in cervical and upper thoracic cord injuries, which is treated with volume resuscitation and, if necessary, vasopressors such as dopamine, norepinephrine, or phenylephrine, making option E correct. Options A, C, and D state the wrong target or duration, and option D wrongly withholds vasopressors in a patient in neurogenic shock. Option B is doubly wrong: the target is a mean arterial pressure rather than a systolic goal, and treatment of acute spinal cord injury with corticosteroids is no longer recommended given the lack of compelling evidence. Early prophylaxis for venous thromboembolism, an aggressive bowel regimen, and nutritional support are also strongly recommended in these patients.

Key point:
after acute spinal cord injury, target a mean arterial pressure above 85 mmHg for seven days with fluids and vasopressors, and do not give steroids.

Source: Practice of Neurocritical Care, 2nd ed., Perioperative Neurosurgical Critical Care, pp. 503-504

Question 165

Perioperative Neurosurgical Critical Care · Postoperative seizures and seizure prophylaxis after craniotomy · moderate

A 72-year-old man is on postoperative day 1 after craniotomy for evacuation of an acute subdural hematoma. He is on levetiracetam started at the time of surgery. Overnight the nurse reports that his level of consciousness has been waxing and waning, without overt convulsive movements; he is intermittently unresponsive for several minutes and then follows commands again. Vital signs are stable and laboratory studies including sodium and glucose are normal. Which of the following is the most appropriate next step?

- **A.** Increase the levetiracetam dose and continue observation
- **B.** Switch from levetiracetam to phenytoin, which is superior for postoperative seizure prophylaxis
- **C.** Obtain an urgent noncontrast head CT and arrange electroencephalography
- **D.** Administer intravenous mannitol empirically for presumed cerebral edema
- **E.** Discontinue antiseizure prophylaxis, since prophylaxis has no proven benefit after craniotomy

Answer: C. Obtain an urgent noncontrast head CT and arrange electroencephalography

Postoperative seizures occur in as many as 15 to 20 percent of patients after cranial neurosurgery, and patients undergoing craniotomy for subdural hematoma evacuation are at particularly high risk, with up to 25 percent demonstrating seizures or epileptiform discharges on electroencephalography in the early postoperative period. Seizures may manifest insidiously as fluctuating alterations in level of consciousness rather than as overt convulsions, exactly as described here. Postoperative seizures should always raise concern for an acute intracranial pathology such as hemorrhage and prompt brain imaging, so urgent head CT together with electroencephalography is the correct next step. Option A treats a presumed seizure without excluding an expanding hematoma. Option B is wrong because a Cochrane review of 10 randomized controlled trials found that most studies showed no significant reduction in postoperative seizures with prophylaxis and no specific antiseizure medication was superior, while phenytoin in some trials was associated with increased adverse outcomes including mortality at 24 months. Option D is wrong because empiric mannitol does not substitute for making the diagnosis with imaging. Option E ignores that this patient may be actively seizing and that a common practice, consistent with the protocol described by Iuchi and colleagues, is levetiracetam for seven days following surgery.

Key point:
fluctuating consciousness after craniotomy, especially after subdural hematoma evacuation, is a seizure until proven otherwise and mandates imaging first to exclude an acute intracranial process.

Source: Practice of Neurocritical Care, 2nd ed., Perioperative Neurosurgical Critical Care, pp. 501-502

Question 166

Perioperative Neurosurgical Critical Care · Vasospasm and delayed cerebral ischemia after aneurysmal SAH · easy

A 49-year-old woman is on postoperative day 10 after craniotomy for clipping of a ruptured anterior communicating artery aneurysm. She has had an external ventricular drain since admission and has been neurologically stable with full strength in all four extremities. Today she develops left arm weakness, and CT angiography shows right middle cerebral artery vasospasm. She has been receiving nimodipine since admission. Which of the following best characterizes the role of nimodipine in this setting?

- **A.** It reliably reverses established angiographic vasospasm and can be given as an intra-arterial rescue therapy
- **B.** It improves outcomes and is a mainstay of treatment even though it has not been shown to directly reduce the incidence of vasospasm
- **C.** It should be discontinued once angiographic vasospasm is documented so that induced hypertension can be achieved
- **D.** It is effective only when administered intrathecally through an external ventricular drain
- **E.** It prevents rebleeding of the secured aneurysm rather than affecting vasospasm

Answer: B. It improves outcomes and is a mainstay of treatment even though it has not been shown to directly reduce the incidence of vasospasm

Nimodipine has not been demonstrated to directly reduce the incidence of cerebral vasospasm, yet it has been shown to improve outcomes and is therefore a mainstay of treatment after aneurysmal subarachnoid hemorrhage, which makes option B correct. Option A is wrong because nimodipine does not reliably reverse established vasospasm; the endovascular rescue option described is local intra-arterial delivery of calcium channel blockers at angiography. Option C is wrong because nimodipine is continued, and the other primary interventions, euvolemia and blood pressure augmentation, are added rather than substituted. Option D is wrong because nimodipine is given systemically, although for patients with an external ventricular drain in place intrathecal administration of nicardipine is a separate potential intervention. Option E is wrong because rebleeding is prevented by early securing of the aneurysm by clipping or coiling together with strict blood pressure control and avoidance of hypertension in the unsecured period, not by nimodipine. The risk of vasospasm and delayed cerebral ischemia is highest between post-bleed days 3 and 14, which is why serial neurologic examinations, vascular imaging, and transcranial Doppler monitoring are used during that window.

Key point:
nimodipine improves outcome after aneurysmal subarachnoid hemorrhage without reducing angiographic vasospasm, and the vasospasm risk window is post-bleed days 3 to 14.

Source: Practice of Neurocritical Care, 2nd ed., Perioperative Neurosurgical Critical Care, pp. 505

Pharmacology in the Neurocritical Care Unit

Question 167

Pharmacology in the Neurocritical Care Unit · Protein binding and therapeutic drug monitoring of phenytoin · easy

A 62-year-old woman is in the neurointensive care unit on hospital day 6 after a right basal ganglia intracerebral hemorrhage. She has been maintained on phenytoin for early post-hemorrhagic seizures. She is hemodynamically stable, afebrile, and has had no clinical or electrographic seizures, but the nurse reports she has become progressively more lethargic and has horizontal nystagmus on examination. Laboratory studies show sodium 138 mmol/L, creatinine 0.9 mg/dL, and albumin 2.1 g/dL. The total phenytoin concentration drawn at trough is 6.5 mcg/mL. Which of the following is the most appropriate next step?

- **A.** Increase the phenytoin maintenance dose until the total serum concentration is 10 to 20 mcg/mL
- **B.** Obtain a free (unbound) phenytoin concentration and target 1 to 2 mcg/mL
- **C.** Administer an additional phenytoin load of 20 mg/kg and recheck the total level in 2 hours
- **D.** Repeat the total phenytoin level immediately after the next intravenous infusion is completed
- **E.** Discontinue phenytoin and obtain a serum osmolar gap to assess for drug accumulation

Answer: B. Obtain a free (unbound) phenytoin concentration and target 1 to 2 mcg/mL

Phenytoin is more than 90% protein bound, and in critical illness hypoalbuminemia and competition for binding sites raise the unbound (free) fraction, so a total concentration in the subtherapeutic range can coexist with a free concentration that is therapeutic or toxic. The chapter states that when assessing whether a highly protein-bound medication is therapeutic, a free concentration is preferred over a total serum concentration because corrective equations may be inaccurate; the therapeutic free phenytoin range is 1 to 2 mcg/mL and the total range is 10 to 20 mcg/mL. This patient's lethargy and nystagmus are listed adverse effects of phenytoin and suggest her free level is already at or above target, so escalating the dose (option A) or reloading 20 mg/kg (option C) risks toxicity, which is amplified because phenytoin shows nonlinear, saturable kinetics in which small dose changes produce large concentration swings. Repeating a total level right after an infusion (option D) samples before distribution is complete and still does not address the altered protein binding. The osmolar gap (option E) is used to monitor propylene glycol accumulation with lorazepam infusions and mannitol clearance, not phenytoin efficacy, and phenytoin infusions actually confound the osmolar gap because the vehicle contains propylene glycol.

Key point:
In hypoalbuminemic critically ill patients, judge phenytoin by the free level of 1 to 2 mcg/mL, not the total level of 10 to 20 mcg/mL.

Source: Practice of Neurocritical Care, 2nd ed., Pharmacology in the Neurocritical Care Unit, pp. 519

Question 168

Pharmacology in the Neurocritical Care Unit · Augmented renal clearance and antiepileptic dosing · moderate

A 24-year-old man is admitted to the neurointensive care unit after a motorcycle collision with a severe traumatic brain injury. He is 4 days into his stay, is hemodynamically stable on no vasopressors, and his sequential organ failure assessment (SOFA) score is 3. Serum creatinine is 0.6 mg/dL and an 8-hour urine collection yields a measured creatinine clearance of 168 mL/min/1.73 m2. He has been receiving levetiracetam 1,000 mg intravenously every 12 hours, and continuous electroencephalography now shows two brief electrographic seizures. Which of the following is the most appropriate adjustment to his levetiracetam regimen?

- **A.** Levetiracetam 500 mg intravenously every 12 hours
- **B.** Levetiracetam 1,000 mg intravenously every 24 hours
- **C.** Levetiracetam 1,500 mg intravenously every 12 hours
- **D.** Levetiracetam 1,000 mg intravenously every 8 hours
- **E.** Levetiracetam 1,000 mg intravenously every 12 hours with a free drug level in 48 hours

Answer: D. Levetiracetam 1,000 mg intravenously every 8 hours

This patient meets criteria for augmented renal clearance (ARC), defined as enhanced elimination of solute compared with an expected baseline and correlating with a glomerular filtration rate greater than 130 mL/min/1.73 m2; his measured clearance of 168 mL/min/1.73 m2 confirms it, and he scores high on the ARC Scoring System with age under 50 (6 points), trauma admission (3 points), and SOFA less than 4 (1 point). Levetiracetam is primarily renally eliminated, and the chapter specifically recommends adjusting levetiracetam to 1,000 mg every 8 hours in ARC to avoid subtherapeutic trough concentrations, making option D correct. Reducing the dose or extending the interval (options A and B) would worsen subtherapeutic troughs. Simply enlarging each dose while keeping a 12-hour interval (option C) does not fix the trough problem, because in a drug with rapid clearance it is the interval, not the peak, that determines the trough. Levetiracetam is not highly protein bound and has low potential for drug interactions, so a free level (option E) is not a recognized monitoring strategy.

Key point:
Augmented renal clearance is a glomerular filtration rate greater than 130 mL/min/1.73 m2, and levetiracetam should be given as 1,000 mg every 8 hours in that setting.

Source: Practice of Neurocritical Care, 2nd ed., Pharmacology in the Neurocritical Care Unit, pp. 520, 527

Question 169

Pharmacology in the Neurocritical Care Unit · Neuromuscular blocking agents after spinal cord injury · moderate

A 38-year-old man sustained a complete C5 spinal cord injury 3 weeks ago and was extubated 5 days ago. He now develops aspiration pneumonia with worsening hypoxemia and requires emergent reintubation. His blood pressure is 104/62 mm Hg, heart rate is 58 beats/min, and serum potassium is 4.6 mmol/L with a creatinine of 0.8 mg/dL. The team plans rapid sequence intubation. Which of the following is the most appropriate neuromuscular blocking agent for this intubation?

- **A.** Rocuronium 1.2 mg/kg intravenously
- **B.** Succinylcholine 1.5 mg/kg intravenously
- **C.** Pancuronium 0.1 mg/kg intravenously
- **D.** Atracurium 0.5 mg/kg intravenously
- **E.** Cisatracurium 0.1 mg/kg intravenously

Answer: A. Rocuronium 1.2 mg/kg intravenously

Succinylcholine, the only available depolarizing agent, causes potassium efflux with a typical rise in serum potassium of 0.5 to 1 mEq/L, but in states of extensive denervation or upper motor neuron injury it can produce lethal hyperkalemia and cardiac arrest; the chapter states it should not be administered between 3 days and 9 months following spinal cord injury, which excludes option B at 3 weeks. Rocuronium at 0.6 to 1.2 mg/kg is a nondepolarizing aminosteroid with an onset suitable for rapid sequence intubation and a duration of about 30 minutes, making it the best choice here. Pancuronium has a duration of 90 to 100 minutes and causes tachycardia through vagal inhibition, which is undesirable for a single intubating dose. Atracurium should be avoided in patients with neurologic injury because its inactive metabolite laudanosine causes central nervous system stimulation and possible seizures. Cisatracurium at 0.1 to 0.2 mg/kg has a 45- to 60-minute duration and a slower onset than rocuronium, so it is a poor rapid sequence agent even though its Hofmann degradation is attractive in organ dysfunction.

Key point:
Avoid succinylcholine between 3 days and 9 months after spinal cord injury because of denervation hyperkalemia; use rocuronium 0.6 to 1.2 mg/kg instead.

Source: Practice of Neurocritical Care, 2nd ed., Pharmacology in the Neurocritical Care Unit, pp. 523-524

Question 170

Pharmacology in the Neurocritical Care Unit · Monitoring scheduled mannitol therapy · moderate

A 58-year-old woman with a large right middle cerebral artery infarct and cerebral edema has been receiving scheduled mannitol 1 g/kg intravenously every 6 hours for 3 days for intracranial pressure control. Her intracranial pressure has been 12 to 16 mm Hg over the past 24 hours. This morning, serum sodium is 146 mmol/L, blood urea nitrogen is 28 mg/dL, glucose is 110 mg/dL, and creatinine has risen from 0.8 to 1.4 mg/dL. Measured serum osmolality is 332 mOsm/kg and calculated serum osmolarity is 308 mOsm/L. Which of the following is the most appropriate next step?

- **A.** Continue the current mannitol regimen and repeat the osmolality in 24 hours
- **B.** Increase the mannitol dose to 1.5 g/kg every 6 hours to overcome tachyphylaxis
- **C.** Hold the next mannitol dose and extend the dosing interval before redosing
- **D.** Continue mannitol and add furosemide to augment the osmotic diuresis
- **E.** Continue mannitol but administer subsequent doses without an in-line filter to speed delivery

Answer: C. Hold the next mannitol dose and extend the dosing interval before redosing

The osmolality gap is the measured serum osmolality minus the calculated serum osmolarity, and here it is 332 minus 308, or 24 mOsm/kg. Although guidelines do not define a validated cutoff for acute kidney injury risk, an osmolar gap greater than 20 mOsm/kg has been used in practice to indicate incomplete clearance of mannitol, which may predispose to renal failure; the chapter instructs that if the gap is elevated, the dose may be held and either the dose reduced or the interval extended to allow elimination before repeat dosing. Her rising creatinine reinforces this. Continuing unchanged or escalating the dose (options A and B) allows further mannitol accumulation in a patient whose intracranial pressure is already controlled at 12 to 16 mm Hg. Adding a loop diuretic (option D) compounds the hypovolemia and electrolyte depletion that already require routine monitoring of potassium, magnesium, phosphorus, and calcium during mannitol therapy. An in-line filter should always be used with mannitol even when no crystals are visible, so option E is unsafe.

Key point:
With scheduled mannitol, calculate the osmolar gap before each dose and hold or space the dose when it exceeds 20 mOsm/kg.

Source: Practice of Neurocritical Care, 2nd ed., Pharmacology in the Neurocritical Care Unit, pp. 525

Question 171

Pharmacology in the Neurocritical Care Unit · Hyperchloremia and acute kidney injury with hypertonic saline · hard

A 30-year-old man with a severe traumatic brain injury has been receiving a continuous infusion of 3% sodium chloride at 50 mL/hr targeting a serum sodium of 145 to 155 mmol/L. Intracranial pressures have remained below 20 mm Hg for the past 24 hours. On day 3, laboratory studies show sodium 154 mmol/L, chloride 119 mmol/L, serum bicarbonate 15 mmol/L, potassium 3.1 mmol/L, and a creatinine that has risen from 0.7 to 2.1 mg/dL with a marked fall in urine output. Which of the following is the most appropriate next step?

- **A.** Continue the 3% sodium chloride infusion and increase the target sodium to 160 to 165 mmol/L
- **B.** Discontinue all hyperosmolar therapy and manage intracranial pressure with sedation alone
- **C.** Discontinue the 3% sodium chloride and begin scheduled mannitol 1 g/kg every 6 hours
- **D.** Discontinue the 3% sodium chloride and begin an equimolar hypertonic sodium acetate infusion
- **E.** Change the infusion to a compounded balanced hypertonic solution of sodium chloride plus sodium acetate

Answer: E. Change the infusion to a compounded balanced hypertonic solution of sodium chloride plus sodium acetate

Hyperchloremia from continuous 3% sodium chloride is associated with acute kidney injury, metabolic acidosis, and prolonged length of stay, and hyperchloremia with hypokalemia occurs in about 50% of patients on continuous infusions; this patient has a chloride of 119 mmol/L, a bicarbonate of 15 mmol/L, hypokalemia, and a tripled creatinine. Guidelines recommend an upper sodium range of 155 to 160 mEq/L and a chloride range of 110 to 115 mEq/L to decrease acute kidney injury risk, so his chloride is above the recommended ceiling. The chapter's specific remedy for hyperchloremia is to compound a balanced hypertonic solution from sodium chloride and sodium acetate, for example 32 mL of 4 mEq/mL sodium chloride, 46 mL of 2 mEq/mL sodium acetate, and 422 mL of sterile water for a 3% balanced solution, which preserves the sodium load while lowering the chloride burden. Pushing sodium to 160 to 165 mmol/L (option A) exceeds the recommended range and worsens renal risk, while abandoning hyperosmolar therapy entirely (option B) risks losing intracranial pressure control. Mannitol (option C) is a poor substitute in established acute kidney injury because it accumulates when clearance falls, and an all-acetate solution (option D) corrects the chloride but may produce metabolic alkalosis.

Key point:
When chloride exceeds 110 to 115 mEq/L on hypertonic saline, switch to a balanced sodium chloride plus sodium acetate solution rather than abandoning or intensifying the infusion.

Source: Practice of Neurocritical Care, 2nd ed., Pharmacology in the Neurocritical Care Unit, pp. 525-526

Question 172

Pharmacology in the Neurocritical Care Unit · Propofol-related infusion syndrome · moderate

A 19-year-old woman with refractory status epilepticus has been maintained on a propofol infusion at 90 mcg/kg per minute for the past 3 days along with intravenous fentanyl. Her body mass index is 18 kg/m2 and she has required low-dose norepinephrine throughout. Overnight she becomes bradycardic to 42 beats/min with new hypotension. Laboratory studies show a lactate of 6.8 mmol/L, potassium 6.1 mmol/L, triglycerides 720 mg/dL, and creatine kinase 9,400 units/L. An echocardiogram shows a new reduction in ejection fraction. Which of the following best explains this presentation?

- **A.** Propylene glycol accumulation from the sedative vehicle
- **B.** Propofol-related infusion syndrome
- **C.** Fentanyl-induced histamine release with vasodilation
- **D.** Valproate-induced hyperammonemic encephalopathy
- **E.** Intensive care unit-acquired weakness from a steroid and aminosteroid combination

Answer: B. Propofol-related infusion syndrome

Propofol-related infusion syndrome is a rare but potentially fatal complication characterized by rapid onset of heart failure, bradycardia, lactic acidosis, hyperlipidemia, hyperkalemia, and rhabdomyolysis, all of which this patient displays. She carries essentially every listed risk factor: a high dose greater than 75 mcg/kg per minute, a prolonged infusion greater than 48 hours, a low body mass index, young age, and concomitant vasopressor use; mortality has been reported as high as 80% despite supportive care, and creatine kinase should be monitored with prolonged use to identify high-risk patients. Propylene glycol accumulation (option A) is a complication of lorazepam infusions, not propofol, and presents with an elevated osmolar gap and a sepsis-like picture with acute kidney injury. Fentanyl does not cause histamine release, which is a property of morphine, so option C is wrong. Valproate hyperammonemic encephalopathy (option D) presents with altered mental status and elevated ammonia rather than rhabdomyolysis and hyperkalemia. Intensive care unit-acquired weakness (option E) follows corticosteroid plus aminosteroid neuromuscular blocker exposure and does not cause acute cardiac failure and lactic acidosis.

Key point:
Bradycardia, heart failure, lactic acidosis, hyperkalemia, hyperlipidemia, and rhabdomyolysis on propofol above 75 mcg/kg per minute for more than 48 hours is propofol-related infusion syndrome.

Source: Practice of Neurocritical Care, 2nd ed., Pharmacology in the Neurocritical Care Unit, pp. 522

Question 173

Pharmacology in the Neurocritical Care Unit · Vasopressor selection in acute spinal cord injury · easy

A 45-year-old man is admitted to the neurointensive care unit after a motor vehicle collision with a complete C3-C4 spinal cord injury. He is intubated and adequately volume resuscitated. His mean arterial pressure is 50 mm Hg, heart rate is 41 beats/min, and temperature is 36.4 degrees Celsius. The team wishes to augment spinal cord perfusion pressure. Which of the following is the most appropriate agent?

- **A.** Phenylephrine infusion
- **B.** Dobutamine infusion
- **C.** Arginine vasopressin at a fixed dose of 0.03 units/min
- **D.** Norepinephrine infusion
- **E.** Oral midodrine

Answer: D. Norepinephrine infusion

This patient has neurogenic shock with both hypotension and bradycardia, and the chapter states that in patients with both bradycardia and hypotension the preferred agents are norepinephrine and dopamine, neither of which should cause reflex bradycardia. Norepinephrine has strong alpha activity along with beta-1 activity, so it raises mean arterial pressure without worsening the heart rate. Phenylephrine (option A) is a pure alpha agonist with no beta-1 effect and will provoke reflex bradycardia in a patient already at 41 beats/min. Dobutamine (option B) is predominantly an inotrope with beta-2 mediated vasodilation and can lower mean arterial pressure further. Arginine vasopressin (option C) is used as an adjunctive agent in vasodilatory shock at a flat dose of 0.03 units/min, for central diabetes insipidus at 20 to 30 milliunits/hr, and in donor hormone replacement, but it has no chronotropic effect and is not a first-line single agent here. Midodrine (option E) is an oral noncatecholamine unsuited to acute resuscitation.

Key point:
When hypotension and bradycardia coexist, choose norepinephrine or dopamine, and avoid phenylephrine because of reflex bradycardia.

Source: Practice of Neurocritical Care, 2nd ed., Pharmacology in the Neurocritical Care Unit, pp. 526

Question 174

Pregnancy in Neurocritical Care · Eclampsia: seizure management and antihypertensive choice · moderate

A 28-year-old woman at 34 weeks of gestation is brought to the emergency department after a witnessed generalized tonic-clonic seizure at home. For the past two days she has had a persistent occipital headache and blurred vision. Blood pressure is 172/112 mm Hg, heart rate 96/min, and urine dipstick shows 3+ protein. She is postictal but arousable, with no focal deficits. Noncontrast head CT is unremarkable; MRI shows symmetric parieto-occipital FLAIR hyperintensity. She has no prior history of epilepsy. Which of the following is the most appropriate treatment to prevent recurrent seizures in this patient?

- **A.** Intravenous lorazepam followed by a levetiracetam maintenance infusion
- **B.** Intravenous fosphenytoin load followed by oral phenytoin
- **C.** Intravenous magnesium infusion
- **D.** Intravenous enalaprilat with an oral valproate load
- **E.** Intravenous diazepam infusion titrated to burst suppression

Answer: C. Intravenous magnesium infusion

This woman has new-onset hypertension plus proteinuria with a seizure, which defines eclampsia, and her MRI shows posterior reversible encephalopathy syndrome, one of the common imaging findings in this setting. Although antiepileptic medications are often given, magnesium infusion is the mainstay of treatment for seizures of preeclampsia and eclampsia, so choice C is correct. Choices A, B, and E use conventional antiseizure drugs, which are first-line only for noneclamptic status epilepticus; in eclampsia the underlying condition must be treated and magnesium is the specific therapy. Choice D is doubly wrong because angiotensin-converting enzyme inhibitors and angiotensin receptor blockers are teratogenic, causing skeletal abnormalities and pulmonary hypoplasia, and are not recommended in pregnancy; blood pressure control here is best obtained with labetalol, nifedipine, or hydralazine. Eclamptic seizures tend to be tonic-clonic and may occur before labor and up to 48 hours afterward, so magnesium is continued through the peripartum period. The only known cure of preeclampsia or eclampsia is delivery, and between 24 and 34 weeks the maternal and fetal risks of expectant management versus delivery must be weighed.

Key point:
magnesium infusion, not a standard antiseizure drug, is the mainstay for eclamptic seizures, and blood pressure is controlled with labetalol, nifedipine, or hydralazine rather than an ACE inhibitor or ARB.

Source: Practice of Neurocritical Care, 2nd ed., Pregnancy in Neurocritical Care, pp. 538

Question 175

Pregnancy in Neurocritical Care · Acute ischemic stroke in pregnancy: thrombolysis and thrombectomy · moderate

A 37-year-old woman at 22 weeks of gestation develops sudden right hemiplegia and global aphasia while at work. She arrives 70 minutes after symptom onset. Blood pressure is 148/86 mm Hg, glucose is 104 mg/dL, and NIH Stroke Scale score is 18. Noncontrast head CT shows no hemorrhage and an early hyperdense left middle cerebral artery sign; CT angiography confirms a left M1 occlusion. Fetal heart tones are normal. Which of the following is the most appropriate management?

- **A.** Administer intravenous tissue plasminogen activator and proceed to mechanical thrombectomy
- **B.** Administer intravenous tissue plasminogen activator alone, because thrombectomy is unsafe in pregnancy
- **C.** Proceed directly to mechanical thrombectomy alone, because tissue plasminogen activator is teratogenic
- **D.** Give oral aspirin only, because thrombolysis and thrombectomy both endanger the fetus
- **E.** Obtain MRI and magnetic resonance angiography before any reperfusion therapy is considered

Answer: A. Administer intravenous tissue plasminogen activator and proceed to mechanical thrombectomy

The overarching principle is that the best treatment for the mother is the best treatment for the fetus, and timely diagnosis and reperfusion should not be withheld because of pregnancy. Recombinant tissue plasminogen activator is too large to cross the placental barrier and therefore carries no teratogenic or hemorrhagic risk to the fetus, which makes choices C and D wrong. Mechanical thrombectomy has been performed safely in pregnant women with large vessel occlusion in case reports and case series with favorable outcomes, so withholding it as in choice B is not supported. Choice E delays reperfusion for imaging that adds nothing after CT and CT angiography have already excluded hemorrhage and demonstrated the occlusion. Pregnant patients were excluded from the randomized thrombolysis trials, so the evidence is observational, but successful use has been reported in several case reports and series. Postpartum hemorrhage is the situation in which thrombolysis must be reconsidered, and stroke risk in pregnancy is about 30 per 100,000 deliveries, roughly three times that of other young adults.

Key point:
treat tPA-eligible and thrombectomy-eligible pregnant stroke patients as you would nonpregnant patients, because tPA does not cross the placenta and thrombectomy has been performed safely.

Source: Practice of Neurocritical Care, 2nd ed., Pregnancy in Neurocritical Care, pp. 538

Question 176

Pregnancy in Neurocritical Care · Cerebral venous thrombosis in the puerperium · easy

A 31-year-old woman who delivered vaginally 10 days ago presents with a 4-day history of progressively worsening headache, nausea, and vomiting, followed this morning by a focal seizure with secondary generalization. Blood pressure is 126/78 mm Hg and she is afebrile. Noncontrast head CT shows a right parietal hemorrhagic lesion that does not conform to an arterial territory, and CT venography demonstrates filling defects in the superior sagittal and right transverse sinuses. Which of the following is the most appropriate treatment?

- **A.** Intravenous tissue plasminogen activator
- **B.** Oral warfarin titrated to an international normalized ratio of 2 to 3
- **C.** Oral apixaban
- **D.** Subcutaneous enoxaparin
- **E.** Oral aspirin plus clopidogrel

Answer: D. Subcutaneous enoxaparin

This is cerebral venous thrombosis, the most common neurovascular complication of pregnancy and the postpartum period, and most cases occur in the first 6 weeks after delivery. Thrombi most often involve the superior sagittal sinus followed by the transverse or sigmoid sinuses, and most cases involve more than one sinus, exactly as seen here; venous infarcts or hemorrhage are expected radiographic findings and do not contraindicate treatment. The standard of care is anticoagulation, and heparin and enoxaparin are the mainstays because they do not cross the placenta and do not increase the risk of fetal complications, making choice D correct. Warfarin is a teratogen and is contraindicated, which excludes choice B, and direct oral anticoagulants such as apixaban have not been studied for cerebral venous thrombosis, excluding choice C. Antiplatelet therapy and systemic thrombolysis are not the standard of care for this condition. Clinical presentation prognosticates long-term outcome: most patients who present with headache alone achieve a symptom-free recovery, whereas those presenting in coma have worse long-term neurological recovery.

Key point:
anticoagulate cerebral venous thrombosis in pregnancy and the puerperium with heparin or low molecular weight heparin, even when there is venous hemorrhage; warfarin is teratogenic and direct oral anticoagulants are unstudied.

Source: Practice of Neurocritical Care, 2nd ed., Pregnancy in Neurocritical Care, pp. 539

Question 177

Pregnancy in Neurocritical Care · Maternal cardiac arrest and perimortem cesarean section · moderate

A 30-year-old woman at 30 weeks of gestation collapses on the labor and delivery floor. She is pulseless and apneic, and the fundus is palpable well above the umbilicus. The team begins chest compressions and calls a maternal cardiac arrest code. Which of the following is the most appropriate approach to her resuscitation?

- **A.** Tilt the patient into a left lateral decubitus position and continue chest compressions in that position
- **B.** Manually displace the uterus upward and to the patient's left, and perform perimortem cesarean section within 4 minutes of starting compressions
- **C.** Establish femoral venous access, halve the epinephrine dose for fetal safety, and reassess at 10 minutes
- **D.** Continue standard supine compressions and defer perimortem cesarean section until 20 minutes of resuscitation have elapsed
- **E.** Perform perimortem cesarean section only after return of spontaneous circulation has been achieved

Answer: B. Manually displace the uterus upward and to the patient's left, and perform perimortem cesarean section within 4 minutes of starting compressions

The aorta and inferior vena cava are compressed by the gravid uterus in the supine position starting at 12 to 14 weeks of gestation, so aortocaval compression must be relieved. Left lateral decubitus positioning improves circulation and is recommended for hypotension, but effective chest compressions cannot be delivered in that position, which makes choice A wrong; manual displacement of the uterus upward and to the patient's left is the most effective maneuver during compressions. Perimortem cesarean section is recommended within 4 minutes of initiating chest compressions when the uterus is above the level of the umbilicus, because it relieves all aortocaval compression and may restore spontaneous circulation, so choices D and E are wrong; a British study showed perimortem cesarean section was performed sooner in mothers who survived, supporting a primarily maternal benefit. Choice C is wrong on both counts: intravenous access must be established above the diaphragm because venous return below the diaphragm is impaired by uterine compression, and advanced life support medications are given at the same doses as in nonpregnant patients. A difficult airway should be anticipated in every pregnant woman and intubation attempted with a smaller endotracheal tube, and hypoxemia develops faster because of reduced functional residual capacity and increased oxygen consumption. Oxytocin and prostaglandins should be available after arrest to reduce hemorrhage.

Key point:
during maternal cardiac arrest, manually displace the uterus up and to the left, use above-the-diaphragm access and standard drug doses, and perform perimortem cesarean section within 4 minutes when the uterus is above the umbilicus.

Source: Practice of Neurocritical Care, 2nd ed., Pregnancy in Neurocritical Care, pp. 541-542

Question 178

Pregnancy in Neurocritical Care · Spontaneous spinal epidural hematoma: gestational age threshold · hard

A 29-year-old woman at 35 weeks of gestation develops sudden severe interscapular pain followed within an hour by dense bilateral leg weakness, a T6 sensory level, and urinary retention. Blood pressure is 132/80 mm Hg and she is afebrile; she takes no anticoagulants. Emergent MRI of the spine shows a spontaneous dorsal epidural hematoma extending from T4 to T7 with severe cord compression. Fetal heart rate tracing is reassuring. Which of the following is the most appropriate next step?

- **A.** Immediate surgical decompression with continuous fetal heart rate monitoring, deferring delivery
- **B.** Administer high-dose intravenous methylprednisolone and repeat MRI in 24 hours
- **C.** Administer intravenous magnesium for fetal neuroprotection and observe for spontaneous resolution
- **D.** Induce vaginal delivery with epidural anesthesia, then decompress the spine after delivery
- **E.** Perform cesarean section with intraoperative neurological monitoring, then proceed to spinal decompression

Answer: E. Perform cesarean section with intraoperative neurological monitoring, then proceed to spinal decompression

For spontaneous spinal epidural hematoma in pregnancy, the gestational age determines the sequence of interventions. When the fetus is greater than 32 weeks of gestation, clinicians perform cesarean section with intraoperative neurological monitoring prior to spinal decompression, which makes choice E correct at 35 weeks. Choice A describes the opposite strategy, spinal decompression with fetal monitoring, which is preferred only when the fetus is less than 32 weeks. Choices B and C delay definitive decompression of an acutely compressed cord, and neither steroids nor magnesium treats the mechanical compression. Choice D is wrong because vaginal delivery and neuraxial instrumentation are inappropriate here, and delivery route in this scenario is cesarean section. Spinal cord injury during pregnancy is rare and reported in limited series; injuries sustained earlier in pregnancy are more likely to be complicated by abnormal fetal development, whereas later injuries tend to coincide with normal development and birth.

Key point:
in spontaneous spinal epidural hematoma, deliver by cesarean section with intraoperative neurological monitoring first if the fetus is greater than 32 weeks, and decompress the spine first with fetal monitoring if the fetus is less than 32 weeks.

Source: Practice of Neurocritical Care, 2nd ed., Pregnancy in Neurocritical Care, pp. 540

Prognostication in Neurocritical Care

Question 179

Prognostication in Neurocritical Care · Properties of prognostic models: discrimination versus calibration · easy

A newly published model predicts 6-month mortality after severe brain injury and reports an area under the receiver operating characteristic curve of 0.82. When the model is applied to a similar but separate cohort, it consistently predicts a 40% risk of death in patients whose observed mortality is only 15%, while performing accurately at the highest illness severities. Which property of this model is most clearly deficient?

- **A.** Discrimination
- **B.** Internal validity
- **C.** Calibration
- **D.** Sensitivity
- **E.** Interobserver reliability

Answer: C. Calibration

Calibration reflects how well a tool's predicted probability of an outcome agrees with the actual observed probability, and a well-calibrated mortality model remains accurate across a range of illness severities, which is exactly what fails here. Discrimination is the ability to separate patients who do and do not have the outcome, and it is reported as sensitivity, specificity, or the area under the curve, also called the c-statistic; an area under the curve of 0.82 shows this model discriminates adequately, so choices A and D are not the deficiency. Internal validity refers to whether a tool is methodologically sound and free of bias and confounding, and nothing in the scenario indicates a flawed derivation, so choice B is wrong. Calibration is assessed with a calibration curve of observed versus predicted probability, where perfect calibration is a diagonal line with slope equal to 1, or with the Hosmer-Lemeshow statistic, which has limited statistical power and gives no information about the degree of miscalibration. Despite its importance, calibration is reported far less often than discrimination, and miscalibration in critical illness can mean systematically overestimating or underestimating a patient's risk of death or disability.

Key point:
discrimination tells you whether a model separates outcome groups, calibration tells you whether its predicted probabilities are actually true, and calibration is the property most often missing from published neurocritical care models.

Source: Practice of Neurocritical Care, 2nd ed., Prognostication in Neurocritical Care, pp. 556-557

Question 180

Prognostication in Neurocritical Care · Decompressive craniectomy in large hemispheric stroke: DESTINY II · moderate

A 66-year-old woman with a right middle cerebral artery infarct is admitted to the neurointensive care unit. On hospital day 2 she becomes progressively less arousable, and repeat CT shows a space-occupying infarct with 8 mm of midline shift. The neurosurgical team offers decompressive craniectomy, and the family asks what the operation would realistically accomplish. Based on the trial data specific to patients in this age group, which of the following statements is most accurate?

- **A.** About half of patients who undergo craniectomy will walk unassisted at 12 months
- **B.** Craniectomy reduces 12-month mortality from 71% to 22% and increases functional independence from 2% to 14%
- **C.** Craniectomy does not reduce mortality in patients older than 60 years
- **D.** Craniectomy reduces 12-month mortality from 76% to 43%, but functional independence is not achieved in either group
- **E.** Craniectomy increases the likelihood of functional independence to about 14% without affecting mortality

Answer: D. Craniectomy reduces 12-month mortality from 76% to 43%, but functional independence is not achieved in either group

DESTINY II was designed specifically to study early decompressive craniectomy in patients older than 60 years, who had largely been excluded from the earlier trials, and it enrolled 112 patients aged 60 to 80 with space-occupying middle cerebral artery infarcts. With maximal medical management, 76% died within 12 months and no patient achieved functional independence; among those who underwent craniectomy, 43% died within 12 months, no patient achieved functional independence, and 6% were able to walk unassisted. Choices A and B describe the pooled analysis of the three earlier multicenter randomized trials, which enrolled 93 patients aged 18 to 60 years: medical management produced 71% mortality at 12 months with 2% functional independence, whereas craniectomy produced 22% mortality and 14% functional independence, with about half of surgical patients walking unassisted at 12 months. Those numbers do not apply to a 66-year-old, so choices A, B, and E are wrong. Choice C is directly contradicted by DESTINY II, which showed a clear survival benefit. Because providers differ substantially in what level of dependence they consider acceptable, and because dominant versus nondominant hemisphere involvement influences clinician impressions more than the outcome literature supports, shared decision-making anchored in the patient's own values is essential.

Key point:
in patients older than 60, decompressive craniectomy for malignant middle cerebral artery infarction improves survival, cutting 12-month mortality from 76% to 43%, but does not produce functional independence.

Source: Practice of Neurocritical Care, 2nd ed., Prognostication in Neurocritical Care, pp. 558

Question 181

Prognostication in Neurocritical Care · Intracerebral hemorrhage: self-fulfilling prophecy and guideline-based counseling · hard

A 51-year-old man with hypertension presents with confusion, left hemiplegia, and dysarthria. He opens his eyes to pain, does not follow commands, and mumbles inappropriate words. Noncontrast head CT shows a right basal ganglia intracerebral hemorrhage of 47 mL by the ABC divided by 2 method, with 4 mm of midline shift and trace intraventricular blood without hydrocephalus. Six hours after admission, the distressed family states that he would never want to live in a vegetative state and asks about his chance of recovery and about a do-not-resuscitate order. Which of the following is the most appropriate approach?

- **A.** Provide an individualized clinical estimate that explicitly conveys uncertainty and defer any new do-not-resuscitate order until after the second full day of hospitalization
- **B.** Quote the 30-day mortality associated with his Original intracerebral hemorrhage score as his personal probability of death
- **C.** Place a do-not-resuscitate order now and limit medical and surgical interventions accordingly
- **D.** Recalculate his prognosis with a newer multivariable scale, which head-to-head studies have shown to be superior to the Original intracerebral hemorrhage score
- **E.** Decline to discuss prognosis until his 90-day functional outcome can be observed

Answer: A. Provide an individualized clinical estimate that explicitly conveys uncertainty and defer any new do-not-resuscitate order until after the second full day of hospitalization

Intracerebral hemorrhage prognostication is uniquely vulnerable to the self-fulfilling prophecy, because withdrawal of life-sustaining therapy frequently occurs within the first 24 hours and the resulting deaths then bias the very scales used to predict death. As a safeguard, the 2015 American Heart Association and American Stroke Association guidelines explicitly discourage new do-not-resuscitate orders before the second full day of hospitalization unless the patient had a pre-existing order, and they warn against limiting medical and surgical interventions simply because a do-not-resuscitate order exists, which makes choice C wrong. Those same guidelines explicitly warn against using cohort outcome data to predict an individual patient's outcome, so choice B is inappropriate. Choice D is wrong because head-to-head studies have not shown predictive superiority of any single intracerebral hemorrhage instrument over another, adding covariates does not necessarily improve discrimination, and the guidelines still recommend the Original intracerebral hemorrhage score as the default severity grading scale because of its worldwide familiarity. Choice E abandons the clinician's duty to counsel, and early family meetings are associated with shorter intensive care stays and better perceived quality of discussion. Notably, subjective 90-day outcome prediction by expert clinicians outperforms the Original intracerebral hemorrhage score even after excluding patients for whom withdrawal was recommended, and when goals-of-care decisions were deferred for the first 5 days, observed mortality was markedly lower than the score predicted, with about 30% of patients reaching a 90-day modified Rankin Scale score of 0 to 3.

Key point:
counsel with individualized clinical judgment plus explicit uncertainty, avoid applying cohort statistics to the individual, and do not write a new do-not-resuscitate order before the second full hospital day.

Source: Practice of Neurocritical Care, 2nd ed., Prognostication in Neurocritical Care, pp. 559-562

Question 182

Prognostication in Neurocritical Care · Aneurysmal subarachnoid hemorrhage radiographic grading · moderate

A 54-year-old woman presents with a thunderclap headache and vomiting. She is drowsy and mildly confused but has no focal deficit, corresponding to Hunt and Hess grade 3. Noncontrast head CT shows thick subarachnoid blood measuring more than 1 mm in the basal cisterns and sylvian fissures, with intraventricular blood in both lateral ventricles. Catheter angiography confirms a ruptured anterior communicating artery aneurysm. Which of the following best characterizes her radiographic risk stratification for symptomatic vasospasm?

- **A.** Modified Fisher grade 2, with an odds ratio of 1.6 compared with grade 1
- **B.** Modified Fisher grade 3, with an odds ratio of 1.6 compared with grade 1
- **C.** Fisher grade 2, with an odds ratio of 1.3 compared with grade 1
- **D.** Fisher grade 4, with an odds ratio of 1.7 compared with grade 1
- **E.** Modified Fisher grade 4, with an odds ratio of 2.2 compared with grade 1

Answer: E. Modified Fisher grade 4, with an odds ratio of 2.2 compared with grade 1

The modified Fisher scale grades focal or diffuse subarachnoid blood by whether it is less than or at least 1 mm thick and by the presence of intraventricular hemorrhage; thick blood of at least 1 mm plus intraventricular hemorrhage is grade 4, which carries the highest symptomatic vasospasm odds ratio of 2.2 relative to grade 1. Grade 2 is thin blood with intraventricular hemorrhage and grade 3 is thick blood without it, both with an odds ratio of 1.6, so choices A and B understate her risk. Choice C describes original Fisher grade 2, a diffuse or thin layer of blood less than 1 mm, which does not match a scan showing thick clot. Choice D is wrong because original Fisher grade 4 denotes intracerebral or intraventricular clot with diffuse or absent blood in the basal cisterns and carries an odds ratio of only 1.7; the modified Fisher scale was derived from a much larger cohort and may predict vasospasm more accurately. The Claassen scale similarly combines clot thickness with bilateral intraventricular hemorrhage, and its grade 4, thick subarachnoid blood with intraventricular hemorrhage in both lateral ventricles, carries a 40% incidence of delayed cerebral ischemia. Her Hunt and Hess grade 3 reflects clinical severity and correlates with mortality, but it uses vague descriptive terms and has substantial interobserver variability, which the World Federation of Neurologic Surgeons scale addresses by grading on objective Glasgow Coma Scale and motor deficit criteria.

Key point:
thick subarachnoid blood of at least 1 mm plus intraventricular hemorrhage is modified Fisher grade 4, the highest vasospasm risk stratum with an odds ratio of 2.2.

Source: Practice of Neurocritical Care, 2nd ed., Prognostication in Neurocritical Care, pp. 563-564

Question 183

Prognostication in Neurocritical Care · Severe traumatic brain injury: withdrawal of life-sustaining therapy and detection of consciousness · moderate

A 24-year-old man is admitted after a motorcycle crash with a Glasgow Coma Scale score of 5 and diffuse traumatic axonal injury with traumatic subarachnoid hemorrhage on CT. On hospital day 2 he remains comatose on fentanyl and propofol infusions, and the family asks whether continued treatment is futile. Which of the following statements about prognosis in severe traumatic brain injury is most accurate?

- **A.** Coma on admission after severe traumatic brain injury invariably portends a poor functional outcome
- **B.** At least 30% of patients hospitalized with severe traumatic brain injury die in hospital, roughly 70% of those deaths are associated with withdrawal of life-sustaining therapy, and most withdrawal occurs within the first 3 days
- **C.** Most withdrawal of life-sustaining therapy occurs after the first week, once prognostic uncertainty has largely resolved
- **D.** Sedation, pain, and medical illness rarely confound bedside assessment of consciousness in this setting
- **E.** The IMPACT prognostic calculator discriminates 6-month outcome with an area under the curve greater than 0.90

Answer: B. At least 30% of patients hospitalized with severe traumatic brain injury die in hospital, roughly 70% of those deaths are associated with withdrawal of life-sustaining therapy, and most withdrawal occurs within the first 3 days

Severe traumatic brain injury is strongly subject to the self-fulfilling prophecy: at least 30% of patients admitted for severe traumatic brain injury die in the hospital, about 70% of those deaths are associated with withdrawal of life-sustaining therapy for apparent poor prognosis, and the majority of withdrawals occur in the first 3 days, precisely when prognosis is most uncertain, which makes choice B correct and choice C wrong. Choice A is wrong because outcomes after severe traumatic brain injury range from death and permanent vegetative state to dramatic return to functional independence, and reemergence of consciousness during hospitalization predicts better long-term functional outcome. Choice D is wrong because accurate detection of consciousness is confounded by sedation, pain, delirium, and medical illness, so that many conscious patients are misdiagnosed as unconscious; standardized tools such as the Coma Recovery Scale Revised improve diagnostic and prognostic accuracy, and a significant portion of patients judged unconscious by that gold standard show volitional responses on advanced MRI and EEG and are termed covertly conscious. Choice E overstates the data, because the IMPACT calculator, which predicts 6-month mortality or unfavorable outcome in adults with an initial Glasgow Coma Scale of 12 or less, has an area under the curve of only 0.66 to 0.80, indicating fair to good discrimination. This patient's ongoing sedation makes any bedside prognostic assessment on day 2 especially unreliable.

Key point:
most in-hospital deaths after severe traumatic brain injury follow early withdrawal of life-sustaining therapy in the first 3 days, so early prognostic pessimism risks becoming self-fulfilling.

Source: Practice of Neurocritical Care, 2nd ed., Prognostication in Neurocritical Care, pp. 565-567

Question 184

Prognostication in Neurocritical Care · Cognitive biases in surrogate decision-making · easy

During a family meeting for a 72-year-old man with a severe dominant hemisphere stroke, his daughter states that if he survives dependent on a wheelchair and unable to speak fluently, he would consider that no life at all. You are aware that patients living with comparable disability consistently rate their own quality of life higher than nondisabled surrogates predict they would. Which cognitive bias does this discrepancy specifically describe?

- **A.** Recall bias
- **B.** Focusing illusion
- **C.** Response shift
- **D.** Disability paradox
- **E.** Optimistic misinterpretation of quantitative prognosis

Answer: D. Disability paradox

The disability paradox is the observation that people with disabilities report greater quality of life than surrogates without disabilities imagine they would report, which is exactly the discrepancy described, so choice D is correct. Recall bias is the tendency of surrogates to remember patients as healthier and more independent than they actually were before hospitalization, so choice A does not fit. The focusing illusion describes surrogates failing to imagine a patient's adaptability because they focus on the aspects of life that will change rather than the aspects that will remain the same, and choice B is therefore a related but distinct bias. Response shift, also called scale recalibration or emotional adaptation, refers to surrogates being unaware that patients adapt to disability over time by adjusting their standards, values, and perception of quality of life, so choice C is also distinct. Choice E describes a separate pitfall in which surrogates interpret even definitive numeric statements optimistically, illustrated by the general intensive care unit study in which surrogates told of a 5% chance of survival reported their own perception of that chance as high as 40%. Awareness of these affective forecasting biases keeps them from exerting disproportionate influence, and clinicians should discuss worst-case scenarios honestly while not underestimating a patient's future ability to adapt.

Key point:
the disability paradox is the specific bias that people living with disability rate their quality of life higher than nondisabled surrogates predict.

Source: Practice of Neurocritical Care, 2nd ed., Prognostication in Neurocritical Care, pp. 568-569

Question 185

Sedation and Analgesia · Sedation depth targets before neuromuscular blockade · easy

A 53-year-old woman is in the neurocritical care unit on day 7 after a motorcycle collision with a traumatic subarachnoid hemorrhage. She is receiving fentanyl 50 mcg/hr and propofol 20 mcg/kg per minute and has been maintained at a Richmond Agitation-Sedation Scale (RASS) score of -1. She now develops bilateral infiltrates and severe hypoxemia with a ratio of arterial oxygen partial pressure to fraction of inspired oxygen of 130. The team plans to start a cisatracurium infusion for 48 hours as part of early management of acute respiratory distress syndrome. Which of the following is the most appropriate sedation target before the neuromuscular blocker is started?

- **A.** RASS 0 to -1
- **B.** RASS 0 to -2
- **C.** RASS -4 to -5
- **D.** RASS -2 to -3
- **E.** RASS -1 to -3

Answer: C. RASS -4 to -5

Neuromuscular blockers provide no sedation or analgesia, so patients must be deeply sedated before paralysis; the chapter states that the one notable exception to targeting light sedation is when neuromuscular blockers are used, and that patients should be sedated to a RASS target of -4 to -5 before these agents are initiated. Her acute respiratory distress syndrome regimen matches the studies cited in the chapter, which used a ratio of arterial oxygen partial pressure to fraction of inspired oxygen below 150 with deep sedation at RASS -4 to -5 and high-dose cisatracurium for 48 hours in the early phase. A RASS of 0 to -2 (option B) is the reasonable light sedation target for the general population, and RASS 0 to -1 (option A) or -2 to -3 (option D and option E) all leave the patient potentially aware while paralyzed. Deep sedation carries real costs, including prolonged neurocritical care unit stay, more delirium, more infection, and delayed awakening, but those are accepted here for the duration of blockade.

Key point:
Light sedation is RASS 0 to -2, but any patient receiving a neuromuscular blocker must first be at RASS -4 to -5.

Source: Practice of Neurocritical Care, 2nd ed., Sedation and Analgesia, pp. 583, 585

Question 186

Sedation and Analgesia · Propofol-related infusion syndrome · hard

A 17-year-old boy is in the intensive care unit after a motor vehicle collision with a severe traumatic brain injury. He has been receiving propofol 90 mcg/kg per minute and midazolam 5 mg/hr for elevated intracranial pressure. On day 4 his blood pressure falls to 68/40 mm Hg despite fluids. Laboratory studies show sodium 149 mmol/L, potassium 5.2 mmol/L, creatinine 3.0 mg/dL, creatine kinase 26,750 units/L, and triglycerides 550 mg/dL. The nurse reports that his urine has turned green. Which of the following is the most likely cause of this presentation?

- **A.** Accumulation of the active metabolite alphahydroxymidazolam
- **B.** Propofol-related infusion syndrome
- **C.** Propylene glycol toxicity from the midazolam infusion
- **D.** Contrast-induced acute tubular necrosis
- **E.** Nonconvulsive status epilepticus with autonomic instability

Answer: B. Propofol-related infusion syndrome

Propofol-related infusion syndrome should be suspected in a patient with risk factors who develops persistent metabolic acidosis, hyperlactatemia, hyperkalemia, refractory hypotension, rhabdomyolysis, or ventricular arrhythmias, and acidosis with rhabdomyolysis on propofol is considered a highly indicative warning sign. This patient carries the classic risk factors of young age and a dose above 80 mcg/kg per minute continued beyond 48 hours, and he shows refractory hypotension, hyperkalemia, renal failure, rhabdomyolysis with a creatine kinase of 26,750 units/L, and hypertriglyceridemia. Green urine is a common red herring: it reflects a phenolic metabolite, is benign and self-limited, and is not a sign of toxicity. Alphahydroxymidazolam accumulation (option A) prolongs arousal in renal dysfunction but does not cause rhabdomyolysis or cardiovascular collapse, and midazolam contains no propylene glycol, so option C is wrong; the propylene glycol-containing parenteral agents are lorazepam, diazepam, and phenobarbital. Management is immediate propofol withdrawal plus cardiac pacing, vasopressor and inotropic support, hemodialysis, and extracorporeal membrane oxygenation if needed.

Key point:
Rhabdomyolysis with acidosis, hyperkalemia, and shock on propofol above 80 mcg/kg per minute for more than 48 hours is propofol-related infusion syndrome, and green urine is not a sign of it.

Source: Practice of Neurocritical Care, 2nd ed., Sedation and Analgesia, pp. 587-588

Question 187

Sedation and Analgesia · Propylene glycol toxicity from benzodiazepine infusions · moderate

A 60-year-old woman is intubated in the intensive care unit for severe alcohol withdrawal and has required high doses of intravenous lorazepam by continuous infusion for 4 days, along with as-needed acetaminophen and oxycodone. After several days of improvement she acutely becomes hypotensive and bradycardic. Laboratory studies show a lactate of 8.5 mmol/L, arterial pH 7.28, creatinine 2.3 mg/dL (baseline 0.8 mg/dL), and an elevated serum osmolar gap. Blood cultures are negative and there is no new imaging finding. Which of the following best explains her deterioration?

- **A.** Propylene glycol toxicity
- **B.** Opioid withdrawal from intermittent oxycodone
- **C.** Acetaminophen-induced hepatotoxicity
- **D.** Relative adrenal insufficiency
- **E.** Recurrent alcohol withdrawal with delirium tremens

Answer: A. Propylene glycol toxicity

Propylene glycol is the diluent in intravenous lorazepam, diazepam, and phenobarbital, and high doses given for prolonged periods, especially as continuous infusions, produce toxicity manifesting as hypotension, bradycardia, lactic acidosis, hyperosmolality, seizures, renal failure, and a sepsis-like syndrome; treatment is hemodialysis to remove the excess propylene glycol. This patient's lactic acidosis, acute kidney injury, elevated osmolar gap, hypotension, and bradycardia after 4 days of high-dose lorazepam infusion fit precisely. Opioid withdrawal (option B) produces hypertension, tachycardia, mydriasis, diarrhea, and diaphoresis rather than bradycardia and lactic acidosis. Acetaminophen hepatotoxicity (option C) requires doses above 4 grams per day and presents with transaminitis rather than an osmolar gap. Adrenal insufficiency (option D) and recurrent withdrawal (option E) do not explain the elevated osmolar gap, which is the discriminating finding here.

Key point:
Lactic acidosis, renal failure, bradycardia, and an elevated osmolar gap during a high-dose lorazepam infusion is propylene glycol toxicity, treated by stopping the drug and considering hemodialysis.

Source: Practice of Neurocritical Care, 2nd ed., Sedation and Analgesia, pp. 587, 590

Question 188

Sedation and Analgesia · Shivering control during targeted temperature management · moderate

A 58-year-old man is comatose after return of spontaneous circulation from an out-of-hospital cardiac arrest and is undergoing targeted temperature management. He is already receiving scheduled acetaminophen, buspirone, and skin counterwarming. The nurse palpates shivering localized to the neck and thorax only, giving a Bedside Shivering Assessment Scale score of 1, and the surface device is struggling to hold the target temperature. Which of the following is the most appropriate next intervention?

- **A.** Meperidine 100 mg intravenously every 6 hours
- **B.** Rocuronium 1 mg/kg intravenous bolus
- **C.** Propofol 50 mcg/kg per minute infusion
- **D.** Meperidine 12.5 mg intravenously every 6 hours
- **E.** Midazolam 5 mg intravenous bolus every 6 hours

Answer: D. Meperidine 12.5 mg intravenously every 6 hours

Shivering combats cooling and increases systemic and cerebral metabolic demand, so it must be controlled, and the Bedside Shivering Assessment Scale is the validated bedside tool used to titrate therapy; a score of 1 is mild shivering confined to the neck and thorax and warrants mild sedation or analgesia rather than escalation to paralysis. Meperidine is the most effective opioid for this purpose, lowering the shiver threshold by 1.2 to 2.2 degrees Celsius, and a dose of 12.5 mg has been shown to be as effective as 100 mg with fewer side effects such as dizziness, somnolence, seizures, and hypotension, which is why option D beats option A. Propofol (option C) is an option if shivering persists and deeper sedation becomes necessary, but it is not the first step for mild shivering. Neuromuscular blockade (option B) is a last-line option to be used with extreme caution, and train-of-four monitoring is unreliable during hypothermia. Midazolam (option E) assists ventilator synchrony but does not meaningfully lower the shiver threshold, unlike dexmedetomidine, which reduces it by about 0.7 to 2 degrees Celsius.

Key point:
For shivering during targeted temperature management, meperidine 12.5 mg is as effective as 100 mg with fewer adverse effects, and paralysis is last line.

Source: Practice of Neurocritical Care, 2nd ed., Sedation and Analgesia, pp. 584, 597, 601

Question 189

Sedation and Analgesia · Sedative choice in the patient approaching extubation · easy

A 45-year-old man was admitted to the intensive care unit 2 days ago for suspected seizure activity, which has since been ruled out on continuous electroencephalography. He is now meeting weaning criteria and the team plans extubation later today, but he has become increasingly agitated and is pulling at his lines, with a Richmond Agitation-Sedation Scale score of +2. His blood pressure is 138/78 mm Hg and heart rate is 88 beats/min. Which of the following is the most appropriate agent?

- **A.** Propofol infusion at 35 mcg/kg per minute
- **B.** Fentanyl infusion at 25 mcg/hr
- **C.** Lorazepam 2 mg intravenously every 4 hours
- **D.** Dexmedetomidine infusion at 1.5 mcg/kg per hour
- **E.** Dexmedetomidine infusion at 0.2 mcg/kg per hour

Answer: E. Dexmedetomidine infusion at 0.2 mcg/kg per hour

Dexmedetomidine is unique among these agents in providing sedation without diminishing central respiratory drive, which makes it well suited to nonintubated patients and to those being weaned toward extubation, and patients remain arousable to stimulation and return to rest when stimulation stops, preserving serial neurologic examinations. The recommended starting dose is 0.2 mcg/kg per hour with upward titration, so option E is correct and option D is wrong because starting at 1.5 mcg/kg per hour increases the risk of hypotension and bradycardia; boluses should likewise be avoided for the same reason. Propofol (option A) causes respiratory depression and requires a controlled airway, and fentanyl (option B) also impairs respiratory drive, so neither is appropriate immediately before extubation. Lorazepam (option C) causes respiratory depression, and benzodiazepines are associated with a longer length of stay, longer mechanical ventilation, and an increased risk of delirium compared with nonbenzodiazepine regimens.

Key point:
Dexmedetomidine started at 0.2 mcg/kg per hour and titrated up is the sedative of choice around extubation because it does not depress respiratory drive.

Source: Practice of Neurocritical Care, 2nd ed., Sedation and Analgesia, pp. 588, 607

Question 190

Sedation and Analgesia · Neuroleptic selection and QT prolongation · moderate

A 71-year-old woman in the neurocritical care unit after evacuation of a subdural hematoma develops hyperactive delirium with visual hallucinations and is repeatedly climbing out of bed. Nonpharmacologic measures have failed and pain, hypoxemia, and hypoglycemia have been excluded. She has no history of Parkinson disease or other movement disorder. Her baseline electrocardiogram shows a QT interval corrected for heart rate of 446 msec, and she is receiving no other QT-prolonging medications. Which of the following is the most appropriate agent?

- **A.** Haloperidol 5 mg intravenously
- **B.** Olanzapine 5 mg intramuscularly
- **C.** Quetiapine 25 mg per tube twice daily
- **D.** Risperidone 1 mg per tube twice daily
- **E.** Midazolam 2 mg intravenously every hour as needed

Answer: B. Olanzapine 5 mg intramuscularly

Neuroleptics are commonly used for hyperactive delirium in the neurocritical care unit and produce sedation with minimal respiratory depression, but QT prolongation is a class effect that limits their use, so a baseline electrocardiogram is recommended. Olanzapine has a lower incidence of QT prolongation than the other neuroleptics and is therefore the best choice in a patient with a mildly prolonged corrected QT interval; the intramuscular route is preferred in most clinical situations because of its rapid onset of 15 to 45 minutes, and intravenous use is off-label and requires close monitoring for respiratory depression. Haloperidol (option A) causes the most QT prolongation along with extrapyramidal symptoms. Quetiapine and risperidone (options C and D) must be given enterally, have a slower onset of 1 to 1.5 hours, and are also associated with QT prolongation and extrapyramidal symptoms; quetiapine at the lowest tolerated dose is specifically the agent reserved for patients with Parkinson disease, which this patient does not have. A benzodiazepine (option E) is the wrong direction, since benzodiazepines are associated with an increased risk of delirium.

Key point:
When a neuroleptic is needed and the corrected QT interval is borderline, olanzapine carries the lowest risk of further QT prolongation.

Source: Practice of Neurocritical Care, 2nd ed., Sedation and Analgesia, pp. 593

Question 191

Sedation and Analgesia · Opioid selection in renal dysfunction · moderate

A 62-year-old man is intubated in the intensive care unit after extensive spinal surgery. His history includes diabetes, hypertension, stage 3 chronic kidney disease with a baseline creatinine of 2.3 mg/dL, and depression. His mean arterial pressure is 90 mm Hg and his heart rate is 120 beats/min, and the Critical Care Pain Observation Tool score is 6. The team believes pain is driving his vital sign changes. Which of the following is the most appropriate analgesic to initiate?

- **A.** Morphine 4 mg intravenously every 2 hours as needed
- **B.** Acetaminophen 1,000 mg by mouth every 4 hours as needed
- **C.** Hydromorphone 0.5 mg intravenously every 2 hours as needed
- **D.** Remifentanil 0.5 mcg/kg intravenously every 4 hours as needed
- **E.** Meperidine 25 mg intravenously every 4 hours as needed

Answer: C. Hydromorphone 0.5 mg intravenously every 2 hours as needed

Morphine is a poor choice in renal dysfunction because its active metabolite morphine-6-glucuronide is at least as potent as the parent drug and accumulates in renal insufficiency, prolonging sedation, while morphine-3-glucuronide is neurotoxic and can cause seizures; risk of accumulation is highest in elderly, obese, and renally impaired patients. Hydromorphone is metabolized by the liver to inactive metabolites, has a quick intravenous onset of 5 to 10 minutes, and is commonly used for intermittent analgesia in the neurocritical care unit, making option C correct. Acetaminophen 1,000 mg every 4 hours (option B) delivers 6,000 mg per day and exceeds the toxic threshold of 4 grams daily, above which N-acetyl-p-benzoquinone imine accumulates and causes hepatic injury. Remifentanil (option D) is rapidly hydrolyzed by nonspecific plasma esterases with a half-life of 3 to 10 minutes, so it must be given as an infusion rather than as an as-needed bolus, although that same property makes it attractive in renal and hepatic dysfunction. Meperidine (option E) is not recommended for routine analgesia and is reserved for shivering.

Key point:
Avoid morphine when the kidneys are impaired because morphine-6-glucuronide accumulates; hydromorphone has only inactive metabolites.

Source: Practice of Neurocritical Care, 2nd ed., Sedation and Analgesia, pp. 597-598

Question 192

Sedation and Analgesia · Sedation interruption in intracranial hypertension · hard

A 34-year-old man with a severe traumatic brain injury has an intracranial pressure monitor and a brain tissue oxygen probe in place. He is receiving propofol and fentanyl infusions, with intracranial pressures running 16 to 19 mm Hg and a brain tissue oxygen tension of 24 mm Hg. The covering team requests that a daily sedation interruption protocol be applied to him as it is for the medical intensive care unit patients. Which of the following is the most appropriate approach?

- **A.** Abruptly interrupt both sedation and analgesia daily until he is awake or appears uncomfortable
- **B.** Interrupt sedation only after starting a neuromuscular blocking agent to prevent agitation
- **C.** Substitute bispectral index monitoring for the clinical examination and continue uninterrupted sedation
- **D.** Wean sedation in a judicious, stepwise fashion with continuous intracranial pressure and brain tissue oxygen monitoring
- **E.** Abandon neurological examination during sedation and rely on serial computed tomography instead

Answer: D. Wean sedation in a judicious, stepwise fashion with continuous intracranial pressure and brain tissue oxygen monitoring

Daily sedation interruption reduces ventilator days and intensive care unit length of stay in general medical patients, but neurologically injured patients become disinhibited when sedation is lifted and are at risk for self-extubation, deleterious changes in intracranial pressure and cerebral perfusion pressure, and cerebral hypoxia and ischemia. In a prospective study of wake-up tests in severe acute neurologic injury, only 2 percent of tests detected a new neurologic deficit while 30 percent had to be aborted for uncontrolled intracranial hypertension above 20 mm Hg, agitation, or systemic or cerebral desaturation, and cerebral hypoxia with a brain tissue oxygen tension below 20 mm Hg occurred in 70 percent of those aborted trials; another study showed significant rises in intracranial pressure and cortisol during liberation trials. The chapter therefore concludes that a more judicious and methodical approach to sedation weaning is appropriate in patients with underlying intracranial hypertension, which is option D. Abrupt daily interruption (option A) is the general intensive care unit protocol that these data argue against here, and interruption should not be performed at all in patients paralyzed or sedated to treat a pathological phenomenon, which also makes option B incoherent. Bispectral index (option C) is unreliable in hypothermia, shock, and shivering and is subject to movement artifact, so its use is generally discouraged in this population.

Key point:
In patients with intracranial hypertension, wean sedation methodically with intracranial pressure and brain tissue oxygen monitoring rather than applying an abrupt daily interruption protocol.

Source: Practice of Neurocritical Care, 2nd ed., Sedation and Analgesia, pp. 583, 594

Seizures and Status Epilepticus

Question 193

Seizures and Status Epilepticus · Definition of status epilepticus · easy

The International League Against Epilepsy defines status epilepticus along two time dimensions: t1, the time beyond which a seizure is unlikely to stop spontaneously and treatment should begin, and t2, the time beyond which long-term consequences are expected. For generalized convulsive status epilepticus, which of the following correctly pairs t1 with t2?

- **A.** t1 of 5 minutes and t2 of 30 minutes
- **B.** t1 of 5 minutes and t2 of 60 minutes
- **C.** t1 of 10 minutes and t2 of 30 minutes
- **D.** t1 of 30 minutes and t2 of 60 minutes
- **E.** t1 of 1 minute and t2 of 10 minutes

Answer: A. t1 of 5 minutes and t2 of 30 minutes

For convulsive status epilepticus the ILAE operational definition sets t1 at 5 minutes and t2 at 30 minutes; for nonconvulsive status epilepticus both t1 and t2 remain undefined. The 5-minute threshold is grounded in the observation that most seizures terminate spontaneously within a few minutes, so a seizure persisting beyond 5 minutes should almost always be treated. Options B and D reflect the older 30-minute and 60-minute cutoffs that were historically used as research definitions but are no longer the operational treatment threshold, and option C misstates t1 for convulsive status epilepticus. Option E is not a recognized pairing. The commonly used clinical corollary is intermittent or continuous seizure activity lasting at least 5 minutes without return to pre-seizure baseline.

Key point:
For convulsive status epilepticus, treat at 5 minutes (t1) and expect long-term consequences beyond 30 minutes (t2).

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 620

Question 194

Seizures and Status Epilepticus · First-line therapy for status epilepticus · easy

A 27-year-old woman with a history of epilepsy is brought to the emergency department after a witnessed generalized tonic-clonic seizure at home. In the department she is observed to have two additional generalized tonic-clonic seizures, each lasting about 2 minutes, without returning to her baseline mental status in between. Her temperature is 37.4 C, heart rate 112 beats per minute, blood pressure 148/86 mmHg, and oxygen saturation 94 percent on 4 liters by nasal cannula. Point-of-care glucose is 106 mg/dL and she has a functioning peripheral intravenous line. Which of the following is the most appropriate initial medication?

- **A.** Levetiracetam 60 mg/kg IV
- **B.** Lorazepam 4 mg IV
- **C.** Fosphenytoin 20 mg PE/kg IV
- **D.** Propofol infusion after endotracheal intubation
- **E.** Observation without medication, since each individual seizure is self-limited

Answer: B. Lorazepam 4 mg IV

This patient meets the definition of status epilepticus because she has had recurrent seizures without return to baseline in between, so she requires immediate treatment and observation is inappropriate. Guidelines recommend a benzodiazepine as emergent initial therapy, and lorazepam is the preferred agent at 0.1 mg/kg IV up to a maximum of 4 mg per dose, given by slow push at 2 mg/min and repeatable in 5 to 10 minutes. Lorazepam had the highest rate of seizure control (65 percent) in the VA cooperative study and has a duration of action against status epilepticus of 4 to 14 hours versus roughly 20 minutes for diazepam. Levetiracetam 60 mg/kg and fosphenytoin 20 mg PE/kg are correct second-line (urgent control) doses but should follow, not replace, the benzodiazepine. Anesthetic infusion with propofol is reserved for refractory status epilepticus after first- and second-line agents have failed.

Key point:
Give a weight-based benzodiazepine first in status epilepticus, and give it as fast as possible.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 624

Question 195

Seizures and Status Epilepticus · Prehospital management and the RAMPART trial · moderate

Paramedics are called to a 55-year-old man who has been having continuous generalized tonic-clonic activity for approximately 8 minutes. He is cyanotic with clenched jaw and copious oral secretions. Two attempts at peripheral intravenous access have failed because of ongoing convulsive movements. His fingerstick glucose is 128 mg/dL. Which of the following is the most appropriate next step?

- **A.** Withhold antiseizure medication until intravenous access is obtained in the emergency department
- **B.** Place an intraosseous line and give fosphenytoin 20 mg PE/kg
- **C.** Administer midazolam 10 mg intramuscularly
- **D.** Administer levetiracetam 60 mg/kg intramuscularly
- **E.** Administer phenobarbital 20 mg/kg intramuscularly

Answer: C. Administer midazolam 10 mg intramuscularly

Guidelines list midazolam 0.2 mg/kg intramuscularly, up to a maximum of 10 mg, as the recommended benzodiazepine for patients without intravenous access, and its intramuscular onset of action is roughly 15 minutes. The RAMPART study found a potential benefit for a 10 mg intramuscular midazolam injection compared with intravenous lorazepam for prehospital status epilepticus, because avoiding the need for intravenous access allowed more rapid initiation of treatment and therefore better outcomes. Withholding treatment (option A) is wrong: in the out-of-hospital benzodiazepine trial, 59 percent of lorazepam and 43 percent of diazepam patients were seizure free on arrival versus 21 percent of placebo patients, and respiratory complications were actually less frequent with benzodiazepines (10 to 11 percent) than with placebo (23 percent). Fosphenytoin and phenobarbital are second-line agents and should not precede a benzodiazepine, and levetiracetam is not given intramuscularly as a loading strategy in this setting.

Key point:
No intravenous access in status epilepticus means intramuscular midazolam 0.2 mg/kg up to 10 mg, not a delay in treatment.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 624

Question 196

Seizures and Status Epilepticus · Second-line therapy and the ESETT trial · moderate

A 44-year-old man with generalized convulsive status epilepticus has received two doses of lorazepam totaling 8 mg IV and continues to convulse. The team is choosing an urgent control (second-line) agent. Which of the following statements about the Established Status Epilepticus Treatment Trial (ESETT) best informs this decision?

- **A.** Levetiracetam was clearly superior, terminating seizures in about 70 percent of patients
- **B.** Fosphenytoin was clearly superior, with significantly less hypotension than the comparators
- **C.** Valproic acid was clearly superior and is now the guideline-preferred second-line agent
- **D.** Valproic acid, fosphenytoin, and levetiracetam were similarly effective, achieving the primary outcome in 46, 45, and 47 percent of patients respectively
- **E.** All three agents failed in more than 90 percent of patients and the trial was stopped for harm

Answer: D. Valproic acid, fosphenytoin, and levetiracetam were similarly effective, achieving the primary outcome in 46, 45, and 47 percent of patients respectively

ESETT was a randomized, blinded comparative-effectiveness trial of valproic acid 40 mg/kg (max 3000 mg), fosphenytoin 20 mg PE/kg (max 1500 mg PE), and levetiracetam 60 mg/kg (max 4500 mg) for benzodiazepine-refractory convulsive status epilepticus, with a primary outcome of termination of clinically apparent seizures plus improved responsiveness at 60 minutes without additional antiseizure medication. All three agents were effective in approximately half of patients: valproic acid 46 percent, fosphenytoin 45 percent, and levetiracetam 47 percent, so options A, B, and C are wrong. The trial enrolled 384 patients for 400 enrollments and was stopped early for futility, not for harm, which excludes option E. Side effects including hypotension, arrhythmia, and intubation were comparable, with fosphenytoin showing a nonsignificantly higher rate of hypotension and intubation. Notably, most ESETT patients received benzodiazepine doses lower than current guidelines recommend, so adequate benzodiazepine dosing remains essential.

Key point:
Valproate, fosphenytoin, and levetiracetam are roughly equivalent second-line agents, each working in about half of benzodiazepine-refractory patients.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 624

Question 197

Seizures and Status Epilepticus · Benzodiazepine tachyphylaxis · moderate

A 62-year-old woman with herpes simplex encephalitis is on continuous EEG for altered mental status, which shows frequent nonconvulsive seizures. The first dose of lorazepam 2 mg IV produces transient electrographic seizure cessation and brief clinical improvement, but a second identical dose given 20 minutes later through the same visibly patent and well-flushed intravenous catheter produces no clinical or electrographic effect. Which of the following best explains the loss of efficacy?

- **A.** Reduced blood-brain barrier permeability to lorazepam
- **B.** Downregulation of N-methyl-D-aspartate receptors on postsynaptic neurons
- **C.** Increased surface expression of gamma-aminobutyric acid type A receptors
- **D.** Induction of hepatic cytochrome P450 enzymes accelerating lorazepam clearance
- **E.** Internalization of benzodiazepine-sensitive gamma-aminobutyric acid type A receptors with upregulation of N-methyl-D-aspartate receptors

Answer: E. Internalization of benzodiazepine-sensitive gamma-aminobutyric acid type A receptors with upregulation of N-methyl-D-aspartate receptors

Tachyphylaxis to benzodiazepines during ongoing status epilepticus is largely mediated by internalization of benzodiazepine-sensitive GABA-A receptors, together with reduced sensitivity of GABA-A receptors in hippocampal dentate granule cells, reduced surface expression from altered receptor trafficking, upregulation of NMDA receptors, and overexpression of drug efflux transporters. This is a time-dependent process, which is why longer status epilepticus durations increase the likelihood of refractoriness and why treatment must be given rapidly. Option C is the opposite of what occurs, and option B is likewise backwards since NMDA receptors are upregulated rather than downregulated. Option A is not a described mechanism of benzodiazepine failure in status epilepticus, and option D would not develop over 20 minutes. Intravenous infiltration can also cause apparent loss of effect, but it is excluded here by inspection of a patent, well-flushed line.

Key point:
Prolonged status epilepticus makes GABA-A receptors disappear from the synapse and NMDA receptors proliferate, which is why benzodiazepines fail with time and why NMDA-directed agents such as ketamine are useful later.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 622

Question 198

Seizures and Status Epilepticus · Nonconvulsive status epilepticus after treated convulsive status · moderate

A 27-year-old woman with no significant medical history presents after right-sided focal motor seizure activity that generalized. Clinical convulsions stop after a total of 6 mg of lorazepam IV, about 18 minutes after onset. Non-contrast head CT, complete blood count, basic metabolic panel, and hepatic function panel are unremarkable. Thirty-five minutes after the convulsions stopped, she remains unresponsive with a Glasgow Coma Scale of 7 (E1, V2, M4). Temperature is 37.6 C, pulse 95 beats per minute, blood pressure 138/57 mmHg, and oxygen saturation 99 percent. Which of the following is the most appropriate next diagnostic step?

- **A.** Continuous EEG monitoring
- **B.** Urine toxicology screen
- **C.** MRI of the brain
- **D.** CT venography
- **E.** Serum paraneoplastic antibody panel

Answer: A. Continuous EEG monitoring

Patients who are adequately treated for status epilepticus should begin to awaken within 15 to 20 minutes of successful seizure termination, and those who do not should be assumed to have entered nonconvulsive status epilepticus until an EEG proves otherwise. Among patients with convulsive status epilepticus whose convulsions stop after benzodiazepines, 48 percent have ongoing nonconvulsive seizures and 14 percent are in nonconvulsive status epilepticus; in the VA cooperative study, 20 percent of patients whose visible convulsions were stopped remained in electrographic status epilepticus. Because nonconvulsive status epilepticus continues to damage the brain without motor activity and independently affects morbidity and mortality, continuous EEG is the most pressing test. Toxicology screening, MRI, CT venography, and a paraneoplastic panel are all reasonable parts of an etiologic workup for new-onset status epilepticus, but none of them will detect ongoing seizures, and delaying EEG risks additional injury.

Key point:
Failure to begin awakening within 15 to 20 minutes after convulsions stop equals nonconvulsive status epilepticus until continuous EEG proves otherwise.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 623

Question 199

Seizures and Status Epilepticus · 2HELPS2B seizure risk score · hard

The 2HELPS2B score developed by Struck and colleagues stratifies the risk of seizures in hospitalized patients undergoing continuous EEG. Which of the following EEG findings is weighted at 2 points rather than 1 point in this score?

- **A.** Sporadic epileptiform discharges
- **B.** Brief ictal rhythmic discharges
- **C.** Lateralized rhythmic delta activity
- **D.** A periodic or rhythmic pattern with a frequency greater than 2 Hz
- **E.** Presence of plus features on a periodic or rhythmic pattern

Answer: B. Brief ictal rhythmic discharges

The 2HELPS2B score assigns 1 point each for a periodic or rhythmic pattern at a frequency greater than 2 Hz, sporadic epileptiform discharges, lateralized rhythmic delta activity or bilateral independent periodic discharges, plus features, and a prior clinical seizure, while brief ictal rhythmic discharges are weighted at 2 points. The resulting risk gradient runs from about 5 percent at the lowest score up to greater than 95 percent at a score of 6 or more, with intermediate values of roughly 12, 27, 50, 73, and 88 percent. Options A, C, D, and E are all genuine components of the score but each contributes only a single point. The score exists because the majority of seizures in critically ill patients are nonconvulsive and therefore require EEG for diagnosis, while monitoring every ICU patient is impractical and older clinical selection algorithms lacked adequate sensitivity and specificity.

Key point:
In 2HELPS2B, brief ictal rhythmic discharges carry double weight, and a maximal score predicts a greater than 95 percent risk of seizures.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 621

Question 200

Seizures and Status Epilepticus · Drug-induced seizures in the ICU · easy

A 72-year-old man with chronic kidney disease is being treated in the medical ICU for hospital-acquired pneumonia with cefepime. On hospital day 3 he becomes progressively encephalopathic with intermittent eyelid twitching. Head CT is unremarkable, electrolytes and glucose are normal, and continuous EEG demonstrates nonconvulsive status epilepticus. Which of the following best explains the mechanism by which his antibiotic precipitated seizures?

- **A.** Direct agonism at N-methyl-D-aspartate receptors
- **B.** Blockade of voltage-gated sodium channels in cortical neurons
- **C.** Antagonism of gamma-aminobutyric acid receptors after crossing the blood-brain barrier
- **D.** Inhibition of astrocytic glutamate reuptake transporters
- **E.** Enhanced dopaminergic transmission in the mesolimbic system

Answer: C. Antagonism of gamma-aminobutyric acid receptors after crossing the blood-brain barrier

Cefepime is commonly used in the ICU and crosses the blood-brain barrier, where it produces seizures by antagonism of gamma-aminobutyric acid receptors, a direct neurotoxic effect rather than an idiosyncratic reaction. This is distinct from the broader group of medications that merely lower the seizure threshold, which includes various antimicrobials, analgesics, antihistamines, antineoplastics, anesthetics, antipsychotics, antidepressants, antiarrhythmics, immunosuppressants, and CNS stimulants, and which typically cause seizures during rapid administration or at high concentrations. Options A, D, and E are not the described mechanism for cefepime. Option B describes the mechanism of several antiseizure medications such as phenytoin and lacosamide and would be expected to suppress rather than provoke seizures. Note also that the patient has a nonconvulsive presentation, which is the rule rather than the exception in the ICU.

Key point:
Cefepime causes seizures by crossing into the CNS and antagonizing GABA receptors, and should be suspected in any encephalopathic ICU patient receiving it.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 620

Question 201

Seizures and Status Epilepticus · Second-line drug selection and drug interactions · hard

An 80-kg 40-year-old man is in the neuro ICU with an intracranial abscess being treated empirically with intravenous meropenem and vancomycin. While awaiting culture results he becomes difficult to arouse. Repeat head CT shows expected postoperative changes only. Continuous EEG shows evolving 3 Hz sharp wave discharges consistent with nonconvulsive status epilepticus. He receives a cumulative 8 mg of lorazepam IV with no change on EEG. His blood pressure is 132/74 mmHg and the QTc and PR intervals are normal. Which of the following is the most appropriate medication to give next?

- **A.** Valproic acid 3000 mg IV
- **B.** Levetiracetam 1000 mg IV
- **C.** Phenobarbital 500 mg IV
- **D.** Fosphenytoin 1600 mg PE IV
- **E.** Lacosamide 50 mg IV

Answer: D. Fosphenytoin 1600 mg PE IV

All of the listed drug classes are acceptable second-line options for benzodiazepine-refractory status epilepticus, so this question turns on the correct dose and on a drug interaction. Fosphenytoin is loaded at 20 mg PE/kg, which for an 80-kg patient is 1600 mg PE, making option D both correctly dosed and safe here. Valproic acid is dosed appropriately within the recommended 20 to 40 mg/kg range (maximum 3000 mg), but it interacts with carbapenems such as meropenem, causing precipitous drops in serum valproate levels, so it should be avoided in this patient. Levetiracetam is loaded at 60 mg/kg (up to 4500 mg), which would be 4800 mg capped at 4500 mg, so 1000 mg is far too low, and phenobarbital is loaded at 20 mg/kg, which would be 1600 mg rather than 500 mg. Lacosamide loading doses are 200 to 600 mg, so 50 mg is a maintenance-level dose that will not control status epilepticus.

Key point:
In benzodiazepine-refractory status epilepticus, give a full weight-based load, and avoid valproate in patients on carbapenems.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 624-625, 639

Question 202

Seizures and Status Epilepticus · Refractory status epilepticus: choice of anesthetic · easy

A 34-year-old intubated woman with cryptogenic encephalitis remains in electrographic status epilepticus despite lorazepam 0.1 mg/kg IV and a full levetiracetam load of 60 mg/kg IV followed by maintenance dosing. Her blood pressure is 118/70 mmHg on no vasopressors. Which of the following is the most appropriate next step?

- **A.** Begin a pentobarbital infusion loaded at 5 mg/kg IV
- **B.** Initiate a ketogenic diet
- **C.** Give perampanel 12 mg by nasogastric tube
- **D.** Repeat lorazepam 0.1 mg/kg IV every 10 minutes until seizures stop
- **E.** Begin a continuous midazolam infusion with a 0.2 mg/kg bolus followed by 0.05 to 2 mg/kg/hr

Answer: E. Begin a continuous midazolam infusion with a 0.2 mg/kg bolus followed by 0.05 to 2 mg/kg/hr

This patient has refractory status epilepticus, defined as status that fails to respond to first- and second-line antiseizure medications, and guidelines recommend treatment with continuous infusions of midazolam or propofol; she is intubated, which is a prerequisite for intravenous anesthetics. Pentobarbital is typically reserved for super-refractory status epilepticus, after non-barbiturate anesthetic infusions such as midazolam, propofol, or ketamine have failed, so option A escalates too far too fast. The ketogenic diet may be useful but is logistically challenging and is not a first move in acute refractory status. Perampanel has been reported to abort seizures in only 17 to 36 percent of critically ill patients, is available only orally, and enteral absorption is unreliable in critical illness because of slow gastric emptying and reduced peristalsis. Repeated benzodiazepine dosing is unlikely to succeed given benzodiazepine tachyphylaxis in prolonged status.

Key point:
Refractory status epilepticus means a continuous midazolam or propofol infusion in an intubated patient, with pentobarbital held in reserve for super-refractory disease.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 625

Question 203

Seizures and Status Epilepticus · Titration and duration of anesthetic infusions · moderate

A patient with refractory status epilepticus has been started on a continuous anesthetic infusion. Which of the following best reflects current guideline recommendations for the endpoint and duration of that infusion?

- **A.** Titrate to cessation of electrographic seizures or burst suppression, and maintain that endpoint for 24 to 48 hours before weaning
- **B.** Titrate to cessation of electrographic seizures only, and wean after 6 to 12 hours
- **C.** Titrate to a completely isoelectric EEG and maintain it for 72 hours
- **D.** Titrate to burst suppression and maintain it for a minimum of 7 days
- **E.** Titrate to resolution of clinical convulsions and wean immediately thereafter

Answer: A. Titrate to cessation of electrographic seizures or burst suppression, and maintain that endpoint for 24 to 48 hours before weaning

Continuous anesthetic infusions in refractory status epilepticus should be titrated to an EEG endpoint, and guidelines accept either cessation of electrographic seizures or burst suppression as that endpoint; supporting data for one target over the other are poor. Guidelines currently recommend maintaining the achieved endpoint for 24 to 48 hours before attempting to wean, which excludes the 6 to 12 hour (option B) and 7 day (option D) durations. A completely isoelectric EEG is not the recommended target, so option C is wrong. Option E is wrong because clinical convulsions frequently cease while electrographic status continues, which is precisely why EEG rather than motor activity guides therapy. The speed of weaning is driven by drug pharmacokinetics: propofol, ketamine, and pentobarbital are typically weaned by 20 percent every 3 hours while midazolam is decreased by 50 percent every 3 hours.

Key point:
Titrate anesthetics to electrographic seizure cessation or burst suppression, hold that endpoint 24 to 48 hours, then wean by pharmacokinetics.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 628-629

Question 204

Seizures and Status Epilepticus · Breakthrough seizures during anesthetic weaning · hard

A 58-year-old man with refractory status epilepticus has been seizure free on a midazolam infusion for 36 hours and is on maintenance levetiracetam. During the wean, continuous EEG shows recurrent electrographic seizures. His serum levetiracetam level is at the low end of the therapeutic range and his blood pressure is stable. Which of the following is the most appropriate management?

- **A.** Continue the wean at the current rate and reassess in 12 hours
- **B.** Bolus midazolam and continue the infusion at a higher maintenance rate, and optimize the levetiracetam dose or add a non-anesthetic antiseizure medication before re-attempting the wean
- **C.** Stop midazolam and transition directly to a pentobarbital infusion
- **D.** Change the midazolam wean rate to 20 percent every 3 hours and make no other changes
- **E.** Add therapeutic hypothermia to a target of 32 to 34 C for 24 hours

Answer: B. Bolus midazolam and continue the infusion at a higher maintenance rate, and optimize the levetiracetam dose or add a non-anesthetic antiseizure medication before re-attempting the wean

Breakthrough seizures during anesthetic therapy are typically managed with boluses of the anesthetic together with either continuation at a higher maintenance dose or the addition of a non-anesthetic antiseizure medication such as lacosamide. Guidelines further specify that before weaning is re-attempted, at least one non-anesthetic antiseizure medication the patient is already receiving should be optimized to therapeutic levels or increased, or a new non-anesthetic agent started, which makes option B the complete answer. Continuing an unchanged wean through active electrographic seizures (option A) allows ongoing seizure-related brain injury. Pentobarbital is reserved for super-refractory status after non-barbiturate infusions fail, and this patient has not yet had his background regimen optimized, so option C is premature. Option D addresses only wean kinetics and, incidentally, misstates the recommended midazolam wean, which is 50 percent every 3 hours; the 20 percent every 3 hours rate applies to propofol, ketamine, and pentobarbital. Hypothermia (option E) improved electrographic seizure control in the HIBERNATUS trial but did not improve functional outcome and is not standard rescue therapy.

Key point:
Breakthrough seizures mean re-bolus and re-escalate the anesthetic, and optimize a non-anesthetic drug before trying to wean again.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 629

Question 205

Seizures and Status Epilepticus · Seizure prophylaxis in the critically ill · moderate

A 58-year-old man is admitted after a motor vehicle collision with a severe traumatic brain injury, a Glasgow Coma Scale of 6, and bifrontal contusions on CT. He has had no clinical or electrographic seizures. Which of the following best reflects guideline recommendations regarding antiseizure medication in this patient?

- **A.** Levetiracetam is recommended over phenytoin on the basis of level 1 evidence in severe traumatic brain injury
- **B.** Prophylaxis should be withheld, since guidelines recommend against antiseizure medication prophylaxis after all forms of acute brain injury
- **C.** Phenytoin for 7 days is recommended to reduce the incidence of early post-traumatic seizures, although it does not change the long-term risk of epilepsy
- **D.** Phenytoin should be continued for 6 months to prevent the development of post-traumatic epilepsy
- **E.** Valproic acid should be started and continued until hospital discharge

Answer: C. Phenytoin for 7 days is recommended to reduce the incidence of early post-traumatic seizures, although it does not change the long-term risk of epilepsy

The Brain Trauma Foundation guideline for severe traumatic brain injury recommends phenytoin to decrease the early incidence of post-traumatic seizures within 7 days when the benefit is judged to outweigh the risk, and the American Academy of Neurology recommends 7 days of phenytoin prophylaxis for moderate to severe closed head injury. This builds on a randomized double-blind trial showing that phenytoin reduced seizures during the first week after severe head injury while leaving the long-term risk of epilepsy unchanged, which makes option D incorrect. Option A is wrong because the guideline explicitly states there is insufficient evidence to recommend levetiracetam over phenytoin. Option B overstates the picture: guidelines recommend against prophylaxis in intracerebral hemorrhage and acute ischemic stroke, but the American Stroke Association states prophylactic anticonvulsants may be considered in the immediate post-hemorrhagic period after aneurysmal subarachnoid hemorrhage, and traumatic brain injury has its own recommendation. Valproic acid is not the guideline-recommended prophylactic agent in this setting.

Key point:
Seven days of phenytoin for severe traumatic brain injury reduces early seizures only, while prophylaxis is recommended against in intracerebral hemorrhage and ischemic stroke.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 631

Question 206

Seizures and Status Epilepticus · Hypothermia for status epilepticus (HIBERNATUS) · moderate

A 29-year-old woman is intubated and mechanically ventilated for convulsive status epilepticus. Her family read about cooling as a treatment for seizures and asks whether it should be added to her regimen. Which of the following best summarizes the findings of the HIBERNATUS multicenter randomized controlled trial of hypothermia as an add-on therapy in this population?

- **A.** Hypothermia improved 3-month functional outcome and is now recommended as adjunctive therapy
- **B.** Hypothermia significantly increased mortality and the trial was stopped early for harm
- **C.** Hypothermia significantly reduced progression to super-refractory status epilepticus
- **D.** Hypothermia improved electrographic seizure control, with nonconvulsive status epilepticus in 11 percent versus 22 percent, but did not improve functional outcome at 3 months
- **E.** Hypothermia worsened electrographic seizure control while improving functional outcome

Answer: D. Hypothermia improved electrographic seizure control, with nonconvulsive status epilepticus in 11 percent versus 22 percent, but did not improve functional outcome at 3 months

HIBERNATUS randomized 270 critically ill, ventilated patients with convulsive status epilepticus to hypothermia at a target temperature of 32 to 34 C for 24 hours as an add-on therapy. Electrographic seizure control improved, with nonconvulsive status epilepticus occurring in 11 percent of cooled patients versus 22 percent of controls (odds ratio 0.40, 95 percent CI 0.20 to 0.79), but the predetermined primary outcome of improved functional outcome at 3 months was not met, which makes option D correct and options A and E wrong. There was no difference in rates of super-refractory status epilepticus or other secondary outcomes, excluding option C. The trial was not stopped for harm and did not show an increase in mortality, excluding option B. The authors did not account for differences in underlying etiology, time of status onset, or the number and choice of antiseizure medications, which limits interpretation.

Key point:
HIBERNATUS showed hypothermia improves electrographic seizure control in convulsive status epilepticus but not 3-month functional outcome.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 629

Question 207

Seizures and Status Epilepticus · Treatment of nonconvulsive seizures (TRENdS) · moderate

A 70-year-old woman is on continuous EEG after cardiac surgery for persistent encephalopathy and is found to have frequent nonconvulsive seizures. She has a first-degree atrioventricular block on telemetry but is hemodynamically stable. The team discusses whether to load lacosamide or fosphenytoin. Which of the following statements is best supported by the only randomized, controlled, prospective trial to date that specifically evaluated treatment of nonconvulsive seizures?

- **A.** Fosphenytoin 20 mg PE/kg was significantly superior to lacosamide for terminating nonconvulsive seizures
- **B.** Lacosamide 400 mg IV was significantly superior to fosphenytoin, establishing it as the standard of care
- **C.** Both agents terminated nonconvulsive seizures in fewer than 20 percent of patients, so anesthetic infusion should be used first
- **D.** The trial was stopped early because of excess cardiac events in the lacosamide arm
- **E.** Lacosamide 400 mg IV terminated nonconvulsive seizures in 63.3 percent of patients versus 50 percent with fosphenytoin, a difference that did not reach statistical significance

Answer: E. Lacosamide 400 mg IV terminated nonconvulsive seizures in 63.3 percent of patients versus 50 percent with fosphenytoin, a difference that did not reach statistical significance

The Treatment of Recurrent Electrographic Nonconvulsive Seizures (TRENdS) trial compared lacosamide 400 mg IV load with fosphenytoin 20 mg PE/kg IV load in patients with frequent nonconvulsive seizures and found no significant difference in electrographic seizure termination or lack of recurrence over 24 hours on continuous EEG. Lacosamide terminated nonconvulsive seizures in 63.3 percent versus 50 percent for fosphenytoin, and seizure burden fell by 98 percent versus 76 percent, but statistical superiority was not demonstrated, which makes options A and B wrong. Response rates well above 20 percent exclude option C, and the trial was not terminated for cardiac harm, excluding option D. TRENdS was the first randomized trial to use electrographic outcomes as the primary endpoint. Relevant to this patient, lacosamide can prolong the PR interval and cause second-degree or complete atrioventricular block, so cardiac monitoring is recommended with higher doses and in patients with cardiac disease.

Key point:
TRENdS showed lacosamide 400 mg IV and fosphenytoin 20 mg PE/kg are statistically comparable for nonconvulsive seizures, with a numerical edge favoring lacosamide that did not reach statistical significance.

Source: Practice of Neurocritical Care, 2nd ed., Seizures and Status Epilepticus, pp. 630

Subarachnoid Hemorrhage

Question 208

Subarachnoid Hemorrhage · Diagnosis: negative CT and lumbar puncture · easy

A 45-year-old woman with a long history of migraine presents 14 hours after the abrupt onset of what she calls the most severe headache of her life, which she says is clearly different from her usual migraine attacks. She has an extensive smoking history and denies recreational drug use. Temperature 36.9 C, blood pressure 148/86 mm Hg, heart rate 88/min. Her neurologic examination, including cranial nerves and funduscopy, is normal. A noncontrast head CT is interpreted as showing no intracranial abnormality. Which of the following is the most appropriate next step?

- **A.** Administer subcutaneous sumatriptan and a migraine cocktail and discharge her from the emergency department
- **B.** Obtain brain MRI with gradient echo sequences
- **C.** Perform a lumbar puncture
- **D.** Perform catheter digital subtraction angiography
- **E.** Repeat the noncontrast head CT in 6 hours

Answer: C. Perform a lumbar puncture

The sensitivity of noncontrast head CT for SAH is 98%-100% within 12 hours of onset, but falls to 90%-95% at 24 hours, 80% at 3 days, and 50% at 7 days, so a negative CT never entirely rules out SAH and lumbar puncture must be performed when clinical suspicion is strong. The classic CSF findings are an elevated opening pressure and protein, red blood cell counts that do not diminish from tube one to tube four, and xanthochromia of the centrifuged supernatant. Xanthochromia may take up to 12 hours to appear because the in vivo enzymatic conversion of heme to biliverdin and then to bilirubin takes 6-12 hours, so at 14 hours from onset the tap is now informative; xanthochromia then persists for about 2 weeks. Treating this as migraine and discharging her risks a missed aneurysm, which is the classic misdiagnosis pattern described in the chapter. MRI with hemosiderin-sensitive sequences is most useful for subacute presentations more than 4 days out, not for excluding acute SAH at 14 hours. DSA is a vascular study performed after SAH is established, and simply repeating the CT does not increase sensitivity.

Key point:
a negative head CT does not exclude SAH, and lumbar puncture is mandatory when suspicion remains.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 652-653

Question 209

Subarachnoid Hemorrhage · Blood pressure management before aneurysm securing · moderate

A 58-year-old man with untreated hypertension presents with sudden severe headache and vomiting. He is awake with a stiff neck and no focal deficit, corresponding to World Federation of Neurological Surgeons grade II. Noncontrast CT shows diffuse blood in the basal cisterns, and CT angiography demonstrates a 7 mm anterior communicating artery aneurysm. Coiling is scheduled for approximately 10 hours from now. His blood pressure is 198/104 mm Hg with a mean arterial pressure of 135 mm Hg; his usual outpatient systolic pressures are in the 150s. Which of the following is the most appropriate blood pressure management?

- **A.** Start a nicardipine infusion at 5 mg/hr and titrate to a systolic blood pressure below 160 mm Hg
- **B.** Give intermittent intravenous hydralazine 10 mg boluses to a systolic blood pressure below 140 mm Hg
- **C.** Start a labetalol infusion titrated to a mean arterial pressure below 70 mm Hg
- **D.** Withhold antihypertensive therapy unless the systolic blood pressure exceeds 220 mm Hg
- **E.** Start a norepinephrine infusion targeting a mean arterial pressure 20 mm Hg above his baseline

Answer: A. Start a nicardipine infusion at 5 mg/hr and titrate to a systolic blood pressure below 160 mm Hg

Between symptom onset and aneurysm obliteration, blood pressure should be controlled with a titratable agent, and a decrease in systolic blood pressure to below 160 mm Hg is reasonable; a mean arterial pressure below 110 mm Hg is also recommended, because systolic pressures above 160 mm Hg are a recognized risk factor for rebleeding. Continuous infusions such as nicardipine 5-15 mg/hr, clevidipine 1-32 mg/hr, or labetalol 0.5-2 mg/min up to a maximum of 10 mg/min are preferred over bolus dosing precisely to avoid wide swings in pressure, which makes intermittent hydralazine boluses a poor choice; hydralazine is also typically avoided because of rebound hypertension. Driving the mean arterial pressure below 70 mm Hg risks cerebral ischemia, and hypotension should be avoided; premorbid baseline pressures should be used to refine the target. Leaving a pressure of 198/104 mm Hg untreated in an unsecured aneurysm ignores the strongest modifiable rebleeding risk factor. Induced hypertension with norepinephrine is the treatment for delayed cerebral ischemia after the aneurysm is secured, not before.

Key point:
before the aneurysm is secured, use a titratable infusion to bring systolic pressure below 160 mm Hg and mean arterial pressure below 110 mm Hg while avoiding hypotension.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 658, 661-662

Question 210

Subarachnoid Hemorrhage · Transcranial Doppler and the Lindegaard ratio · hard

A 47-year-old woman is on postbleed day 6 after coiling of a ruptured basilar apex aneurysm. She develops new right face, arm, and leg weakness with mild word-finding difficulty. Transcranial Doppler shows a left middle cerebral artery mean flow velocity of 201 cm/sec at a depth of 55 mm with a left extracranial internal carotid artery mean velocity of 40 cm/sec, and a right middle cerebral artery mean flow velocity of 122 cm/sec with a right extracranial internal carotid artery mean velocity of 52 cm/sec. Which of the following best characterizes these findings?

- **A.** Vasospasm of both middle cerebral arteries
- **B.** Vasospasm of the right middle cerebral artery only
- **C.** Hyperemia of both middle cerebral arteries
- **D.** A normal study, since velocities of this magnitude are expected on postbleed day 6
- **E.** Vasospasm of the left middle cerebral artery with hyperemia of the right middle cerebral artery

Answer: E. Vasospasm of the left middle cerebral artery with hyperemia of the right middle cerebral artery

Mean cerebral blood flow velocities above 120 cm/sec with a Lindegaard ratio above 3 are worrisome for vasospasm, and velocities above 200 cm/sec with a ratio above 6 suggest severe vasospasm. The Lindegaard ratio is the middle cerebral artery mean velocity divided by the ipsilateral extracranial internal carotid artery mean velocity, and it is what distinguishes true vasospasm from global hyperemia, which also raises absolute velocities. On the left, 201 divided by 40 gives a ratio of about 5, which together with a velocity of 201 cm/sec indicates vasospasm and correlates with the patient's new right-sided deficit. On the right, the velocity of 122 cm/sec is mildly elevated but the ratio is only about 2.3, which is below 3 and therefore indicates hyperemia rather than spasm, so options A and B are wrong. Calling the study normal ignores that a left middle cerebral velocity above 200 cm/sec has a high positive predictive value for angiographic spasm, whereas velocities below 120 cm/sec have a high negative predictive value.

Key point:
always normalize an elevated middle cerebral artery velocity by the ipsilateral extracranial internal carotid artery velocity, since a Lindegaard ratio above 3 means vasospasm and below 3 means hyperemia.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 665, 673

Question 211

Subarachnoid Hemorrhage · Delayed cerebral ischemia refractory to induced hypertension · moderate

The same 47-year-old woman with a left middle cerebral artery mean velocity of 201 cm/sec and a Lindegaard ratio of 5 is treated for delayed cerebral ischemia. She is maintained euvolemic on isotonic crystalloid and a norepinephrine infusion is titrated upward until her mean arterial pressure is 40 mm Hg above baseline, with a systolic pressure of 200 mm Hg. Her aphasia and right hemiparesis fail to improve after this trial. Which of the following is the most appropriate next step?

- **A.** Institute prophylactic hypervolemia and hemodilution to complete triple-H therapy
- **B.** Proceed to cerebral angioplasty and/or selective intra-arterial vasodilator therapy
- **C.** Perform prophylactic balloon angioplasty of the right middle cerebral artery because its velocity is also elevated
- **D.** Start an intravenous magnesium sulfate infusion targeting a supranormal serum magnesium level
- **E.** Add simvastatin 80 mg daily for the remainder of the vasospasm window

Answer: B. Proceed to cerebral angioplasty and/or selective intra-arterial vasodilator therapy

Cerebral angioplasty and selective intra-arterial vasodilator therapy with agents such as nicardipine, milrinone, or verapamil is reasonable in patients with symptomatic vasospasm, particularly those who are not responding to hypertensive therapy, and is the guideline-supported next step once an adequate trial of induced hypertension has failed. Triple-H therapy, meaning hypertension with hypervolemia and hemodilution, is associated with adverse outcomes and is no longer recommended; euvolemia rather than hypervolemia is the target, and there is evidence of harm from aggressive fluid administration aimed at hypervolemia. Prophylactic balloon angioplasty before the development of angiographic spasm is explicitly not recommended, a Class III Level B recommendation, and the right middle cerebral artery findings represent hyperemia rather than spasm. Intravenous magnesium was tested in two phase 3 trials, IMASH and MASH-2, and showed no difference in clinical outcomes or vasospasm incidence. Simvastatin was tested in the phase 3 STASH trial with no short-term or long-term benefit, and meta-analyses show no improvement in delayed cerebral ischemia, mortality, or favorable outcome.

Key point:
when symptomatic vasospasm does not respond to induced hypertension, escalate to endovascular angioplasty or intra-arterial vasodilators, not to hypervolemia, magnesium, or statins.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 660, 664-666, 673

Question 212

Subarachnoid Hemorrhage · Prophylaxis against delayed cerebral ischemia · easy

A 52-year-old woman is admitted after aneurysmal subarachnoid hemorrhage and undergoes successful surgical clipping on hospital day 1. She is neurologically intact and hemodynamically stable. Which of the following interventions has the strongest evidence for reducing delayed cerebral ischemia and should be started as soon as possible?

- **A.** Intravenous magnesium sulfate infusion for 14 days
- **B.** Simvastatin 40 mg daily for 21 days
- **C.** Prophylactic hypervolemia with albumin and crystalloid
- **D.** Enteral nimodipine 60 mg every 4 hours for 21 days
- **E.** Prophylactic balloon angioplasty of the proximal intracranial vessels on day 3

Answer: D. Enteral nimodipine 60 mg every 4 hours for 21 days

Calcium channel blockade with nimodipine and maintenance of euvolemia have the strongest evidence as prophylactic interventions for delayed cerebral ischemia, and oral or enteral nimodipine 60 mg every 4 hours should be given to all patients with subarachnoid hemorrhage for 21 days, a Class I Level A and high quality strong recommendation. Nimodipine is thought to work through neuroprotection by limiting calcium entry into ischemic neurons, reducing inflammation, and improving microcirculatory flow, and it reduces infarction and improves outcome even though it does not reverse angiographic vasospasm. A common side effect is hypotension, which can be managed by reducing the dose and increasing the frequency to 30 mg every 2 hours rather than stopping the drug. Magnesium was negative in the IMASH and MASH-2 phase 3 trials and simvastatin was negative in the STASH trial, so neither is recommended for prophylaxis or therapy. Prophylactic hypervolemia is not recommended and can cause cardiopulmonary complications, and prophylactic balloon angioplasty before angiographic spasm develops is a Class III Level B recommendation against.

Key point:
nimodipine 60 mg every 4 hours for 21 days plus euvolemia is the evidence-based prophylaxis for delayed cerebral ischemia.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 660, 664-665, 667

Question 213

Subarachnoid Hemorrhage · Hyponatremia and cerebral salt wasting · moderate

A 60-year-old man is on postbleed day 6 after coiling of a ruptured anterior communicating artery aneurysm. He has become mildly lethargic. Serum sodium is 128 mEq/L, serum osmolality 265 mOsm/kg, urine sodium 60 mmol/L, and urine osmolality 450 mOsm/kg. His net fluid balance is negative 2.5 L over 48 hours, his weight is down 3 kg, and he is tachycardic with flat neck veins. Which of the following is the most appropriate management?

- **A.** Restrict total oral and intravenous fluid intake to 1 L per day
- **B.** Infuse 5% dextrose in water at 100 mL/hr and recheck sodium in 6 hours
- **C.** Infuse 0.9% sodium chloride to restore intravascular volume and add fludrocortisone 0.2 to 0.4 mg per day
- **D.** Start a 3% sodium chloride infusion together with free water restriction
- **E.** Discontinue nimodipine and observe

Answer: C. Infuse 0.9% sodium chloride to restore intravascular volume and add fludrocortisone 0.2 to 0.4 mg per day

About 30% of patients with subarachnoid hemorrhage develop hyponatremia from either the syndrome of inappropriate antidiuretic hormone secretion or cerebral salt wasting, and both share the same laboratory profile of serum sodium below 134 mEq/L, serum osmolality below 274 mOsm/kg, urine sodium above 20 mmol/L, and urine osmolality above 100 mOsm/kg, so volume status is what separates them. This patient is volume depleted with a markedly negative fluid balance, weight loss, and tachycardia, which identifies cerebral salt wasting; the treatment of choice is infusion of 0.9% sodium chloride to restore intravascular volume, with fludrocortisone 0.2 to 0.4 mg per day added when active diuresis prevents maintenance of adequate fluid balance. Fluid restriction, whether alone or combined with hypertonic saline, is unacceptable in subarachnoid hemorrhage because intravascular volume contraction increases the risk of delayed cerebral ischemia. Giving 5% dextrose in water provides free water and would worsen the hyponatremia. Nimodipine should be continued for the full 21 days; when it causes hypotension the dose is split to 30 mg every 2 hours rather than stopped. Patients on fludrocortisone can develop hypokalemia, so potassium must be followed closely.

Key point:
never fluid restrict for hyponatremia after subarachnoid hemorrhage; treat cerebral salt wasting with isotonic saline and add fludrocortisone 0.2 to 0.4 mg per day if diuresis continues.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 667-668

Question 214

Subarachnoid Hemorrhage · Antifibrinolytic therapy for rebleeding prevention · moderate

A 63-year-old woman presents to a rural emergency department 2 hours after sudden severe headache and is found to have Hunt and Hess grade 3 subarachnoid hemorrhage on noncontrast CT. Her systolic blood pressure has been brought below 160 mm Hg with a nicardipine infusion. Helicopter transfer to a comprehensive stroke center is arranged, and the earliest that her aneurysm can be secured is approximately 12 hours from now. She has no thrombotic history and no contraindication to antifibrinolytic therapy. Which of the following regimens is most appropriate?

- **A.** Tranexamic acid 1 g now, then 1 g every 6 hours until the aneurysm is secured, for a maximum of 72 hours
- **B.** Aminocaproic acid 4 to 10 g load then 1 to 2 g/hr, continued for 14 days after coiling
- **C.** Withhold antifibrinolytic therapy and begin it only if clinical rebleeding occurs
- **D.** Withhold antifibrinolytic therapy, since nimodipine 60 mg every 4 hours prevents rebleeding
- **E.** Aminocaproic acid begun 72 hours after ictus and continued until hospital discharge

Answer: A. Tranexamic acid 1 g now, then 1 g every 6 hours until the aneurysm is secured, for a maximum of 72 hours

For patients with an unavoidable delay in aneurysm obliteration, a significant risk of rebleeding, and no compelling contraindication, short-term therapy of less than 72 hours with tranexamic acid or epsilon-aminocaproic acid is reasonable, a Class IIa Level B recommendation; tranexamic acid is given as 1 g at diagnosis followed by 1 g every 6 hours until the aneurysm is secured, to a maximum of 72 hours. The rationale is that rebleeding risk is highest within the first 24 hours, at 4% to 15%, and particularly within the first 6 hours, and rebleeding carries a mortality reported as high as 70%. Aminocaproic acid is dosed as a 4 to 10 g intravenous load followed by 1 to 2 g/hr until the time of aneurysm treatment, so option B has the right drug but the wrong duration: delayed therapy started more than 48 hours after ictus or prolonged therapy beyond 3 days exposes patients to side effects when rebleeding risk has already fallen sharply and should be avoided, which also excludes option E. Waiting for clinical rebleeding defeats the purpose of prophylaxis. Nimodipine prevents delayed cerebral ischemia, not rebleeding; the definitive prevention is early securing of the aneurysm plus blood pressure control. Patients on these agents should be monitored for deep venous thrombosis, cardiac ischemia, and ischemic stroke.

Key point:

antifibrinolytics for subarachnoid hemorrhage are a short bridge, started at diagnosis and stopped when the aneurysm is secured or at 72 hours, whichever comes first.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 658, 661-662

Question 215

Subarachnoid Hemorrhage · Acute hydrocephalus after subarachnoid hemorrhage · moderate

A 47-year-old woman with Fisher grade IV subarachnoid hemorrhage underwent coil embolization of a basilar apex aneurysm on admission day 1. Two days after admission she becomes confused with a declining level of consciousness, impaired upward gaze, sluggish pupillary reactions, and a rising mean arterial pressure. Which of the following diagnostic tests is most appropriate to identify the likely cause of these changes?

- **A.** Continuous electroencephalography
- **B.** Serum sodium and osmolality
- **C.** Catheter digital subtraction angiography
- **D.** Transcranial Doppler ultrasonography
- **E.** Noncontrast head CT

Answer: E. Noncontrast head CT

The combination of declining consciousness, impaired upward gaze from pressure on the dorsal midbrain, sluggish pupils, and rising blood pressure is the classic picture of acute symptomatic hydrocephalus, which occurs in 15% to 20% of patients with subarachnoid hemorrhage and is diagnosed by noncontrast head CT showing ventricular dilation. Electroencephalography can reveal nonconvulsive seizures, which do occur in up to 20% of poor-grade patients, but seizures would not produce a sustained blood pressure elevation with impaired upward gaze. Hyponatremia can depress the level of consciousness but does not cause persistent limitation of upgaze or a rise in arterial pressure. Angiography for vasospasm is not the immediate next step in this context, and delayed cerebral ischemia typically begins 3 to 5 days after the bleed with a peak between days 5 and 7, making it unlikely on day 2. Transcranial Doppler has the same timing problem and does not evaluate the ventricles. If hydrocephalus is confirmed in a stuporous or comatose patient, an external ventricular drain should be placed, and it can be removed once intracranial pressure has been stable for more than 24 hours with successful weaning and clamping.

Key point:
new impaired upward gaze with declining consciousness early after subarachnoid hemorrhage is acute hydrocephalus until a noncontrast head CT proves otherwise.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 662-663, 671, 673

Question 216

Subarachnoid Hemorrhage · ISAT and choice of aneurysm securing modality · easy

A 55-year-old woman presents with World Federation of Neurological Surgeons grade 2 subarachnoid hemorrhage from a ruptured 6 mm anterior communicating artery aneurysm. The multidisciplinary team judges the aneurysm to be equally amenable to surgical clipping and endovascular coiling. Which of the following statements about the International Subarachnoid Aneurysm Trial best informs the treatment decision?

- **A.** Surgical clipping produced a lower rate of death or dependency at 1 year
- **B.** Endovascular coiling produced a lower rate of death or dependency at 1 year, 23.7% versus 30.6%, and a lower risk of epilepsy
- **C.** The two modalities produced identical rates of death or dependency at 1 year
- **D.** Endovascular coiling produced both lower rebleeding and lower rates of incomplete aneurysm occlusion
- **E.** The advantage of coiling seen at 1 year had disappeared by 10 years of follow-up

Answer: B. Endovascular coiling produced a lower rate of death or dependency at 1 year, 23.7% versus 30.6%, and a lower risk of epilepsy

ISAT randomized 2,134 patients with predominantly small, under 10 mm, anterior circulation aneurysms judged suitable for either treatment, exactly the situation described here. Patients allocated to endovascular coiling were less likely to be dead or disabled at 1 year, with 190 of 801 patients, or 23.7%, at modified Rankin Scale 3 to 6 versus 243 of 793, or 30.6%, after clipping, with a P value of 0.0019, and coiling also carried a lower risk of epilepsy and less cognitive impairment at 12 months. Even 10 years after the hemorrhage the probability of disability-free survival remained greater in the endovascular group, so option E is wrong. Option D reverses the trade-off: the risks of partial aneurysm occlusion and of rebleeding were actually lower in the surgical clipping group. Coiling is therefore the first choice whenever feasible, although many aneurysms, such as middle cerebral artery aneurysms, those with a wide neck-to-body ratio, those with normal branches arising from the dome, and those with a large parenchymal hematoma, remain better suited to clipping.

Key point:
in ISAT, coiling of small anterior circulation ruptured aneurysms reduced death or dependency at 1 year from 30.6% to 23.7%, while clipping gave more durable occlusion.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 656-657, 669

Question 217

Subarachnoid Hemorrhage · Seizure prophylaxis after subarachnoid hemorrhage · moderate

A 44-year-old man with World Federation of Neurological Surgeons grade 1 subarachnoid hemorrhage was started on levetiracetam in the emergency department and underwent coil embolization on hospital day 1. He has had no clinical seizures, and his neurologic examination is normal and reliable. Which of the following is the most appropriate management of his antiseizure medication?

- **A.** Continue levetiracetam for 12 months
- **B.** Change to phenytoin and continue for 6 months
- **C.** Continue indefinitely until an electroencephalogram shows no epileptiform activity
- **D.** Discontinue now that the aneurysm is secured; if prophylaxis is continued at all, limit it to a short course of 3 to 7 days
- **E.** Continue through hospital day 14 to cover the entire period of delayed cerebral ischemia risk

Answer: D. Discontinue now that the aneurysm is secured; if prophylaxis is continued at all, limit it to a short course of 3 to 7 days

Prophylactic anticonvulsant therapy in a non-comatose patient without clinical seizures can be stopped once the aneurysm is secured, and if prophylaxis is used at all it should be a short course of 3 to 7 days; antiseizure medications can be discontinued on the first day after the procedure in low-grade patients who have not seized. Routine long-term anticonvulsant use is not recommended, a Class III Level B recommendation, and the long-term risk of epilepsy after subarachnoid hemorrhage is low, at roughly 5% and mostly in high-grade patients, which excludes options A and E. Routine prophylaxis with phenytoin specifically is not recommended because phenytoin has been associated with worse clinical outcomes and more medication-related complications; levetiracetam has equivalent efficacy for prophylaxis with fewer adverse effects, so switching to phenytoin is the wrong direction. A negative electroencephalogram is not the trigger for stopping prophylaxis; continuous electroencephalography is instead reserved for poor-grade patients who fail to improve or deteriorate without explanation, in whom nonconvulsive seizures or status epilepticus are found in up to 20%. In high-grade patients prophylaxis may reasonably be continued for 1 week.

Key point:
stop prophylactic antiseizure medication once the aneurysm is secured in a good-grade patient who has not seized, keep any course to 3 to 7 days, and avoid phenytoin.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 659, 667, 674

Question 218

Subarachnoid Hemorrhage · Perimesencephalic subarachnoid hemorrhage · moderate

A 51-year-old man presents with sudden severe headache and neck stiffness. His neurologic examination is entirely normal. Noncontrast head CT shows blood centered in the prepontine and perimesencephalic cisterns immediately anterior to the midbrain and pons, without extension into the sylvian fissures or over the convexities. CT angiography is negative, and catheter digital subtraction angiography performed the same day shows no aneurysm or vascular malformation. Which of the following statements about this patient is most accurate?

- **A.** Repeat catheter angiography in 7 to 14 days will identify an aneurysm in about one quarter of such patients
- **B.** His risk of rebleeding, delayed cerebral ischemia, and hydrocephalus is comparable to that of aneurysmal subarachnoid hemorrhage
- **C.** The source of bleeding is presumed venous, and delayed cerebral ischemia, rebleeding, and hydrocephalus are very rare in this population
- **D.** This bleeding pattern accounts for approximately 80% of all cases of spontaneous subarachnoid hemorrhage
- **E.** Nimodipine is contraindicated with this pattern of hemorrhage

Answer: C. The source of bleeding is presumed venous, and delayed cerebral ischemia, rebleeding, and hydrocephalus are very rare in this population

Approximately 10% of patients with nontraumatic subarachnoid hemorrhage, and about 60% of those with a negative angiogram, have blood centered in the perimesencephalic cisterns close to the midbrain and pons. These patients typically have a normal neurologic examination and a favorable prognosis, the source of bleeding is presumed to be venous, and delayed cerebral ischemia, rebleeding, and hydrocephalus are all very rare, which makes option B incorrect. Because of this benign natural history, a negative angiogram in the setting of a perimesencephalic pattern does not warrant repeat or follow-up angiography; option A quotes the yield of up to 24% that applies to repeat angiography in patients with a non-perimesencephalic hemorrhage and an initially negative study. The 80% figure in option D is the proportion of spontaneous subarachnoid hemorrhage caused by aneurysmal rupture, not the perimesencephalic proportion. Nimodipine is not contraindicated in any pattern of subarachnoid hemorrhage.

Key point:
a perimesencephalic pattern with a negative angiogram implies a presumed venous source, a benign course, and no need for repeat angiography.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 651, 653

Question 219

Subarachnoid Hemorrhage · Focal signs and aneurysm localization · easy

A 49-year-old woman presents with abrupt severe headache, vomiting, and neck stiffness. On examination she is drowsy but follows commands. The left pupil is 6 mm and nonreactive, there is left ptosis, and the left eye cannot adduct, elevate, or depress. Noncontrast head CT shows diffuse blood in the basal cisterns. Which of the following aneurysm locations is most likely responsible?

- **A.** Left posterior communicating artery at its junction with the internal carotid artery
- **B.** Anterior communicating artery
- **C.** Left middle cerebral artery bifurcation
- **D.** Left internal carotid artery within the cavernous sinus
- **E.** Left vertebral artery at the origin of the posterior inferior cerebellar artery

Answer: A. Left posterior communicating artery at its junction with the internal carotid artery

A third nerve palsy in the setting of subarachnoid hemorrhage points to an aneurysm of the posterior communicating artery, and less commonly of the posterior cerebral or superior cerebellar artery; the junction of the internal carotid and posterior communicating artery is also one of the three most common aneurysm sites, along with the anterior communicating artery and the first major branch point of the middle cerebral artery, and 85% to 90% of intracranial aneurysms lie in the anterior circulation. An anterior communicating artery aneurysm classically produces bilateral leg weakness and abulia rather than a pupil-involving third nerve palsy. A middle cerebral artery aneurysm produces hemiparesis with aphasia or visuospatial neglect. A cavernous internal carotid artery aneurysm impinging on the cavernous sinus causes ophthalmoplegia, typically with multiple cranial nerve involvement, and it is an extradural location that is not a typical source of subarachnoid blood in the basal cisterns. A vertebral artery aneurysm at the posterior inferior cerebellar artery origin is a posterior circulation lesion that produces brainstem signs rather than an isolated third nerve palsy. Note that a sixth nerve palsy in this setting is a false localizing sign of raised intracranial pressure, not a localizer.

Key point:
a pupil-involving third nerve palsy with subarachnoid hemorrhage means a posterior communicating artery aneurysm until proven otherwise.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 650-651

Question 220

Subarachnoid Hemorrhage · Modified Fisher scale and delayed cerebral ischemia risk · hard

A 60-year-old woman presents with sudden severe headache and a Glasgow Coma Scale score of 14. Noncontrast head CT shows thick subarachnoid blood filling the basal cisterns and both sylvian fissures, together with layered blood in both lateral ventricles. Which of the following correctly gives her modified Fisher grade and the associated risk of delayed cerebral ischemia?

- **A.** Grade 1, with a delayed cerebral ischemia risk of about 12%
- **B.** Grade 2, with a delayed cerebral ischemia risk of about 21%
- **C.** Grade 3, with a delayed cerebral ischemia risk of about 19%
- **D.** Grade 4, with a delayed cerebral ischemia risk of about 12%
- **E.** Grade 4, with a delayed cerebral ischemia risk of about 40%

Answer: E. Grade 4, with a delayed cerebral ischemia risk of about 40%

The modified Fisher scale is the most commonly used system for estimating the risk of delayed cerebral ischemia, and it grades on two variables only: whether the subarachnoid blood is thick or thin, and whether intraventricular hemorrhage is present. Thick subarachnoid hemorrhage with intraventricular hemorrhage is grade 4 and carries the highest risk, with delayed cerebral ischemia in 40% and infarction in 28%. Thick subarachnoid hemorrhage without intraventricular blood is grade 3, with delayed cerebral ischemia in 19% and infarction in 12%, which is why option C misassigns this patient. Minimal or thin subarachnoid hemorrhage without intraventricular blood is grade 1, with 12% delayed cerebral ischemia and 6% infarction, and thin blood with intraventricular hemorrhage is grade 2, with 21% and 14%. Grade 0, with no subarachnoid or intraventricular blood, carries essentially no risk. Note that this scale is distinct from the Hunt and Hess and World Federation of Neurological Surgeons scales, which grade clinical status rather than blood burden.

Key point:
thick subarachnoid blood plus intraventricular hemorrhage is modified Fisher grade 4, the highest-risk group, with roughly a 40% rate of delayed cerebral ischemia.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 655

Question 221

Subarachnoid Hemorrhage · Neurogenic cardiac and pulmonary complications · moderate

A 57-year-old woman is admitted with Hunt and Hess grade 4 subarachnoid hemorrhage. Within hours of admission she becomes hypoxemic with pink frothy endotracheal secretions and bilateral pulmonary infiltrates, and her blood pressure falls to 88/50 mm Hg. Troponin is 4.2 ng/mL and B-type natriuretic peptide is markedly elevated. The electrocardiogram shows deep symmetric T-wave inversions with a prolonged QT interval. Transthoracic echocardiography shows apical akinesis with a hyperkinetic base and an ejection fraction of 30%. Which of the following best explains her presentation?

- **A.** Acute coronary plaque rupture producing an anterior ST-elevation myocardial infarction
- **B.** Catecholamine-mediated neurogenic stunned myocardium with neurogenic pulmonary edema
- **C.** Nimodipine-induced hypotension with secondary demand ischemia
- **D.** Septic cardiomyopathy from early ventriculostomy-related infection
- **E.** Massive pulmonary embolism with acute right heart strain

Answer: B. Catecholamine-mediated neurogenic stunned myocardium with neurogenic pulmonary edema

Cardiac and pulmonary disturbances are among the most common complications of subarachnoid hemorrhage and are thought to result from the surge in catecholamines and sympathetic tone at the moment of aneurysm rupture, which causes direct myocardial injury. Transient electrocardiographic abnormalities occur in about 50% to 100% of patients and include sinus tachycardia, peaked T waves, T-wave inversions, QT prolongation, and ST segment depression or elevation, and about 30% of patients have elevated serum troponin, so an elevated troponin alone does not imply coronary occlusion. The regional wall motion abnormality with apical ballooning and a hyperkinetic base in a high-grade patient is the classic description of stunned myocardium or Takotsubo cardiomyopathy, which can present with sudden hypoxemia and cardiogenic shock with or without pulmonary edema; neurogenic pulmonary edema from increased pulmonary vascular permeability can occur in isolation or together with the cardiac syndrome. Nimodipine does cause hypotension but not troponin elevation with apical ballooning, and this severity of illness within hours of admission is not explained by a ventriculostomy infection or by pulmonary embolism, which would produce right rather than apical dysfunction. Management centers on euvolemia, lung-protective ventilation with tidal volumes less than 7 cc/kg of ideal body weight, cardiac output monitoring, and inotropes such as milrinone or dobutamine in patients with delayed cerebral ischemia.

Key point:
apical ballooning with elevated troponin and QT prolongation after subarachnoid hemorrhage is catecholamine-mediated neurogenic stunned myocardium, not acute coronary syndrome.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 667-668, 674

Question 222

Subarachnoid Hemorrhage · Screening for unruptured intracranial aneurysms · easy

In which of the following groups do the American Heart Association and American Stroke Association guidelines recommend noninvasive screening for unruptured intracranial aneurysms with CT angiography or magnetic resonance angiography?

- **A.** Any patient with a single first-degree relative who has a known intracranial aneurysm
- **B.** Patients with autosomal recessive polycystic kidney disease
- **C.** All patients who both smoke and have hypertension
- **D.** Patients with two or more family members who have an intracranial aneurysm or subarachnoid hemorrhage
- **E.** All patients younger than 15 years of age

Answer: D. Patients with two or more family members who have an intracranial aneurysm or subarachnoid hemorrhage

The 2015 American Heart Association and American Stroke Association guideline on unruptured intracranial aneurysms recommends that patients with two or more family members with intracranial aneurysms or subarachnoid hemorrhage be offered screening with CT angiography or magnetic resonance angiography, a Class I Level B recommendation; within such families, hypertension, smoking, and female sex predict a particularly high risk. A single affected first-degree relative is not by itself an indication for screening. The renal indication is autosomal dominant, not autosomal recessive, polycystic kidney disease, particularly with a family history of intracranial aneurysms, also Class I Level B. It is reasonable, at Class IIa Level B, to offer angiographic screening to patients with coarctation of the aorta and to patients with microcephalic osteodysplastic primordial dwarfism, and type IV Ehlers-Danlos syndrome is also listed. Smoking and hypertension are modifiable risk factors for aneurysm formation and rupture but are not standalone screening indications, and there is no age-based screening recommendation.

Key point:
screen for unruptured aneurysms when two or more family members are affected, or in autosomal dominant polycystic kidney disease.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 674

Question 223

Subarachnoid Hemorrhage · Angiogram-negative subarachnoid hemorrhage · hard

A 54-year-old man presents with sudden severe headache and photophobia. Noncontrast head CT shows diffuse thick subarachnoid blood in the basal cisterns extending into both sylvian fissures and the anterior interhemispheric fissure. CT angiography shows no aneurysm, and catheter digital subtraction angiography performed on the day of admission is also negative. He remains clinically stable. Which of the following is the most appropriate next step?

- **A.** Repeat catheter angiography 7 to 14 days after the initial presentation
- **B.** No further vascular imaging, since a negative digital subtraction angiogram excludes an aneurysm
- **C.** Repeat noncontrast head CT in 24 hours, with no further vascular imaging
- **D.** Perform a lumbar puncture to confirm the diagnosis of subarachnoid hemorrhage
- **E.** Substitute a CT perfusion study on day 4 for repeat angiography

Answer: A. Repeat catheter angiography 7 to 14 days after the initial presentation

This is a diffuse, non-perimesencephalic bleeding pattern, so an aneurysm must still be presumed until excluded. A digital subtraction angiogram can be falsely negative because of small aneurysm size, a compressive hematoma, local thrombosis, vasospasm, or poor technique, and such patients should have a repeat study 7 to 14 days after the initial presentation, in which up to 24% will be found to have an aneurysm. Option B is therefore wrong: digital subtraction angiography is the gold standard but is not infallible, and a negative study in this pattern does not end the workup. Repeating the head CT does not address the vascular question. Lumbar puncture is superfluous when subarachnoid blood is already visible on CT, and it also has no role in evaluating suspected rebleeding since xanthochromia persists for 2 weeks or more. CT perfusion is a tool for assessing delayed cerebral ischemia between days 4 and 8, not a substitute for finding the bleeding source. Brain and cervical spine MRI should also be considered in this setting to look for an arteriovenous malformation of the brain, brainstem, or spinal cord; by contrast, a perimesencephalic pattern with a negative angiogram requires no repeat study.

Key point:
after a negative angiogram in non-perimesencephalic subarachnoid hemorrhage, repeat catheter angiography at 7 to 14 days, which finds an aneurysm in up to 24% of cases.

Source: Practice of Neurocritical Care, 2nd ed., Subarachnoid Hemorrhage, pp. 653, 661

Traumatic Brain Injury

Question 224

Traumatic Brain Injury · Cerebral hemodynamics and autoregulation · moderate

A 28-year-old man is admitted to the neurointensive care unit after a motorcycle crash with a severe traumatic brain injury (GCS 5). An intraparenchymal intracranial pressure (ICP) monitor is placed. Over the first 24 hours the team notes that every time mean arterial pressure (MAP) is raised with norepinephrine, ICP rises in parallel, and the moving correlation between MAP and ICP (pressure reactivity index) is strongly positive, indicating that pressure autoregulation is absent. Which of the following best describes the expected relationship between cerebral perfusion pressure (CPP) and cerebral blood flow (CBF) in this patient?

- **A.** CBF will vary directly with CPP because cerebral vessel radius is effectively fixed
- **B.** CBF will remain constant across CPPs of approximately 50 to 150 mmHg through changes in vessel diameter
- **C.** CBF will be determined chiefly by vessel radius, with CPP having little impact
- **D.** CBF will vary inversely with CPP because of compensatory vasoconstriction
- **E.** CBF will become independent of blood viscosity once autoregulation is lost

Answer: A. CBF will vary directly with CPP because cerebral vessel radius is effectively fixed

CPP is calculated as MAP minus ICP, and if cerebral blood flow is modeled as flow of a Newtonian fluid through a rigid tube, the Hagen-Poiseuille relationship makes CBF proportional to CPP and to vessel radius raised to the fourth power, and inversely proportional to blood viscosity. When pressure autoregulation is intact, vessel diameter changes to keep CBF constant across a wide range of CPPs, so the primary determinant of flow is vessel radius and CPP has little impact; this is what options B and C describe, and they apply to the patient with preserved autoregulation, not this one. When autoregulation is lost, vessel radius remains constant and flow becomes 'pressure passive,' varying directly with CPP, which is exactly why ICP and CPP are used as surrogates for CBF only under the assumption that autoregulation is disturbed. Option D is wrong because there is no mechanism producing an inverse relationship; loss of autoregulation means loss of the vasoconstrictor response. Option E is wrong because viscosity remains an inverse determinant of CBF regardless of autoregulatory status, which is the basis of the rheological effect of osmotic agents.

Key point:
when pressure autoregulation is lost after traumatic brain injury, cerebral blood flow becomes pressure passive and varies directly with cerebral perfusion pressure.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 687

Question 225

Traumatic Brain Injury · Surgical management of epidural hematoma · easy

A 22-year-old man is struck in the left temple during an assault. He was briefly unconscious, then awake and conversant with paramedics, and is now drowsy but follows commands, with a GCS of 14 and no focal deficit. Blood pressure is 138/78 mmHg and pulse is 72 beats per minute. Noncontrast head CT shows a left temporal lens-shaped hyperdense extra-axial collection that does not cross the coronal suture, measuring 45 cm3 in volume and 18 mm in maximal thickness, with 6 mm of midline shift. Which of the following is the most appropriate next step in management?

- **A.** Admission for observation with a repeat head CT in 6 hours
- **B.** Immediate surgical evacuation via craniotomy
- **C.** Placement of an intracranial pressure monitor and medical management
- **D.** Bolus mannitol with a repeat head CT in 24 hours
- **E.** Bedside burr hole drainage in the intensive care unit

Answer: B. Immediate surgical evacuation via craniotomy

Guideline recommendations for the surgical management of traumatic brain injury state that an epidural hematoma greater than 30 cm3 should be surgically evacuated regardless of the patient's GCS score, and this lesion is 45 cm3. Craniotomy is the preferred approach because it provides a more complete evacuation of the hematoma than other methods, which makes option E inadequate. Nonoperative management with serial CT scanning and close neurological observation (options A and D) is reserved for an epidural hematoma less than 30 cm3 with less than 15 mm thickness and less than 5 mm of midline shift in a patient with a GCS greater than 8 and no focal deficit; this lesion exceeds every one of those limits. An ICP monitor (option C) does not address a surgically evacuable mass lesion and would delay definitive therapy in a hematoma that characteristically expands rapidly and causes herniation. The reassuring examination is misleading: the classic lucid interval, in which a patient is initially unconscious, wakes up without obvious deficit, and then deteriorates, occurs in approximately 47% of patients with epidural hematoma.

Key point:
an epidural hematoma greater than 30 cubic centimeters should be evacuated regardless of the GCS score.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 691, 696

Question 226

Traumatic Brain Injury · Surgical management of subdural hematoma · hard

A 61-year-old woman falls down a flight of stairs. Her GCS at the scene was 10 (E3 V3 M4). On arrival in the emergency department 45 minutes later her GCS is 7 (E1 V2 M4). Pupils are 3 mm and briskly reactive bilaterally. Blood pressure is 142/80 mmHg and oxygen saturation is 98% after intubation. Noncontrast head CT shows a right convexity crescentic hyperdense extra-axial collection with a maximal thickness of 8 mm and 4 mm of midline shift, with patent basal cisterns. An intraparenchymal monitor is placed and reads an ICP of 14 mmHg. Which of the following is the most appropriate next step?

- **A.** Continued medical management with a repeat head CT in 6 hours
- **B.** Bolus hyperosmolar therapy and repeat imaging once the ICP exceeds 22 mmHg
- **C.** Placement of an external ventricular drain for cerebrospinal fluid diversion
- **D.** Surgical evacuation of the hematoma
- **E.** Observation with treatment triggered only if the midline shift exceeds 5 mm

Answer: D. Surgical evacuation of the hematoma

An acute subdural hematoma with a thickness greater than 10 mm or a midline shift greater than 5 mm should generally be evacuated regardless of GCS, and this lesion meets neither of those size criteria. However, a comatose patient (GCS less than 9) with an acute subdural hematoma less than 10 mm thick and less than 5 mm of midline shift should still undergo surgical evacuation if the GCS decreases by more than 2 points between the time of injury and hospital admission, and/or the pupils are asymmetric or fixed and dilated, and/or the ICP is greater than 20 mmHg. This patient fell from a GCS of 10 to a GCS of 7, a decrease of 3 points, so surgery is indicated even though her pupils are reactive and her ICP is 14 mmHg; when surgery is indicated it should be performed as soon as possible. Options A, B and E all apply a size or pressure trigger that this patient is not required to reach, and delay an operation that is already indicated by the neurological deterioration criterion. An external ventricular drain (option C) treats hydrocephalus and allows cerebrospinal fluid drainage but does not remove an extra-axial mass lesion, and it carries a higher rate of bleeding and infection than an intraparenchymal monitor.

Key point:
a comatose patient with a small acute subdural hematoma still needs evacuation if the GCS drops by more than 2 points, the pupils become asymmetric or fixed and dilated, or the ICP exceeds 20 mmHg.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 696

Question 227

Traumatic Brain Injury · Basilar skull fracture and cerebrospinal fluid leak · moderate

A 19-year-old man is brought to the emergency department after a motor vehicle collision. He is alert with a GCS of 15. Examination shows retroauricular ecchymosis and hemotympanum on the right, and clear fluid dripping from the right nostril that tests positive for beta-2 transferrin. CT of the head demonstrates a nondisplaced right temporal bone fracture with a small amount of pneumocephalus and no intracranial mass lesion. The trauma team asks how to reduce his risk of meningitis. Which of the following is the most appropriate management?

- **A.** Begin intravenous ceftriaxone and continue until the leak resolves
- **B.** Begin intravenous vancomycin plus cefepime for 7 days
- **C.** Withhold prophylactic antibiotics, observe the leak, and pursue operative repair if it persists beyond 7 days
- **D.** Place a lumbar drain to divert cerebrospinal fluid and lower the leak pressure
- **E.** Proceed to immediate operative repair of the skull base defect

Answer: C. Withhold prophylactic antibiotics, observe the leak, and pursue operative repair if it persists beyond 7 days

Cerebrospinal fluid leak is common after skull base injury, occurring in 28% of patients with penetrating traumatic brain injury in one series, and it is associated with an increased risk of infection. Nonetheless, prophylactic antibiotics do not appear to reduce the incidence of infection, and the Infectious Diseases Society of America therefore does not recommend prophylactic antibiotics for patients with basilar skull fractures and resultant cerebrospinal fluid leak, which makes options A and B incorrect. What that society does recommend is surgical intervention to repair a prolonged leak, defined as lasting more than 7 days, together with pneumococcal vaccination. Immediate operative repair (option E) is not indicated because most traumatic leaks close spontaneously within the first week. Lumbar cerebrospinal fluid drainage (option D) is specifically listed among the therapeutic maneuvers to avoid in traumatic brain injury by the Seattle International Severe Traumatic Brain Injury Consensus Conference.

Key point:
do not give prophylactic antibiotics for a basilar skull fracture with cerebrospinal fluid leak; observe, vaccinate against pneumococcus, and repair a leak that persists beyond 7 days.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 692, 699

Question 228

Traumatic Brain Injury · Blood pressure targets · easy

According to Brain Trauma Foundation guidelines, what is the minimum systolic blood pressure that should be maintained in a 24-year-old patient with severe traumatic brain injury?

- **A.** 80 mmHg
- **B.** 90 mmHg
- **C.** 100 mmHg
- **D.** 105 mmHg
- **E.** 110 mmHg

Answer: E. 110 mmHg

The Brain Trauma Foundation recommends maintaining systolic blood pressure at 100 mmHg or greater for patients who are 50 to 69 years old, and at 110 mmHg or greater for patients who are 15 to 49 years old or older than 70 years; a 24-year-old therefore falls into the 110 mmHg group, making option E correct and option C the threshold for the wrong age band. The older literature figure of 90 mmHg (option B) is the classic definition of hypotension rather than a treatment target, and a single episode of hypotension by that definition is strongly associated with worse outcome independent of age, admission GCS, or pupillary function, and with a two-fold increase in mortality. The American College of Surgeons simply recommends maintaining systolic blood pressure above 100 mmHg. Options A and D correspond to no recommended threshold. These thresholds apply in hospital and it is reasonable to generalize them to the prehospital setting, where avoidance of hypotension and hypoxia is the primary goal.

Key point:
the Brain Trauma Foundation systolic blood pressure floor is 100 mmHg for ages 50 to 69 and 110 mmHg for ages 15 to 49 or over 70.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 695, 698

Question 229

Traumatic Brain Injury · Tranexamic acid and CRASH-3 · moderate

A 47-year-old man falls from a ladder and arrives in the emergency department 90 minutes after injury. His GCS is 12, blood pressure is 128/72 mmHg, and pulse is 88 beats per minute. Noncontrast head CT shows a small right temporal contusion and a thin layer of traumatic subarachnoid hemorrhage. INR is 1.0, platelet count is 240,000 per microliter, and he takes no anticoagulant or antiplatelet drugs. Which of the following interventions has been shown in a large randomized controlled trial to reduce head injury-related mortality in patients such as this one?

- **A.** Recombinant activated factor VII
- **B.** Intravenous methylprednisolone
- **C.** Prophylactic platelet transfusion
- **D.** Intravenous tranexamic acid
- **E.** Prophylactic hypothermia to 33 degrees Celsius

Answer: D. Intravenous tranexamic acid

CRASH-3 was an international multicenter randomized placebo-controlled trial of the antifibrinolytic tranexamic acid in 12,737 patients with traumatic brain injury, examining head injury-related in-hospital mortality within 28 days. The trial's primary endpoint was negative overall, but in the prespecified subgroup of patients with mild-to-moderate traumatic brain injury, and not in those with severe injury, head injury-related mortality was significantly reduced; this patient has a GCS of 12, which is moderate injury, so tranexamic acid might be considered. Methylprednisolone (option B) is contraindicated because the MRC CRASH trial showed that corticosteroids increase death and severe disability, and avoidance of steroids is the only Brain Trauma Foundation recommendation supported by high-quality evidence. Prophylactic hypothermia (option E) failed in the National Acute Brain Injury Study Hypothermia I and II trials and in the POLAR trial. Recombinant factor VIIa and prophylactic platelet transfusion (options A and C) are not supported; this patient has no coagulopathy to correct, although coagulation abnormalities when present should be corrected rapidly.

Key point:
CRASH-3 was negative overall but showed reduced head injury-related mortality with tranexamic acid in mild-to-moderate traumatic brain injury.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 696

Question 230

Traumatic Brain Injury · Intracranial pressure thresholds · easy

According to the current Brain Trauma Foundation guidelines, treatment of intracranial hypertension after severe traumatic brain injury should be initiated when the intracranial pressure exceeds which of the following values?

- **A.** 15 mmHg
- **B.** 20 mmHg
- **C.** 22 mmHg
- **D.** 25 mmHg
- **E.** 30 mmHg

Answer: C. 22 mmHg

Although it is common practice to maintain intracranial pressure below 20 mmHg, the Brain Trauma Foundation, based on low-quality evidence, now recommends initiating treatment when the intracranial pressure exceeds 22 mmHg. That threshold derives from a retrospective study of 459 patients in which statistical analysis defined the value that best discriminated between survivors and nonsurvivors and between good and poor functional outcomes. The American College of Surgeons instead uses 20 mmHg as an alert threshold and institutes therapy for an intracranial pressure between 20 and 25 mmHg, so options B and D describe that different framework rather than the Brain Trauma Foundation number. Options A and E correspond to no guideline threshold; 15 mmHg is within the normal range and waiting until 30 mmHg would forgo established therapy. It is worth remembering that while intracranial hypertension is associated with increased mortality, it has not been established that lowering intracranial pressure improves outcome, and interventions such as craniectomy, hypothermia, and pharmacological coma reduce pressure without proven outcome benefit.

Key point:
the Brain Trauma Foundation treatment threshold is an intracranial pressure above 22 mmHg, whereas the American College of Surgeons alerts at 20 and treats between 20 and 25 mmHg.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 698

Question 231

Traumatic Brain Injury · Osmotherapy and mannitol monitoring · moderate

A 34-year-old woman with severe traumatic brain injury and bifrontal contusions has received repeated 1 g/kg boluses of 20% mannitol over the past 24 hours for intracranial pressures in the high 20s. Current labs are sodium 152 mEq/L, BUN 28 mg/dL, glucose 126 mg/dL, creatinine 1.4 mg/dL (baseline 0.8), and a measured serum osmolality of 345 mOsm/kg. Blood pressure is 104/60 mmHg and urine output has been 400 mL/hr. Her intracranial pressure is again 26 mmHg. Which of the following is the most appropriate next step?

- **A.** Continue mannitol boluses because the serum sodium remains below 160 mEq/L
- **B.** Change to a continuous mannitol infusion to smooth the intracranial pressure profile
- **C.** Give furosemide to accelerate mannitol clearance and enhance the osmotic effect
- **D.** Withhold further mannitol and use hypertonic saline if additional osmotherapy is required
- **E.** Continue mannitol but reduce the dose to 0.25 g/kg per bolus

Answer: D. Withhold further mannitol and use hypertonic saline if additional osmotherapy is required

With repeated mannitol dosing, the serum osmolar gap, calculated as measured osmolality minus calculated osmolality where calculated osmolality equals 2 times sodium plus BUN divided by 2.8 plus glucose divided by 18, is the better gauge of mannitol accumulation than serum osmolarity alone, because serum osmolarity also rises from the attendant hypernatremia. Here the calculated osmolality is 2(152) plus 10 plus 7, which is 321 mOsm/kg, giving an osmolar gap of 24 mOsm/kg, above the recommended ceiling of 20 mOsm/kg, and the patient additionally has a rising creatinine, brisk osmotic diuresis, and a falling blood pressure. Further mannitol risks renal failure, intravascular volume depletion with hypotension, and rebound intracranial hypertension, so it should be withheld; hypertonic saline is the reasonable alternative because it is a plasma expander and is preferred in a volume-depleted patient, whereas mannitol as a diuretic suits the volume-overloaded patient. Options A and E are wrong because the sodium ceiling of 160 mEq/L governs hypertonic saline rather than mannitol accumulation, and simply lowering the dose does not address the accumulation or the renal injury. Continuous mannitol infusion (option B) is explicitly avoided because of the risk of renal failure and rebound cerebral edema, and it appears on the consensus list of maneuvers to avoid, as does furosemide (option C).

Key point:
monitor repeated mannitol dosing with the serum osmolar gap, keeping it below 20 mOsm/kg, and switch to hypertonic saline when the patient is volume depleted.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 699-700

Question 232

Traumatic Brain Injury · Herniation and hyperventilation · moderate

A 40-year-old man with a severe traumatic brain injury and an external ventricular drain has an abrupt rise in intracranial pressure to 45 mmHg. His right pupil dilates to 6 mm and becomes nonreactive, and he develops extensor posturing. Blood pressure is 178/95 mmHg with a pulse of 52 beats per minute. Cerebrospinal fluid drainage produces no improvement. A stat head CT is ordered and the neurosurgical team is mobilizing for the operating room. In addition to a bolus of a hyperosmolar agent, which of the following is the most appropriate temporizing measure?

- **A.** Start a continuous infusion of mannitol
- **B.** Place a lumbar drain for cerebrospinal fluid drainage
- **C.** Administer intravenous dexamethasone
- **D.** Begin surface cooling to a core temperature of 33 degrees Celsius
- **E.** Transiently hyperventilate to an end-tidal carbon dioxide of 30 to 35 mmHg

Answer: E. Transiently hyperventilate to an end-tidal carbon dioxide of 30 to 35 mmHg

Hyperventilation produces blood and cerebrospinal fluid alkalosis, which causes cerebral vasoconstriction, reduced cerebral blood volume, and therefore a lower intracranial pressure; sustained vigorous hyperventilation risks cerebral ischemia and is not recommended as routine therapy, but in emergencies such as acute herniation it may be used transiently as a bridge to more definitive therapy such as surgery or an osmotic agent. This patient has the Cushing response with hypertension and bradycardia plus a blown pupil and posturing, exactly the situation in which empiric head of bed elevation, hyperventilation, and a hyperosmolar agent are indicated. Routinely decreasing the arterial carbon dioxide below 30 mmHg is on the consensus list of maneuvers to avoid, which is why the target here stays at 30 to 35 mmHg. Continuous mannitol infusion (option A) and lumbar cerebrospinal fluid drainage (option B) are both on that same avoid list, and lumbar drainage additionally risks herniation. Steroids (option C) are associated with increased mortality after traumatic brain injury and should not be used, and hypothermia (option D) has failed in randomized trials and is too slow to serve as a rescue maneuver.

Key point:
hyperventilation is a transient bridge for acute herniation only, never routine therapy, and the arterial carbon dioxide should not be driven routinely below 30 mmHg.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 695-696, 699-700

Question 233

Traumatic Brain Injury · Refractory intracranial hypertension and tiered therapy · hard

A 26-year-old man with severe diffuse traumatic brain injury and no surgically evacuable mass lesion has intracranial pressures of 30 to 35 mmHg for the past 4 hours. He is already receiving deep sedation and analgesia, has the head of bed elevated with the neck in neutral position, is normocapnic with an arterial carbon dioxide of 38 mmHg, has had cerebrospinal fluid drained through an external ventricular drain, and has received repeated boluses of hypertonic saline with a serum sodium now of 158 mEq/L. A trial of neuromuscular blockade lowered the intracranial pressure only transiently. Cerebral perfusion pressure is 62 mmHg. Which of the following is the most appropriate next step?

- **A.** Induce therapeutic hypothermia to 33 degrees Celsius
- **B.** Administer intravenous methylprednisolone
- **C.** Give furosemide together with scheduled mannitol infusions
- **D.** Place a lumbar drain for additional cerebrospinal fluid drainage
- **E.** Proceed to decompressive craniectomy

Answer: E. Proceed to decompressive craniectomy

This patient has exhausted tier 1 measures (head of bed elevation, neutral neck position, normocapnia, sedation and analgesia, cerebrospinal fluid drainage, osmotherapy) and tier 2 measures (neuromuscular blockade, mild hypocapnia), so a tier 3 therapy is appropriate, and the tier 3 options are barbiturate coma, secondary decompressive craniectomy, and mild hypothermia. Craniectomy is the most effective way to lower intracranial pressure, and it is the only one of the listed answers that is an accepted late-tier intervention. Therapeutic hypothermia (option A) should be avoided in traumatic brain injury: prophylactic hypothermia failed in the National Acute Brain Injury Study Hypothermia I and II and POLAR trials, a pediatric trial of hypothermia for intracranial hypertension suggested harm, and a multinational adult trial of hypothermia at 32 to 35 degrees Celsius for intracranial hypertension was stopped early for suggestion of harm. Steroids (option B) increase death and severe disability, as shown by the MRC CRASH trial, and their avoidance is the only guideline recommendation based on high-quality evidence. Furosemide, scheduled hyperosmolar infusions, nonbolus continuous mannitol, and lumbar cerebrospinal fluid drainage (options C and D) all appear explicitly on the Seattle International Severe Traumatic Brain Injury Consensus Conference list of maneuvers to avoid.

Key point:

for medically refractory intracranial hypertension, escalate to a tier 3 therapy such as decompressive craniectomy, and avoid hypothermia, steroids, furosemide, scheduled hyperosmolar infusions, and lumbar drainage.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 699-701

Question 234

Traumatic Brain Injury · Decompressive craniectomy trials · hard

The family of a 30-year-old woman with severe traumatic brain injury and medically refractory intracranial hypertension asks what decompressive craniectomy is likely to accomplish. The neurosurgeon bases the discussion on the RESCUEicp trial, a multicenter randomized trial of 408 patients comparing last-tier decompressive craniectomy with continued medical management for refractory traumatic intracranial hypertension. Which of the following best describes the 6-month results of that trial?

- **A.** Both mortality and the rate of good functional outcome were significantly improved by surgery
- **B.** Mortality was unchanged but functional outcome was significantly better with surgery
- **C.** Mortality was significantly lower with surgery without a significant increase in good functional outcome
- **D.** Mortality was significantly higher with surgery
- **E.** Neither mortality nor functional outcome differed between the groups

Answer: C. Mortality was significantly lower with surgery without a significant increase in good functional outcome

In RESCUEicp, at 6 months surgery was associated with significantly lower mortality, 26.9% versus 48.9% in the medical group, but there was no significant difference in good functional outcome, 42.8% for surgery versus 34.6% for medical management. The roughly 22% absolute reduction in mortality translated into a larger proportion of survivors in a vegetative state and with lower severe disability, as well as a higher proportion with upper severe disability, with no difference in rates of moderate disability or good recovery. At 12 months, a secondary outcome, surgery was associated with both lower mortality, 30.4% versus 52%, and better functional outcome, 45.4% versus 32.5% with p equal to 0.01, so option A describes the 12-month rather than the 6-month result. Option D matches neither trial: RESCUEicp showed lower, not higher, mortality with surgery, and in DECRA, where 155 patients with diffuse injury were randomized to early bifrontotemporoparietal craniectomy within 72 hours, mean intracranial pressures were lower and mortality was similar, with only 6-month functional outcome worse, at an odds ratio of 2.21 and a 95% confidence interval of 1.14 to 4.26. Options B and E misstate both trials.

Key point:
RESCUEicp showed that last-tier decompressive craniectomy cut 6-month mortality roughly in half but converted those deaths largely into severe disability and vegetative state rather than good recovery.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 701

Question 235

Traumatic Brain Injury · Cerebral perfusion pressure targets · moderate

A 52-year-old man with severe traumatic brain injury has an intracranial pressure of 16 mmHg and a mean arterial pressure of 95 mmHg with the arterial line transducer zeroed at the phlebostatic axis. He is euvolemic and off vasopressors. A trainee proposes starting norepinephrine and giving additional crystalloid to drive the cerebral perfusion pressure to 85 to 90 mmHg in order to improve cerebral oxygen delivery. Which of the following is the most appropriate management?

- **A.** Raise the cerebral perfusion pressure above 90 mmHg with norepinephrine
- **B.** Maintain the cerebral perfusion pressure between 60 and 70 mmHg and avoid pressor-driven elevation above 70 mmHg
- **C.** Maintain the cerebral perfusion pressure above 70 mmHg with aggressive fluid loading
- **D.** Lower the mean arterial pressure so that the cerebral perfusion pressure falls below 60 mmHg
- **E.** Titrate the cerebral perfusion pressure solely to keep the jugular venous oxygen saturation above 50%

Answer: B. Maintain the cerebral perfusion pressure between 60 and 70 mmHg and avoid pressor-driven elevation above 70 mmHg

The Brain Trauma Foundation currently recommends maintaining cerebral perfusion pressure between 60 and 70 mmHg, and elevating it above 70 mmHg with intravenous fluids and vasopressors should be avoided because of the risk of lung injury. A randomized controlled trial that compared cerebral perfusion pressure-targeted therapy, keeping cerebral perfusion pressure above 70 mmHg, with intracranial pressure-targeted therapy, keeping cerebral perfusion pressure above 50 mmHg and intracranial pressure below 20 mmHg, found no difference in outcome but a five-fold increased incidence of acute respiratory distress syndrome and more frequent vasopressor use in the aggressive group; options A and C therefore invite harm without benefit, and routinely raising cerebral perfusion pressure above 90 mmHg is on the consensus list of maneuvers to avoid. Option D is wrong because while raising cerebral perfusion pressure is not advantageous, lowering it below a critical threshold is clearly deleterious, and optimization in a normotensive patient should begin with lowering intracranial pressure rather than manipulating blood pressure. Jugular venous oxygen saturation above 50% and brain tissue oxygen tension above 15 mmHg are level III monitoring targets, not the sole determinant of cerebral perfusion pressure titration, making option E incorrect. Note that a transducer zeroed at the phlebostatic axis overestimates true cerebral perfusion pressure when the head of bed is elevated, by as much as 20 mmHg with elevation beyond 30 degrees.

Key point:
target a cerebral perfusion pressure of 60 to 70 mmHg and do not push it higher with fluids and vasopressors, which multiplies the risk of acute respiratory distress syndrome five-fold.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 699, 702, 707

Question 236

Traumatic Brain Injury · Posttraumatic seizure prophylaxis · easy

A 35-year-old man with a severe traumatic brain injury and bilateral frontal contusions was loaded with phenytoin in the emergency department and is now on hospital day 1 in the intensive care unit. He has had no clinical seizures, and continuous electroencephalography shows no electrographic seizures. The team asks how long to continue the prophylactic anticonvulsant. Which of the following is the most appropriate plan?

- **A.** Discontinue the phenytoin now, since prophylaxis is not recommended
- **B.** Continue prophylaxis for a total of 1 week after the injury
- **C.** Continue prophylaxis for 1 month after the injury
- **D.** Continue prophylaxis for 3 months after the injury
- **E.** Continue prophylaxis indefinitely because of the risk of posttraumatic epilepsy

Answer: B. Continue prophylaxis for a total of 1 week after the injury

Brain Trauma Foundation guidelines recommend the use of an anticonvulsant, specifically phenytoin, for 1 week following traumatic brain injury and recommend against longer durations of prophylactic therapy, so option B is correct and options C, D and E all extend treatment beyond the recommendation. Option A is wrong because prophylaxis for the first week is recommended: seizures occur in 10% to 30% of patients with traumatic brain injury, and up to 25% of comatose patients may have nonconvulsive seizures, which theoretically worsen outcome by raising cerebral metabolic rate and intracranial pressure. The reason for stopping at a week is that prophylactic anticonvulsants reduce the incidence of early posttraumatic seizures but do not lessen the odds of developing posttraumatic epilepsy, so continuing offers no late benefit. Many centers use alternative agents such as fosphenytoin or levetiracetam, generalizing the phenytoin data, although the impact of prophylaxis on outcome and the comparative efficacy of these agents are unknown.

Key point:
give anticonvulsant prophylaxis, classically phenytoin, for 1 week after traumatic brain injury and no longer, because it prevents early seizures but not posttraumatic epilepsy.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 702

Question 237

Traumatic Brain Injury · Paroxysmal sympathetic hyperactivity · moderate

A 24-year-old man is on hospital day 8 after a severe traumatic brain injury with diffuse axonal injury. As sedation is weaned he develops episodes lasting 20 to 40 minutes of heart rate 140 beats per minute, blood pressure 190/100 mmHg, respiratory rate 34 breaths per minute, temperature 38.9 degrees Celsius, profuse diaphoresis, and extensor posturing. The episodes occur several times a day and are reliably triggered by endotracheal suctioning and repositioning. Blood and urine cultures, chest radiograph, and a repeat head CT are unrevealing, and electroencephalography during an episode shows no seizure activity. Intravenous morphine aborts the episodes. Which of the following is the most appropriate maintenance medication to add?

- **A.** Propranolol
- **B.** Metoprolol
- **C.** Haloperidol
- **D.** Phenytoin
- **E.** Dantrolene

Answer: A. Propranolol

This is paroxysmal sympathetic hyperactivity, a syndrome of paroxysmal concurrent increases in sympathetic and motor activity that occurs in approximately 8% to 13% of patients with severe traumatic brain injury, typically begins about 1 week after injury once sedation is weaned, and is provoked by triggers such as endotracheal suctioning, turning, bathing, constipation, and bladder distension. Propranolol is considered the beta-blocker of choice because its lipophilicity allows it to cross the blood brain barrier, and beta-blockers attenuate the effects of excessive circulating catecholamines; typical dosing is 1 to 5 mg intravenously every hour as needed or 10 to 20 mg by mouth every 6 to 8 hours, with a maximum of 320 mg per day. Metoprolol (option B) is beta-1 selective and appears to be ineffective in this syndrome. Intravenous morphine is the prototypical opiate and the intravenous opiate of choice for aborting episodes, and combination therapy with a maintenance agent is usually needed, so adding propranolol to morphine is rational. Dantrolene (option E) acts directly on skeletal muscle and is useful for clonus, spasticity, and posturing, but it is not first line and carries potentially irreversible hepatotoxicity requiring liver function monitoring. Haloperidol and phenytoin (options C and D) treat agitation and seizures respectively, neither of which is the process here, since the electroencephalogram showed no seizures.

Key point:
for paroxysmal sympathetic hyperactivity, intravenous morphine is the opiate of choice and lipophilic propranolol is the beta-blocker of choice, while beta-1 selective metoprolol is ineffective.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Brain Injury, pp. 702-704

Traumatic Spinal Cord Injury

Question 238

Traumatic Spinal Cord Injury · Neurogenic shock and blood pressure targets · moderate

A 38-year-old man is admitted to the neurocritical care unit after a motorcycle collision. Imaging shows a C6 fracture-dislocation, and his examination is consistent with a motor-complete cervical spinal cord injury. He is intubated and normothermic. Heart rate is 48 beats/min, blood pressure 82/48 mmHg (mean arterial pressure 59 mmHg), and central venous pressure is 10 mmHg after 2 L of crystalloid. Hemoglobin is 12.9 g/dL, lactate 1.4 mmol/L, and focused ultrasound shows a hyperdynamic left ventricle with no free fluid. An arterial catheter is placed and a norepinephrine infusion is started. Which of the following is the most appropriate hemodynamic goal for the next 7 days?

- **A.** Systolic blood pressure greater than 90 mmHg
- **B.** Mean arterial pressure of 65 to 70 mmHg
- **C.** Mean arterial pressure of 85 to 90 mmHg
- **D.** Mean arterial pressure of 100 to 110 mmHg
- **E.** Central venous pressure of 12 to 15 mmHg with no specific pressure target

Answer: C. Mean arterial pressure of 85 to 90 mmHg

This patient has neurogenic shock, a distributive shock from sympathetic denervation that produces bradycardia from unopposed vagal tone and hypotension from loss of vascular tone; it typically occurs with lesions at or above T4. Although the usual resuscitation target for shock is a systolic blood pressure above 90 mmHg or a mean arterial pressure above 65 mmHg, the chapter states that this is insufficient after spinal cord injury and that the current recommendation is to maintain a mean arterial pressure of 85 to 90 mmHg for the first 7 days, based on the anticipated duration of the secondary injury period. Options A and B are the generic shock targets that the guideline explicitly considers inadequate here. Option D exceeds any recommended target and adds vasopressor risk without supporting data. Option E is wrong because volume status alone does not define the goal; euvolemia plus alpha-agonist support to a defined pressure target is the described approach, with very frequent monitoring best achieved by an indwelling arterial catheter. Importantly, this recommendation rests only on class III evidence, and some data suggest augmentation may not improve outcome and may increase complications.

Key point:
after acute spinal cord injury, target a mean arterial pressure of 85 to 90 mmHg for 7 days, recognizing this is a class III recommendation.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Spinal Cord Injury, pp. 723, 725, 730

Question 239

Traumatic Spinal Cord Injury · Airway management in cervical spinal cord injury · hard

A 24-year-old man is brought to the emergency department after a diving accident. Computed tomography shows a C4 burst fracture, and examination reveals no motor or sensory function below C4, including absent sacral sensation and rectal tone. Over 20 minutes he becomes increasingly agitated with paradoxical abdominal movement, a respiratory rate of 34 breaths/min, and oxygen saturation falling to 86% on a nonrebreather mask. He is in a rigid cervical collar. Fiberoptic equipment is not immediately at the bedside. Which of the following is the most appropriate approach to securing the airway?

- **A.** Noninvasive bilevel positive-pressure ventilation with reassessment in 1 hour
- **B.** Awake nasotracheal intubation with the cervical collar left in place
- **C.** Awake fiberoptic intubation once equipment is obtained from another unit
- **D.** Rapid sequence intubation with the rigid cervical collar left in place
- **E.** Rapid sequence intubation with the collar opened and manual in-line stabilization

Answer: E. Rapid sequence intubation with the collar opened and manual in-line stabilization

This is a complete injury at C4, within the C3 to C5 segments that give rise to the phrenic nerve, so diaphragmatic function is severely compromised and immediate ventilatory support is required; the chapter states that all patients with a complete cervical injury above C5 should be intubated as soon as possible. In an urgent or emergent situation the recommended approach is rapid sequence intubation with the anterior portion of the collar removed and manual in-line stabilization carefully maintained, taking extreme care not to hyperextend or rotate the neck. Option C is the preferred technique when time permits, because awake fiberoptic intubation is usually preferred electively, but delaying an emergent airway to retrieve equipment is unsafe, and in-line stabilization is explicitly the alternative when fiberoptic equipment is unavailable. Option A is wrong because intubation is preferred over noninvasive ventilation, since recovery of adequate ventilatory function after high cervical injury is uncertain and slow when it occurs. Option D is wrong because leaving the rigid collar in place markedly impairs laryngoscopy; the collar is opened while an assistant maintains in-line stabilization. Option B is not an appropriate emergent rescue in a hypoxemic, agitated patient.

Key point:
emergent intubation after high cervical spinal cord injury is rapid sequence intubation with manual in-line stabilization, with awake fiberoptic reserved for the nonemergent setting.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Spinal Cord Injury, pp. 722-723, 730

Question 240

Traumatic Spinal Cord Injury · Pharmacologic therapy and steroids · easy

A 47-year-old woman is admitted 4 hours after a fall from a ladder with an ASIA B injury at T5. She is hemodynamically supported with norepinephrine and is scheduled for decompression and fusion. The referring surgeon asks whether high-dose intravenous methylprednisolone should be given because the patient is still within the early treatment window described in the National Acute Spinal Cord Injury Study (NASCIS) trials. Which of the following is the most appropriate response?

- **A.** Withhold methylprednisolone, because steroids are not supported by evidence and are associated with harm
- **B.** Give methylprednisolone as a bolus followed by a 24-hour infusion, since she is within 8 hours of injury
- **C.** Give methylprednisolone as a bolus followed by a 48-hour infusion, since her injury is motor complete
- **D.** Give dexamethasone instead, because it has fewer infectious complications than methylprednisolone
- **E.** Give methylprednisolone only after surgical decompression is completed

> **Answer: A.** Withhold methylprednisolone, because steroids are not supported by evidence and are associated with harm
>
> The chapter states that intravenous steroids in the treatment of spinal cord injury are contraindicated and no longer supported by evidence. The NASCIS trials of the 1990s suggested that early continuous methylprednisolone improved functional outcome, but subsequent analyses found little to no statistical validity in those results and uncovered harmful effects, and no later randomized trial has replicated the findings. Reported harms include increased pneumonia, septic shock, and death. The American Association of Neurological Surgeons, the Neurocritical Care Society, and the Congress of Neurological Surgeons all recommend against steroid use in spinal cord injury, which makes options B and C incorrect regardless of the time window or completeness of injury. Option D is incorrect because no corticosteroid is recommended, and substituting an agent does not address the lack of benefit. Option E is incorrect because the timing relative to surgery does not create an indication that otherwise does not exist.
>
> **Key point:**
> do not give corticosteroids for acute traumatic spinal cord injury; the NASCIS results did not hold up and steroids increase pneumonia, septic shock, and death.
>
> Source: Practice of Neurocritical Care, 2nd ed., Traumatic Spinal Cord Injury, pp. 726

Question 241

Traumatic Spinal Cord Injury · Timing of surgical decompression · moderate

A 55-year-old man arrives 6 hours after a motor vehicle collision with a C5-C6 fracture-dislocation and jumped facets. His examination shows 1/5 strength in the upper extremities, 0/5 in the lower extremities, and preserved perianal sensation. Blood pressure is 88/50 mmHg with a heart rate of 52 beats/min. Computed tomography angiography shows no cerebrovascular injury. He is hemodynamically stabilized with norepinephrine. Which of the following is the most appropriate next step in management?

- **A.** Continue closed traction only and plan definitive fixation at 5 to 7 days once edema resolves
- **B.** Proceed to decompression and stabilization within 24 hours of injury
- **C.** Maintain a rigid cervical collar and obtain flexion and extension radiographs before any operation
- **D.** Delay operative decompression until the mean arterial pressure has been maintained above 85 mmHg for 72 hours
- **E.** Place a prophylactic inferior vena cava filter before considering operative decompression

Answer: B. Proceed to decompression and stabilization within 24 hours of injury

The 2017 consensus guideline recommends early surgery, defined as within 24 hours of injury, for adults with acute spinal cord injury requiring surgical treatment, and the Surgical Timing in Acute Spinal Cord Injury Study (STASCIS) showed that early decompression increases the odds of improving at least two grades on the ASIA scale by nearly threefold. An unstable dislocation with jumped facets should also be reduced as soon as possible, with radiographic guidance to avoid overdistraction if the ligaments are incompetent, and a detailed neurological examination should be performed before and after reduction. Options A and D are wrong because deliberate delay forfeits the benefit demonstrated for surgery within 24 hours; hemodynamic augmentation and operative decompression proceed together, not sequentially. Option C is wrong because dynamic flexion and extension films are not indicated in a patient with a known unstable injury and a neurological deficit, and a collar alone without reduction of the facets is unacceptable. Option E is wrong because routine prophylactic vena cava filtration is not a standard need in spinal cord injury and is reserved for high-risk patients or those with contraindications to pharmacologic prophylaxis.

Key point:
for acute spinal cord injury needing surgery, decompress and stabilize within 24 hours; STASCIS showed nearly a threefold increase in the odds of a two-grade ASIA improvement.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Spinal Cord Injury, pp. 725, 726, 730

Question 242

Traumatic Spinal Cord Injury · ASIA impairment scale and prognosis · easy

A 44-year-old man is admitted with a C6 spinal cord injury after a fall. On examination he has partial loss of movement and sensation below C6. Voluntary motor function is present below the neurologic level, and most of the key muscle groups below the level of injury have a muscle grade of 3 or better, allowing movement against gravity. Sacral sensation and voluntary anal contraction are intact. How should this injury be classified?

- **A.** C6 ASIA A
- **B.** C6 ASIA B
- **C.** C6 ASIA C
- **D.** C6 ASIA D
- **E.** C6 ASIA E

Answer: D. C6 ASIA D

The International Standards for Neurological Classification of Spinal Cord Injury define ASIA D as motor incomplete injury in which motor function is preserved below the neurologic level and at least 50% of key muscle functions below the level of injury have a muscle grade of 3 or greater. This patient has preserved voluntary motor function below C6 with most key muscles at grade 3 or better, which is ASIA D. ASIA C is the mirror image, with more than 50% of key muscles below the level graded less than 3, so option C is wrong. ASIA A requires no motor or sensory function in the sacral segments S4-S5, and this patient has intact sacral sensation and anal contraction, so option A is wrong. ASIA B is sensory incomplete, meaning sensory but not motor function is preserved below the neurologic level including the sacral segments, and this patient has motor function, so option B is wrong. ASIA E is normal function, which does not apply. Prognostically this matters: incomplete motor loss, ASIA C or D, occurs in about 40% of patients and 86% of those regain the ability to ambulate, whereas nearly 90% of ASIA A patients do not regain function.

Key point:
ASIA C versus D is decided by whether at least half the key muscles below the level of injury are grade 3 or better.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Spinal Cord Injury, pp. 724, 731

Question 243

Traumatic Spinal Cord Injury · Autonomic dysreflexia · hard

A 29-year-old man with a motor-complete spinal cord injury at C7 sustained 3 weeks earlier develops a pounding headache, flushing and sweating above the clavicles, and nasal congestion during a routine turn. Blood pressure rises from a baseline of 104/62 mmHg to 212/118 mmHg, and heart rate falls from 78 to 46 beats/min. He is afebrile. His indwelling urinary catheter has drained no urine for several hours, and the bladder is percussed to the umbilicus. Which of the following is the most appropriate initial step?

- **A.** Administer atropine 0.3 mg intravenously for the bradycardia
- **B.** Start a nicardipine infusion and titrate to a systolic pressure below 140 mmHg
- **C.** Obtain computed tomography of the head to evaluate for intracranial hemorrhage
- **D.** Administer intravenous methylprednisolone for presumed cord edema
- **E.** Relieve the bladder outflow obstruction by flushing or replacing the urinary catheter

Answer: E. Relieve the bladder outflow obstruction by flushing or replacing the urinary catheter

This is autonomic dysreflexia, which typically occurs in motor-complete injuries above T6, usually in the subacute to chronic phase but occasionally acutely. A noxious stimulus below the level of injury, most classically bladder distention or fecal impaction but also lower extremity pain, drives sympathetic outflow below the lesion, and the resulting vasoconstriction produces a hypertensive emergency with reflex bradycardia and flushing above the lesion. Treatment begins with avoidance and removal of the trigger, so relieving the obstructed catheter is the correct first action, and nursing measures such as digital stimulation, stool softeners, laxatives, and catheterization are specifically described as reducing occurrence. Option B is not the first step; intravenous antihypertensives such as nicardipine or nitroprusside, or ganglionic blockers, are used if blood pressure remains elevated after the trigger is addressed. Option A is wrong because atropine 0.3 mg intravenously is used for symptomatic reflex bradycardia provoked by vagal stimulation such as suctioning or turning, not for the bradycardia that is a baroreflex response to severe hypertension. Options C and D do not address the mechanism and delay definitive treatment.

Key point:
in autonomic dysreflexia above T6, find and remove the noxious trigger first, most often a distended bladder or impacted stool, before reaching for antihypertensives.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Spinal Cord Injury, pp. 726

Question 244

Traumatic Spinal Cord Injury · Venous thromboembolism prophylaxis · moderate

A 61-year-old woman is 36 hours out from posterior decompression and fusion for a T4 fracture with an ASIA A injury. She has no intracranial hemorrhage, no ongoing surgical bleeding, hemoglobin is stable at 10.8 g/dL, and platelets are 210,000/microliter. The neurosurgical team has cleared her for pharmacologic prophylaxis. Which of the following is the most appropriate venous thromboembolism prophylaxis strategy?

- **A.** Mechanical prophylaxis with pneumatic compression devices alone until she is mobilized
- **B.** Placement of a prophylactic inferior vena cava filter, because complete injuries carry an especially high risk
- **C.** Therapeutic anticoagulation with a direct oral anticoagulant started on postoperative day 7
- **D.** Low molecular weight heparin started now, combined with pneumatic compression devices
- **E.** Daily lower extremity duplex ultrasound surveillance with treatment only if thrombus is detected

Answer: D. Low molecular weight heparin started now, combined with pneumatic compression devices

The risk of venous thromboembolism after spinal cord injury is extremely high, described as essentially omnipresent without prophylaxis, and pulmonary embolism remains one of the common causes of death in this population. Guidelines recommend initiating prophylaxis within 72 hours of injury in the absence of contraindications, using low molecular weight heparin or low-dose unfractionated heparin combined with mechanical prophylaxis such as pneumatic compression stockings, which is exactly option D and fits this patient at 36 hours with no bleeding contraindication. Option A is inadequate because mechanical prophylaxis alone is not the recommended regimen when pharmacologic prophylaxis can be given. Option B is wrong because routine prophylactic vena cava filtration should not be considered a standard need in spinal cord injury and is reserved for patients at high risk or with contraindications to standard prophylaxis. Option C is wrong on both dose and timing, since prophylactic rather than therapeutic dosing is indicated and waiting until day 7 exceeds the 72-hour window. Option E replaces prophylaxis with surveillance and leaves the patient unprotected during the highest-risk period.

Key point:
start low molecular weight heparin or low-dose unfractionated heparin plus mechanical prophylaxis within 72 hours of spinal cord injury; inferior vena cava filters are not routine.

Source: Practice of Neurocritical Care, 2nd ed., Traumatic Spinal Cord Injury, pp. 726

Part III: Administration

Question 245

Administrative and Managerial Principles · ICU staffing models and outcomes · moderate

A 68-year-old woman with a large left basal ganglia intracerebral hemorrhage is admitted to a community hospital ICU where any admitting practitioner may direct critical care and an intensivist consultation is optional. Over 3 days she requires intubation, an external ventricular drain, and blood pressure control, and her ICU stay is prolonged by a ventilator-associated pneumonia. Hospital leadership asks the newly hired neurointensivist to redesign the unit's physician coverage to improve outcomes. Which of the following changes is best supported by the evidence described for ICU staffing?

- **A.** Convert to an open unit with 24-hour in-house hospitalist coverage and optional intensivist consultation
- **B.** Retain the current model but add a daily telemedicine review by an off-site intensivist as the sole change
- **C.** Convert to high-intensity staffing, in which patients are managed primarily by an intensivist or an intensivist consultation is mandatory
- **D.** Retain low-intensity staffing but increase the nurse-to-patient ratio to two nurses for each patient
- **E.** Convert to a mixed medical-surgical generalist ICU so that all critically ill patients are cared for in one unit

Answer: C. Convert to high-intensity staffing, in which patients are managed primarily by an intensivist or an intensivist consultation is mandatory

The chapter distinguishes high-intensity staffing, in which patients are either managed primarily by an intensivist or an intensivist consultation is mandatory, from low-intensity staffing, in which there is no intensivist or the consultation is optional; the unit described is low intensity. Studies show that high-intensity staffing compared with low-intensity staffing leads to improved survival and shorter length of stay, and intensivist-directed care also improves resource use and patient and staff satisfaction. Intensivist-staffed units have lower resource utilization because these physicians reduce inappropriate admissions, prevent complications that prolong stay, and identify discharge opportunities. Options A and B leave the intensivist consultation optional, which is the definition of low-intensity staffing, so neither corrects the deficiency. Option D addresses nursing ratios, which do affect outcomes, but the deficiency described in this unit is physician staffing intensity, and the chapter reserves 1:1 nursing for the highest-acuity patients rather than making it a unit-wide standard. Option E is wrong because specialist units, including neurocritical care units, have positive effects on patient outcomes, and neurocritical care in units is associated with improved outcome and decreased length of stay for intracerebral hemorrhage, aneurysmal subarachnoid hemorrhage, traumatic brain injury, and ischemic stroke.

Key point:
high-intensity staffing, meaning intensivist-managed care or a mandatory intensivist consult, improves survival and shortens length of stay compared with low-intensity staffing.

Source: Practice of Neurocritical Care, 2nd ed., Administrative and Managerial Principles, pp. 739-740

Question 246

Administrative and Managerial Principles · EMTALA and interhospital transfer · hard

A 57-year-old man is brought to a rural critical access hospital emergency department with a Glasgow Coma Scale score of 8, blood pressure 198/104 mmHg, and a noncontrast head computed tomography scan showing a 60 mL right putaminal hemorrhage with intraventricular extension. The hospital participates in Medicare, has no neurosurgical service, and has no ICU capable of external ventricular drain management. He is intubated and started on a nicardipine infusion, but he cannot be fully stabilized locally. A comprehensive stroke center 90 minutes away has an available neurocritical care bed. Under the Emergency Medical Treatment and Active Labor Act, which of the following is required of the transferring physician?

- **A.** Obtain written consent from the patient's surrogate before the receiving facility may be contacted
- **B.** Delay transfer until the emergency medical condition has been fully stabilized, regardless of local capability
- **C.** Send the patient with any available transport, since qualified personnel and equipment are the receiving facility's responsibility
- **D.** Withhold further treatment once transfer is arranged so that care is not duplicated at the receiving hospital
- **E.** Certify that the benefits of transfer outweigh the risks and document the name of the accepting individual and facility

Answer: E. Certify that the benefits of transfer outweigh the risks and document the name of the accepting individual and facility

Under the Emergency Medical Treatment and Active Labor Act of 1986, a patient with an emergency medical condition should ideally be stabilized before transfer, but when that is not possible the transferring physician must certify that the benefits of transfer outweigh the risks, and when transferring must document the name of the accepting individual and the accepting facility. Option B is therefore wrong, because the law explicitly accommodates transfer of an unstabilized patient when local capability is exceeded and the risk-benefit certification is made. Option A is wrong because although patients or legally authorized surrogates should be informed of the risks and benefits, surrogate consent is not the trigger for transfer, and in fact a patient or surrogate may request transfer whether or not the condition has been stabilized. Option C is wrong because qualified transport personnel with all appropriate and necessary equipment must accompany the patient during transfer. Option D is wrong because the transferring hospital is obligated to provide all medical treatment within its capacity in order to minimize risk to the patient. The receiving facility, for its part, must accept the transfer as long as it has the capacity and space, the implication being that it has a service or capability the transferring facility lacks.

Key point:
to transfer an unstabilized emergency patient, certify that benefits outweigh risks, name the accepting physician and hospital, send the pertinent records and imaging, and use qualified transport.

Source: Practice of Neurocritical Care, 2nd ed., Administrative and Managerial Principles, pp. 743, 750

Question 247

Administrative and Managerial Principles · Neurocritical care unit design and bed planning · easy

A neurointensivist is recruited to build a new neurocritical care unit at a 450-bed public hospital that is both a Level 1 trauma center and a comprehensive stroke center. Neurotrauma patients will continue to be admitted to the existing trauma unit. As a first estimate for the initial business plan, how many neurocritical care beds should be proposed?

- **A.** 2 to 4 beds
- **B.** 8 to 12 beds
- **C.** 18 to 24 beds
- **D.** 30 to 36 beds
- **E.** 45 to 55 beds

Answer: B. 8 to 12 beds

Few published studies exist on which to base neurocritical care bed calculations, but the chapter states that 8 to 12 neurocritical care beds per 400 to 500 total hospital beds is a reasonable first estimate for a comprehensive stroke center and trauma center, which fits this 450-bed hospital. That number must then be modified for local factors; in particular, if neurotrauma patients were to be admitted to the neurocritical care unit rather than the trauma unit, the bed number would need to be higher, so with neurotrauma remaining in the trauma unit the base estimate stands. Options A and C to E fall outside the cited range and would either underserve the population or commit resources without a supporting needs analysis. The bed number is only one element of unit design, which also includes physical location near imaging suites, the emergency department, interventional suites, and operating rooms, room design that includes natural light and noise reduction to lessen delirium, and equipment such as electroencephalography, electronic ICU, and multimodality neuromonitoring systems. In the United States 8% to 12% of hospital beds are devoted to critical care compared with 3% in other developed countries, so bed planning should be driven by demonstrated need rather than by demand alone.

Key point:
plan roughly 8 to 12 neurocritical care beds per 400 to 500 hospital beds at a comprehensive stroke and trauma center, adjusting upward if neurotrauma patients are admitted to the unit.

Source: Practice of Neurocritical Care, 2nd ed., Administrative and Managerial Principles, pp. 739-740

Quality Measurement in Neurocritical Care

Question 248

Quality Measurement in Neurocritical Care · Donabedian classification of quality measures · easy

A neurocritical care unit launches a quality improvement project to shorten ICU length of stay for patients with aneurysmal subarachnoid hemorrhage by standardizing transfer to the step-down unit once vasospasm surveillance is complete. To make sure the project does not simply shift harm downstream, the team also tracks the rate of unplanned readmission to the ICU within 48 hours of transfer. In the classification of quality measures, the readmission rate as used here is best described as which type of measure?

- **A.** Balance measure
- **B.** Structure measure
- **C.** Process measure
- **D.** Outcome measure
- **E.** Patient-reported experience measure

Answer: A. Balance measure

A balance measure assesses the unintended consequences of an improvement intervention, and the textbook example is exactly this one, monitoring readmission rate during a project to reduce length of stay. A structure measure captures infrastructure such as equipment, personnel, or policies, with examples including an external ventricular drain insertion bundle, a status epilepticus order set, and a designated stroke unit, so option B is wrong. A process measure captures how well a care process is followed, such as the use of nimodipine in subarachnoid hemorrhage or standardized operating room to ICU handoffs, so option C is wrong. An outcome measure captures the results of care, such as the modified Rankin Scale at 6 months or the Thrombolysis in Cerebral Infarction score after thrombectomy, and although readmission is an outcome in the abstract, here it is deliberately tracked as a counterbalance to the length of stay intervention. Option E is wrong because patient-reported measures capture the patient's or caregiver's perceived health and experience, such as Hospital Consumer Assessment of Healthcare Providers and Systems scores. The Donabedian model classifies measures as structure, process, and outcome, with structure and process driving outcome, and outcome measures are generally considered the gold standard.

Key point:
a metric tracked specifically to detect harm caused by a quality improvement intervention is a balance measure.

Source: Practice of Neurocritical Care, 2nd ed., Quality Measurement in Neurocritical Care, pp. 755-756, 762-763

Question 249

Quality Measurement in Neurocritical Care · CMS value-based payment programs · moderate

A 550-bed academic hospital reviews its neurocritical care unit data and finds elevated rates of central line-associated bloodstream infection and catheter-associated urinary tract infection, placing the hospital in the lowest performing quartile nationally for hospital-acquired conditions. The quality officer is asked to quantify the financial exposure from the relevant Centers for Medicare and Medicaid Services program. Which of the following best describes the penalty this hospital faces?

- **A.** A 2% withholding of inpatient payments that may be partially or fully recouped for good performance
- **B.** A maximum 3% penalty on inpatient payments applied across six clinical cohorts
- **C.** A 1% reduction in inpatient payments under the Hospital-Acquired Condition Reduction Program
- **D.** Loss of eligibility for the American College of Surgeons Trauma Quality Improvement Program registry
- **E.** A 6% reduction in inpatient payments applied to all Medicare discharges for the reporting year

Answer: C. A 1% reduction in inpatient payments under the Hospital-Acquired Condition Reduction Program

The Hospital-Acquired Condition Reduction Program applies a 1% payment reduction on inpatient payments to the 25% of hospitals scoring in the lowest performing quartile for hospital-acquired conditions, which is precisely this hospital's situation; central line-associated bloodstream infection and catheter-associated urinary tract infection are among the Patient Safety Indicators, a set of 26 evidence-based indicators developed by the Agency for Healthcare Research and Quality. Option A describes the Hospital Value-Based Purchasing program, which withholds 2% of inpatient revenue in a revenue-neutral fashion across four equally weighted domains of clinical outcomes, patient and community engagement, safety, and efficiency and cost. Option B describes the Hospital Readmission Reduction Program, which applies a maximum 3% penalty and penalizes half of hospitals for each of six clinical cohorts with higher than median readmissions. Option E confuses the aggregate exposure with a single program; up to 6% of Medicare reimbursement is at risk across all three programs combined, not from one. Option D is wrong because the Trauma Quality Improvement Program is a voluntary American College of Surgeons registry, not a Centers for Medicare and Medicaid Services pay-for-performance program.

Key point:
the Hospital-Acquired Condition Reduction Program cuts inpatient payments by 1% for the worst-performing quartile, Value-Based Purchasing puts 2% at stake, and Readmission Reduction penalizes up to 3%, for a combined exposure of up to 6%.

Source: Practice of Neurocritical Care, 2nd ed., Quality Measurement in Neurocritical Care, pp. 757, 763

Question 250

Quality Measurement in Neurocritical Care · Risk adjustment and the mortality index · hard

Two comprehensive stroke centers care for similar populations of patients with high-grade aneurysmal subarachnoid hemorrhage, and chart review of matched cases shows comparable adherence to nimodipine administration, vasospasm screening, and external ventricular drain insertion bundles. Nevertheless, Hospital B reports a substantially higher mortality index than Hospital A on publicly reported rankings. Review shows that Hospital A consistently documents comorbidities and complications present on admission such as shock, coma, brain compression, and cerebral edema, whereas Hospital B does not. Which of the following best explains Hospital B's worse mortality index?

- **A.** Hospital B's observed mortality is genuinely higher because of unmeasured differences in process of care
- **B.** Hospital B's patients have a higher case mix complexity that the mortality index is designed to ignore
- **C.** Hospital B is penalized because the mortality index counts only deaths occurring within 30 days of admission
- **D.** Hospital B's index is inflated because publicly reported ranking systems weight reputation more heavily than mortality
- **E.** Hospital B's expected mortality is underestimated because incomplete coding weakens risk adjustment

Answer: E. Hospital B's expected mortality is underestimated because incomplete coding weakens risk adjustment

The mortality index is defined as observed mortalities divided by a calculated risk-adjusted estimate of expected mortalities, so anything that lowers the estimate of expected deaths raises the index even when care is identical. When Hospital B fails to document and code major comorbidities and complications present on admission, its patients appear less sick than they are, expected mortality falls, and the mortality index rises, which is the exact scenario the chapter describes. Option A is contradicted by the stem, which establishes comparable adherence to process measures. Option B misstates the purpose of risk adjustment, which exists precisely to account for case mix complexity rather than ignore it. Option C invents a time window that is not part of the mortality index definition. Option D is wrong because although publicly reported ranking systems such as Leapfrog, the Centers for Medicare and Medicaid Services Star Rating, and US News and World Report differ in methodology and weighting, that does not explain a difference in the mortality index itself. Inaccurate risk adjustment makes it difficult to separate true performance signal from noise and can harm an institution's financial solvency and reputation, and there is legitimate concern that hospitals caring for more uninsured and underinsured patients face greater pay-for-performance penalties that could perpetuate disparities.

Key point:
the mortality index is observed divided by expected deaths, so incomplete documentation of admission comorbidities lowers expected mortality and falsely worsens reported quality.

Source: Practice of Neurocritical Care, 2nd ed., Quality Measurement in Neurocritical Care, pp. 757-758